Lie Groups and Their Representation Theory

Global structure and compact representations

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Studying this course

This is a self-reading course in two parts of fourteen lectures. The division into lectures is for convenience, so lectures differ in length. Study at your own pace. A lecture may need several passes. The exercise sheet is separate. Allow another two to four hours for its core problems.

The prerequisites are *Linear Algebra for Physics*, *Finite Groups and Their Representations*, *Lie Algebras and Their Representations*, and basic category language. We use their algebraic results throughout. Ordinary multivariable calculus and elementary ordinary differential equations are also assumed.

The course in one page

This course asks what a group determines that its Lie algebra does not. For example, $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ have isomorphic Lie algebras. The module V_1, of spin one half, is a representation of $\mathrm{SU}(2)$ and not of $\mathrm{SO}(3)$. The course also asks how the representations of a group describe the functions on it.

Part I treats compact groups. It can be read alone. Lectures 1 to 4 build Lie groups, the exponential map, quotients and covering groups. Lecture 4 decides which finite-dimensional representations of a Lie algebra come from a given group. Lectures 5 to 9 treat the structure of a compact group: the invariant integral, maximal tori and their lattices, and highest weights. Lecture 8 lists the irreducible representations of a compact connected group. Lectures 10 to 14 treat functions on a compact group. Lecture 10 proves the Peter–Weyl theorem, which contains Fourier series. Lectures 11 and 12 apply it to convolution and to functions on homogeneous spaces. Lecture 13 treats spin groups and projective representations. Lecture 14 recovers the group from its representations.

Part II removes compactness. Lectures 15 to 17 rebuild the tools of Part I: generators of unitary representations, Haar measure of infinite total mass, and induction. Lectures 18 and 19 treat the Heisenberg group and coadjoint orbits. Lectures 20 to 23 treat $\mathrm{SL}(2,\mathbb{R})$ and end with the list of its irreducible unitary representations. Lectures 24 and 25 treat the Lorentz and Poincaré groups. Lectures 26 and 27 return to compact groups and realise their representations on line bundles and on coadjoint orbits. Lecture 28 compares the function algebra of a group with its enveloping algebra, and it prepares for a later course on quantum groups.

Reading a lecture

Start with the objectives and outcomes. Work through the displayed calculations before reading the next paragraph. Then answer the self-checks without looking at their answers.

Attempt the core exercises before opening the worked solutions. Compare the reasoning as well as the final result. Some sheets include an optional extension. It is marked explicitly and is not needed for the next lecture.

The notes introduce the geometry and analysis needed at each stage. They do not replace courses on differential geometry, topology or functional analysis. Long existence theorems are stated with their hypotheses and a source. The main calculations are carried out here.

A cited theorem is marked as cited, by the word *cited* or a source in its title, or by a sentence next to it. Where a proof is only sketched, the text names the omitted step. The mark [C: source] gives the source of a cited statement, and the mark [I] means that a result comes from a prerequisite course. A cited theorem is an input to the argument that follows it. Longer checks that a source's hypotheses apply are gathered under *Sources and hypotheses* at the end of a lecture.

Conventions

Lie groups are finite-dimensional real Lie groups. Their manifolds are Hausdorff and second countable. Compactness and connectedness are stated when they are needed. Representations are continuous and act on complex vector spaces. A Hilbert-space unitary representation is strongly continuous. Inner products are antilinear in the first slot.

A real Lie algebra and its complexification are distinct objects. In particular, $\mathfrak{k} = \mathrm{Lie}(K)$ is real, while highest-weight theory applies to $\mathfrak{k}_{\mathbb{C}}$. For a torus with Lie algebra $\mathfrak{t}$, our period lattice is

$$\Lambda = \ker(\exp\colon \mathfrak{t} \to T).$$

We write a character as $\chi(\exp H) = e^{i\ell(H)}$. Thus its period condition is $\ell(\Lambda) \subseteq 2\pi\mathbb{Z}$.

The module V_m of $\mathrm{SU}(2)$ has dimension $m + 1$ and spin $m/2$. The operator of a group representation is $\Pi(g)$, and its differential is $d\Pi(X)$. When a basis is chosen, D denotes an operator matrix. Every change of convention is explained where it is used.

Books used in the notes

The reading lists give a route into the books. The following abbreviations are used in some lectures.

H Brian C. Hall, *Lie Groups, Lie Algebras, and Representations*, second edition, Springer, 2015. This is the main route for matrix groups, compact groups and integration of representations.

BD Theodor Bröcker and Tammo tom Dieck, *Representations of Compact Lie Groups*, Springer, 1985. This supplies the general compact-group theory and reconstruction.

S Mark R. Sepanski, *Compact Lie Groups*, Springer, 2007. This is a second route through compact structure and Peter–Weyl theory.

Knapp Beyond
Anthony W. Knapp, *Lie Groups Beyond an Introduction*, second edition. This supplies general Lie-group structure.

Knapp Representation Theory
Anthony W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001 edition. This supplies noncompact examples, smooth vectors and induced representations.

Kirillov Jr.
Alexander Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Cambridge University Press, 2008.

Kirillov Elements
A. A. Kirillov, *Elements of the Theory of Representations*, Springer, 1976.

Kirillov Orbit Method
A. A. Kirillov, *Lectures on the Orbit Method*, American Mathematical Society, 2004.

Warner
Frank W. Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Springer edition, 1983.

Further primary sources are named in the relevant lectures. Knapp's two books serve different purposes; their titles are distinguished throughout.

Part I

Global structure and compact representations

Lecture 1

Lie groups and their tangent algebras

Objectives. *A Lie group has both a group structure and a smooth geometry. We describe that geometry near the identity and recover its Lie algebra. The group* $\mathrm{SU}(2)$ *will give an example in which every step can be calculated.*

By the end of this lecture you will be able to:

- Identify the real dimension of a matrix Lie group.
- Find tangent spaces by differentiating a defining condition and exhibiting curves.
- Recover a matrix Lie bracket from left translations.
- Distinguish a real Lie algebra from its complexification.

The Lie algebras in the earlier course were complex. The tangent algebra of a real Lie group is a real Lie algebra. We will always state the field of scalars.

1.1 Smooth coordinates

A sphere cannot be described by one set of Euclidean coordinates without omitting a point or introducing a singularity. It can be described by several overlapping coordinate systems. A smooth manifold formalises this local description.

A topology specifies which subsets are open. A *neighbourhood* of a point is a set containing an open set around that point. A topological space is *Hausdorff* if distinct points have disjoint open neighbourhoods. It is *second countable* if its topology has a countable *basis*, namely a countable collection of open sets such that every open set is a union of some of them.

A map between topological spaces is *continuous* if the inverse image of every open set is open. A *homeomorphism* is a continuous bijection whose inverse is continuous.

Definition 1.1 (Smooth manifold)**.** A smooth real manifold of dimension d is a topological space M covered by coordinate charts

$$\varphi : U \longrightarrow V \subseteq \mathbb{R}^d.$$

Each chart has open domain $U \subseteq M$ and is a homeomorphism onto an open set V. Where the domains of two charts φ and ψ overlap, the coordinate change $\psi \circ \varphi^{-1}$ is infinitely differentiable. We require M to be Hausdorff and second countable.

A subset of a topological space has the *subspace topology*: its open sets are its intersections with open sets of the space. For subsets of Euclidean space we use this topology. For example, an open arc is open in the circle.

Subsets of Euclidean space are Hausdorff and second countable, so all the matrix groups in this lecture satisfy both conditions. Lecture 2 uses the Hausdorff condition for uniqueness of solutions of differential equations on a manifold. Lecture 3 uses second countability for homogeneous spaces. A smooth map between manifolds is a continuous map whose coordinate expressions are infinitely differentiable on their open domains. A *diffeomorphism* is a smooth bijection whose inverse is smooth.

Example 1.2 (A coordinate on the circle)**.** The unit circle is

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

On the open right semicircle, use y as the coordinate. Its inverse chart is

$$y \longmapsto \left(\sqrt{1 - y^2}, y\right), \qquad -1 < y < 1.$$

The other three semicircles have similar charts. On the upper semicircle use x. On the open quarter circle where both charts apply, y runs over $(0, 1)$, and the coordinate change and its inverse are

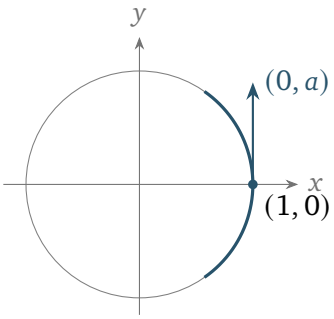
$$x = \sqrt{1 - y^2}, \qquad y = \sqrt{1 - x^2}.$$

These are smooth because $1 - y^2 > 0$ and $1 - x^2 > 0$ there. The other overlaps are similar. Thus the circle is a one-dimensional real manifold, although it lies in a two-dimensional ambient space.

The coordinate y can also describe a velocity. For a real number a, the curve on the circle

$$\gamma_a(t) = \left(\sqrt{1 - a^2 t^2}, at\right)$$

is defined near zero. It starts at $(1, 0)$ and has ambient derivative $(0, a)$. The derivative of its y-coordinate is a. Thus a one-dimensional tangent space can appear as a line inside a larger ambient space.



The coloured arc lies in the domain of one chart, in which y is the local coordinate. The tangent arrow is the velocity at $(1, 0)$ for positive a. Its drawn length is schematic.

1.2 Smooth groups

Product charts $\varphi \times \psi$ on $U \times V$ give $M \times N$ a smooth manifold structure of dimension $\dim M + \dim N$. A finite product of Hausdorff, second-countable spaces has both properties.

Definition 1.3 (Lie group)**.** A real Lie group is a group G that is also a smooth real manifold, with smooth multiplication and inversion:

$$G \times G \longrightarrow G, \quad (g,h) \longmapsto gh, \qquad G \longrightarrow G, \quad g \longmapsto g^{-1}.$$

A Lie group homomorphism is a smooth group homomorphism.

The additive group $\mathbb{R}^d$ is a Lie group. Any finite group with the discrete topology is also a Lie group, since each one-point subset is the domain of a chart modelled on $\mathbb{R}^0$. The tangent space of a finite group therefore has dimension zero. Hence a Lie algebra cannot distinguish finite groups.

Example 1.4 (General linear groups)**.** The group of invertible real matrices is

$$\mathrm{GL}_n(\mathbb{R}) = \{g \in M_{n\times n}(\mathbb{R}) : \det g \neq 0\}.$$

The determinant is continuous, so $\mathrm{GL}_n(\mathbb{R})$ is open in $\mathbb{R}^{n^2}$. Hence its dimension is n^2. The entries of a matrix product are polynomials in the entries of the factors, so matrix multiplication is smooth. Inversion is smooth because

$$g^{-1} = \frac{\operatorname{adj}(g)}{\det g}.$$

The same argument applies to $\mathrm{GL}_n(\mathbb{C})$. The real and imaginary parts of the entries give its coordinates, so its dimension as a real manifold is $2n^2$.

A *complex Lie group* has complex charts and holomorphic group operations. The group $\mathrm{GL}_n(\mathbb{C})$ has this extra structure. Its complex dimension is n^2. A group of complex matrices need not be a complex Lie group. Its defining conditions may involve complex conjugation.

For example, the unitary and special unitary groups are

$$\mathrm{U}(n) = \{U \in M_{n\times n}(\mathbb{C}) : U^*U = I\}, \qquad \mathrm{SU}(n) = \{U \in \mathrm{U}(n) : \det U = 1\}.$$

Here U^* denotes conjugate transpose. Their real Lie-group structures are a standard classical-group result, cited from Bröcker–tom Dieck, Chapter I, Section 1. We verify the manifold structure of $\mathrm{SU}(2)$ directly; Exercise Sheet 1 treats $\mathrm{U}(n)$.

A manifold is *connected* if it cannot be written as the union of two nonempty disjoint open subsets. For example, the orthogonal group is

$$\mathrm{O}(n) = \{g \in M_{n\times n}(\mathbb{R}) : g^{\mathsf{T}}g = I\}, \qquad n \geq 1.$$

Its determinant takes values in $\{1,-1\}$. Both values occur. The identity has determinant one, and $\operatorname{diag}(-1,1,\ldots,1)$ has determinant minus one. The inverse images of these two values are disjoint nonempty open subsets that cover the group. Thus $\mathrm{O}(n)$ is disconnected. Its subgroup with determinant one is $\mathrm{SO}(n)$.

1.3 Tangent vectors at the identity

Definition 1.5 (Tangent vector). Let $\varphi : U \to V$ be a chart of M with $p \in U$. A *smooth curve through* p is a smooth map γ from an open interval containing 0 to M with $\gamma(0) = p$. Two such curves γ and δ are *equivalent* when $(\varphi \circ \gamma)'(0) = (\varphi \circ \delta)'(0)$. A *tangent vector* at p is an equivalence class $[\gamma]$. The set of tangent vectors at p is denoted T_pM. The map $[\gamma] \mapsto (\varphi \circ \gamma)'(0)$ identifies T_pM with $\mathbb{R}^d$. Vector addition and real scalar multiplication on T_pM are transferred through this identification.

The class $[\gamma]$ is the *velocity* of γ, also written $\gamma'(0)$. The chain rule makes the relation between curves independent of the chart. The derivative of a coordinate change is linear, so the vector-space structure on T_pM is also independent of the chart.

A smooth map $f : M \to N$ sends tangent vectors to tangent vectors. If v is represented by γ, the image of v is the velocity of $f \circ \gamma$. This defines the *differential*

$$df_p : T_pM \longrightarrow T_{f(p)}N, \qquad df_p\big(\gamma'(0)\big) = (f \circ \gamma)'(0).$$

In charts this is the usual derivative, so it is real linear. The chain rule gives $d(f \circ h)_p = df_{h(p)} \circ dh_p$.

Our groups lie inside matrix spaces, and we calculate with matrix entries. An *embedded submanifold* is the notion that justifies this. It has the subspace topology. Near each of its points, a smooth change of ambient coordinates turns it into a coordinate subspace. For example, near $(1, 0)$ the coordinates $(r-1, \theta)$ turn the circle into the line $r-1 = 0$. We import the standard properties of an embedded submanifold. Its inclusion is smooth with injective differential. Products of embedded submanifolds are embedded in the corresponding product. Thus smooth ambient maps restrict smoothly to these submanifolds. A smooth ambient map whose image lies in an embedded submanifold is smooth as a map into that submanifold. The differential of the inclusion identifies its tangent vectors with ambient velocities.

We also use one result from multivariable calculus. Let $U \subseteq \mathbb{R}^N$ be open, and suppose a smooth map $F : U \to \mathbb{R}^r$ has surjective differential at every point of $F^{-1}(c)$. The *regular level-set theorem* says that $F^{-1}(c)$ is an embedded submanifold of U of dimension $N - r$, with the subspace topology. The tangent space of $F^{-1}(c)$ at a point p is $\ker dF_p$. The regular level-set theorem and the embedded-submanifold properties are cited from Warner, Chapter 1, sections “Submanifolds, Diffeomorphisms, and the Inverse Function Theorem” and “Implicit Function Theorems” (Theorem 1.38). For example, the circle is $F^{-1}(1)$ with $F(x, y) = x^2 + y^2$. Here $dF_{(x,y)} = (2x, 2y)$ is nonzero on the circle, so the dimension is $2 - 1 = 1$.

The kernel clause of the regular level-set theorem now follows from the chain rule and rank–nullity. Put $P = F^{-1}(c)$ and let $i : P \to U$ be the inclusion. For $p \in P$, since $F \circ i$ is constant, $dF_p \circ di_p = 0$. Thus $di_p(T_pP) \subseteq \ker dF_p$. The inclusion is an embedding, so di_p is injective. Both spaces have dimension $N - r$, because dF_p is surjective. Hence they are equal.

For an embedded manifold in a matrix space, we compute velocities by differentiating the matrix entries of curves. The identity element of a group is denoted e, or I for a matrix group. For a Lie group G, the vector space of interest is

$$\mathfrak{g} = T_eG.$$

For $\mathrm{GL}_n(\mathbb{R})$, every matrix X is a tangent vector at I. Indeed, the curve $\gamma(t) = I + tX$ is invertible for sufficiently small t, and $\gamma'(0) = X$.

Example 1.6 (Tangent vectors to the orthogonal group)**.** Let $g(t)$ be a smooth curve in $\mathrm{O}(n)$ with $g(0) = I$. Put $X = g'(0)$. Differentiate the defining identity at zero by the product rule:

$$g(t)^{\mathsf{T}} g(t) = I \quad \Longrightarrow \quad X^{\mathsf{T}} + X = 0.$$

Thus every tangent vector at I is skew-symmetric. Differentiating a condition gives a necessary condition. We must also show that every such matrix occurs.

We could use the exponential series, which Lecture 2 develops. The Cayley curve needs no convergence argument. For a skew-symmetric matrix X, consider the Cayley curve

$$g_X(t) = \left(I + \frac{tX}{2}\right)\left(I - \frac{tX}{2}\right)^{-1}.$$

It is defined near zero. Set $A = I + tX/2$ and $B = I - tX/2$. Then $A^{\mathsf{T}} = B$, $B^{\mathsf{T}} = A$, and $AB = BA$. Therefore

$$g_X(t)^{\mathsf{T}} g_X(t) = A^{-1} B A B^{-1} = I.$$

The determinant is continuous, takes values in $\{1, -1\}$, and equals one at zero. It remains one near zero. Hence this curve lies in $\mathrm{SO}(n)$. The product rule gives

$$g_X'(0) = \frac{X}{2} + \frac{X}{2} = X.$$

Consequently both orthogonal groups have the same tangent space at I:

$$T_I \,\mathrm{SO}(n) = T_I \,\mathrm{O}(n) = \{X : X^{\mathsf{T}} = -X\}.$$

The example used that $\mathrm{O}(n)$ is a smooth manifold. To justify this assertion, apply the regular level-set theorem to $F(g) = g^{\mathsf{T}} g$, with target the real symmetric matrices. At an orthogonal matrix g,

$$dF_g(H) = H^{\mathsf{T}} g + g^{\mathsf{T}} H.$$

For any symmetric matrix S, choose $H = gS/2$. Since S is symmetric and g is orthogonal, $dF_g(H) = S$. Hence the surjectivity hypothesis holds. Subtracting the dimension of the space of real symmetric matrices from the ambient dimension gives

$$\dim_{\mathbb{R}} \mathrm{O}(n) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

The kernel clause gives the tangent space again: $dF_I(H) = H^{\mathsf{T}} + H$, so $T_I \,\mathrm{O}(n)$ consists of the skew-symmetric matrices. The Cayley curve showed the same thing by hand. The subgroup $\mathrm{SO}(n)$ is open in $\mathrm{O}(n)$, so it has the same local dimension.

Multiplication and inversion by transposition are restrictions of smooth matrix maps. By the embedded-submanifold properties, both orthogonal groups are Lie groups. Equal tangent spaces do not imply equal groups.

1.4 Translations and the bracket

A tangent space at one point is only a vector space, and it has no product. Vector fields do have a commutator, because they act on functions by differentiation. Left translation carries a tangent vector at e to a vector field on G. We therefore obtain a bracket on the tangent space. Lecture 2 shows that this bracket measures how far small group elements fail to commute.

Left multiplication by $g \in G$ is the smooth map

$$L_g : G \longrightarrow G, \qquad L_g(h) = gh.$$

Since L_g has inverse $L_{g^{-1}}$, the differential of L_g at e identifies tangent spaces:

$$(dL_g)_e : T_eG \longrightarrow T_gG.$$

For a matrix group this differential sends X to gX. For example, in $\mathrm{SO}(2)$ take the rotation and the tangent matrix

$$g = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Left translation gives the tangent vector at g:

$$gX = \begin{pmatrix} -\sin\theta & -\cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix}.$$

This is the derivative of the rotation matrix with angle $\theta + t$ at $t = 0$.

A *vector field* assigns a tangent vector to each point. It is smooth if its components in each coordinate chart are smooth functions. The field determined by $X \in T_eG$ is

$$X^L(g) = (dL_g)_e X.$$

A vector field V on G is *left-invariant* if

$$(dL_h)_g\big(V(g)\big) = V(hg) \qquad \text{for all } g, h \in G.$$

The field X^L is left-invariant. Indeed, $L_h \circ L_g = L_{hg}$, so the chain rule gives $(dL_h)_g\big((dL_g)_e X\big) = (dL_{hg})_e X$.

Smooth vector fields act on smooth functions by directional differentiation. For a smooth vector field V and a smooth function f,

$$(Vf)(p) = df_p(V(p)).$$

Their commutator as differentiation operators is

$$[V, W]f = V(Wf) - W(Vf).$$

Theorem 1.7 (The tangent Lie algebra; cited). *The fields X^L obtained by left translation are smooth. The commutator of two smooth vector fields is again a vector field. The commutator of left-invariant fields is left-invariant. The value of this commutator at the identity defines a real Lie bracket on T_eG. For smooth vector fields V and W, in coordinates,*

$$[V, W](p) = DW_p(V(p)) - DV_p(W(p)).$$

For an embedded submanifold, this formula may be evaluated using smooth ambient extensions of tangent fields. The result is tangent to the submanifold.

The construction of the bracket from left-invariant fields is cited from Bröcker–tom Dieck, Chapter I, Section 2. The coordinate formula and the statement for embedded submanifolds are cited from Warner, Chapter 1, sections “Vector Fields” and “Submanifolds, Diffeomorphisms, and the Inverse Function Theorem”. The Jacobi identity is the commutator identity for the corresponding differentiation operators.

For an embedded matrix group, use the ambient extensions $X^L(g) = gX$ and $Y^L(g) = gY$. Since $Y^L(h) = hY$ is linear in h, its derivative in direction H is HY. Thus ambient calculus gives

$$D(Y^L)_g(gX) = gXY,$$
$$D(X^L)_g(gY) = gYX.$$

By the coordinate formula in the cited theorem, the vector-field bracket is the first of these derivatives minus the second. At the identity, this gives

$$[X, Y] = XY - YX.$$

This is the bracket studied in the earlier course. The cited submanifold compatibility shows that this ambient calculation is the intrinsic bracket on G. Hence $XY - YX$ belongs to T_IG. For skew-symmetric matrices, closure can also be seen directly:

$$(XY - YX)^{\mathsf{T}} = YX - XY = -(XY - YX).$$

The real tangent Lie algebra of $\mathrm{SO}(n)$ is denoted $\mathfrak{so}(n)$. For $\mathrm{GL}_n(\mathbb{C})$, viewed as a real group, the tangent algebra is all complex matrices with real scalars and the same commutator.

1.5 The group SU(2) and its real algebra

The two rows of a matrix in $\mathrm{SU}(2)$ are orthonormal. Write the first row as (α, β). The unit vectors orthogonal to this row are the multiples of $(-\overline{\beta}, \overline{\alpha})$ by complex numbers of modulus one. The determinant then equals that multiple, so determinant one forces it to be one. Hence every element of $\mathrm{SU}(2)$ has the form

$$U = \begin{pmatrix} \alpha & \beta \\ -\overline{\beta} & \overline{\alpha} \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Conversely, a matrix of the displayed form satisfies

$$\begin{pmatrix} \overline{\alpha} & -\beta \\ \overline{\beta} & \alpha \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ -\overline{\beta} & \overline{\alpha} \end{pmatrix} = \begin{pmatrix} |\alpha|^2 + |\beta|^2 & 0 \\ 0 & |\alpha|^2 + |\beta|^2 \end{pmatrix}.$$

The matrix U has determinant $|\alpha|^2 + |\beta|^2$. Thus this formula identifies $\mathrm{SU}(2)$ with the unit sphere S^3 in $\mathbb{C}^2 = \mathbb{R}^4$. The regular level-set theorem gives real dimension three. The displayed map from (α, β) to U is a real-linear injection from $\mathbb{R}^4$ into $M_{2\times 2}(\mathbb{C})$. It is a linear coordinate isomorphism onto its image, so it carries the embedded sphere to an embedded submanifold of the full matrix space. Multiplication and inversion are smooth restrictions of the matrix operations.

Proposition 1.8 (The tangent algebra of SU(2)). *The tangent Lie algebra of* $\mathrm{SU}(2)$ *is the real space*

$$\mathfrak{su}(2) = \left\{ \begin{pmatrix} ix & y + iz \\ -y + iz & -ix \end{pmatrix} : x, y, z \in \mathbb{R} \right\},$$

with the matrix commutator.

Proof. First we prove the necessary conditions. Let $U(t)$ be a smooth curve in $\mathrm{SU}(2)$ with $U(0) = I$, and put $X = U'(0)$. Differentiating $U(t)^* U(t) = I$ at zero gives $X^* = -X$.

The determinant condition requires a separate derivative. Write $X = (x_{jk})$. To first order in t,

$$U(t) = \begin{pmatrix} 1 + tx_{11} + o(t) & tx_{12} + o(t) \\ tx_{21} + o(t) & 1 + tx_{22} + o(t) \end{pmatrix}.$$

Its determinant has expansion

$$\det U(t) = 1 + t(x_{11} + x_{22}) + o(t).$$

Here $o(t)$ denotes a quantity whose ratio to t tends to zero. The products of two off-diagonal entries have order t^2, so they do not contribute to the derivative. Differentiating $\det U(t) = 1$ now gives $\operatorname{tr} X = 0$.

A skew-Hermitian matrix has purely imaginary diagonal entries. Its entry below the diagonal is minus the conjugate of the entry above it. Since the trace is zero, the diagonal entries are ix and $-ix$ for some $x \in \mathbb{R}$. Writing the entry above the diagonal as $y + iz$ gives the displayed form.

Next we prove the converse. Every matrix of the displayed form is a real linear combination of

$$A = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Let $X = xA + yB + zC$ and $r^2 = x^2 + y^2 + z^2$. Direct multiplication gives

$$X^2 = -r^2 I, \qquad \det(I + tX) = 1 + t^2 r^2.$$

These identities provide a curve with the required velocity:

$$U_X(t) = \frac{I + tX}{\sqrt{1 + t^2 r^2}}.$$

The scalar factor appears squared in a 2×2 determinant. Hence the determinant of $U_X(t)$ is one. Since $X^* = -X$ and $X^2 = -r^2 I$,

$$U_X(t)^* U_X(t) = \frac{(I - tX)(I + tX)}{1 + t^2 r^2} = I.$$

Also $U_X(0) = I$ and $U_X'(0) = X$. Hence every matrix of the displayed form is tangent to $\mathrm{SU}(2)$ at I.

Finally, $\mathrm{SU}(2)$ is an embedded matrix group. By the calculation with ambient extensions in the previous section, its bracket is the matrix commutator. ∎

Each group has so far needed its own curve. Lecture 2 gives a single one, $\exp(tX)$.

The multiplication of the basis matrices starts with $AB = C$ and $BA = -C$. The next products confirm a cyclic pattern:

$$BC = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = A, \qquad CB = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = -A.$$

Similarly, $CA = B$ and $AC = -B$. The bracket table is

$$[A, B] = 2C, \qquad [B, C] = 2A, \qquad [C, A] = 2B.$$

The scalars in this basis expansion are real. Multiplying A by i gives a Hermitian matrix, which is not in $\mathfrak{su}(2)$. The fact that its elements have complex entries does not make $\mathfrak{su}(2)$ a complex vector space. In physics, with the Pauli matrices $\sigma_1, \sigma_2, \sigma_3$ of *Linear Algebra for Physics*, one has $A = i\sigma_3$, $B = i\sigma_2$ and $C = i\sigma_1$.

The weight calculations of the earlier course use the matrices e and f given below. These matrices are not skew-Hermitian, so they are not in $\mathfrak{su}(2)$. To use them we must allow complex combinations of the real generators. We now relate this real algebra to the earlier complex algebra $\mathfrak{sl}_2(\mathbb{C})$.

Definition 1.9 (Complexification)**.** The complexification of a real Lie algebra $\mathfrak{k}$ is the complex vector space

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{k} \otimes_{\mathbb{R}} \mathbb{C},$$

with the complex-bilinear extension of its bracket. Thus $[X \otimes a, Y \otimes b] = [X, Y] \otimes ab$.

Let $\iota\colon \mathfrak{su}(2)_{\mathbb{C}} \to \mathfrak{sl}_2(\mathbb{C})$ be the complex-linear map given on pure tensors by $\iota(X \otimes a) = aX$. It extends the inclusion of $\mathfrak{su}(2)$ into complex matrices. Its values are traceless because the elements of $\mathfrak{su}(2)$ are traceless. The images of A, B, C span all traceless complex matrices, since

$$h = -iA, \qquad e = \frac{B - iC}{2}, \qquad f = -\frac{B + iC}{2}$$

give the three matrices

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

They are the usual basis matrices of $\mathfrak{sl}_2(\mathbb{C})$. These three matrices are complex linearly independent. The map ι preserves brackets by complex bilinearity. Its domain and image both have complex dimension three, so ι is a complex Lie algebra isomorphism:

$$\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}_2(\mathbb{C}).$$

The real algebra has dimension three over $\mathbb{R}$. Its complexification has dimension three over $\mathbb{C}$. The real algebra underlying $\mathfrak{sl}_2(\mathbb{C})$ has dimension six over $\mathbb{R}$; it is a different object.

Self-checks

1. Why is $\mathrm{GL}_n(\mathbb{C})$ open in $\mathbb{C}^{n^2}$ but $\mathrm{U}(n)$ is not?
2. Does differentiating $g(t)^{\mathsf{T}} g(t) = I$ prove that every skew-symmetric matrix is tangent to $\mathrm{SO}(n)$?
3. Is $\mathrm{diag}(1, -1)$ in $\mathfrak{su}(2)$? Is it in the complex span of $\mathfrak{su}(2)$?

Answers. (1) The determinant is continuous, so the set where it is nonzero is open in $\mathbb{C}^{n^2}$. For real ε with $0 < |\varepsilon| < 1$, a unitary matrix U satisfies

$$((1+\varepsilon)U)^*((1+\varepsilon)U) = (1+\varepsilon)^2 I \neq I.$$

These non-unitary matrices approach U, so no ambient ball about U lies in $\mathrm{U}(n)$. (2) No. Differentiation proves necessity. The Cayley curve proves sufficiency. (3) No. The matrix $\mathrm{diag}(1, -1)$ is traceless but Hermitian, so it is not skew-Hermitian. It does lie in the complex span, since it equals $-iA$.

Summary and reading

A Lie group has smooth multiplication and inversion. Its tangent space at the identity is a real Lie algebra. Left translation determines the bracket. For embedded matrix groups, the bracket is the matrix commutator. The algebra records local information. Distinct groups can have the same tangent algebra.

Read Hall, second edition, Sections 1.1–1.5 and 3.3–3.6, and Bröcker–tom Dieck, Chapter I, Sections 1–2. The regular level-set theorem and the embedded-submanifold properties are the geometric imports, cited from Warner, Chapter 1. Lecture 2 develops the exponential map and connects these tangent vectors to one-parameter subgroups.

Lecture 2

Exponential maps and local group structure

Objectives. *A tangent vector at the identity generates a one-parameter subgroup. The exponential map collects these subgroups into one map from the Lie algebra to the group. We ask how much of the multiplication is determined by the bracket. Near the identity, the bracket determines the whole multiplication. Far from the identity, it does not.*

By the end of this lecture you will be able to:

- Compute exponentials of rotation and boost generators and of elements of $\mathfrak{sl}_2(\mathbb{R})$.
- Identify continuous one-parameter subgroups of $\mathrm{GL}_n(\mathbb{C})$ with exponentials.
- Differentiate a homomorphism, and use $\exp(\mathrm{Ad}_g X) = g\exp(X)g^{-1}$ and $\mathrm{Ad}(\exp X) = e^{\mathrm{ad}\,X}$.
- Compute the second-order BCH term and recognise non-exponential elements.

2.1 The matrix exponential

For a complex $n \times n$ matrix X, let $\|X\|$ be the operator norm for the standard Hermitian norm on $\mathbb{C}^n$. The operator norm satisfies $\|XY\| \le \|X\|\,\|Y\|$.

Definition 2.1 (Matrix exponential)**.** For $X \in M_{n\times n}(\mathbb{C})$, the exponential of X is

$$\exp X = e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}.$$

Since $\|X^k\| \le \|X\|^k$, the series is dominated by the scalar series of $e^{\|X\|}$. Hence the series converges absolutely, and uniformly on bounded sets. Each entry of $\exp X$ is an everywhere convergent power series in the entries of X. So exp is smooth, and its series may be differentiated term by term.

Proposition 2.2 (Properties of the matrix exponential)**.** *Let $X, Y \in M_{n\times n}(\mathbb{C})$ and $g \in \mathrm{GL}_n(\mathbb{C})$.*

(i) *If $XY = YX$, then $\exp(X+Y) = \exp X \exp Y$.*

(ii) *The matrix $\exp X$ is invertible, with inverse $\exp(-X)$.*

(iii) *The curve $t \mapsto \exp(tX)$ has derivative $X\exp(tX) = \exp(tX)X$.*

(iv) $g\exp(X)g^{-1} = \exp(gXg^{-1})$.

(v) $\det\exp X = e^{\mathrm{tr}\,X}$.

Proof. (i) Absolute convergence lets us multiply the series and collect terms of total degree m. The binomial theorem holds because $XY = YX$:

$$\exp X \exp Y = \sum_{m=0}^{\infty} \sum_{k+l=m} \frac{X^k Y^l}{k!\,l!} = \sum_{m=0}^{\infty} \frac{(X+Y)^m}{m!}.$$

(ii) Apply the product rule (i) to the commuting pair X and $-X$. (iii) Differentiate $\sum_k t^k X^k/k!$ term by term and factor out X on either side. (iv) We have $(gXg^{-1})^k = gX^k g^{-1}$. Conjugation by g is linear and continuous, so it commutes with the sum of the series. (v) Schur triangularisation gives $X = PTP^{-1}$ with T upper triangular. Let t_{jj} be the diagonal entries of T. Then $\exp T$ is upper triangular with diagonal entries $e^{t_{jj}}$. By the conjugation formula (iv),

$$\det \exp X = \det \exp T = \prod_j e^{t_{jj}} = e^{\operatorname{tr} T} = e^{\operatorname{tr} X}.$$

∎

The hypothesis $XY = YX$ in the product rule, Proposition 2.2(i), cannot be dropped. Exercise 2.3 gives nilpotent matrices E and F with $\exp E \exp F \neq \exp(E+F)$. The Baker–Campbell–Hausdorff formula in § 2.5 shows that the bracket governs the correction.

2.2 One-parameter subgroups

Definition 2.3 (One-parameter subgroup)**.** A one-parameter subgroup of a Lie group G is a continuous homomorphism $\varphi\colon \mathbb{R} \to G$ from the additive group of real numbers.

For every matrix X, the curve $t \mapsto \exp(tX)$ is a one-parameter subgroup of $\mathrm{GL}_n(\mathbb{C})$, by the product rule, Proposition 2.2(i). Conversely, continuity alone forces this form.

Theorem 2.4 (One-parameter subgroups of $\mathrm{GL}_n(\mathbb{C})$)**.** *Let $\varphi\colon \mathbb{R} \to \mathrm{GL}_n(\mathbb{C})$ be a one-parameter subgroup. Then φ is differentiable, and $\varphi(t) = \exp(tX)$ for the unique matrix $X = \varphi'(0)$. If φ takes values in $\mathrm{GL}_n(\mathbb{R})$, then X is a real matrix.*

Proof. We first show that φ is differentiable. Averaging a continuous function over a moving interval gives a differentiable function. The homomorphism law writes φ as its own moving average times a fixed matrix.

Step 1: differentiability. By continuity of φ, choose $\varepsilon > 0$ with $\|\varphi(s) - I\| \le \tfrac{1}{2}$ for $0 \le s \le \varepsilon$. Integrate entrywise and set

$$M = \frac{1}{\varepsilon} \int_0^{\varepsilon} \varphi(s)\, ds.$$

Then $\|M - I\| \le \tfrac{1}{2}$, so the Neumann series $\sum_{k\ge 0} (I-M)^k$ inverts M. The homomorphism property gives

$$\varphi(t) M = \frac{1}{\varepsilon} \int_0^{\varepsilon} \varphi(t+s)\, ds = \frac{1}{\varepsilon} \int_t^{t+\varepsilon} \varphi(u)\, du.$$

The right side is continuously differentiable in t, by the fundamental theorem of calculus. Multiplication by the constant matrix M^{-1} preserves this property, so φ is continuously differentiable too.

Step 2: the equation. Put $X = \varphi'(0)$. Differentiate $\varphi(t+s) = \varphi(t)\varphi(s)$ in s at $s = 0$. This gives $\varphi'(t) = \varphi(t) X$.

Step 3: uniqueness. Let $\psi(t) = \varphi(t)\exp(-tX)$. By the Leibniz rule and the derivative formula, Proposition 2.2(iii),

$$\psi'(t) = \varphi(t)X\exp(-tX) - \varphi(t)X\exp(-tX) = 0.$$

Hence $\psi(t) = \psi(0) = I$, and $\varphi(t) = \exp(tX)$. Any Y with $\varphi(t) = \exp(tY)$ satisfies $Y = \varphi'(0) = X$. If φ takes values in $\mathrm{GL}_n(\mathbb{R})$, then its derivative $X = \varphi'(0)$ is a real matrix, as a limit of real difference quotients. ∎

Example 2.5 (Rotations)**.** The algebra $\mathfrak{so}(2)$ is spanned by

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I.$$

Separating even and odd powers in the series gives

$$\exp(\theta J) = \cos\theta\, I + \sin\theta\, J = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The first column of an element of $\mathrm{SO}(2)$ is a unit vector, so it is $(\cos\theta, \sin\theta)$ for some θ. Orthogonality and determinant one then fix the second column. So $\exp\colon \mathfrak{so}(2) \to \mathrm{SO}(2)$ is onto. It is not injective, since $\exp(2\pi J) = I$.

Example 2.6 (Boosts)**.** In special relativity on a line, with the speed of light set to 1, an event has a time coordinate x_0 and a position x_1. Observers in uniform relative motion use coordinates related by linear maps preserving $x_0^2 - x_1^2$. With $\eta = \mathrm{diag}(1, -1)$, these maps form the group

$$\mathrm{O}(1,1) = \{g \in M_{2\times 2}(\mathbb{R}) : g^{\mathsf{T}}\eta g = \eta\}.$$

Let $g(t)$ be a smooth curve in $\mathrm{O}(1,1)$ with $g(0) = I$ and $g'(0) = X$. Differentiating $g(t)^{\mathsf{T}}\eta g(t) = \eta$ at $t = 0$ gives the necessary condition $X^{\mathsf{T}}\eta + \eta X = 0$. In terms of the entries of X,

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad X^{\mathsf{T}}\eta + \eta X = \begin{pmatrix} 2a & b - c \\ b - c & -2d \end{pmatrix}.$$

The condition therefore forces $a = d = 0$ and $b = c$. So X is a multiple of

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad K^2 = I.$$

The even and odd parts of the series now give hyperbolic functions:

$$B(u) = \exp(uK) = \begin{pmatrix} \cosh u & \sinh u \\ \sinh u & \cosh u \end{pmatrix}.$$

Since $\cosh^2 u - \sinh^2 u = 1$, we get $B(u)^{\mathsf{T}}\eta B(u) = \eta$. So K is the velocity of a curve in the group, and the tangent algebra is exactly $\mathbb{R}K$.

The parameter u is the rapidity. The boost $B(u)$ sends the events $(\tau, 0)$ of an observer at rest to $(\tau\cosh u, \tau\sinh u)$, a motion with velocity $\tanh u$. Rapidities add, since $B(u)B(u') = B(u+u')$. Velocities do not add. Unlike the parametrisation of rotations, the map $u \mapsto B(u)$ is injective. Every boost has trace $2\cosh u \geq 2$.

2.3 The exponential map of a Lie group

So far our groups are matrix groups. Not every Lie group is one, and the reading of Lecture 4 cites an example. On a general Lie group, the exponential power series is not defined. Instead we use the left-invariant vector fields $X^L(g) = (dL_g)_e X$ of Lecture 1, with $X \in \mathfrak{g} = T_eG$.

Definition 2.7 (Integral curve). An integral curve of a vector field V on a manifold M is a smooth curve $\gamma : I \to M$, on an open interval I, with $\gamma'(t) = V(\gamma(t))$ for all $t \in I$.

In a chart, this condition is a first-order system of ODEs with smooth coefficients. The Picard–Lindelöf theorem of the ODE background gives local existence and uniqueness. Two integral curves with the same initial point agree on their common interval. The agreement set is open by local uniqueness, closed because limits in a Hausdorff manifold are unique, and nonempty. An interval is connected, so it is the whole interval.

Left translation maps integral curves of X^L to integral curves of X^L. For such a curve γ and $g \in G$, the chain rule and $L_g \circ L_h = L_{gh}$ give

$$(L_g \circ \gamma)'(t) = (dL_g)_{\gamma(t)}(dL_{\gamma(t)})_e X = (dL_{g\gamma(t)})_e X = X^L\big(g\gamma(t)\big).$$

Theorem 2.8 (Exponential map of a Lie group; cited). *Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $X \in \mathfrak{g}$. The integral curve γ_X of X^L with $\gamma_X(0) = e$ is defined on all of $\mathbb{R}$. The map*

$$\exp : \mathfrak{g} \longrightarrow G, \qquad \exp X = \gamma_X(1),$$

is smooth, and $\gamma_X(t) = \exp(tX)$ for all real t.

This is proved in Kirillov Jr., Chapter 3, in the section on the exponential map, and in Warner, Chapter 3. We sketch the algebraic part. The omitted, cited step is that γ_X exists for all t and that $\gamma_X(1)$ depends smoothly on X.

For fixed s, the curves $t \mapsto \gamma_X(s+t)$ and $t \mapsto \gamma_X(s)\gamma_X(t)$ are integral curves of X^L starting at $\gamma_X(s)$. By uniqueness,

$$\gamma_X(s+t) = \gamma_X(s)\gamma_X(t).$$

For a real number c, the curve $t \mapsto \gamma_X(ct)$ is the integral curve of $(cX)^L$ through e. So $\gamma_{cX}(t) = \gamma_X(ct)$. At $t = 1$ this gives $\exp(cX) = \gamma_X(c)$.

Let $G \subseteq \mathrm{GL}_n(\mathbb{C})$ be an embedded matrix Lie group, as in Lecture 1. Left translation is matrix multiplication, so γ_X solves $\gamma' = \gamma X$ with $\gamma(0) = I$ in the matrix space. Step 3 of the proof of the one-parameter subgroup theorem, Theorem 2.4, shows that $\exp(tX)$ is the only solution. Hence the two exponentials agree, and $\exp(tX) \in G$ for every $X \in \mathfrak{g}$.

Lemma 2.9 (Smooth one-parameter subgroups). *Let $\beta : \mathbb{R} \to G$ be a smooth homomorphism with $\beta'(0) = X$. Then $\beta(t) = \exp(tX)$ for all t.*

Proof. Differentiate $\beta(t+s) = L_{\beta(t)}\beta(s)$ in s at $s = 0$. This gives $\beta'(t) = X^L(\beta(t))$. A homomorphism sends the identity to the identity, so $\beta(0) = e$. Hence β and γ_X are both integral curves of X^L through e. By uniqueness of integral curves, they agree. ∎

Proposition 2.10 (Local invertibility)**.** *The differential of* exp *at* 0 *is the identity of* $\mathfrak{g}$*. Consequently there are open sets* $U \ni 0$ *in* $\mathfrak{g}$ *and* $V \ni e$ *in* G *such that* exp *restricts to a diffeomorphism* $U \to V$*. We may take* $U = -U$*.*

Proof. Identify $T_0\mathfrak{g}$ with $\mathfrak{g}$. The line $t \mapsto tX$ has velocity X. Its image under exp is γ_X, whose velocity at 0 is $X^L(e) = X$. Hence $d\exp_0(X) = X$.

Read exp in a chart around e. The inverse function theorem of multivariable calculus, which we cite, says the following. A smooth map between open subsets of $\mathbb{R}^d$ with invertible derivative at a point restricts to a diffeomorphism between open neighbourhoods of that point and its image. This gives U and V. Replace U by $U \cap (-U)$, and replace V by the image of $U \cap (-U)$ under exp. Both sets remain open. ∎

Write $\log\colon V \to U$ for the inverse. Local invertibility, Proposition 2.10, is a local statement. Indeed, the rotation example, Example 2.5, shows that exp need not be injective on all of $\mathfrak{g}$.

Proposition 2.11 (Generation by exponentials)**.** *If* G *is connected, every element of* G *is a finite product* $\exp X_1 \exp X_2 \cdots \exp X_k$*.*

Proof. Let $V = \exp U$ be the neighbourhood given by local invertibility, Proposition 2.10. Since $\exp(-X) = (\exp X)^{-1}$ and $U = -U$, we have $V^{-1} = V$. Let H be the set of finite products of elements of V. It is a subgroup. It is open, since $H = \bigcup_{h \in H} hV$. Its complement is a union of cosets gH, so it is open too. As $e \in H$ and G is connected, $H = G$. ∎

The proof uses only that V is open, contains e and satisfies $V^{-1} = V$.

A group that is not connected still has a connected part around e. The next definition describes it by paths.

Definition 2.12 (Identity component)**.** The *identity component* G_0 of a Lie group G is the set of elements of G joined to e by a continuous path in G.

For example, let $G = \mathrm{O}(2)$. The determinant is continuous with values ± 1, so it is constant along a path. Hence $G_0 \subseteq \mathrm{SO}(2)$. Conversely, every element of $\mathrm{SO}(2)$ is $\exp(\theta J)$ by the rotation example, Example 2.5, and $t \mapsto \exp(t\theta J)$ joins it to I. So $G_0 = \mathrm{SO}(2)$.

Proposition 2.13 (Properties of the identity component)**.** *Let* G *be a Lie group with Lie algebra* $\mathfrak{g}$*.*

(a) G_0 *is a normal subgroup of* G*, and it is both open and closed.*

(b) G_0 *has Lie algebra* $\mathfrak{g}$*, and it contains* $\exp(tX)$ *for all* $X \in \mathfrak{g}$ *and* $t \in \mathbb{R}$*.*

(c) G_0 *is connected, and it contains every connected subset of* G *through* e*. Hence* G *is connected exactly when* $G = G_0$*.*

Proof. (a) Let γ and δ be continuous paths from e to g and from e to h. Then $t \mapsto \gamma(t)\delta(t)$ joins e to gh, and $t \mapsto \gamma(t)^{-1}$ joins e to g^{-1}. For $x \in G$, the path $t \mapsto x\gamma(t)x^{-1}$ joins e to xgx^{-1}. So G_0 is a normal subgroup.

Next, let $V = \exp U$ be the neighbourhood given by local invertibility, Proposition 2.10. For $g \in G_0$ and $X \in U$, the path $t \mapsto g\exp(tX)$, $0 \le t \le 1$, joins g to $g\exp X$. Hence the open set

gV lies in G_0, so G_0 is open. Its complement is a union of cosets gG_0. These cosets are open, so G_0 is closed.

(b) An open subset containing e has the same tangent space at e as G. The curve $s \mapsto \exp(sX)$ joins e to $\exp(tX)$.

(c) Suppose $G_0 = O_1 \cup O_2$ with O_1 and O_2 disjoint, open in G_0 and nonempty, and with $e \in O_1$. Choose $g \in O_2$ and a path γ from e to g in G. Every point of this path is joined to e along γ, so γ lies in G_0. Then $\gamma^{-1}(O_1)$ and $\gamma^{-1}(O_2)$ would disconnect $[0,1]$. So G_0 is connected.

Let C be a connected subset of G containing e. By (a), $C \cap G_0$ is open and closed in C, and it is nonempty. So $C \subseteq G_0$. Thus G_0 is the connected component of e. Finally, if G is connected, then taking $C = G$ gives $G = G_0$. The converse is the first statement of this part. ∎

2.4 Conjugation and homomorphisms

Conjugation $C_g(h) = ghg^{-1}$ is a smooth automorphism of G fixing e. The adjoint action of g is the linear map $\mathrm{Ad}_g = (dC_g)_e \colon \mathfrak{g} \to \mathfrak{g}$, also written $\mathrm{Ad}(g)$. For a matrix group, differentiating $g\exp(tX)g^{-1}$ at $t = 0$ gives $\mathrm{Ad}_g X = gXg^{-1}$. Hence $gXg^{-1} \in \mathfrak{g}$.

Theorem 2.14 (Differentiating homomorphisms). *Let $\Phi \colon G \to H$ be a Lie group homomorphism and $\phi = d\Phi_e$.*

(i) $\Phi(\exp X) = \exp(\phi(X))$ *for all* $X \in \mathfrak{g}$.

(ii) $\exp(\mathrm{Ad}_g X) = g\exp(X)g^{-1}$ *for all* $g \in G$ *and* $X \in \mathfrak{g}$.

(iii) If G is connected, then Φ is determined by ϕ.

(iv) If G and H are matrix Lie groups, then $\phi([X,Y]) = [\phi(X), \phi(Y)]$.

Proof. (i) The curve $\beta(t) = \Phi(\exp tX)$ is a smooth homomorphism $\mathbb{R} \to H$ with velocity $\phi(X)$ at 0. Apply the lemma on smooth one-parameter subgroups, Lemma 2.9, in H and set $t = 1$.

(ii) Apply the intertwining formula (i) to $\Phi = C_g$, whose differential is Ad_g.

(iii) By generation by exponentials, Proposition 2.11, each element of G is a product $\exp X_1 \cdots \exp X_k$. By the intertwining formula (i), the image of this product under Φ is $\exp \phi(X_1) \cdots \exp \phi(X_k)$, which depends only on ϕ.

(iv) For $g \in G$ and $Y \in \mathfrak{g}$, the matrix $gYg^{-1} \in \mathfrak{g}$ is the velocity of $g\exp(tY)g^{-1}$ at $t = 0$. Differentiate $\Phi(g\exp(tY)g^{-1}) = \Phi(g)\Phi(\exp tY)\Phi(g)^{-1}$ at $t = 0$:

$$\phi(gYg^{-1}) = \Phi(g)\,\phi(Y)\,\Phi(g)^{-1}.$$

Put $g = \exp(sX)$ and use the intertwining formula (i) on the right:

$$\phi\big(e^{sX}Ye^{-sX}\big) = e^{s\phi(X)}\phi(Y)e^{-s\phi(X)}.$$

Differentiate at $s = 0$, using the derivative formula, Proposition 2.2(iii). Since ϕ is linear, it commutes with differentiation. The result is $\phi(XY - YX) = \phi(X)\phi(Y) - \phi(Y)\phi(X)$. This is the bracket identity $\phi([X,Y]) = [\phi(X), \phi(Y)]$. ∎

The bracket identity, Theorem 2.14(iv), holds for every Lie group homomorphism. The general proof uses the vector-field bracket of Lecture 1. We cite Kirillov Jr., Chapter 3.

The most important target is $H = \mathrm{GL}(V)$, for a finite-dimensional complex vector space V. A representation $\Pi\colon G \to \mathrm{GL}(V)$ has a differential $d\Pi$ at e, and the intertwining formula (i) reads $\Pi(\exp X) = \exp(d\Pi(X))$.

We now apply the theorem to the adjoint action itself. Here $\mathrm{GL}(\mathfrak{g})$ is the group of invertible linear maps of $\mathfrak{g}$, and $\mathrm{End}(\mathfrak{g})$ is its tangent space at the identity.

Proposition 2.15 (The adjoint representation)**.** *Let G be a Lie group with Lie algebra $\mathfrak{g}$. The map $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$ is a smooth homomorphism. Write $\mathrm{ad}\colon \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ for its differential at e. Then, for all $X, Y \in \mathfrak{g}$,*

$$\mathrm{Ad}(\exp X) = e^{\mathrm{ad}\,X}, \qquad \mathrm{ad}\,X(Y) = [X, Y].$$

Proof. Since $C_{gh} = C_g \circ C_h$, the chain rule gives $\mathrm{Ad}_{gh} = \mathrm{Ad}_g\,\mathrm{Ad}_h$. The map $(g, h) \mapsto ghg^{-1}$ is smooth, so its derivative in h at $h = e$ depends smoothly on g. Hence Ad is a smooth homomorphism.

The intertwining formula, Theorem 2.14(i), with $\Phi = \mathrm{Ad}$ and $H = \mathrm{GL}(\mathfrak{g})$, gives $\mathrm{Ad}(\exp X) = \exp(\mathrm{ad}\,X)$. This is the exponential of the operator $\mathrm{ad}\,X$, which we write as $e^{\mathrm{ad}\,X}$.

By the definition of a differential, $\mathrm{ad}\,X(Y)$ is the derivative of $\mathrm{Ad}(\exp sX)Y$ at $s = 0$. For a matrix group, $\mathrm{Ad}(\exp sX)Y = e^{sX}Ye^{-sX}$, and the derivative formula, Proposition 2.2(iii), gives $\mathrm{ad}\,X(Y) = XY - YX = [X, Y]$. For a general Lie group we cite $\mathrm{ad}\,X(Y) = [X, Y]$ from Warner, Chapter 3, Section 3.46 and Proposition 3.47. ∎

For example, in $\mathrm{GL}_2(\mathbb{R})$ take

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad [h, E] = 2E.$$

So $\mathrm{ad}\,h$ multiplies E by 2, and $e^{t\,\mathrm{ad}\,h}E = e^{2t}E$. Direct multiplication with $\exp(th) = \mathrm{diag}(e^t, e^{-t})$ gives the same result: $\exp(th)E\exp(-th) = e^{2t}E$. Exercise 2.5 computes the operator $\mathrm{ad}\,A$ on $\mathfrak{su}(2)$.

The adjoint representation gives the product rule for commuting elements of any Lie group.

Proposition 2.16 (Commuting exponentials)**.** *Let G be a Lie group, and let $X, Y \in \mathfrak{g}$ with $[X, Y] = 0$. Then $\exp(tX)$ and $\exp(sY)$ commute for all $s, t \in \mathbb{R}$, and*

$$\exp(X + Y) = \exp X \exp Y.$$

Proof. Since $\mathrm{ad}\,X(Y) = [X, Y] = 0$, we get $\mathrm{Ad}(\exp tX)Y = e^{t\,\mathrm{ad}\,X}Y = Y$. Apply Theorem 2.14(ii) with $g = \exp(tX)$ to the element sY:

$$\exp(tX)\exp(sY)\exp(-tX) = \exp\bigl(s\,\mathrm{Ad}(\exp tX)Y\bigr) = \exp(sY).$$

So $\exp(tX)$ and $\exp(sY)$ commute. Hence $\beta(t) = \exp(tX)\exp(tY)$ is a smooth homomorphism $\mathbb{R} \to G$.

Its velocity at 0 is $X+Y$. Indeed, let $m(g,h)=gh$. Since $m(g,e)=g$ and $m(e,h)=h$, the differential of m at (e,e) sends $(X,0)$ to X and $(0,Y)$ to Y. By linearity it sends (X,Y) to $X+Y$. The lemma on smooth one-parameter subgroups, Lemma 2.9, gives $\beta(t)=\exp\bigl(t(X+Y)\bigr)$. Now put $t=1$. ∎

The same differential shows that the velocity at 0 of a product of curves through e is the sum of their velocities.

2.5 The Baker–Campbell–Hausdorff formula

By local invertibility, Proposition 2.10, a product of exponentials of small elements has a logarithm in $\mathfrak{g}$. We compute its leading terms in $\mathrm{GL}_n(\mathbb{C})$. Write $O(t^k)$ for a quantity bounded in norm by a constant times $|t|^k$ for small $|t|$.

Proposition 2.17 (Second-order term)**.** *Let $X,Y\in M_{n\times n}(\mathbb{C})$. For small $|t|$, put $Z(t)=\log\bigl(\exp(tX)\exp(tY)\bigr)$. Then*

$$Z(t)=t(X+Y)+\frac{t^2}{2}[X,Y]+O(t^3).$$

Proof. The map Z is smooth with $Z(0)=0$. Taylor's theorem gives matrices A,B with $Z(t)=tA+t^2B+O(t^3)$. Hence $\|Z(t)\|=O(t)$, so the tail $\sum_{k\ge3}Z^k/k!$ has norm at most $\|Z\|^3e^{\|Z\|}=O(t^3)$. Therefore

$$\exp Z(t)=I+Z+\tfrac12Z^2+O(t^3)=I+tA+t^2\bigl(B+\tfrac12A^2\bigr)+O(t^3).$$

Multiplying the two exponential series to second order gives

$$\exp(tX)\exp(tY)=I+t(X+Y)+t^2\bigl(\tfrac12X^2+XY+\tfrac12Y^2\bigr)+O(t^3).$$

The two left sides are equal. If $tP+t^2Q=O(t^3)$, dividing by t and letting $t\to0$ gives $P=0$, and then $Q=0$. Comparing coefficients,

$$A=X+Y,\qquad B=\tfrac12X^2+XY+\tfrac12Y^2-\tfrac12(X+Y)^2=\tfrac12(XY-YX).$$ ∎

The second-order term also shows what the bracket means. Multiplying out the series gives

$$\exp(tX)\exp(tY)\exp(-tX)\exp(-tY)=I+t^2[X,Y]+O(t^3),$$

as Exercise 2.6 shows. So the bracket measures, to leading order, how far $\exp(tX)$ and $\exp(tY)$ fail to commute. On a general Lie group, the flow of X^L is right multiplication by $\exp(tX)$. So this product is the end point of the path from e that follows X, Y, $-X$ and $-Y$ in turn. Figure 2.1 draws this path.

Exercise 2.7 computes the third-order term. The general statement is the following.

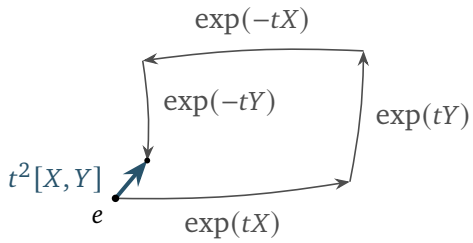


Figure 2.1: The path from e with steps $\exp(tX)$, $\exp(tY)$, $\exp(-tX)$ and $\exp(-tY)$, each multiplying on the right. The coloured arrow goes from e to the end point. In the coordinates given by log, this displacement is $t^2[X,Y] + O(t^3)$. The drawing is schematic.

Theorem 2.18 (Convergent BCH formula; cited)**.** *Let G be a Lie group with finite-dimensional real Lie algebra $\mathfrak{g}$. There is a neighbourhood W of 0 in $\mathfrak{g}$ with the following property. For $X, Y \in W$, the series grouped by total degree,*

$$H(X,Y) = X + Y + \tfrac{1}{2}[X,Y] + \tfrac{1}{12}\big([X,[X,Y]] + [Y,[Y,X]]\big) + \cdots,$$

converges in $\mathfrak{g}$, depends continuously on (X,Y), and satisfies $\exp X \exp Y = \exp H(X,Y)$. The degree-$m$ term is a combination of iterated brackets of m entries from X and Y, with universal rational coefficients independent of G.

Knapp, *Lie Groups Beyond an Introduction*, Theorem B.22, proves this convergent series for connected groups. The identity component reduces a general G to that case, by Proposition 2.13. The section on sources at the end of this lecture records what Hall, Knapp and Kirillov Jr. each cover.

The theorem answers the main question near the identity. As before, $\log\colon V \to U$ is the local inverse of exp on G.

Proposition 2.19 (Local determination)**.** *Let G and G' be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{g}'$. Write $\exp'$ and e' for the exponential and the identity of G'. Let $\psi\colon \mathfrak{g} \to \mathfrak{g}'$ be a homomorphism of real Lie algebras, and put $f = \exp' \circ \psi \circ \log$ on V. Then f is smooth and $df_e = \psi$. There is an open neighbourhood $V_1 \subseteq V$ of e such that, for all $a, b \in V_1$,*

$$ab \in V \quad \text{and} \quad f(ab) = f(a)f(b).$$

If ψ is an isomorphism, then f restricts to a diffeomorphism from an open neighbourhood of e onto an open neighbourhood of e'.

Proof. Since ψ is linear and preserves brackets, it maps each term of $H(X,Y)$ to the corresponding term of $H(\psi X, \psi Y)$. Since it is continuous, it maps the convergent sum to the sum.

Write W' for the BCH neighbourhood of G'. Choose V_1 so close to e that, for all $a, b \in V_1$, the elements $X = \log a$ and $Y = \log b$ satisfy three conditions:

(a) $X, Y \in W$;
(b) $\psi X, \psi Y \in W'$;
(c) $H(X,Y) \in U$, the set on which exp is inverted.

Continuity of log, ψ and H, with $H(0,0)=0$, makes this possible. Since $X, Y \in W$ by condition (a), $ab = \exp H(X,Y)$. Since $H(X,Y) \in U$ by condition (c), $ab \in V$ and $\log(ab) = H(X,Y)$. Therefore

$$f(ab) = \exp'\big(\psi H(X,Y)\big) = \exp'\big(H(\psi X, \psi Y)\big) = \exp'(\psi X)\exp'(\psi Y) = f(a)f(b).$$

The third equality uses condition (b), $\psi X, \psi Y \in W'$.

The map f is smooth, as a composite of smooth maps. We have $f(e) = \exp'(\psi(0)) = \exp'(0) = e'$. By local invertibility, Proposition 2.10, $d\exp_0$ and $d\exp'_0$ are identity maps, so $d\log_e$ is the identity too. The chain rule gives

$$df_e = d\exp'_0 \circ \psi \circ d\log_e = \psi.$$

If ψ is an isomorphism, it is invertible. The inverse function theorem then gives an open neighbourhood $V_0 \subseteq V$ of e that f maps diffeomorphically onto an open neighbourhood of e'. ∎

So isomorphic Lie algebras give the same multiplication near the identity. The multiplicative part of the proposition does not need ψ to be injective. Lecture 4 uses the same argument for a homomorphism that need not be injective.

The bracket does not determine the global group. For example, $\mathrm{O}(n)$ and $\mathrm{SO}(n)$ have the same Lie algebra. The next section shows a subtler effect.

2.6 Exponentials in $\mathrm{SL}(2,\mathbb{R})$

We now show that exp need not be onto, even on a connected group.

For $g \in \mathrm{SL}(2,\mathbb{R})$, expanding a 2×2 determinant gives

$$\det(g + tM) = 1 + t\,\mathrm{tr}(g^{-1}M) + t^2 \det(g^{-1}M).$$

Hence $d\det_g(M) = \mathrm{tr}(g^{-1}M)$. This differential is onto $\mathbb{R}$, since $d\det_g(g) = \mathrm{tr}\, I = 2$. By the regular level-set theorem of Lecture 1, $\mathrm{SL}(2,\mathbb{R})$ is an embedded matrix Lie group. Its algebra $\mathfrak{sl}_2(\mathbb{R})$ is the kernel of $d\det_I$, so it consists of the real traceless 2×2 matrices.

Let $X \in \mathfrak{sl}_2(\mathbb{R})$. Cayley–Hamilton gives $X^2 = \delta I$ with $\delta = -\det X$. Separating even and odd powers, as for rotations and boosts,

$$\exp X = c(\delta) I + s(\delta) X, \qquad c(\delta) = \sum_{k\ge 0} \frac{\delta^k}{(2k)!}, \quad s(\delta) = \sum_{k\ge 0} \frac{\delta^k}{(2k+1)!}.$$

Comparison with the hyperbolic and circular series gives

$$c(\delta) = \begin{cases} \cosh\sqrt{\delta}, & \delta > 0, \\ 1, & \delta = 0, \\ \cos\sqrt{-\delta}, & \delta < 0, \end{cases} \qquad s(\delta) = \begin{cases} \sinh\sqrt{\delta}/\sqrt{\delta}, & \delta > 0, \\ 1, & \delta = 0, \\ \sin\sqrt{-\delta}/\sqrt{-\delta}, & \delta < 0. \end{cases}$$

Proposition 2.20 (A trace bound). *For every $X \in \mathfrak{sl}_2(\mathbb{R})$, $\operatorname{tr}\exp X = 2c(\delta) \geq -2$. Equality holds only when $\exp X = -I$. Consequently, an element of $\mathrm{SL}(2,\mathbb{R})$ is not an exponential if its trace is less than -2, or if its trace is -2 and it differs from $-I$.*

Proof. Since $\operatorname{tr} X = 0$, we have $\operatorname{tr}\exp X = 2c(\delta)$. For $\delta \geq 0$, $c(\delta) \geq 1$. For $\delta < 0$, $c(\delta)$ is a cosine, so it is at least -1.

Suppose $c(\delta) = -1$. Then $\delta < 0$, and $\sqrt{-\delta}$ is an odd multiple of π. Hence $\sin\sqrt{-\delta} = 0$ and $s(\delta) = 0$. Therefore $\exp X = c(\delta)I + s(\delta)X = -I$. ∎

Example 2.21 (A non-exponential). Let $a > 0$ with $a \neq 1$, and set

$$g_a = \begin{pmatrix} -a & 0 \\ 0 & -1/a \end{pmatrix} \in \mathrm{SL}(2,\mathbb{R}).$$

Its eigenvalues are negative and unequal, and

$$\operatorname{tr} g_a + 2 = -\left(a + \frac{1}{a} - 2\right) = -\frac{(a-1)^2}{a} < 0.$$

By the trace bound, Proposition 2.20, the element g_a is not $\exp X$ for any $X \in \mathfrak{sl}_2(\mathbb{R})$. For $a = 2$, the trace is $-5/2$.

The element g_a is a product of two exponentials. With $h = \operatorname{diag}(1,-1)$,

$$\exp(\pi J) = -I, \qquad \exp\big((\log a)h\big) = \begin{pmatrix} a & 0 \\ 0 & 1/a \end{pmatrix},$$

so $g_a = \exp(\pi J)\exp\big((\log a)h\big)$.

Exercise 2.4 proves that $\mathrm{SL}(2,\mathbb{R})$ is connected. So generation by exponentials, Proposition 2.11, applies, but one exponential does not suffice. Local invertibility, Proposition 2.10, is not contradicted. For a near 1, the element g_a is near $-I$, far from I. Lecture 6 shows that in a compact connected group every element is a single exponential.

Self-checks

1. Is $\exp\colon \mathfrak{so}(2) \to \mathrm{SO}(2)$ injective? Is $-I \in \mathrm{O}(1,1)$ of the form $\exp(uK)$?
2. Can a real matrix with negative determinant be $\exp X$ for a real matrix X? Does this contradict generation by exponentials, Proposition 2.11?
3. If $X, Y \in W$ and $[X,Y] = 0$, what is $H(X,Y)$? Is $\log(\exp X \exp Y) = X + Y$ for all commuting X, Y?

Answers. (1) No, since $\exp(2\pi J) = I = \exp 0$. Also no: $\operatorname{tr}(-I) = -2$, while $\operatorname{tr}\exp(uK) = 2\cosh u \geq 2$. (2) No: for real X, $\det\exp X = e^{\operatorname{tr} X} > 0$. There is no contradiction, because $\mathrm{GL}_n(\mathbb{R})$ is disconnected by the determinant sign. (3) Every later term is an iterated bracket, so $H(X,Y) = X + Y$. Still, take $G = \mathrm{SO}(2)$ and $X = Y = \pi J$. The product $\exp X \exp Y$ is I. For the local inverse $\log\colon V \to U$ of exp on $\mathrm{SO}(2)$, $\log I = 0 \neq 2\pi J$. Here $2\pi J \notin U$, since $\exp(2\pi J) = \exp 0$, $0 \in U$ and exp is injective on U.

Summary

Every continuous one-parameter subgroup of $\mathrm{GL}_n(\mathbb{C})$ is $t \mapsto \exp(tX)$. For a general Lie group, flows of left-invariant fields define exp, a local diffeomorphism at 0. Homomorphisms intertwine exponentials, and connected groups are generated by exponentials. The identity component is an open and closed normal subgroup with the same Lie algebra. The adjoint representation satisfies $\mathrm{Ad}(\exp X) = e^{\operatorname{ad} X}$ and $\operatorname{ad} X(Y) = [X, Y]$, and elements with $[X, Y] = 0$ satisfy $\exp(X + Y) = \exp X \exp Y$. Near the identity, the BCH series computes multiplication from the bracket. Globally, exp need not be injective or surjective, and the Lie algebra $\mathfrak{g}$ does not determine the components of G.

Sources and hypotheses

This section records what the sources prove for the convergent BCH formula, Theorem 2.18.

Hall, Sections 5.3–5.5, proves the integral form of the formula for matrix Lie groups and X, Y near 0. Hall, Section 5.6, computes the series only through degree three.

Knapp, *Lie Groups Beyond an Introduction*, Theorem B.22, proves the convergent series for connected groups. This suffices, because the identity component of G is open, has Lie algebra $\mathfrak{g}$ and contains each curve $t \mapsto \exp(tX)$. Knapp's proof reduces to matrices with $\|X\| + \|Y\| < \log 2$. There the degree-n term of his logarithm series (B.24) has norm at most $c_n(\|X\| + \|Y\|)^n$, where $-\log(2 - e^s) = \sum_n c_n s^n$ and $c_n \geq 0$. Hence the series converges uniformly for $\|X\| + \|Y\| \leq r < \log 2$, and H is continuous. Knapp's Proposition B.43 converts the rational coefficients of (B.24) into rational bracket coefficients. Kirillov Jr., Section 3.7, Theorem 3.37, states the same series.

Reading

Read Hall, second edition, Chapter 2 on the matrix exponential, Chapter 3 on matrix Lie algebras and homomorphisms, and Sections 5.3–5.7 on BCH and group versus algebra homomorphisms. Kirillov Jr., Chapter 3, treats the general exponential map and the Campbell–Hausdorff formula. Warner, Chapter 3, Sections 3.46–3.47, treats the adjoint representation on a general Lie group. Lecture 3 uses the exponential to study subgroups and quotients.

Lecture 3

Subgroups, quotients and homogeneous spaces

Objectives. *A subgroup of a Lie group need not be a submanifold, and a coset space need not be a manifold. Closedness of the subgroup makes the coset space a manifold. Normality is a separate condition, needed for a group law. We test both conditions on an irrational line in a torus and on two descriptions of the sphere S^2.*

By the end of this lecture you will be able to:

- State the closed-subgroup theorem and the subgroup–subalgebra correspondence with their hypotheses.
- Prove that an irrational line in the two-torus is dense and not embedded.
- Identify a homogeneous space with a coset space by computing a stabiliser.
- Compute tangent spaces and isotropy actions for $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$.

3.1 Closed and immersed subgroups

A subset S of a manifold M carries the subspace topology of Lecture 1: its open sets are the sets $U \cap S$ with U open in M. The subset S is *closed* if $M \setminus S$ is open. In a manifold, S is closed exactly when S contains the limit of each convergent sequence of its points.

The *closure* $\overline{S}$ consists of the points whose every open neighbourhood meets S. It is the smallest closed set containing S. The set S is *dense* if $\overline{S} = M$. By the definition of closure, S is dense exactly when S meets every nonempty open subset of M.

In a manifold, every point of $\overline{S}$ is a limit of a sequence in S. To obtain the sequence, choose points of S in chart balls of radii tending to zero.

We also need compactness, in the form of finite subcovers. Lecture 5 uses this form to prove strict positivity of the Haar integral.

Definition 3.1 (Compact space)**.** A topological space is *compact* if every cover by open sets has a finite subcover.

A continuous image of a compact space is compact. Indeed, the inverse images of an open cover of the image form an open cover of the space. A finite subfamily of them covers the space, so the corresponding open sets cover the image.

A subset of $\mathbb{R}^n$ is compact exactly when it is closed and bounded (Heine–Borel). For subsets of $\mathbb{R}^n$ this agrees with the definition by sequences in *Linear Algebra for Physics*. For a set of matrices, closed and bounded must be read in the matrix space. For example, the set $\{tI : 0 < t \leq 1\}$ is bounded and it is closed in $\mathrm{GL}_n(\mathbb{R})$. It is not compact, because its closure in the matrix space contains the zero matrix.

Definition 3.2 (Immersions and embeddings)**.** A smooth map $f\colon N \to M$ is an *immersion* if df_p is injective for every $p \in N$. It is an *embedding* if it is an injective immersion and a homeomorphism from N onto $f(N)$ with the subspace topology.

The regular level sets of Lecture 1 are images of embeddings. An injective immersion can fail to be an embedding. The reason is that the topology carried over from N to $f(N)$ can be finer than the subspace topology, that is, it can have more open sets.

Definition 3.3 (Immersed and embedded subgroups)**.** Let G be a Lie group. An *immersed subgroup* of G is a subgroup H with a Lie group structure for which the inclusion $\iota\colon H \to G$ is an immersion. It is an *embedded subgroup* if ι is an embedding.

The differential $d\iota_e$ identifies $\mathrm{Lie}(H)$ with a subspace $\mathfrak{h} \subseteq \mathfrak{g}$. By Lecture 2, the differential of a Lie group homomorphism preserves brackets. Hence $\mathfrak{h}$ is a real Lie subalgebra of $\mathfrak{g}$.

Theorem 3.4 (Closed-subgroup theorem; cited)**.** *Let G be a Lie group and let $H \subseteq G$ be a subgroup that is a closed subset. Then H is an embedded submanifold of G, and with this structure H is a Lie group. Its Lie algebra is*

$$\mathfrak{h} = \{X \in \mathfrak{g} : \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}.$$

The hypothesis concerns only the subset H. Smoothness of H is a conclusion. For example, $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ are closed in $\mathrm{GL}_n(\mathbb{C})$, because they are defined by continuous equations.

A related regularity theorem removes a hypothesis from homomorphisms.

Theorem 3.5 (Automatic smoothness; cited)**.** *Every continuous homomorphism between finite-dimensional real Lie groups is smooth.*

See Bröcker–tom Dieck, Chapter I, Section 3, Proposition (3.12). In particular, finite-dimensional continuous representations have differentials. We use this fact from Lecture 4 onwards. In particular, by the lemma on smooth one-parameter subgroups in Lecture 2, every continuous one-parameter subgroup of a Lie group has the form $t \mapsto \exp(tX)$.

Theorem 3.6 (Subgroups and subalgebras; cited)**.** *Let G be a Lie group and let $\mathfrak{h} \subseteq \mathfrak{g}$ be a real Lie subalgebra. There is exactly one connected immersed subgroup $H \subseteq G$ with $\mathrm{Lie}(H) = \mathfrak{h}$. As a set, H is the subgroup generated by $\exp(\mathfrak{h})$. Every connected immersed subgroup arises in this way from its own Lie algebra.*

Warner, Chapter 3, proves Theorems 3.4 and 3.6 for general Lie groups. Hall, Section 5.9, proves Theorem 3.6 for matrix Lie groups. Kirillov Jr., Chapters 2 and 3, states these two subgroup theorems.

The two theorems run in opposite directions. The first starts with a subset and concludes that it is a manifold. The second starts with a subalgebra and constructs a group. That group need not be closed or embedded, so the word “immersed” in Theorem 3.6 cannot be replaced by “embedded”. The next section gives an example of a connected immersed

subgroup that is neither closed nor embedded.

3.2 An irrational line in the two-torus

The two-torus $T^2 = \mathrm{U}(1) \times \mathrm{U}(1)$ is a compact abelian Lie group. Identify its Lie algebra $i\mathbb{R} \oplus i\mathbb{R}$ with $\mathbb{R}^2$ by $(2\pi i a, 2\pi i b) \mapsto (a, b)$. The exponential map then becomes the surjective homomorphism

$$E\colon \mathbb{R}^2 \longrightarrow T^2, \qquad E(a,b) = (e^{2\pi i a}, e^{2\pi i b}),$$

with kernel $\mathbb{Z}^2$. For a real number α, the subalgebra $\mathfrak{h}_\alpha = \mathbb{R}(1,\alpha)$ gives the homomorphism and image

$$\varphi_\alpha(t) = E(t, \alpha t) = (e^{2\pi i t}, e^{2\pi i \alpha t}), \qquad L_\alpha = \varphi_\alpha(\mathbb{R}).$$

The velocity of φ_α at t is the left translate of $(2\pi i, 2\pi i\alpha) \neq 0$, so φ_α is an immersion. For rational α the image is a closed circle subgroup (Exercise Sheet 3). Irrational α behaves differently.

We look at the line only where it crosses one vertical circle. At times $x + k$ with k an integer, the first coordinate stays at $e^{2\pi i x}$, and the second moves by α modulo one at each step. Figure 3.1 shows this. So density reduces to one claim: the multiples of α, taken modulo one, come close to every number.

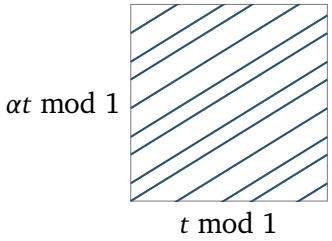


Figure 3.1: The line L_α for $0 \le t \le 8$ and $\alpha = (\sqrt{5} - 1)/2$, drawn in the unit square with opposite sides identified. Each new unit-time piece is shifted upwards by α modulo one.

Lemma 3.7 (Small multiples). *Let α be irrational and $n \ge 1$ an integer. There are integers q and m with $1 \le q \le n$ and*

$$0 < |q\alpha - m| < \frac{1}{n}.$$

Proof. For $j = 0, 1, \ldots, n$, put $f_j = j\alpha - \lfloor j\alpha \rfloor \in [0,1)$. These $n+1$ numbers lie in the n intervals $[s/n, (s+1)/n)$. By the pigeonhole principle, two of them, f_{j_1} and f_{j_2} with $j_1 < j_2$, lie in one interval. Set $q = j_2 - j_1$ and $m = \lfloor j_2\alpha \rfloor - \lfloor j_1\alpha \rfloor$. Then $q\alpha - m = f_{j_2} - f_{j_1}$, which has absolute value less than $1/n$. It is nonzero, since otherwise $\alpha = m/q$ would be rational. ∎

Proposition 3.8 (The irrational line). *Let α be irrational. Then:*

(a) φ_α is injective;
(b) L_α is dense in T^2;
(c) $L_\alpha \neq T^2$, so L_α is not closed;
(d) φ_α is not a homeomorphism onto L_α with the subspace topology.

Proof. (a) If $\varphi_\alpha(t) = \varphi_\alpha(s)$, then $t - s$ and $\alpha(t - s)$ are integers. If $t \neq s$, then α is the ratio of these two integers, so it is rational. Hence $t = s$.

(b) Fix a point $(e^{2\pi i x}, e^{2\pi i y})$ and an integer $n \geq 1$. First, take q and m from the small-multiples lemma, Lemma 3.7, and put $\delta = q\alpha - m \neq 0$. Let k be an integer nearest to $(y - \alpha x)/\delta$. Multiplying by $|\delta|$,

$$|k\delta - (y - \alpha x)| \leq \tfrac{1}{2}|\delta|.$$

Next, put $t = x + kq$. Here kq is an integer, and $\alpha kq = k\delta + km$ with km an integer. Therefore

$$\varphi_\alpha(t) = \left(e^{2\pi i x},\, e^{2\pi i(\alpha x + k\delta)}\right).$$

Finally, compare coordinates. The first coordinates of $\varphi_\alpha(t)$ and the chosen point agree. A chord of the unit circle is no longer than its arc, so $|e^{2\pi i u} - e^{2\pi i v}| \leq 2\pi|u - v|$ for all $u \in \mathbb{R}$ and $v \in \mathbb{R}$. The second coordinates therefore differ by at most $\pi|\delta| < \pi/n$. As n is arbitrary, L_α meets every neighbourhood of the chosen point. Since the chosen point is arbitrary, L_α is dense.

(c) Suppose $\varphi_\alpha(t) = (1, -1)$. Then t is an integer r, and $r\alpha = m + \frac{1}{2}$ for an integer m. Here $r \neq 0$, so $\alpha = (2m + 1)/(2r)$ would be rational. Thus $(1, -1) \notin L_\alpha$. A dense closed subset is the whole space, so L_α is not closed.

(d) For each n, take q_n and m_n from the small-multiples lemma, Lemma 3.7. Then

$$\varphi_\alpha(q_n) = \left(1, e^{2\pi i(q_n\alpha - m_n)}\right) \longrightarrow (1, 1) = \varphi_\alpha(0),$$

but $q_n \geq 1$ for every n. If the inverse map $L_\alpha \to \mathbb{R}$ were continuous at $(1, 1)$, then q_n would converge to 0. This is impossible. Hence the inverse map is not continuous at $(1, 1)$, and φ_α is not a homeomorphism onto L_α. ∎

So points close together in the torus can be far apart along the line. Figure 3.2 shows this near the identity. This is why the subgroup is not embedded.

Give L_α the Lie group structure transported from $\mathbb{R}$ by the bijection φ_α. It is then the connected immersed subgroup with Lie algebra $\mathfrak{h}_\alpha$ given by the subgroup–subalgebra correspondence, Theorem 3.6. By the irrational-line proposition, Proposition 3.8, the map φ_α is not a homeomorphism onto L_α with the subspace topology, so this subgroup is not embedded. The same proposition shows that L_α is not closed, so the closed-subgroup theorem, Theorem 3.4, does not apply to it. Since L_α is dense by the same proposition, its closure is the closed subgroup T^2. The Lie algebra $\mathbb{R}^2$ of T^2 is strictly larger than $\mathfrak{h}_\alpha$.

3.3 Coset spaces, actions and quotient groups

Let H be a subgroup of a Lie group G. The *coset space* $G/H = \{gH : g \in G\}$ has the quotient map π given by $\pi(g) = gH$. Its *quotient topology* declares $V \subseteq G/H$ open when $\pi^{-1}(V)$ is open in G. The group G acts on G/H by left translation, $\ell_g(xH) = gxH$.

Closedness is necessary for G/H to be Hausdorff. In a Hausdorff space each point is closed, because every other point has an open neighbourhood avoiding it. So if G/H is Hausdorff, then $H = \pi^{-1}(eH)$ is closed. For irrational α, the space T^2/L_α is therefore not a manifold.

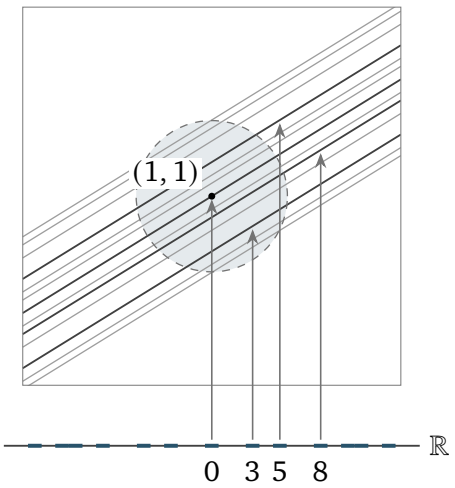


Figure 3.2: The square $[-\frac{1}{2}, \frac{1}{2}]^2$ is T^2, with opposite sides identified and the identity $(1,1)$ at the centre. Here $\alpha = (\sqrt{5}-1)/2$. The strands are the pieces of L_α for t in the heavy unit intervals around integers k of the line below. They are all the pieces with $|k| \le 13$ that meet the small dashed disc. The arrows show where the strands for $k = 0, 3, 5, 8$ come from.

Theorem 3.9 (Quotients by closed subgroups; cited)**.** *Let G be a Lie group and H a closed subgroup with Lie algebra $\mathfrak{h}$. Then:*

(1) G/H is Hausdorff and second countable. It has a unique smooth manifold structure for which π is a submersion*, that is, smooth with surjective differential at every point;*

(2) $\dim G/H = \dim G - \dim H$, and $d\pi_e$ induces an isomorphism $\mathfrak{g}/\mathfrak{h} \cong T_{eH}(G/H)$;

(3) the action $G \times G/H \to G/H$ is smooth;

(4) a map f from G/H to a manifold is smooth exactly when $f \circ \pi$ is smooth.

This theorem is in Bröcker–tom Dieck, Chapter I, Section 4, and in Warner, Chapter 3.

Definition 3.10 (Smooth actions)**.** A *smooth action* of G on a manifold M is a smooth map $G \times M \to M$, $(g,p) \mapsto g \cdot p$, with $e \cdot p = p$ and $g \cdot (g' \cdot p) = (gg') \cdot p$. The *stabiliser* of p is $G_p = \{g \in G : g \cdot p = p\}$, and the *orbit* of p is $G \cdot p$. The action is *transitive* if it has only one orbit. A manifold with a transitive smooth action is a *homogeneous space*.

Proposition 3.11 (Stabilisers and orbit maps)**.** *Let G act smoothly on M, fix $p \in M$, and put $H = G_p$.*

(a) H is a closed subgroup, and $G_{g\cdot p} = gHg^{-1}$.

(b) The orbit map $\beta(gH) = g \cdot p$ is a well-defined, injective, smooth map $G/H \to M$ with image $G \cdot p$. It is equivariant: $\beta(\ell_{g'}(gH)) = g' \cdot \beta(gH)$.

Proof. (a) Since $e \cdot p = p$, we have $e \in H$. If $g, g' \in H$, then $(gg') \cdot p = g \cdot (g' \cdot p) = p$ and $g^{-1} \cdot p = g^{-1} \cdot (g \cdot p) = p$. The point p is closed, because M is Hausdorff. So H is closed, as the preimage of p under the continuous map $g \mapsto g \cdot p$. Finally, $x \cdot (g \cdot p) = g \cdot p$ exactly when $(g^{-1}xg) \cdot p = p$, that is, when $x \in gHg^{-1}$.

(b) We have $g \cdot p = g' \cdot p$ exactly when $g^{-1}g' \in H$, that is, when $gH = g'H$. One direction gives

well-definedness, the other injectivity. Since H is closed by part (a), the quotient theorem, Theorem 3.9, applies to G/H. The composite $\beta \circ \pi(g) = g \cdot p$ is smooth, so β is smooth by the smoothness criterion of Theorem 3.9(4). Equivariance is the identity $(g'g) \cdot p = g' \cdot (g \cdot p)$. ∎

Theorem 3.12 (Homogeneous spaces; cited). *Let a Lie group G act smoothly and transitively on a manifold M, and let $p \in M$. Then the orbit map $\beta : G/G_p \to M$ is a G-equivariant diffeomorphism.*

The proof uses the second countability of G, which is part of the definition of a manifold in Lecture 1; see Warner, Chapter 3, or Kirillov Jr., Chapter 2.

We now compute T_pM from $\mathfrak{g}$. An element X of $\mathfrak{g}$ moves p with a velocity, namely the velocity of $t \mapsto \exp(tX) \cdot p$ at 0. For a transitive action, the elements with zero velocity form $\mathfrak{h}$, and the velocities fill T_pM. So $\mathfrak{g}/\mathfrak{h} \cong T_pM$. Next, the stabiliser H fixes p, so it acts on T_pM. This action tells the two sphere models below apart.

Two derivatives of the orbit map are used below. First, for $X \in \mathfrak{g}$, differentiate $\beta(\pi(\exp tX)) = \exp(tX) \cdot p$ at $t = 0$. This shows that $d\beta_{eH}(d\pi_e X)$ is the velocity of $t \mapsto \exp(tX) \cdot p$. This holds for every smooth action.

Second, assume now that the action is transitive. By the homogeneous-space theorem, Theorem 3.12, $d\beta_{eH}$ is an isomorphism $T_{eH}(G/H) \to T_pM$. Combining this with the tangent-space isomorphism of Theorem 3.9(2), we see that the composite $d\beta_{eH} \circ d\pi_e$ has kernel $\mathfrak{h}$ and induces an isomorphism $\mathfrak{g}/\mathfrak{h} \cong T_pM$.

As before, $H = G_p$. Let $k \in H$. The translation ℓ_k fixes eH. Its differential $d(\ell_k)_{eH}$ is called the *isotropy action* of k. The map $a_k : M \to M$, given by $a_k(p') = k \cdot p'$ for $p' \in M$, fixes p. Equivariance of β says $\beta \circ \ell_k = a_k \circ \beta$. The chain rule at eH therefore gives

$$d\beta_{eH} \circ d(\ell_k)_{eH} = d(a_k)_p \circ d\beta_{eH}.$$

So, transported by the isomorphism $d\beta_{eH}$, the isotropy action of k is the differential of a_k at p.

A coset space is not automatically a group.

Proposition 3.13 (Normality). *The rule $(xH)(yH) = xyH$ is well defined on G/H exactly when H is normal. It then makes G/H a group.*

Proof. Suppose the rule is well defined. Let $k \in H$ and $y \in G$. The representatives e and k of the coset H give the same product with yH, so $kyH = yH$. Hence $y^{-1}ky \in H$, and H is normal. Conversely, if H is normal, then for $k, k' \in H$,

$$xk\,yk' = xy\,(y^{-1}ky)\,k' \in xyH.$$

The group axioms are inherited from G. ∎

Theorem 3.14 (Quotient groups; cited). *Let H be a closed normal subgroup of a Lie group G. With the structure of Theorem 3.9, G/H is a Lie group and π is a homomorphism. The subalgebra $\mathfrak{h}$ is an ideal, and $\mathrm{Lie}(G/H) \cong \mathfrak{g}/\mathfrak{h}$ as Lie algebras.*

This theorem is in Warner, Chapter 3. Closedness of H makes G/H a manifold, and normality of H makes G/H a group. Since T^2 is abelian, L_α is normal, so T^2/L_α is a group. For irrational

α it is not a manifold. The two models of the sphere below are manifolds without a coset group law.

3.4 The sphere as SO(3)/SO(2)

The group $\mathrm{SO}(3)$ acts smoothly on $S^2 = \{v \in \mathbb{R}^3 : |v| = 1\}$ by matrix multiplication, with base point e_3. Let $J_1, J_2, J_3 \in \mathfrak{so}(3)$ be the matrices of Exercise Sheet 1, so that $J_k v = e_k \times v$. The rotations about the third axis are

$$R_\theta = \exp(\theta J_3) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Proposition 3.15 (Stabiliser of the north pole)**.** *The action of* $\mathrm{SO}(3)$ *on* S^2 *is transitive. The stabiliser of* e_3 *is* $H = \{R_\theta : \theta \in \mathbb{R}\} \cong \mathrm{SO}(2)$.

Proof. Let $v \in S^2$. Extend v to an orthonormal basis (u, w, v) of $\mathbb{R}^3$. The matrix g with columns u, w, v is orthogonal and $ge_3 = v$. Replacing u by $-u$ if necessary, we get $\det g = 1$. Thus the action is transitive.

Now let $ge_3 = e_3$. The third column of g is e_3. The rows of g are also orthonormal, because $gg^{\mathsf{T}} = I$. The third row is a unit vector with third entry one, so its other entries vanish. Hence

$$g = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}, \qquad A^{\mathsf{T}} A = I, \qquad \det A = \det g = 1.$$

The first column of A is a unit vector, so it equals $(\cos\theta, \sin\theta)$ for some θ. The second column is a unit vector orthogonal to the first, and $\det A = 1$ selects $(-\sin\theta, \cos\theta)$. So $g = R_\theta$. Conversely, $R_\theta e_3 = e_3$. ∎

By the homogeneous-space theorem, Theorem 3.12, the map $gH \mapsto ge_3$ is an equivariant diffeomorphism $\mathrm{SO}(3)/\mathrm{SO}(2) \to S^2$. The dimensions agree: $3 - 1 = 2$.

The sphere is the level set $|v|^2 = 1$, so $T_{e_3} S^2 = e_3^\perp$ by Lecture 1. By the first derivative computation of the previous section, $d\beta_{eH} \circ d\pi_e$ sends $X \in \mathfrak{so}(3)$ to the velocity of $\exp(tX)e_3$, which is Xe_3. On the basis,

$$J_1 e_3 = e_1 \times e_3 = -e_2, \qquad J_2 e_3 = e_2 \times e_3 = e_1, \qquad J_3 e_3 = 0.$$

So $X \mapsto Xe_3$ has kernel $\mathbb{R}J_3 = \mathrm{Lie}(H)$ and image $e_3^\perp$. It induces the isomorphism $\mathfrak{so}(3)/\mathbb{R}J_3 \cong T_{e_3} S^2$, which is $d\beta_{eH}$ composed with the tangent-space isomorphism of Theorem 3.9(2).

For the isotropy action, the map $a_{R_\theta}(v) = R_\theta v$ is the restriction to S^2 of a linear map of $\mathbb{R}^3$. Its differential at e_3 is therefore R_θ restricted to $e_3^\perp$. In the coordinates (x, y) for $xe_1 + ye_2$, this is the rotation of the tangent plane by θ. Only $R_\theta = I$ acts trivially.

The stabiliser H is not normal. Take the rotation by $\pi/2$ about e_1,

$$g_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad g_0 e_3 = -e_2.$$

By the conjugation formula of Proposition 3.11(a), $g_0 H g_0^{-1}$ is the stabiliser of $-e_2$. But $R_{\pi/2}(-e_2) = e_1$. Hence $R_{\pi/2} \in H$ and $R_{\pi/2} \notin g_0 H g_0^{-1}$. By the normality proposition, Proposition 3.13, there is no coset multiplication on $\mathrm{SO}(3)/\mathrm{SO}(2)$.

3.5 The sphere as SU(2)/U(1)

Let V be the real vector space of traceless Hermitian 2×2 matrices. It has the coordinates

$$\xi(x,y,z) = \begin{pmatrix} z & x-iy \\ x+iy & -z \end{pmatrix}, \qquad -\det \xi(x,y,z) = x^2+y^2+z^2.$$

The group $\mathrm{SU}(2)$ acts on V by $U \cdot \xi = U\xi U^*$. The matrix $U\xi U^*$ is Hermitian. Since $U^* = U^{-1}$, it has the same trace and determinant as ξ. Thus $\mathrm{SU}(2)$ acts smoothly on $S = \{\xi \in V : \det \xi = -1\}$, which the coordinates identify with S^2. In physics, with the Pauli matrices of Lecture 1, one writes $\xi(x,y,z) = x\sigma_1 + y\sigma_2 + z\sigma_3$, and S is the Bloch sphere of a two-level system. The base point is $N = \xi(0,0,1) = \mathrm{diag}(1,-1)$. With $A = \mathrm{diag}(i,-i)$ from Lecture 1, put $k_\theta = \exp(\theta A) = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$ and $T = \{k_\theta : \theta \in \mathbb{R}\} \cong \mathrm{U}(1)$.

Proposition 3.16 (Stabiliser in SU(2)). *The action of* $\mathrm{SU}(2)$ *on* S *is transitive, and the stabiliser of* N *is* T.

Proof. Let $\xi \in S$. Its characteristic polynomial is $s^2 - (\operatorname{tr}\xi)s + \det\xi = s^2 - 1$, so its eigenvalues are ± 1. By the spectral theorem, $\xi = WNW^*$ for a unitary W whose columns are unit eigenvectors of ξ for the eigenvalues 1 and -1. Since W is unitary, its determinant has modulus one. Choose $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ and $\lambda^2 = \overline{\det W}$. Then $U = \lambda W$ has determinant one, and $UNU^* = |\lambda|^2 WNW^* = \xi$.

For the stabiliser, $UNU^* = N$ is equivalent to $UN = NU$. Write U with entries a, b, c, d. Then

$$UN = \begin{pmatrix} a & -b \\ c & -d \end{pmatrix}, \qquad NU = \begin{pmatrix} a & b \\ -c & -d \end{pmatrix}.$$

Equality forces $b = c = 0$. A diagonal unitary matrix has $|a| = 1$, and $ad = 1$ gives $d = \overline{a}$. Thus $U = k_\theta$ for some θ. Conversely, k_θ commutes with N. ∎

By the homogeneous-space theorem, Theorem 3.12, the map $UT \mapsto UNU^*$ is an equivariant diffeomorphism $\mathrm{SU}(2)/T \to S \cong S^2$. Again the dimensions are $3 - 1 = 2$.

The map $d\beta_{eT} \circ d\pi_e$ sends $X \in \mathfrak{su}(2)$ to the velocity of $\exp(tX)N\exp(tX)^*$ at $t = 0$. By the product rule and since $X^* = -X$, this velocity is $XN + NX^* = [X, N]$. Use the basis A, B, C of $\mathfrak{su}(2)$ from Lecture 1, where

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Direct multiplication gives $[A, N] = 0$ and

$$[B,N] = \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix} = \xi(-2,0,0), \qquad [C,N] = \begin{pmatrix} 0 & -2i \\ 2i & 0 \end{pmatrix} = \xi(0,2,0).$$

The kernel of $X \mapsto [X, N]$ is $\mathbb{R}A = \mathrm{Lie}(T)$. The image of this map is $\{\xi(x, y, 0)\}$. In the coordinates, S is the unit sphere and N is its north pole, so this image is $T_N S$. Thus $X \mapsto [X, N]$ induces the isomorphism $\mathfrak{su}(2)/\mathbb{R}A \cong T_N S$, which is $d\beta_{eT}$ composed with the tangent-space isomorphism of Theorem 3.9(2).

For the isotropy action, the map $a_{k_\theta}(\xi) = k_\theta \xi k_\theta^*$ is the restriction to S of a linear map of V. Its differential at N is the same conjugation, acting on $T_N S$:

$$k_\theta\, \xi(x, y, 0)\, k_\theta^* = \begin{pmatrix} 0 & e^{2i\theta}(x - iy) \\ e^{-2i\theta}(x + iy) & 0 \end{pmatrix}.$$

Thus $x + iy \mapsto e^{-2i\theta}(x + iy)$, a rotation by -2θ. It is trivial exactly when $\theta \in \pi\mathbb{Z}$, that is, when $k_\theta = \pm I$. In physics, the element $k_{-\varphi/2} = \exp(-i\varphi\sigma_3/2)$ therefore rotates the tangent plane by φ, which is the half angle of spin one half.

The two models have the same space, circle stabilisers and tangent dimension. They differ in the isotropy action. Run each stabiliser circle once. Both R_θ and k_θ do this as θ runs over $[0, 2\pi]$. In the coordinates (x, y) used above, the tangent plane then turns once for SO(3), and twice in the opposite sense for SU(2).

The sense of rotation depends on the choice of parametrisation, because replacing k_θ by $k_{-\theta}$ reverses it. The invariant statement is the kernel of the isotropy action, which is trivial for SO(3) and $\{\pm I\}$ for SU(2). Moreover, $-I$ acts trivially on all of V, since $(-I)\xi(-I)^* = \xi$ for every $\xi \in V$. Lecture 4 uses the element $-I$ to relate the two groups.

The subgroup T is not normal in SU(2) (Exercise 3.6(ii)), so SU(2)/U(1) has no coset group law.

Self-checks

1. For which real α is L_α closed in T^2?
2. Is $(gH)(g'H) = gg'H$ a well-defined product on SO(3)/SO(2)?
3. Why is the stabiliser in SU(2) of every point of S a compact subgroup conjugate to T?

Answers. (1) Exactly for rational α. Let $\alpha = p/q$ with integers p and $q \geq 1$. Since q and $\alpha q = p$ are integers, $\varphi_\alpha(q) = (1, 1)$. So φ_α has period q, and $L_\alpha = \varphi_\alpha([0, q])$. This is a continuous image of a compact interval, hence compact in $\mathbb{C}^2$. It is therefore closed in $\mathbb{C}^2$, and so closed in T^2. For irrational α, the subgroup L_α is not closed by the irrational-line proposition, Proposition 3.8(c). (2) No. The stabiliser SO(2) is not normal, as g_0 shows. By the normality proposition, Proposition 3.13, the product is therefore not well defined. (3) By transitivity each point is $U \cdot N$. Its stabiliser is UTU^{-1} by the conjugation formula of Proposition 3.11(a). This is the image of the compact circle T under the continuous map $k \mapsto UkU^{-1}$, so it is compact.

Summary and reading

A closed subgroup of a Lie group is an embedded subgroup. A real Lie subalgebra determines a connected immersed subgroup, which need not be closed or embedded. For example,

the irrational line is dense and not embedded. For closed H, the coset space G/H is a manifold whose tangent space at eH is isomorphic to $\mathfrak{g}/\mathfrak{h}$. If a Lie group G acts smoothly and transitively on a manifold M, then M is equivariantly diffeomorphic to G/G_p for each $p \in M$. Normality is a further condition, needed for G/H to be a group. The sphere is diffeomorphic to both $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$. The two models have different isotropy actions.

Read Bröcker–tom Dieck, Chapter I, Section 4; Hall, Section 5.9; and Kirillov Jr., Chapter 2. Warner, Chapter 3, is the reference for the cited closed-subgroup, correspondence, quotient and homogeneous-space theorems. Lecture 4 studies the element $-I$ of $\mathrm{SU}(2)$ through covering groups.

Lecture 4

Covering groups, integration and descent

Objectives. *We ask when a finite-dimensional Lie algebra representation is the differential of a representation of a given connected Lie group. On a simply connected group it always is. On other connected groups we integrate on the universal covering group and then test a central kernel. For example, spin one half is a representation of* $\mathrm{SU}(2)$ *and not of* $\mathrm{SO}(3)$.

By the end of this lecture you will be able to:

- Describe the universal covering group by path classes and use the lifting theorem.
- Prove that a discrete normal subgroup of a connected group is central.
- State Lie's integration theorem and Lie III with their hypotheses, and explain path transport.
- Prove the descent criterion and apply it to $\mathbb{R} \to \mathrm{U}(1)$ and $\mathrm{SU}(2) \to \mathrm{SO}(3)$.

Lie groups are finite-dimensional, real, Hausdorff and second countable. From Lecture 1–Lecture 3 we use three facts. The map exp is a diffeomorphism near 0. A connected Lie group is generated by any neighbourhood of e. A continuous homomorphism Φ of Lie groups is smooth and satisfies $\Phi(\exp X) = \exp d\Phi(X)$. Hence two continuous homomorphisms on a connected Lie group with equal differentials are equal.

4.1 Loops, homotopy and covering groups

Definition 4.1. A *path* in a space X is a continuous map $\gamma\colon [0,1] \to X$. It is a *loop at* x if $\gamma(0) = \gamma(1) = x$. Paths γ_0, γ_1 from x to y are *homotopic* if there is a continuous map $F\colon [0,1]^2 \to X$ such that

$$F(0,t) = \gamma_0(t), \qquad F(1,t) = \gamma_1(t), \qquad F(s,0) = x, \qquad F(s,1) = y.$$

The space X is *simply connected* if it is path connected and every loop is homotopic to a constant loop.

A space is *path connected* if any two of its points are joined by a path. It is *locally path connected* if every neighbourhood of a point contains a path-connected neighbourhood of that point. Every manifold is locally path connected, and $[0,1]$ and $[0,1]^2$ are both path connected and locally path connected. A *topological group* is a group with a topology for which multiplication and inversion are continuous.

If σ starts where γ ends, the path $\gamma * \sigma$ traverses γ on $[0, \frac{1}{2}]$ and then σ on $[\frac{1}{2}, 1]$, each at double speed. Homotopy classes of loops at x form the *fundamental group* $\pi_1(X, x)$ under this product.

Definition 4.2. Let X and Y be path connected. A continuous surjection $p\colon Y \to X$ is a *covering map* if each point of X has an open neighbourhood U such that $p^{-1}(U)$ is a disjoint union of open sets, each mapped homeomorphically onto U by p.

The basic example is the map $p\colon \mathbb{R} \to \mathrm{U}(1)$ given by $p(t) = e^{it}$. Take a point $z \in \mathrm{U}(1)$ and a short open arc U around z. Then $p^{-1}(U)$ is a disjoint union of open intervals, spaced 2π apart, and p maps each interval homeomorphically onto U. So p is a covering map. Figure 4.1 draws the line as a helix, so that p becomes the vertical projection.

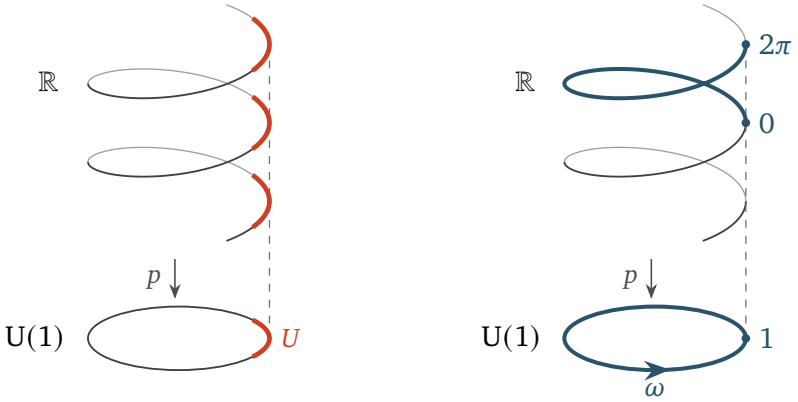


Figure 4.1: The map p projects the helix $\mathbb{R}$ vertically onto the circle $\mathrm{U}(1)$. In the first panel the coloured arcs of the helix lie over the arc U; only three of the infinitely many arcs over U are drawn. In the second panel the heavy path lies over the loop $\omega(t) = e^{2\pi i t}$ and runs from 0 to 2π, so it does not close.

Theorem 4.3 (Lifting theorem; cited). *Let $p\colon Y \to X$ be a covering map.*

(a) Let Z be path connected, locally path connected and simply connected. Let $f\colon Z \to X$ be continuous, and let $p(y_0) = f(z_0)$. Then there is a unique continuous $\tilde{f}\colon Z \to Y$ with $p \circ \tilde{f} = f$ and $\tilde{f}(z_0) = y_0$.

(b) Let γ_0, γ_1 be homotopic paths from x to x', and let $p(y_0) = x$. Then the lifts of γ_0 and γ_1 that start at y_0 have the same endpoint.

We cite this theorem from H §13.1 and S §1.2. The endpoint statement (b) follows from the lifting statement (a), applied to a homotopy F with $Z = [0,1]^2$ and $\tilde{F}(0,0) = y_0$. Each fibre of p is discrete. So if $\tilde{F}$ maps a connected set into one fibre, then $\tilde{F}$ maps that set to a single point.

The lift $\tilde{F}$ maps the edge $t = 0$ into the fibre over x, so $\tilde{F}$ maps this edge to y_0. By uniqueness of lifts in (a) with $Z = [0,1]$, the restrictions of $\tilde{F}$ to the edges $s = 0$ and $s = 1$ are the lifts of γ_0 and γ_1 from y_0.

The lift $\tilde{F}$ maps the edge $t = 1$ into the fibre $p^{-1}(x')$, so $\tilde{F}$ also maps this edge to a single point. Hence the two lifts have the same endpoint.

By the endpoint statement (b), a loop whose lift has distinct endpoints is not homotopic to a constant loop.

In the example of Figure 4.1, a path γ in $\mathrm{U}(1)$ starting at 1 lifts to a path $\tilde{\gamma}$ in $\mathbb{R}$ starting at 0. The endpoint $\tilde{\gamma}(1)$ is the *total angle* through which γ turns. The second panel shows a loop with total angle 2π.

Definition 4.4. A *universal covering group* of a connected Lie group G is a connected, simply connected Lie group $\tilde{G}$ with a smooth homomorphism $p\colon \tilde{G} \to G$ that is a covering map.

Path classes. Let G be connected. Let $\tilde{G}$ be the set of homotopy classes $[\gamma]$ of paths in G starting at e. Define

$$p[\gamma] = \gamma(1), \qquad [\gamma][\delta] = [\, t \mapsto \gamma(t)\delta(t)\,], \qquad [\gamma]^{-1} = [\, t \mapsto \gamma(t)^{-1}\,].$$

If F is a homotopy from γ to γ' and E one from δ to δ', then $(s,t) \mapsto F(s,t)E(s,t)$ is a homotopy from $\gamma\delta$ to $\gamma'\delta'$. So the product is well defined. The group axioms hold pointwise in t. The kernel of p consists of the classes of loops at e.

The pointwise product agrees with concatenation up to homotopy. Let c_x be the constant path at x. Replacing γ by $\gamma * c_{\gamma(1)}$ and δ by $c_e * \delta$ does not change their classes, since each is a reparametrisation. The pointwise product of the new paths is $\gamma * (\gamma(1)\delta)$, where $\gamma(1)\delta$ is the path $t \mapsto \gamma(1)\delta(t)$. For loops at e this is $\gamma * \delta$.

We compute this group for $G = \mathrm{U}(1)$. Two paths from 1 to the same point are homotopic exactly when they have the same total angle. If they are homotopic, the endpoint statement of the lifting theorem, Theorem 4.3(b), gives equal total angles. Conversely, if $\tilde{\gamma}_0(1) = \tilde{\gamma}_1(1)$, then

$$F(s,t) = e^{i((1-s)\tilde{\gamma}_0(t) + s\tilde{\gamma}_1(t))}$$

is a homotopy from γ_0 to γ_1, and both endpoints stay fixed. The pointwise product adds total angles, and every real number is the total angle of the path $t \mapsto e^{iat}$. So the path-class group of $\mathrm{U}(1)$ is $\mathbb{R}$ with addition. Fix θ with $e^{i\theta} = \gamma(1)$. The total angle of γ is $\theta + 2\pi k$ for an integer k. The class of γ records this integer, and the endpoint of γ does not.

It remains to give $\tilde{G}$ a topology. We sketch the construction. The details are cited after the theorem on smooth consequences, Theorem 4.5. Take a simply connected open neighbourhood U of $\gamma(1)$, for example a translate of a small exponential ball.

The basic open set $\mathcal{U}([\gamma], U)$ consists of the classes $[\gamma * \sigma]$, where σ is a path in U starting at $\gamma(1)$. The map p sends this set bijectively onto U, because paths in U with the same endpoints are homotopic in U.

The cited details show that p is a covering map, that the group operations are continuous and that $\tilde{G}$ is simply connected. Charts on G pulled back through p give the smooth structure on $\tilde{G}$.

An *isomorphism over G* from a universal covering group $p\colon \tilde{G} \to G$ to another one $p'\colon \tilde{G}' \to G$ is an isomorphism of Lie groups $\alpha\colon \tilde{G} \to \tilde{G}'$ with $p' \circ \alpha = p$.

Theorem 4.5 (Smooth consequences; cited)**.** *Let G be a connected Lie group.*

(a) The path-class construction gives a universal covering group $p\colon \tilde{G} \to G$, unique up to isomorphism over G. Its kernel is discrete and isomorphic to $\pi_1(G, e)$, and dp is an isomorphism.

> (b) *Let G' be a connected Lie group and $q\colon G' \to G$ a surjective smooth homomorphism with dq bijective. Then q is a covering map and* $\ker q$ *is discrete.*

We cite this from H §5.8 and from Knapp Beyond, Section I.11, *Covering groups*. Hall assumes that G is a matrix group, but $\tilde{G}$ need not be one.

> **Proposition 4.6.** *Let G be a connected topological group. Let N be a normal subgroup of G that is discrete in the subspace topology. Then N lies in the centre of G.*

Proof. Fix $n \in N$ and put $c(g) = gng^{-1}$. The map $c\colon G \to G$ is continuous, because multiplication and inversion are continuous. Normality gives $c(G) \subseteq N$. Since G is connected, $c(G)$ is connected. It contains $c(e) = n$.

Since N is discrete, every point of N is open in N. In particular, $\{n\}$ is open in N. Its complement $N \setminus \{n\}$ is a union of open singletons, so $\{n\}$ is also closed in N. Hence $\{n\}$ is open and closed in the subset $c(G)$ of N. Since $c(G)$ is connected and contains n, $c(G) = \{n\}$. In other words, $gng^{-1} = n$ for every $g \in G$. ∎

By the theorem on smooth consequences, Theorem 4.5(a), $\ker p$ is a discrete normal subgroup of the connected group $\tilde{G}$. By the proposition on discrete normal subgroups, Proposition 4.6, $\ker p$ is central. Hence $\pi_1(G, e)$, which is isomorphic to $\ker p$, is abelian. Exercise 4.1 shows that connectedness cannot be dropped.

4.2 Lie's integration theorem and Lie III

> **Theorem 4.7** (Lie's integration theorem; cited, with a sketch)**.** *Let G and H be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$. Assume that G is connected and simply connected. Let $\varphi\colon \mathfrak{g} \to \mathfrak{h}$ be a homomorphism of real Lie algebras. Then there is a unique smooth homomorphism $\Phi\colon G \to H$ with $d\Phi = \varphi$. It satisfies $\Phi(\exp X) = \exp \varphi(X)$.*

We cite this from H §5.7 for matrix groups. For general Lie groups we cite Knapp Beyond, Sections I.10–11, *Elementary theory of Lie groups* and *Covering groups*. For representations take $H = \mathrm{GL}(V)$ with $\dim_{\mathbb{C}} V < \infty$.

Path transport. Near e the map we want is given by exp. To reach a far element g, we take a path in small steps and multiply the images of the steps in order. Two paths could give different answers, and simple connectivity excludes this. The existence proof has three steps and a conclusion. Only the conclusion uses simple connectivity.

Step 1: a local homomorphism. Choose a neighbourhood B of 0 on which exp is a diffeomorphism. Define $f(\exp X) = \exp \varphi(X)$ for $X \in B$. For small X, Y we have $\exp X \exp Y = \exp Z(X, Y)$. The Baker–Campbell–Hausdorff series is the series of iterated brackets

$$Z(X, Y) = X + Y + \frac{1}{2}[X, Y] + \cdots .$$

This is the series $H(X,Y)$ of Lecture 2. We write Z here because H is the target group. Since φ preserves brackets, it sends $Z(X,Y)$ to $Z(\varphi X, \varphi Y)$. Hence, for some neighbourhood W of e,

$$f(xy) = f(x)f(y) \qquad (x, y \in W). \tag{4.1}$$

Hall proves the series for matrix groups in H §§5.3–5.6. In general we cite Knapp Beyond, Appendix B.4, Theorem B.22, *Campbell–Baker–Hausdorff formula*.

Step 2: transport along a path. Let γ start at e. Choose $0 = t_0 < \cdots < t_n = 1$ with $\gamma(s)^{-1}\gamma(t) \in W$ whenever s, t lie in one interval $[t_{j-1}, t_j]$. Define

$$\Phi_\gamma = f\big(\gamma(t_0)^{-1}\gamma(t_1)\big) \cdots f\big(\gamma(t_{n-1})^{-1}\gamma(t_n)\big).$$

Inserting a point s into $[t_{j-1}, t_j]$ replaces one factor $f(ab)$ by $f(a)f(b)$, where $a = \gamma(t_{j-1})^{-1}\gamma(s)$ and $b = \gamma(s)^{-1}\gamma(t_j)$. By the local homomorphism property (4.1), applied to a and b, the product is unchanged. Any two such partitions have a common refinement. Hence Φ_γ depends only on γ, not on the partition.

Step 3: homotopy invariance. We sketch the topological step. Let F be a homotopy from γ to γ', both from e to g. Cut $[0,1]^2$ into squares so small that $F(q)^{-1}F(q') \in W$ for any two points q, q' of one square. Give an edge from q to q' the factor $f(F(q)^{-1}F(q'))$. The transport of an edge path is the product of its factors in order. Let one square have corners a, b, c, d in cyclic order, with images x_a, x_b, x_c, x_d. By the local homomorphism property (4.1),

$$f(x_a^{-1}x_b)\, f(x_b^{-1}x_c) = f(x_a^{-1}x_c) = f(x_a^{-1}x_d)\, f(x_d^{-1}x_c).$$

So the two routes $a \to b \to c$ and $a \to d \to c$ around one square have equal transport. Figure 4.2 shows such a move.

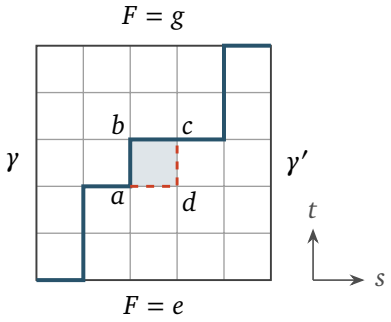


Figure 4.2: A homotopy F on a grid. The heavy edge path runs from $(0,0)$ to $(1,1)$. One move replaces its route $a \to b \to c$ by the dashed route $a \to d \to c$ across the shaded square.

Each edge path we use runs in the grid from $(0,0)$ to $(1,1)$. Start with the column $s = 0$ followed by the edge $t = 1$. End with the edge $t = 0$ followed by the column $s = 1$. Pass from the first to the last one square at a time. Each move swaps the routes $a \to b \to c$ and $a \to d \to c$ around one square, so it keeps the transport. The edges at $t = 0$ and $t = 1$ contribute $f(e) = e$, because F is constant there. By Step 2 the two columns give Φ_γ and $\Phi_{\gamma'}$. Hence $\Phi_\gamma = \Phi_{\gamma'}$. This step holds in every Lie group.

Conclusion. As G is simply connected, any two paths from e to g are homotopic. So $\Phi(g) = \Phi_\gamma$ does not depend on the path γ. The *increments* of a path δ are the elements $\delta(s)^{-1}\delta(t)$ that

enter the factors of Step 2. They do not change when δ is replaced by its left translate $g\delta$, given by $t \mapsto g\delta(t)$. So if δ runs from e to h, then

$$\Phi(gh) = \Phi_{\gamma * g\delta} = \Phi_\gamma \, \Phi_\delta = \Phi(g)\,\Phi(h).$$

Extending a path from e to g by a single short step gives $\Phi(g \exp X) = \Phi(g) \exp \varphi(X)$ for small X. Hence Φ is smooth and $d\Phi = \varphi$.

Without simple connectivity, Steps 1–3 still hold, but transport may depend on the homotopy class of the path. Self-check 2 gives an example.

Theorem 4.8 (Lie III; cited)**.** *Every finite-dimensional real Lie algebra is the Lie algebra of a connected, simply connected Lie group. This group is unique up to isomorphism.*

We cite existence from H §5.10. There Ado's theorem and H §5.9 give a connected Lie subgroup of a general linear group with the given Lie algebra. This subgroup need not be closed. The universal covering group of this subgroup is the required group. Uniqueness follows by integrating an isomorphism and its inverse with Lie's integration theorem, Theorem 4.7. The universal covering group of $\mathrm{SL}(2,\mathbb{R})$ is not isomorphic to any matrix Lie group (H §4.8). Hence Lie III needs abstract Lie groups.

4.3 Descent through a covering kernel

Theorem 4.9 (Descent)**.** *Let $p\colon \tilde{G} \to G$ be a surjective smooth homomorphism of Lie groups with dp bijective, and let $\Gamma = \ker p$. Let H be a topological group and $\tilde{\Phi}\colon \tilde{G} \to H$ a continuous homomorphism.*

(a) A homomorphism $\Phi\colon G \to H$ with $\Phi \circ p = \tilde{\Phi}$ exists if and only if $\Gamma \subseteq \ker \tilde{\Phi}$.

(b) Such Φ is unique and continuous. If H is a Lie group and $\tilde{\Phi}$ is smooth, then Φ is smooth and $d\Phi = d\tilde{\Phi} \circ (dp)^{-1}$.

Proof. Local sections. Since p is a homomorphism, left translations satisfy $p \circ L_x = L_{p(x)} \circ p$. Differentiating at e gives

$$dp_x \circ (dL_x)_e = (dL_{p(x)})_e \circ dp.$$

Both translations have invertible derivatives, and dp is bijective. Hence dp_x is bijective at every $x \in \tilde{G}$. By the inverse function theorem, each x has an open neighbourhood V_x that p maps diffeomorphically onto an open set. Let $s_x\colon p(V_x) \to V_x$ be the inverse of this restriction of p.

Necessity. If $\Phi \circ p = \tilde{\Phi}$ and $\gamma \in \Gamma$, then $\tilde{\Phi}(\gamma) = \Phi(e) = e$.

Sufficiency. Assume $\Gamma \subseteq \ker \tilde{\Phi}$. Let $g \in G$. Since p is surjective, we can write $g = p(x)$. Set $\Phi(g) = \tilde{\Phi}(x)$. If also $g = p(y)$, then $x^{-1}y \in \Gamma$ and

$$\tilde{\Phi}(y) = \tilde{\Phi}(x)\,\tilde{\Phi}(x^{-1}y) = \tilde{\Phi}(x).$$

Hence Φ is well defined and $\Phi \circ p = \tilde{\Phi}$. Moreover, Φ is a homomorphism, because

$$\Phi\big(p(x)p(y)\big) = \Phi\big(p(xy)\big) = \tilde{\Phi}(xy) = \tilde{\Phi}(x)\tilde{\Phi}(y).$$

Regularity. On $p(V_x)$ we have $\Phi = \Phi \circ p \circ s_x = \tilde{\Phi} \circ s_x$. This is continuous, and smooth when $\tilde{\Phi}$ is smooth. The open sets $p(V_x)$ cover G. Differentiating $\Phi \circ p = \tilde{\Phi}$ at e gives $d\Phi \circ dp = d\tilde{\Phi}$. Uniqueness holds because p is surjective. ∎

The proof uses only that H is a topological group. For example, the unitary group of a Hilbert space is a topological group for pointwise norm convergence. Lecture 19 uses this case.

Corollary 4.10. *Let G be a connected Lie group with universal covering group $p\colon \tilde{G} \to G$, and let $\Gamma = \ker p$. Let $\varphi\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a representation with $\dim_{\mathbb{C}} V < \infty$. Let $\tilde{\Pi}$ be the representation of $\tilde{G}$ with $d\tilde{\Pi} = \varphi \circ dp$. Then $\varphi = d\Pi$ for a representation Π of G if and only if $\tilde{\Pi}(\Gamma) = \{I\}$. In that case Π is unique and $\Pi \circ p = \tilde{\Pi}$.*

Proof. Since $\tilde{G}$ is simply connected, Lie's integration theorem, Theorem 4.7, gives $\tilde{\Pi}$, and $\tilde{\Pi}$ is unique. If Π exists, then $\Pi \circ p$ has differential $\varphi \circ dp$. Hence, by the uniqueness of $\tilde{\Pi}$, $\Pi \circ p = \tilde{\Pi}$, and $\tilde{\Pi}(\gamma) = \Pi(e) = I$ for $\gamma \in \Gamma$. Conversely, if $\tilde{\Pi}(\Gamma) = \{I\}$, then the Descent Theorem, Theorem 4.9, gives Π with $d\Pi = \varphi \circ dp \circ (dp)^{-1} = \varphi$. Uniqueness holds because G is connected. ∎

For disconnected groups the differential need not determine the homomorphism. For example, Exercise 4.6 gives two characters of $\mathrm{O}(2)$ with the same differential.

4.4 The line over the circle

Let the surjection $p\colon \mathbb{R} \to \mathrm{U}(1)$ be given by $p(t) = e^{it}$. Here $\operatorname{Lie} \mathrm{U}(1) = i\mathbb{R}$ and $\exp(i\theta) = e^{i\theta}$. Then

$$p(s+t) = p(s)\,p(t), \qquad dp(s) = \frac{d}{du}\Big|_{u=0} e^{ius} = is, \qquad \Gamma = \ker p = 2\pi\mathbb{Z}.$$

The line is simply connected, because $F(s,t) = (1-s)\gamma(t) + s\gamma(0)$ contracts any loop γ. By the theorem on smooth consequences, Theorem 4.5(b), p is a covering map, so it is a universal covering group. By the uniqueness and kernel statements of Theorem 4.5(a), $\pi_1(\mathrm{U}(1), 1) \cong 2\pi\mathbb{Z} \cong \mathbb{Z}$.

Concretely, the loop $\omega(t) = e^{2\pi i t}$ lifts from 0 to $t \mapsto 2\pi t$, which ends at $2\pi \neq 0$. By the endpoint statement of the lifting theorem, Theorem 4.3(b), ω is not homotopic to a constant loop.

A *scalar infinitesimal weight* on $\mathbb{R}$ is a number $\lambda \in \mathbb{C}$, viewed as the representation $\varphi_\lambda(s) = \lambda s$ on $\mathbb{C}$. It integrates to $\tilde{\chi}_\lambda(t) = e^{\lambda t}$.

The *period* of λ is $e^{2\pi\lambda} = \tilde{\chi}_\lambda(2\pi)$, the value at the generator 2π of Γ.

Say that λ *descends* if $\varphi_\lambda \circ (dp)^{-1}$ is the differential of a character of $\mathrm{U}(1)$. A *character* of $\mathrm{U}(1)$ here is a continuous homomorphism to the nonzero complex numbers. It is a one-dimensional representation, and its value is its trace. The descent criterion, Corollary 4.10, gives

$$\lambda \text{ descends} \iff e^{2\pi\lambda} = 1 \iff \lambda \in i\mathbb{Z}.$$

For $\lambda = in$ the descended character is $\chi(z) = z^n$, because $\chi(e^{it}) = e^{int}$. Its differential is $d\chi(i\theta) = in\theta$. In the course convention $d\chi = i\ell$ with $\ell(i\theta) = n\theta$. On $\Lambda = \ker\exp = 2\pi i\mathbb{Z}$ we get $\ell(2\pi ik) = 2\pi nk$, so $\ell(\Lambda) \subseteq 2\pi\mathbb{Z}$.

Success. For $\lambda = 3i$ the period is $e^{6\pi i} = 1$. The character $\chi(z) = z^3$ satisfies $\chi(p(t)) = e^{3it} = \tilde{\chi}_{3i}(t)$.

Failure. For $\lambda = i/2$ the period is $e^{i\pi} = -1$. Suppose a character χ had $d\chi(i\theta) = i\theta/2$. The corollary gives $\chi \circ p = \tilde{\chi}_{i/2}$, so

$$1 = \chi(1) = \chi\big(p(2\pi)\big) = \tilde{\chi}_{i/2}(2\pi) = -1.$$

This contradiction shows that $\lambda = i/2$ does not descend.

4.5 SU(2) over SO(3)

Let $\sigma_1, \sigma_2, \sigma_3$ be the *Pauli matrices*

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Put $X_j = -\frac{i}{2}\sigma_j$ for $j = 1, 2, 3$. Then $[X_1, X_2] = X_3$, with cyclic permutations.

Each element of SU(2) has the form

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

So SU(2) is homeomorphic to the unit sphere in $\mathbb{C}^2$, which is connected and simply connected (H §13.2).

The covering. The basis (X_1, X_2, X_3) is orthonormal for the form $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$. Each $\mathrm{Ad}(g)$, given by $\mathrm{Ad}(g)X = gXg^{-1}$, is a linear map of $\mathfrak{su}(2)$ to itself. It preserves this form, because the trace is invariant under conjugation. Hence the matrix of $\mathrm{Ad}(g)$ in the basis (X_1, X_2, X_3) lies in O(3). We identify $\mathrm{Ad}(g)$ with this matrix, so Ad maps SU(2) into O(3). Its image lies in SO(3), because $\det \circ \mathrm{Ad}$ is continuous on the connected group SU(2), takes values ± 1, and equals 1 at I.

The operator $\mathrm{ad}\, X_3$ of Lecture 2 sends $X_1 \mapsto X_2$, $X_2 \mapsto -X_1$ and $X_3 \mapsto 0$. Hence

$$\mathrm{Ad}(\exp\theta X_3) = e^{\theta\,\mathrm{ad}\,X_3} = R_3(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The differential $d\,\mathrm{Ad} = \mathrm{ad}$ (Lecture 2) is injective, since $[X_1, X] = [X_2, X] = 0$ forces $X = 0$. By dimension it is bijective onto $\mathfrak{so}(3)$.

Lecture 3 let SU(2) act on the traceless Hermitian matrices V by $\xi \mapsto U\xi U^*$. This action is Ad. Indeed, $\xi(x, y, z) = x\sigma_1 + y\sigma_2 + z\sigma_3$, so $-\frac{i}{2}\xi(x, y, z) = xX_1 + yX_2 + zX_3$. This linear isomorphism from V to $\mathfrak{su}(2)$ commutes with conjugation by U. The basis of Lecture 1 is

$A = -2X_3$, $B = -2X_2$ and $C = -2X_1$. The factor $-\frac{1}{2}$ gives $[X_1, X_2] = X_3$ with no factor two. Hence $k_\theta = \exp(\theta A) = \exp(-2\theta X_3)$, and

$$\operatorname{Ad}(k_\theta) = R_3(-2\theta).$$

This is the rotation by -2θ found in Lecture 3. The matrix g above is the U of Lecture 1, with $a = \alpha$ and $b = -\bar{\beta}$. Exercise 1.3 chose a different isomorphism onto $\mathfrak{so}(3)$. It differs from this one by conjugation with an element of $\mathrm{SO}(3)$.

If $\operatorname{Ad}(g) = I$, then g commutes with σ_3 and σ_1. The first condition makes g diagonal. The second makes it scalar. So $g = zI$ with $z^2 = \det g = 1$, and $\ker \operatorname{Ad} = \{\pm I\}$.

The image of Ad contains exp of a neighbourhood of 0 in $\mathfrak{so}(3)$, because $d\operatorname{Ad}$ is bijective. Hence the image of Ad is an open subgroup of $\mathrm{SO}(3)$. It is also closed, because its complement is a union of open cosets.

Every rotation has the form $kR_3(\theta)k^{-1}$ with $k \in \mathrm{SO}(3)$. Indeed, let $g \in \mathrm{SO}(3)$. Then $\det(g - I) = \det g \det(I - g^{\mathsf{T}}) = \det(I - g) = -\det(g - I)$, so $\det(g - I) = 0$ and g fixes a unit vector u. By transitivity on S^2 (Lecture 3), there is $k \in \mathrm{SO}(3)$ with $ke_3 = u$. Then $k^{-1}gk$ fixes e_3, so it is some $R_3(\theta)$ by the stabiliser computation of Lecture 3. The path $t \mapsto kR_3(t\theta)k^{-1}$ joins this rotation to I. Hence $\mathrm{SO}(3)$ is connected. The image of Ad is open, closed and nonempty, so Ad is onto.

Let $\Gamma = \ker \operatorname{Ad}$. By the theorem on smooth consequences, Theorem 4.5(b), Ad is a covering map. Since $\mathrm{SU}(2)$ is connected and simply connected, Ad is a universal covering group. By the uniqueness and kernel statements of Theorem 4.5(a), $\pi_1(\mathrm{SO}(3), I) \cong \Gamma = \{\pm I\}$.

The loop $t \mapsto R_3(2\pi t)$ lifts to $t \mapsto \exp(2\pi t X_3) = \operatorname{diag}(e^{-i\pi t}, e^{i\pi t})$, which ends at $-I$. By the endpoint statement of the lifting theorem, Theorem 4.3(b), this loop is not homotopic to a constant loop.

Figure 4.3 shows this loop. The axis-angle form draws $\mathrm{SO}(3)$ as a ball. The point θn, with n a unit vector and $0 \le \theta \le \pi$, stands for the rotation by θ about n. Opposite points of the boundary are one rotation, since a half turn about n equals a half turn about $-n$.

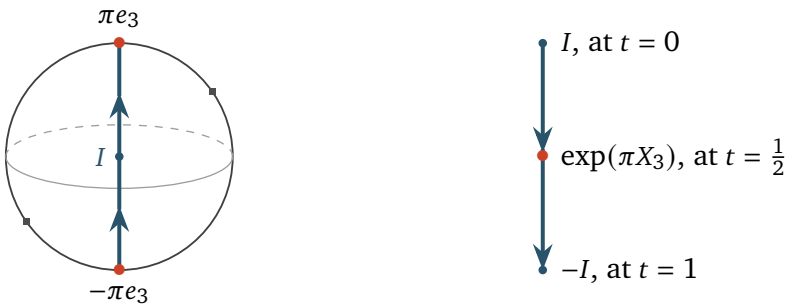


Figure 4.3: On the left, $\mathrm{SO}(3)$ as the ball of radius π. The coloured points are one rotation, and so are the square points. The heavy path is the loop $R_3(2\pi t)$: it rises from the centre to πe_3, reappears at $-\pi e_3$ and returns to the centre. On the right, the lift $t \mapsto \exp(2\pi t X_3)$ of this loop to $\mathrm{SU}(2)$ runs from I to $-I$, so it does not close.

Representations. Let V_m be the irreducible $\mathfrak{sl}(2,\mathbb{C})$-module of dimension $m+1$ and spin $m/2$. It has a basis $v_0,\dots,v_m$ with $\sigma_3 v_k = (m-2k)v_k$. Restriction to $\mathfrak{su}(2)$ gives φ_m, which integrates to Π_m on SU(2). Since $X_3 = -\frac{i}{2}\sigma_3$,

$$\Pi_m(\exp tX_3)v_k = e^{\lambda_k t}v_k, \qquad \lambda_k = -\tfrac{i}{2}(m-2k), \qquad \Pi_m(-I)v_k = e^{2\pi\lambda_k}v_k = (-1)^m v_k.$$

By the descent criterion, Corollary 4.10, $\varphi_m \circ (d\,\mathrm{Ad})^{-1}$ is the differential of a representation of SO(3) exactly when m is even. The homomorphism $t \mapsto R_3(t) = \mathrm{Ad}(\exp tX_3)$ from $\mathbb{R}$ has kernel $2\pi\mathbb{Z}$, and $\exp(2\pi X_3) = -I$. So the test is the scalar test of § 4.4 for each weight λ_k: its period $e^{2\pi\lambda_k}$ must equal 1.

Success: $m = 2$. Here $\Pi_2(-I) = I$. Hence there is a representation Π of SO(3) with $\Pi_2 = \Pi\circ\mathrm{Ad}$. This Π is equivalent to the defining representation. To see this, note that the matrices $d\,\mathrm{Ad}(X)$ also describe $\mathrm{ad}\,X$ on $\mathfrak{sl}(2,\mathbb{C})$ in the complex basis (X_1, X_2, X_3). This adjoint module has σ_3-weights $2, 0, -2$. Hence there is an invertible $E\colon \mathbb{C}^3 \to V_2$ with

$$E\,d\,\mathrm{Ad}(X)E^{-1} = \varphi_2(X).$$

The representation $g \mapsto E\,\mathrm{Ad}(g)E^{-1}$ of SU(2) has differential φ_2. By uniqueness on the connected group SU(2), it equals Π_2. Since Ad is onto, $\Pi(R) = ERE^{-1}$.

Failure: $m = 1$. Here $V_1 = \mathbb{C}^2$ and $\Pi_1(g) = g$. Suppose a representation Π of SO(3) had $d\Pi = \varphi_1 \circ (d\,\mathrm{Ad})^{-1}$. The corollary gives $\Pi \circ \mathrm{Ad} = \Pi_1$, so

$$I = \Pi\big(R_3(2\pi)\big) = \Pi\big(\mathrm{Ad}(-I)\big) = \Pi_1(-I) = -I.$$

Hence no representation of SO(3) has differential $\varphi_1 \circ (d\,\mathrm{Ad})^{-1}$.

For a spin-$\frac{1}{2}$ particle, $\mathbb{C}^2$ describes the internal spin degree of freedom. Rotations act on this space only up to sign. Lecture 13 treats such projective representations.

Self-checks

Check 1. Does the character $t \mapsto e^{i\sqrt{2}\,t}$ of $\mathbb{R}$ descend through $p(t) = e^{it}$ to U(1)?

Answer. No. Its period is $e^{2\pi i\sqrt{2}} \neq 1$, because $\sqrt{2} \notin \mathbb{Z}$.

Check 2. Take $G = \mathrm{U}(1)$ and $\varphi(i\theta) = i\theta/2$, so $f(e^{i\theta}) = e^{i\theta/2}$ for $|\theta| < \pi$. Use $W = \{e^{i\theta} : |\theta| < \pi/2\}$. Compute the transport around $\omega(t) = e^{2\pi i t}$. Which part of the path-transport argument fails?

Answer. The partition $t_j = j/5$ is admissible, since $2\pi/5 < \pi/2$. So $\Phi_\omega = f(e^{2\pi i/5})^5 = (e^{\pi i/5})^5 = -1$. The constant loop gives 1. Steps 1–3 remain valid. The Conclusion fails, because ω is not homotopic to the constant loop, by the lift computed in § 4.4. So $\Phi(1)$ is not well defined.

Check 3. Is φ_3 the differential of a representation of SU(2)? Is $\varphi_3 \circ (d\,\mathrm{Ad})^{-1}$ the differential of a representation of SO(3)?

Answer. Yes for SU(2), by Lie's integration theorem, Theorem 4.7. No for SO(3), since $\Pi_3(-I) = -I$. Equivalently, the weight $\lambda_0 = -\frac{3i}{2}$ has period $e^{-3\pi i} = -1$.

Summary

A connected Lie group G has a universal covering group $p\colon \tilde{G} \to G$. Its kernel $\Gamma \cong \pi_1(G, e)$ is discrete and central. Let φ be a representation of the Lie algebra of G on a finite-dimensional complex vector space. Then $\varphi \circ dp$ integrates to a representation $\tilde{\Pi}$ of $\tilde{G}$. The representation φ is the differential of a representation of G exactly when Γ acts trivially through $\tilde{\Pi}$. For $\mathbb{R} \to \mathrm{U}(1)$ this is the period condition $\lambda \in i\mathbb{Z}$. For $\mathrm{SU}(2) \to \mathrm{SO}(3)$ the condition holds exactly for even m, that is, for integer spin.

Reading

H is Hall, *Lie Groups, Lie Algebras, and Representations*, second edition, 2015. S is Sepanski, *Compact Lie Groups*, 2007. Knapp Beyond is Knapp, *Lie Groups Beyond an Introduction*, digital second edition, 2023.

- H §§4.7–4.8 and 5.7–5.10: group versus Lie algebra homomorphisms, universal covers, subgroups and subalgebras, Lie's third theorem, and a nonmatrix Lie group.
- H §§13.1–13.2: the fundamental group and the fundamental groups of the compact classical groups.
- S §§1.2, 4.2: topological background and the related material for compact groups.
- Knapp Beyond, Sections I.10–11, *Elementary theory of Lie groups* and *Covering groups*; Appendix B.4, Theorem B.22, *Campbell–Baker–Hausdorff formula*: general versions of the cited results.

Lecture 5

Haar Measure, Averaging and Orthogonality

Objectives. *For a finite group G, averaging with $|G|^{-1}\sum_{g\in G}$ proves unitarisability, complete reducibility and orthogonality of matrix coefficients. In this lecture we build an integral on a compact Lie group that replaces this average. It is linear and positive, has total mass 1, does not change when the integrand is shifted on either side, and is strictly positive on nonzero nonnegative functions. Sections 5.1 and 5.2 construct it. Sections 5.3 and 5.4 use no other property of it, and they record which property each argument uses.*

By the end of this lecture you will be able to:

- Integrate continuous functions on a compact manifold against a density.
- Construct the normalised Haar integral of a compact Lie group, and prove that it is right invariant and strictly positive.
- Average inner products and operators to prove unitarisability, complete reducibility and Schur orthogonality for finite-dimensional representations of compact Lie groups.
- Compute Haar integrals, an invariant-vector projector and a coefficient norm on $\mathrm{U}(1)$ and $\mathrm{SU}(2)$.

Lie groups are real, finite-dimensional, Hausdorff and second countable. Representations are complex, finite-dimensional and continuous. Inner products are antilinear in the first slot.

5.1 Densities and integration on manifolds

Let M be a smooth n-manifold with $n \geq 1$. Charts and tangent spaces are as in Lecture 1. In a chart, the Euclidean integral $\int f\,du$ changes by a Jacobian factor when the chart changes. A density supplies a factor that cancels this change. We cite the routine analytic parts of this construction.

Definition 5.1. Let V be a real vector space of dimension n. A *density* on V is a function $\delta\colon V^n \to \mathbb{R}$ such that

$$\delta(Tv_1,\dots,Tv_n) = |\det T|\,\delta(v_1,\dots,v_n) \quad \text{for all linear } T\colon V\to V \text{ and all } v_j \in V.$$

It is *positive* if it is positive on every basis.

A positive density assigns a volume to the box with edges $v_1, \dots, v_n$. The factor $|\det T|$ says how a linear map rescales volume.

Fix a basis $b = (b_j)$ and write $v_j = Tb_j$. Then $\delta(v_1, \dots, v_n) = |\det T|\,\delta(b_1, \dots, b_n)$, so δ is determined by one value. Conversely, let $[v]_b$ be the matrix whose jth column holds the coordinates of v_j. Then

$$\delta_b(v_1, \dots, v_n) = |\det [v]_b|$$

is a density, since $[Tv]_b = [T]_b[v]_b$. Hence the densities on V form the real line $\mathbb{R}\,\delta_b$, and $c\,\delta_b$ is positive exactly when $c > 0$.

In a chart x, let $\partial_{x_j}|_p \in T_pM$ be the velocity of the coordinate curve $t \mapsto x^{-1}(x(p) + te_j)$. These vectors form a basis of T_pM. A tangent vector with coordinate velocity (c_j) is $\sum_j c_j \partial_{x_j}|_p$.

Definition 5.2. A *continuous density* δ on M is a density δ_p on each T_pM such that, for every chart $x\colon U \to \Omega \subset \mathbb{R}^n$, the function ρ_x given by

$$\rho_x(x(p)) = \delta_p\big(\partial_{x_1}|_p, \dots, \partial_{x_n}|_p\big), \qquad p \in U,$$

is continuous on Ω.

Let $y\colon U' \to \Omega'$ be another chart, and put $\phi = y \circ x^{-1}$ on $x(U \cap U')$. The chain rule gives $\partial_{x_j} = \sum_i (\partial y_i/\partial x_j)\,\partial_{y_i}$. The defining property of a density, applied to this change of basis, gives

$$\rho_x(u) = |\det D\phi(u)|\,\rho_y(\phi(u)), \qquad u \in x(U \cap U'). \tag{5.1}$$

Let $f\colon M \to \mathbb{C}$ be continuous and vanish outside a compact subset of U. Define

$$\int_M f\,\delta = \int_\Omega f(x^{-1}(u))\,\rho_x(u)\,du. \tag{5.2}$$

If f vanishes outside a compact subset of $U \cap U'$, then the change-of-variables theorem and the transformation rule (5.1) show that the chart y gives the same value. For example, take spherical angles on S^2, with $p = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, and the density $\delta_p(v_1, v_2) = |\det[\,p\ v_1\ v_2\,]|$. Then the coefficient ρ_x is $\sin\theta$.

Now let M be compact. A finite *partition of unity* is a finite family of continuous functions $\chi_k \geq 0$ with $\sum_k \chi_k = 1$, each vanishing outside a compact subset of a chart domain U_k. We cite the existence of a finite partition of unity on M. For continuous f put

$$\int_M f\,\delta = \sum_k \int_M \chi_k f\,\delta.$$

Let (ψ_l) be a second partition of unity, with chart domains U_l'. Each $\chi_k\psi_l f$ vanishes outside a compact subset of $U_k \cap U_l'$, so its integral is the same in the charts on U_k and on U_l'. Since each partition sums to 1, linearity of the chart integral (5.2) shows that both sums equal $\sum_{k,l} \int_M \chi_k\psi_l f\,\delta$. Hence the integral is well defined. Moreover, it is linear, and it is nonnegative when $f \geq 0$ and δ is positive.

For a diffeomorphism $F\colon M \to M$, the *pullback* density is

$$(F^*\delta)_p(v_1, \dots, v_n) = \delta_{F(p)}(dF_p v_1, \dots, dF_p v_n).$$

The chain rule gives $(F_1 \circ F_2)^* = F_2^* F_1^*$.

Lemma 5.3. *$F^*\delta$ is a continuous density, positive if δ is. For $f \in C(M)$,*

$$\int_M (f \circ F)\, F^*\delta = \int_M f\, \delta.$$

Proof. Let x be a chart on U. Then $x \circ F$ is a chart on $F^{-1}(U)$, and dF sends its coordinate vectors to those of x. So the coefficient of $F^*\delta$ in the chart $x \circ F$ is the continuous function ρ_x. The transformation rule (5.1) then gives continuity in every chart. If f vanishes outside a compact subset of U, both integrals are the same Euclidean integral (5.2). For general f, apply this to each $\chi_k f$ and use the partition of unity $(\chi_k \circ F)$. ∎

In examples we integrate in one chart. Spherical angles on S^2 miss a half great circle, and we ignore it because a curve has zero area. The next lemma states this rule in general. Its proof is cited and uses Lebesgue theory.

One chart often covers M apart from a set contained in finitely many submanifolds of lower dimension. A set $Z \subset \mathbb{R}^n$ is *Lebesgue-null* if, for every $\varepsilon > 0$, countably many boxes of total volume less than ε cover Z. Changing a function on a Lebesgue-null set does not change its Lebesgue integral. We cite the following consequence of Lebesgue theory from Knapp Beyond, Section VIII.1. The integrability statement in the lemma, and the term *measurable* below, belong to that cited theory.

Lemma 5.4. *Let M be compact with a positive continuous density δ. Let $x : U \to \Omega$ be a chart such that $M \setminus U$ lies in a finite union of smooth submanifolds of dimension less than n. Then for $f \in C(M)$ the function $(f \circ x^{-1})\rho_x$ is Lebesgue integrable on Ω, and*

$$\int_M f\, \delta = \int_\Omega f(x^{-1}(u))\, \rho_x(u)\, du.$$

The lemma rests on two cited facts. A submanifold of dimension less than n has Lebesgue-null image in every chart. The change-of-variables formula holds for every nonnegative measurable function. For $f \geq 0$ one applies both facts to each $\chi_k f$, using the transformation rule (5.1), and sums over k. A general $f \in C(M)$ is a linear combination of four nonnegative continuous functions, namely the positive and negative parts of the real and imaginary parts of f.

5.2 The Haar integral

Let G be a compact Lie group with Lie algebra $\mathfrak{g} = T_eG$. If $\dim G = 0$, then G is discrete and compact, hence finite. Until the definition of the Haar integral, Definition 5.6, we assume $n = \dim G \geq 1$. Write $L_h(g) = hg$ and $R_h(g) = gh$. We choose a volume at e and carry it to every point by left translation, as Figure 5.1 shows. So fix a positive density δ_e on $\mathfrak{g}$ and define

$$\delta_g(v_1, \ldots, v_n) = \delta_e(dL_{g^{-1}}v_1, \ldots, dL_{g^{-1}}v_n), \qquad v_j \in T_gG. \tag{5.3}$$

Lemma 5.5. *δ is a positive continuous density, and $L_h^*\delta = \delta$ for all $h \in G$. Every continuous density δ' with $L_h^*\delta' = \delta'$ for all h is a constant multiple of δ.*

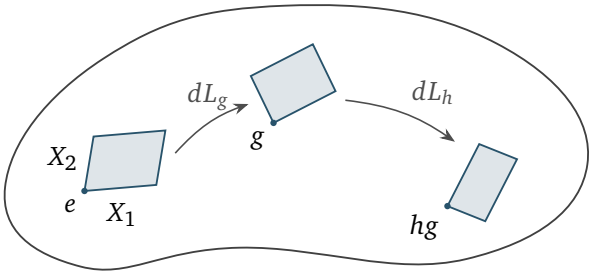


Figure 5.1: The box at e has edges X_1 and X_2. The boxes at g and hg are its images under dL_g and $dL_{hg} = dL_h \circ dL_g$. The density δ takes the same value on all three, although their shapes and areas in a chart may differ.

Proof. Each $dL_{g^{-1}} : T_gG \to \mathfrak{g}$ is a linear isomorphism, so δ_g is positive. Fix a basis (X_j) of $\mathfrak{g}$. The fields $X_j^L(g) = dL_g X_j$ are smooth and form a basis of each tangent space (Lecture 1). In a chart write $\partial_{x_i} = \sum_j a_{ji} X_j^L$. The matrix (a_{ji}) is the inverse of the coefficient matrix of the fields X_j^L in this chart. That coefficient matrix is continuous, hence so is (a_{ji}). Since $dL_{g^{-1}} X_j^L(g) = X_j$, we get $\rho_x = |\det(a_{ji})|\, \delta_e(X_1, \dots, X_n)$, which is continuous. With $v = (v_1, \dots, v_n)$,

$$(L_h^* \delta)_g(v) = \delta_{hg}(dL_h v) = \delta_e(dL_{(hg)^{-1}} dL_h v) = \delta_e(dL_{g^{-1}} v) = \delta_g(v).$$

Now let δ' be left invariant. Taking $h = g^{-1}$ gives $\delta'_g(v) = \delta'_e(dL_{g^{-1}} v)$. The densities on $\mathfrak{g}$ form a real line, so $\delta'_e = c\, \delta_e$ for a real constant c. Hence $\delta' = c\, \delta$. ∎

Definition 5.6. The *normalised Haar integral* of a compact Lie group G is the functional on $C(G)$ given by

$$\int_G f(g)\, dg = \begin{cases} \dfrac{\int_G f\, \delta}{\int_G 1\, \delta}, & \dim G \geq 1, \\[2ex] \dfrac{1}{|G|} \sum_{g \in G} f(g), & G \text{ finite}. \end{cases}$$

The denominator is positive because every ρ_x is positive. Rescaling δ_e does not change the quotient.

Theorem 5.7. *Let G be a compact Lie group, $f \in C(G)$ and $h \in G$.*

(a) The integral is linear, $\int_G f(g)\, dg \geq 0$ if $f \geq 0$, and $\int_G 1\, dg = 1$.

(b) Left invariance: $\int_G f(hg)\, dg = \int_G f(g)\, dg$.

(c) Right invariance: $\int_G f(gh)\, dg = \int_G f(g)\, dg$.

(d) Strict positivity: if $f \geq 0$ and $f(g_0) > 0$ for some g_0, then $\int_G f(g)\, dg > 0$.

Proof. For finite G all parts are clear, since translations permute G. Let $n \geq 1$. Linearity, positivity and normalisation are immediate. Left invariance follows from the pullback lemma, Lemma 5.3, with $F = L_h$, because $L_h^* \delta = \delta$ by the left-invariant density lemma, Lemma 5.5.

(c) By the pullback lemma, Lemma 5.3, $R_h^* \delta$ is a continuous density. Left and right translations commute, so

$$L_k^* R_h^* \delta = (R_h \circ L_k)^* \delta = (L_k \circ R_h)^* \delta = R_h^* L_k^* \delta = R_h^* \delta.$$

By the uniqueness part of the left-invariant density lemma, Lemma 5.5, $R_h^*\delta = c(h)\,\delta$ for a real constant $c(h)$. We compute it at e. The map $L_{h^{-1}} \circ R_h$ is $g \mapsto h^{-1}gh$, whose differential at e is $\mathrm{Ad}(h^{-1})$ (Lecture 2). So for $X = (X_1, \dots, X_n)$ in $\mathfrak{g}$,

$$(R_h^*\delta)_e(X) = \delta_h(dR_h X) = \delta_e(dL_{h^{-1}} dR_h X) = \delta_e(\mathrm{Ad}(h^{-1})X) = j(h)^{-1}\delta_e(X),$$

where $j(h) = |\det \mathrm{Ad}(h)|$. Thus $R_h^*\delta = j(h)^{-1}\delta$. Since Ad is a continuous homomorphism, so is $j\colon G \to (0,\infty)$. Its image is a compact subgroup. If it contained some $t \neq 1$, the powers t^m, $m \in \mathbb{Z}$, would be unbounded. Hence $j = 1$ and $R_h^*\delta = \delta$. Now apply the pullback lemma, Lemma 5.3, with $F = R_h$.

(d) By continuity of f, there exist $c > 0$ and an open set $U \ni g_0$ with $f > c$ on U. The translates aU with $a \in G$ cover G, so compactness (Lecture 3) gives $G = a_1U \cup \cdots \cup a_NU$. Every $g \in G$ has some $a_i^{-1}g \in U$. Since $f > c$ on U and $f \geq 0$ everywhere, we get

$$\Phi(g) = \sum_{i=1}^{N} f(a_i^{-1}g) \geq c \qquad \text{for all } g \in G.$$

By linearity and left invariance, $N\int_G f(g)\,dg = \int_G \Phi(g)\,dg$. By positivity and normalisation, $\int_G \Phi(g)\,dg \geq c > 0$. Hence $\int_G f(g)\,dg > 0$. ∎

We cite uniqueness of the normalised Haar integral from BD I.5. Let $J\colon C(G) \to \mathbb{C}$ be linear with $J(f) \geq 0$ for $f \geq 0$, $J(1) = 1$ and $J(f \circ L_h) = J(f)$ for all $f \in C(G)$ and $h \in G$. Then J is the normalised Haar integral. The left-invariant density lemma, Lemma 5.5, says that the left-invariant density is unique up to a constant factor. This theorem is stronger: it concerns any positive invariant functional on $C(G)$, which is not assumed to come from a density.

We also cite the Riesz representation theorem for positive linear functionals on $C(X)$, with X compact Hausdorff. It shows that the Haar integral is integration against a unique regular Borel probability measure. This is the *normalised Haar measure*. We integrate only continuous functions here, and this lecture does not use this measure. Lecture 9 uses Haar measure for null sets.

5.3 Averaging inner products and operators

Let W be a finite-dimensional complex vector space, and let $F\colon G \to W$ be continuous. We integrate F coordinatewise in a basis of W. The result does not depend on the basis. It satisfies $A\int_G F\,dg = \int_G AF\,dg$ for linear A. It also inherits left and right invariance from the Haar integral, Theorem 5.7.

> **Theorem 5.8** (Unitarisability). *Every representation (Π, V) of a compact Lie group G admits an inner product for which all $\Pi(g)$ are unitary.*

Proof. Choose any inner product $\langle\cdot,\cdot\rangle_0$ on V and set

$$\langle v, w\rangle = \int_G \langle \Pi(g)v, \Pi(g)w\rangle_0\,dg.$$

The integrand is continuous because Π is continuous. By linearity of the integral, the form is antilinear in v and linear in w. It is Hermitian, since the integral commutes with complex

conjugation. For $v \neq 0$, the function $g \mapsto \|\Pi(g)v\|_0^2$ is nonnegative and positive at e. Strict positivity gives $\langle v, v\rangle > 0$. Right invariance, applied to $g \mapsto \langle \Pi(g)v, \Pi(g)w\rangle_0$, gives

$$\langle \Pi(h)v, \Pi(h)w\rangle = \int_G \langle \Pi(gh)v, \Pi(gh)w\rangle_0 \, dg = \langle v, w\rangle. \qquad \blacksquare$$

Compactness cannot be dropped. The boost representation $u \mapsto B(u)$ of $\mathbb{R}$ from Lecture 2 has eigenvalues e^u and e^{-u}, so no inner product makes it unitary. The translation-invariant integral on $\mathbb{R}$ is $\int dt$, and it gives the constant 1 infinite value. The average in the proof is not available.

Corollary 5.9 (Complete reducibility)**.** *Every representation of a compact Lie group is a direct sum of irreducible subrepresentations.*

Proof. Let (Π, V) be a representation, and induct on $\dim V$. If V is irreducible, there is nothing to prove. Otherwise let U be a proper nonzero invariant subspace. Use the invariant inner product of unitarisability, Theorem 5.8. For $u \in U$ and $w \in U^\perp$,

$$\langle u, \Pi(g)w\rangle = \langle \Pi(g)^* u, w\rangle = \langle \Pi(g^{-1})u, w\rangle = 0.$$

So $U^\perp$ is invariant and $V = U \oplus U^\perp$. Apply induction to both summands. $\blacksquare$

The *invariant vectors* of a representation (Π, V) form the subspace

$$V^G = \{v \in V : \Pi(g)v = v \text{ for all } g \in G\}.$$

Proposition 5.10 (Invariant-vector projector)**.** *Let (Π, V) be a representation of a compact Lie group G and $P = \int_G \Pi(g)\, dg$. Then $\Pi(h)P = P = P\Pi(h)$ for all h, and P is a projection with image V^G. If all $\Pi(g)$ are unitary, P is the orthogonal projection onto V^G.*

Proof. Left invariance gives $\Pi(h)P = \int_G \Pi(hg)\, dg = P$. Right invariance gives $P\Pi(h) = P$. The first identity shows $\operatorname{im} P \subset V^G$. For $v \in V^G$ we have $Pv = \int_G v\, dg = v$. Hence $P^2 = P$ and $\operatorname{im} P = V^G$. Now let Π be unitary. As in the proof of complete reducibility, Corollary 5.9, the subspace $U = (V^G)^\perp$ is invariant. Since $\Pi(g)U \subset U$ for all g, also $PU \subset U \cap V^G = 0$. So $U \subset \ker P$. By rank–nullity, $\dim \ker P = \dim V - \dim V^G = \dim U$. Hence $\ker P = (V^G)^\perp$. $\blacksquare$

Let (σ, W) be a second representation. Then $\operatorname{Hom}(V, W)$ carries the representation in which g acts by $E \mapsto \sigma(g)E\Pi(g)^{-1}$. The invariant vectors of this representation are the intertwiners. By the invariant-vector projector, Proposition 5.10, applied to this representation, for every linear $E\colon V \to W$,

$$E^\natural = \int_G \sigma(g)E\Pi(g)^{-1}\, dg \in \operatorname{Hom}_G(V, W). \tag{5.4}$$

5.4 Orthogonality of matrix coefficients

Definition 5.11. Let Π act by unitary operators on a finite-dimensional inner product space V. The *matrix coefficient* of $v, w \in V$ is the function $c^{\Pi}_{v,w}(g) = \langle v, \Pi(g)w\rangle$ on G.

It is antilinear in v and linear in w. Let $D(g)$ be the matrix of $\Pi(g)$ in an orthonormal basis (e_i), so $\Pi(g)e_j = \sum_i D_{ij}(g)e_i$. Then $D_{ij} = c^{\Pi}_{e_i,e_j}$. On $C(G)$ we use

$$\langle f_1, f_2\rangle_{L^2} = \int_G \overline{f_1(g)}\, f_2(g)\, dg.$$

This inner product is also antilinear in the first slot. It is positive definite by strict positivity.

We import Schur's lemma from finite group theory. Its proof is algebraic and holds for any group. For irreducible finite-dimensional complex representations, $\operatorname{Hom}_G(V,W) = 0$ if $V \not\cong W$, and $\operatorname{Hom}_G(V,V) = \mathbb{C}\,\mathrm{id}$.

Theorem 5.12 (Schur orthogonality). *Let (Π, V) and (σ, W) be irreducible unitary representations of a compact Lie group G, with $d = \dim V$.*

(a) If $\Pi \not\cong \sigma$, then $\langle c^{\Pi}_{v,w}, c^{\sigma}_{v',w'}\rangle_{L^2} = 0$ for $v, w \in V$ and $v', w' \in W$.

(b) For $v, w, v', w' \in V$,

$$\int_G \overline{\langle v, \Pi(g)w\rangle}\, \langle v', \Pi(g)w'\rangle\, dg = \frac{1}{d}\, \langle v', v\rangle\, \langle w, w'\rangle.$$

Proof. The key step writes the integrand as a coefficient of an averaged operator. In case (b) put $\sigma = \Pi$ and $W = V$. Define $E\colon V \to W$ by $Eu = \langle w, u\rangle\, w'$. Here $w \in V$ and $w' \in W$, so $E \in \operatorname{Hom}(V,W)$. Since $\Pi(g)^{-1} = \Pi(g)^{*}$,

$$\langle v', \sigma(g) E \Pi(g)^{-1} v\rangle = \langle w, \Pi(g)^{*} v\rangle \langle v', \sigma(g) w'\rangle = \overline{\langle v, \Pi(g)w\rangle}\, \langle v', \sigma(g)w'\rangle.$$

The map $u \mapsto \langle v', u\rangle$ is linear, so it commutes with the integral. Integrating over G therefore gives

$$\int_G \langle v', \sigma(g) E \Pi(g)^{-1} v\rangle\, dg = \langle v', E^{\natural} v\rangle,$$

with the averaged operator $E^{\natural}$ from (5.4). So the left sides in both parts equal $\langle v', E^{\natural} v\rangle$. In case (a), where $\Pi \not\cong \sigma$, Schur's lemma gives $E^{\natural} = 0$. In case (b), where $\sigma = \Pi$, Schur's lemma gives $E^{\natural} = \lambda\,\mathrm{id}$ for some $\lambda \in \mathbb{C}$. Trace is linear and conjugation invariant, so in an orthonormal basis (e_i)

$$d\lambda = \int_G \operatorname{tr}\!\big(\Pi(g) E \Pi(g)^{-1}\big)\, dg = \operatorname{tr} E = \sum_i \langle w, e_i\rangle \langle e_i, w'\rangle = \langle w, w'\rangle.$$

Hence $\langle v', E^{\natural} v\rangle = d^{-1} \langle w, w'\rangle \langle v', v\rangle$. ∎

Corollary 5.13. *Let (Π, V) and (σ, W) be as in Schur orthogonality, Theorem 5.12, with $d = \dim V$. Let D^{Π} and D^{σ} be their matrices in orthonormal bases of V and W. Then*

$$\int_G \overline{D^{\Pi}_{ij}(g)}\, D^{\Pi}_{kl}(g)\, dg = \frac{\delta_{ik}\delta_{jl}}{d}, \qquad \int_G \overline{D^{\Pi}_{ij}(g)}\, D^{\sigma}_{kl}(g)\, dg = 0 \quad (\Pi \not\cong \sigma).$$

In particular $\|c^{\Pi}_{v,w}\|^2_{L^2} = \|v\|^2 \|w\|^2 / d$ for $v, w \in V$.

Proof. Take $v = e_i$, $w = e_j$, $v' = e_k$ and $w' = e_l$ in Schur orthogonality, Theorem 5.12. For the norm identity, take $v' = v$ and $w' = w$ in the relation for a single representation. ∎

In physics, for $G = \mathrm{SU}(2)$ and the module V_m, the entries in an orthonormal weight basis are the Wigner D-functions of spin $m/2$, with $d = m + 1$.

Put $j = i$ and $l = k$, and sum over i and k. The character $\chi_\Pi = \operatorname{tr} \Pi$ is the sum of the diagonal entries D^{Π}_{ii}. Hence the characters of inequivalent irreducible representations are orthonormal. For irreducible unitary Π and σ,

$$\int_G \overline{\chi_\Pi(g)}\, \chi_\sigma(g)\, dg = \begin{cases} 1, & \Pi \cong \sigma, \\ 0, & \Pi \not\cong \sigma. \end{cases}$$

The table records which properties of the integral each argument uses. There, “yes” means that the proof uses the property. Complete reducibility has the same column as unitarisability, since its proof uses only the invariant inner product.

Property	Unitarisability	Projector	Schur orthogonality
Linearity	yes	yes	yes
$\int_G 1\, dg = 1$	only via strict positivity	yes ($Pv = v$)	yes (trace step)
Left invariance	only via strict positivity	yes ($\Pi(h)P = P$)	yes ($E^\natural$ intertwines)
Right invariance	yes	yes ($P\Pi(h) = P$)	no
Strict positivity	yes	no	no

5.5 Examples: the groups U(1) and SU(2)

In each example the single-chart lemma, Lemma 5.4, lets us integrate over one chart.

Example 5.14 (The circle)**.** For $\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\}$ use the parametrisation $\theta \mapsto e^{i\theta}$, $0 < \theta < 2\pi$. Its inverse is a chart, whose domain misses only the point 1. At $z = e^{i\theta}$ we have $\partial_\theta = iz$ and $dL_{z^{-1}}(iz) = i$. Taking $\delta_e(i) = 1$ gives $\rho = 1$, so the integral of 1 against δ is 2π. Hence

$$\int_{\mathrm{U}(1)} f(z)\, dz = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta})\, d\theta.$$

The characters $\chi_n(z) = z^n$, $n \in \mathbb{Z}$, are irreducible unitary representations, and

$$\int_{\mathrm{U}(1)} \overline{z^m}\, z^n\, dz = \frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)\theta}\, d\theta = \delta_{mn}.$$

Example 5.15 (SU(2))**.** Every $g \in \mathrm{SU}(2)$ has the form

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

So $g \mapsto (a, b)$ identifies $\mathrm{SU}(2)$ with $S^3 \subset \mathbb{C}^2 = \mathbb{R}^4$. This map and its inverse are restrictions of real linear maps, so the identification is a diffeomorphism. Differentiating $|\omega|^2 = 1$ gives $T_\omega S^3 \subset \omega^\perp$, and both have dimension 3. So $T_\omega S^3 = \omega^\perp$.

The first column of hg is $h(a,b)^T$. Thus L_h acts on S^3 by the unitary matrix h, which is orthogonal on $\mathbb{R}^4$. For $\omega \in S^3$ and $v_j \in \omega^\perp$ set

$$\nu_\omega(v_1,v_2,v_3) = \bigl|\det[\,\omega\ v_1\ v_2\ v_3\,]\bigr|,$$

a real 4×4 determinant. Extending $T\colon \omega^\perp \to \omega^\perp$ by $T\omega = \omega$ shows that ν_ω is a positive density. In a chart with inverse $u \mapsto \omega(u)$, the coefficient of ν is $|\det[\,\omega\ \partial_1\omega\ \partial_2\omega\ \partial_3\omega\,]|$, which is continuous.

For orthogonal A we get $\nu_{A\omega}(Av) = |\det A|\,\nu_\omega(v) = \nu_\omega(v)$. Hence $L_h^*\nu = \nu$. By the uniqueness part of the left-invariant density lemma, Lemma 5.5, $\nu = c\,\delta$ for the left-invariant density δ of (5.3), with $c > 0$. The density ν is the usual volume of the round sphere S^3, and left multiplication rotates this sphere. So the Haar integral on SU(2) is the uniform average over S^3.

Use Hopf coordinates $a = \cos\eta\, e^{i\xi_1}$ and $b = \sin\eta\, e^{i\xi_2}$, with $0 < \eta < \pi/2$ and $0 < \xi_1, \xi_2 < 2\pi$. This parametrisation is injective. The points it misses lie in the circles $\{a = 0\}$ and $\{b = 0\}$ and in the 2-spheres $\{\operatorname{Im} a = 0\}$ and $\{\operatorname{Im} b = 0\}$. In $\mathbb{R}^4$,

$$\begin{aligned}
\omega &= (\cos\eta\cos\xi_1,\ \cos\eta\sin\xi_1,\ \sin\eta\cos\xi_2,\ \sin\eta\sin\xi_2),\\
\partial_\eta\omega &= (-\sin\eta\cos\xi_1,\ -\sin\eta\sin\xi_1,\ \cos\eta\cos\xi_2,\ \cos\eta\sin\xi_2),\\
\partial_{\xi_1}\omega &= (-\cos\eta\sin\xi_1,\ \cos\eta\cos\xi_1,\ 0,\ 0),\\
\partial_{\xi_2}\omega &= (0,\ 0,\ -\sin\eta\sin\xi_2,\ \sin\eta\cos\xi_2).
\end{aligned}$$

These vectors are mutually orthogonal, with lengths $1, 1, \cos\eta, \sin\eta$. So the parametrisation is an injective immersion of an open box. By the inverse function theorem its inverse is a chart. The coefficient of ν in this chart is the product of these lengths, $\cos\eta\sin\eta$. So the integral of 1 against ν is $\frac{1}{2}(2\pi)^2 = 2\pi^2$, the volume of the unit 3-sphere. Therefore

$$\int_{\mathrm{SU}(2)} f(g)\,dg = \frac{1}{2\pi^2}\int_0^{2\pi}\!\!\int_0^{2\pi}\!\!\int_0^{\pi/2} f\,\cos\eta\sin\eta\,d\eta\,d\xi_1\,d\xi_2. \tag{5.5}$$

The monomial $a^p\bar a^q b^r \bar b^s$ carries the phase $e^{i(p-q)\xi_1}e^{i(r-s)\xi_2}$. So its integral vanishes unless $p = q$ and $r = s$. Also

$$\int_{\mathrm{SU}(2)} |a|^2\,dg = 2\int_0^{\pi/2}\cos^3\eta\sin\eta\,d\eta = 2\Bigl[-\frac{\cos^4\eta}{4}\Bigr]_0^{\pi/2} = \frac{1}{2}, \qquad \int_{\mathrm{SU}(2)} |b|^2\,dg = 1 - \frac{1}{2} = \frac{1}{2}.$$

A coefficient norm. The defining representation $V_1 = \mathbb{C}^2$ is irreducible of dimension 2. In the standard basis $D(g) = g$, so $D_{11} = a$ and $D_{12} = -\bar b$. The matrix-entry relations of Corollary 5.13 predict $\int |D_{11}|^2\,dg = \frac{1}{2}$ and $\int \overline{D_{11}}D_{12}\,dg = 0$. Both agree with the direct computation, since $\overline{D_{11}}D_{12} = -\bar a\bar b$ has a nonzero phase.

An invariant-vector projector. Let SU(2) act on $V_1 \otimes V_1$ by $g \otimes g$. Use the orthonormal basis $e_1\otimes e_1$, $e_1\otimes e_2$, $e_2\otimes e_1$, $e_2\otimes e_2$. The entry of $g\otimes g$ in row (i,k) and column (j,l) is $g_{ij}g_{kl}$. The entries are quadratic monomials in $a, \bar a, b, \bar b$. By the vanishing rule for monomials above, only the products $a\bar a$ and $b\bar b$ have nonzero integral. These products occur only in rows and columns $(1,2)$ and $(2,1)$:

$$g_{11}g_{22} = |a|^2, \qquad g_{12}g_{21} = -|b|^2, \qquad g_{21}g_{12} = -|b|^2, \qquad g_{22}g_{11} = |a|^2.$$

Let P be the averaging projector for this action. Writing D for matrices in this basis, we obtain

$$D(P) = \int_{\mathrm{SU}(2)} g \otimes g \, dg = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

With $\varepsilon = e_1 \otimes e_2 - e_2 \otimes e_1$ we get $Px = \frac{1}{2}\langle \varepsilon, x \rangle \, \varepsilon$. This is the orthogonal projection onto $\mathbb{C}\varepsilon$. In physics, ε is proportional to the singlet state of two spin-one-half particles. By the invariant-vector projector, Proposition 5.10, $(V_1 \otimes V_1)^{\mathrm{SU}(2)} = \mathbb{C}\varepsilon$.

Self-checks

1. Which property of the Haar integral makes the form in the proof of unitarisability, Theorem 5.8, positive definite? Which makes it invariant?

   Answer. Strict positivity gives $\langle v, v \rangle > 0$ for $v \neq 0$. Right invariance gives invariance. Left invariance and normalisation enter only through the proof of strict positivity.

2. Let Π be a nontrivial irreducible representation of a compact Lie group. What is $\int_G \Pi(g) \, dg$?

   Answer. It is 0. By the invariant-vector projector, Proposition 5.10, its image is the subrepresentation V^G. Irreducibility gives $V^G = 0$ or $V^G = V$. If $V^G = V$, then every $\Pi(g)$ is the identity, so every line in V is invariant. Irreducibility then forces $\dim V = 1$, so Π is the trivial representation, contrary to the hypothesis. Hence $V^G = 0$, and the integral is 0.

3. Explain $\int_{\mathrm{SU}(2)} a \, dg = 0$ in two ways.

   Answer. First, in Hopf coordinates the integrand carries the phase $e^{i\xi_1}$, so the integral over ξ_1 vanishes. Second, $a = D^{V_1}_{11}$, and the constant 1 is the matrix coefficient of the trivial representation V_0. Since $V_1 \not\cong V_0$, the relation for inequivalent representations in Corollary 5.13 gives $\int_G \overline{1} \, a \, dg = 0$.

Summary

On a compact Lie group G of positive dimension, left translation of one positive density on $\mathfrak{g}$ gives the Haar integral. Compactness forces $|\det \mathrm{Ad}| = 1$, which gives right invariance. A covering argument by translates gives strict positivity. With these properties, the finite-group proofs of unitarisability, complete reducibility and Schur orthogonality carry over with the sum replaced by the integral. By the matrix-entry relations, Corollary 5.13, the matrix entries of an irreducible unitary representation of dimension d in an orthonormal basis are pairwise L^2-orthogonal, and each has squared norm $1/d$. Completeness of matrix coefficients is the Peter–Weyl theorem of Lecture 10.

Reading

- Bröcker and tom Dieck, *Representations of Compact Lie Groups* (BD), Sections I.5 and II.4.

- Sepanski, *Compact Lie Groups*, §1.4, §2.2 and §3.1.
- Knapp, *Lie Groups Beyond an Introduction* (Knapp Beyond), digital second edition, 2023, §VIII.1, for integration and sets of measure zero.

Lecture 6

Compact Structure and Maximal Tori

Objectives. *This lecture explains why conjugacy in a compact connected group can be studied on a torus. First, an averaged inner product splits the Lie algebra. Next, maximal tori meet every conjugacy class. Finally, a finite Weyl group records which elements of the torus are conjugate.*

By the end of this lecture you will be able to:

- Average an inner product and prove $\mathfrak{k} = \mathfrak{z}(\mathfrak{k}) \oplus [\mathfrak{k}, \mathfrak{k}]$ with $[\mathfrak{k}, \mathfrak{k}]$ semisimple.
- Prove that maximal tori are conjugate, explain the strategy of the torus theorem, and deduce that exp is surjective for compact connected groups.
- Compute $N_K(T)/T$ for $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ and match it with the algebraic Weyl group.
- Locate where these statements fail for $\mathrm{SL}(2, \mathbb{R})$.

Throughout, K is a compact Lie group with Lie algebra $\mathfrak{k}$. Its Haar probability measure dg is left and right invariant (Lecture 5). Connectedness of K is assumed only where a statement says so.

6.1 An invariant inner product and the splitting

Lemma 6.1. *There is an inner product $\langle \cdot, \cdot \rangle$ on $\mathfrak{k}$ such that for all $g \in K$ and $X, Y, Z \in \mathfrak{k}$*

$$\langle \mathrm{Ad}(g)X, \mathrm{Ad}(g)Y \rangle = \langle X, Y \rangle, \qquad \langle [Z, X], Y \rangle = -\langle X, [Z, Y] \rangle. \tag{6.1}$$

Proof. Choose any inner product $\langle \cdot, \cdot \rangle_0$ on $\mathfrak{k}$ and set

$$\langle X, Y \rangle = \int_K \langle \mathrm{Ad}(g)X, \mathrm{Ad}(g)Y \rangle_0 \, dg.$$

The integrand is continuous in g. So the form is well defined, symmetric and bilinear. For $X \neq 0$ the continuous function $g \mapsto \| \mathrm{Ad}(g)X \|_0^2$ is positive on the compact set K. It has a positive minimum m, so $\langle X, X \rangle \geq m > 0$. Right invariance of dg gives

$$\langle \mathrm{Ad}(h)X, \mathrm{Ad}(h)Y \rangle = \int_K \langle \mathrm{Ad}(gh)X, \mathrm{Ad}(gh)Y \rangle_0 \, dg = \langle X, Y \rangle.$$

For the second identity, recall $\mathrm{Ad}(\exp tZ) = e^{t \, \mathrm{ad} Z}$ from Lecture 2. By the first identity, the function $t \mapsto \langle e^{t \, \mathrm{ad} Z} X, e^{t \, \mathrm{ad} Z} Y \rangle$ is constant. By the product rule, its derivative at $t = 0$ is

$$0 = \langle [Z, X], Y \rangle + \langle X, [Z, Y] \rangle.$$

∎

In other words, every $\operatorname{ad} Z$ is skew-symmetric for $\langle \cdot, \cdot \rangle$. From now on we fix such an inner product.

Proposition 6.2. *Let $\mathfrak{z} = \mathfrak{z}(\mathfrak{k})$ be the centre and $\mathfrak{s} = [\mathfrak{k}, \mathfrak{k}]$.*

(a) $\mathfrak{s}^{\perp} = \mathfrak{z}$. Hence $\mathfrak{k} = \mathfrak{z} \oplus \mathfrak{s}$, an orthogonal sum of ideals.

(b) The Killing form B of $\mathfrak{s}$ is negative definite. Hence $\mathfrak{s}$ is semisimple.

Proof. (a) For $Z, X, Y \in \mathfrak{k}$, the skew-symmetry identity in (6.1) gives

$$\langle Z, [X, Y] \rangle = -\langle [X, Z], Y \rangle.$$

So $Z \perp \mathfrak{s}$ exactly when $[X, Z] \perp Y$ for all X, Y. This means $[X, Z] = 0$ for all X, that is, $Z \in \mathfrak{z}$. Both $\mathfrak{z}$ and $\mathfrak{s}$ are ideals. Since $\mathfrak{k}$ is finite-dimensional, $\mathfrak{k} = \mathfrak{s}^{\perp} \oplus \mathfrak{s} = \mathfrak{z} \oplus \mathfrak{s}$.

(b) First, $\mathfrak{z}(\mathfrak{s}) = 0$. An element of $\mathfrak{z}(\mathfrak{s})$ commutes with $\mathfrak{s}$ and with $\mathfrak{z}$, hence with $\mathfrak{k}$ by the splitting in (a). So it lies in $\mathfrak{z} \cap \mathfrak{s} = 0$.

Now let $X \in \mathfrak{s}$ and $A = \operatorname{ad} X|_{\mathfrak{s}}$. Since $\mathfrak{s}$ is an ideal, A maps $\mathfrak{s}$ to itself. The map A is skew-symmetric, since every $\operatorname{ad} X$ is. So in an orthonormal basis $e_1, \dots, e_m$ of $\mathfrak{s}$ the matrix of A satisfies $A^{\mathsf{T}} = -A$. Hence

$$B(X, X) = \operatorname{tr}(A^2) = -\operatorname{tr}(A^{\mathsf{T}} A) = -\sum_{i=1}^{m} \|Ae_i\|^2 \le 0.$$

Equality forces $A = 0$, so $X \in \mathfrak{z}(\mathfrak{s}) = 0$. Thus B is negative definite.

The complexification of B is the Killing form of $\mathfrak{s}_{\mathbb{C}}$, and it is nondegenerate. By the complex Cartan criterion from the Lie Algebras course, $\mathfrak{s}_{\mathbb{C}}$ is semisimple. A nonzero solvable real ideal of $\mathfrak{s}$ would complexify to a nonzero solvable ideal of $\mathfrak{s}_{\mathbb{C}}$. Hence $\mathfrak{s}$ is semisimple. ∎

For example, let $\mathfrak{k} = \mathfrak{u}(n)$. The form $-\operatorname{tr}(XY)$ satisfies both identities in (6.1) without averaging (Exercise 6.1). The proof of the splitting proposition, Proposition 6.2, uses only these identities, so it applies to this form. Here $\mathfrak{z} = i\mathbb{R}I$ and $\mathfrak{s} = \mathfrak{su}(n)$. The splitting of $X \in \mathfrak{u}(n)$ is

$$X = \tfrac{1}{n}(\operatorname{tr} X)\, I + \Big(X - \tfrac{1}{n}(\operatorname{tr} X)\, I\Big) \in i\mathbb{R}I \oplus \mathfrak{su}(n).$$

6.2 Maximal tori and their conjugacy

Definition 6.3. A *torus* is a compact connected abelian Lie group. A *maximal torus* of K is a torus subgroup $T \subseteq K$ not properly contained in another torus subgroup of K.

For example, the diagonal unitary matrices are a maximal torus of $\mathrm{U}(n)$, and the circle $T = \{k_\theta\}$ of Lecture 3 is a maximal torus of $\mathrm{SU}(2)$. § 6.5 proves this.

Let T be a torus with Lie algebra $\mathfrak{t}$. For $t \in T$ and $Y \in \mathfrak{t}$, differentiating $t \exp(sY) t^{-1} = \exp(sY)$ at $s = 0$ gives $\operatorname{Ad}(t)Y = Y$. Putting $t = \exp(rZ)$ with $Z \in \mathfrak{t}$ and differentiating at $r = 0$ gives $[Z, Y] = 0$. So $\mathfrak{t}$ is abelian, and $\exp(X + Y) = \exp X \exp Y$ on $\mathfrak{t}$ (Lecture 2).

Thus $\exp(\mathfrak{t})$ is a subgroup containing a neighbourhood of e (Lecture 2). Such a subgroup is open. It is also closed, because its complement is a union of open cosets. As T is connected, $\exp\colon \mathfrak{t} \to T$ is surjective.

We cite two further facts (H §11.1; S Ch. 5), and Lecture 7 proves the first:

$$T \cong \mathfrak{t}/\Lambda \text{ with } \Lambda = \ker \exp \text{ a lattice}; \qquad \overline{\exp(\mathbb{R}H)} = T \text{ for some } H \in \mathfrak{t}. \tag{6.2}$$

The second is Kronecker's theorem. We call such an H *generic*. Kronecker's theorem also gives $t_0 \in T$ whose powers are dense in T. For example, on the two-torus T^2 of Lecture 3 the irrational line L_α is dense, by the irrational-line proposition. So a nonzero element of $\mathfrak{t}$ tangent to L_α is generic.

Lemma 6.4. *A torus $T \subseteq K$ is maximal if and only if $\mathfrak{t}$ is a maximal abelian subalgebra of $\mathfrak{k}$. Maximal tori exist.*

Proof. Let $\mathfrak{t}$ be maximal abelian, and let $T \subseteq T'$ with T' a torus. Then $\mathfrak{t} \subseteq \mathfrak{t}'$ with $\mathfrak{t}'$ abelian, so $\mathfrak{t} = \mathfrak{t}'$. Hence $T' = \exp(\mathfrak{t}') = \exp(\mathfrak{t}) = T$.

Conversely, let T be maximal, and let $\mathfrak{a}$ be an abelian subalgebra with $\mathfrak{t} \subseteq \mathfrak{a}$. Since $\mathfrak{a}$ is abelian, $\exp(\mathfrak{a})$ is a subgroup, as for $\mathfrak{t}$ above. This subgroup is connected and abelian. Its closure A is a Lie group by the closed subgroup theorem (Lecture 3). The group A is compact, connected and abelian, so it is a torus containing T. Since T is maximal, $A = T$. Hence $\mathfrak{a} \subseteq \mathrm{Lie}(A) = \mathfrak{t}$.

For existence, take a maximal abelian subalgebra $\mathfrak{a}$. As in the second paragraph, $A = \overline{\exp(\mathfrak{a})}$ is a torus. Its Lie algebra is abelian and contains $\mathfrak{a}$, so it equals $\mathfrak{a}$. By the first paragraph, A is a maximal torus. ∎

Theorem 6.5 (Torus theorem; BD §IV.1; S Ch. 5; H §§11.2–11.5)**.** *Let K be compact and connected, and let T be a maximal torus.*

(a) Every $g \in K$ lies in xTx^{-1} for some $x \in K$.

(b) Every maximal torus of K equals xTx^{-1} for some $x \in K$.

Take $K = \mathrm{U}(n)$ with T the diagonal matrices. Then part (a) is the spectral theorem for unitary matrices (Self-check 2). Part (b) says that a maximal torus is diagonal after a change of orthonormal basis. So the theorem extends diagonalisation to every compact connected group.

Hall (H) proves this for compact matrix Lie groups, and Bröcker–tom Dieck (BD) for general compact connected Lie groups. Part (b), the conjugacy of maximal tori, and a Lie algebra form of part (a) have a short complete proof.

Proposition 6.6. *Let T be a maximal torus of K. For every $X \in \mathfrak{k}$ there is $x \in K$ with $\mathrm{Ad}(x)X \in \mathfrak{t}$. Moreover, any two maximal tori of K are conjugate.*

We rotate X by $\mathrm{Ad}(K)$ until it is as close as possible to a generic element $H \in \mathfrak{t}$, using the compactness of K. At the best position the first derivative vanishes, and this says that the rotated element commutes with H.

Proof. Choose a generic $H \in \mathfrak{t}$. Define

$$f \colon K \to \mathbb{R}, \qquad f(x) = \langle \mathrm{Ad}(x)X, H \rangle.$$

It is continuous, so it attains its maximum on the compact group K at some x_0. Put $X_0 = \mathrm{Ad}(x_0)X$ and fix $Y \in \mathfrak{k}$. By Lecture 2,

$$f\big(\exp(sY)x_0\big) = \langle \mathrm{Ad}(\exp sY)X_0, H\rangle = \langle e^{s\,\mathrm{ad}\,Y}X_0, H\rangle.$$

Since f is maximal at x_0, this function of s is maximal at $s = 0$. Hence its derivative $\langle [Y, X_0], H\rangle$ at $s = 0$ vanishes. Using the second identity in (6.1) twice, together with antisymmetry of the bracket, we get

$$0 = \langle [Y, X_0], H\rangle = -\langle X_0, [Y, H]\rangle = \langle X_0, [H, Y]\rangle = -\langle [H, X_0], Y\rangle.$$

Since Y is arbitrary, $[H, X_0] = 0$.

Next we pass from the single element H to all of $\mathfrak{t}$. This step uses genericity of H. Consider

$$S = \{t \in T : \mathrm{Ad}(t)X_0 = X_0\}.$$

It is a closed subgroup of T, since $t \mapsto \mathrm{Ad}(t)X_0$ is continuous. From $[H, X_0] = 0$, every term after the first in the exponential series vanishes. So

$$\mathrm{Ad}(\exp sH)X_0 = e^{s\,\mathrm{ad}\,H}X_0 = X_0 \qquad (s \in \mathbb{R}).$$

Thus S contains $\exp(\mathbb{R}H)$, and hence its closure T. Now let $Z \in \mathfrak{t}$. Differentiating $\mathrm{Ad}(\exp sZ)X_0 = X_0$ at $s = 0$ gives $[Z, X_0] = 0$. So $\mathfrak{t} + \mathbb{R}X_0$ is an abelian subalgebra containing $\mathfrak{t}$. By the characterisation of maximal tori in Lemma 6.4, $\mathfrak{t}$ is maximal abelian, so $X_0 \in \mathfrak{t}$.

Now let T' be a maximal torus with generic element H'. Choose x with $\mathrm{Ad}(x)H' \in \mathfrak{t}$. Then

$$x \exp(sH')x^{-1} = \exp\big(s\,\mathrm{Ad}(x)H'\big) \in T \qquad (s \in \mathbb{R}).$$

Conjugation by x is a homeomorphism and T is closed. Taking closures, $xT'x^{-1} \subseteq T$. The left side is a maximal torus, so equality holds. ∎

Definition 6.7. Let $T \subseteq K$ be a maximal torus. Its *normaliser* is $N_K(T) = \{n \in K : nTn^{-1} = T\}$. The *Weyl group* is $W(K, T) = N_K(T)/T$.

The group $W(K, T)$ acts on T by $nT \cdot t = ntn^{-1}$ and on $\mathfrak{t}$ by $\mathrm{Ad}(n)$. For $n \in N_K(T)$ and $H \in \mathfrak{t}$, the curve $s \mapsto n\exp(sH)n^{-1}$ lies in T. Differentiating it at $s = 0$ gives $\mathrm{Ad}(n)H \in \mathfrak{t}$. Since $\mathrm{Ad}(n)$ is invertible and $\mathfrak{t}$ is finite-dimensional, $\mathrm{Ad}(n)$ maps $\mathfrak{t}$ onto itself. These actions are well defined on cosets because T is abelian.

Lemma 6.8. *Let T be a maximal torus of K and $N = N_K(T)$. Then the identity component N_0 equals T, and $W(K, T)$ is finite.*

Proof. The group N is closed in K, so it is a compact Lie group (Lecture 3). Let $n \in N$ and $H \in \mathfrak{t}$. Then

$$n\exp(sH)n^{-1} = \exp\big(s\,\mathrm{Ad}(n)H\big) \in T \qquad (s \in \mathbb{R}).$$

Differentiating at $s = 0$ gives $\mathrm{Ad}(n)H \in \mathfrak{t}$. If $\exp H = e$, the display gives $\exp(\mathrm{Ad}(n)H) = e$. So $A_n = \mathrm{Ad}(n)|_{\mathfrak{t}}$ maps the lattice Λ into itself.

First, N_0 centralises T. Choose a basis $\lambda_1, \dots, \lambda_r$ of $\mathfrak{t}$ inside Λ. Each map $n \mapsto A_n\lambda_i$ is continuous with values in the discrete set Λ. Since N_0 is connected, this map is constant on

N_0, equal to its value λ_i at e. So $A_n = I$ for $n \in N_0$. Then $n \exp(H) n^{-1} = \exp H$ for all $H \in \mathfrak{t}$. Since $\exp(\mathfrak{t}) = T$, every $n \in N_0$ commutes with every element of T.

Second, $\mathrm{Lie}(N) = \mathfrak{t}$. Let $X \in \mathrm{Lie}(N)$. Then $\exp(sX) \in N_0$ for all s. By the first step, $\mathrm{Ad}(\exp sX)H = H$ for $H \in \mathfrak{t}$. Differentiating at $s = 0$ gives $[X, H] = 0$. So $\mathfrak{t} + \mathbb{R}X$ is abelian, and $X \in \mathfrak{t}$ because $\mathfrak{t}$ is maximal abelian (Lemma 6.4). The connected group N_0 is generated by $\exp(\mathrm{Lie}\, N)$ (Lecture 2). Hence N_0 is generated by $\exp(\mathfrak{t}) = T$, so $N_0 = T$.

Finally, N_0 is open in N (Lecture 2). The cosets of N_0 form a cover of the compact group N by disjoint open sets. Hence there are finitely many cosets. Since $N_0 = T$, the group $W(K, T) = N/N_0$ is finite. ∎

6.3 An outline of the torus theorem via mapping degree

Since $x \exp(X) x^{-1} = \exp(\mathrm{Ad}(x)X)$, the conjugacy statement of Proposition 6.6 shows that every element of $\exp(\mathfrak{k})$ lies in a conjugate of T. Conversely, every conjugate of an element of $T = \exp(\mathfrak{t})$ is an exponential. So part (a) of the torus theorem, Theorem 6.5(a), is equivalent to surjectivity of exp. Surjectivity of exp needs a global argument. We outline the proof in Hall (H §§11.3–11.5).

For connected K define

$$\Phi \colon K/T \times T \to K, \qquad \Phi(xT, t) = xtx^{-1}.$$

This is well defined because T is abelian. The source $K/T \times T$ and the target K are compact connected manifolds of dimension $\dim K$ (Lecture 3).

We count how many times Φ covers a point, with signs. A point y is a *regular value* of a smooth map if the differential of the map is invertible at every point of the inverse image of y. For a smooth map between compact connected oriented manifolds of equal dimension, the signed count of preimages is the same at every regular value. A point outside the image would have count zero. So it is enough to show that the count is nonzero at one regular value. On a circle, an orientation is a direction of travel. The general definition is part of the cited theory.

Figure 6.1 shows the count on the circle, for the maps $f(x) = x + 0.9 \sin 2x$ and $g(x) = 1.5 + \sin x$. A preimage counts $+1$ where the graph rises and -1 where it falls. The map f has count 1 at both levels drawn, and the map g has count 0 at both.

We now carry out the count.

1. Let $t_0 \in T$ be an element whose powers are dense in T, as after (6.2). Then

$$\Phi^{-1}(t_0) = \{(nT,\ n^{-1} t_0 n) : nT \in W(K, T)\}.$$

   Each listed point maps to t_0. Conversely, let $xtx^{-1} = t_0$. Then $x^{-1} t_0^m x = t^m \in T$ for all $m \in \mathbb{Z}$. Taking closures, $x^{-1} T x \subseteq T$. Both are maximal tori, so equality holds and $x \in N_K(T)$. The set is finite by the finiteness of the Weyl group, Lemma 6.8, which was proved without the torus theorem.

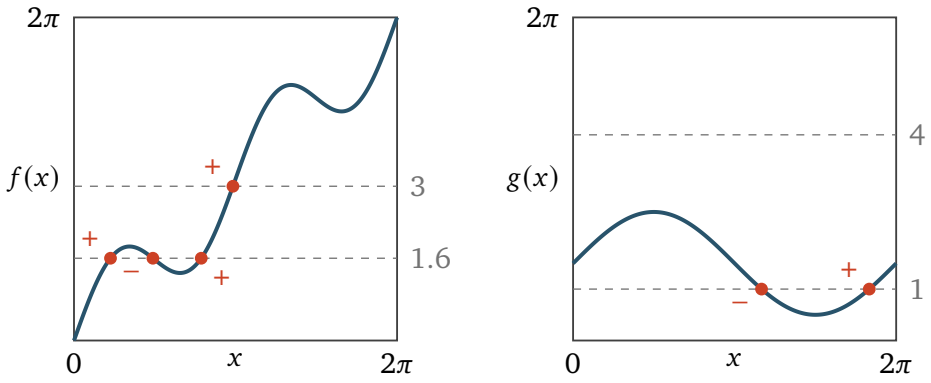


Figure 6.1: Two smooth maps of the circle, drawn on a square whose opposite sides are identified modulo 2π. The dashed lines are levels, numbered at the right, and the marked points are their preimages with the sign of the slope. The map f has signs $+, -, +$ at level 1.6 and $+$ at level 3. The map g has signs $-, +$ at level 1 and no preimage at level 4.

2. The quotient K/T is orientable. Fix orientations of K/T, T and K, and give $K/T \times T$ the product orientation. At each point of $\Phi^{-1}(t_0)$ the differential of Φ is invertible. Its determinant, computed in oriented bases, has the same sign at all of these points.
3. By step 2, t_0 is a regular value of Φ. A smooth map between compact connected oriented manifolds of equal dimension has an integer degree. It is the number of preimages of any regular value, each counted with the sign of the determinant of $d\Phi$ there.
4. So $\deg \Phi = \pm|W(K,T)| \neq 0$. A point outside the image would be a regular value with no preimages, giving degree 0. Hence Φ is surjective.

The omitted step is the degree theory in steps 2–4. It consists of the definition of degree, its independence of the regular value, the orientation of K/T, and the sign comparison at all preimages.

Step 2 rests on a local computation, shown here at (eT, t). By Lecture 3, $Y \mapsto \exp(Y)T$ identifies $\mathfrak{t}^\perp$ with the tangent space of K/T at eT. We identify the tangent spaces of T and of K at t with $\mathfrak{t}$ and $\mathfrak{k}$ by left translation by t. So $Z \in \mathfrak{t}$ corresponds to the curve $s \mapsto t\exp(sZ)$. For $Y \in \mathfrak{t}^\perp$ and $Z \in \mathfrak{t}$,

$$t^{-1}\Phi\big(\exp(sY)T,\ t\exp(sZ)\big) = \exp\big(s\,\mathrm{Ad}(t^{-1})Y\big)\exp(sZ)\exp(-sY).$$

By the product rule, the derivative at $s = 0$ is $\mathrm{Ad}(t^{-1})Y - Y + Z$. The orthogonal map $\mathrm{Ad}(t^{-1})$ fixes $\mathfrak{t}$ pointwise, since T is abelian. Hence it preserves $\mathfrak{t}^\perp$. In these identifications, the differential is block diagonal in $\mathfrak{t}^\perp \oplus \mathfrak{t}$, with blocks $\mathrm{Ad}(t^{-1})|_{\mathfrak{t}^\perp} - I$ and I. Hence

$$\det d\Phi_{(eT,t)} = \det\big(\mathrm{Ad}(t^{-1}) - I\big)\big|_{\mathfrak{t}^\perp}. \tag{6.3}$$

Hall (H §11.5) shows that this determinant is positive at $t = t_0$, and that the sign is the same at the other preimages of t_0. We do not reproduce that part. The same determinant returns in Lecture 9, as the weight in the Weyl integration formula.

6.4 Surjectivity of exp and the Weyl group

Corollary 6.9. *If K is compact and connected, then $\exp\colon \mathfrak{k} \to K$ is surjective.*

Proof. Let $g \in K$. By the torus theorem, Theorem 6.5(a), $g = xtx^{-1}$ with $x \in K$ and $t \in T$. As $\exp\colon \mathfrak{t} \to T$ is surjective, $t = \exp H$ with $H \in \mathfrak{t}$. Hence

$$g = x \exp(H) x^{-1} = \exp\big(\mathrm{Ad}(x)H\big). \qquad \blacksquare$$

Connectedness is needed. The path $s \mapsto \exp(sX)$ joins e to $\exp X$, so $\exp(\mathfrak{k})$ lies in the identity component. For example, no reflection in $\mathrm{O}(2)$ is an exponential.

The *centraliser* of T is $Z_K(T) = \{g \in K : gt = tg \text{ for all } t \in T\}$. It contains T, since T is abelian.

Theorem 6.10 (H §§11.2–11.5; BD §§IV.1–2)**.** *Let K be compact and connected with maximal torus T. Then $Z_K(T) = T$. Two elements of T are conjugate in K if and only if they lie in one $W(K,T)$-orbit.*

Together with the torus theorem, Theorem 6.5(a), we obtain a bijection

$$T/W(K,T) \longrightarrow \{\text{conjugacy classes of } K\}, \qquad W(K,T)\cdot t \longmapsto \text{class of } t.$$

Surjectivity is the torus theorem. Injectivity is the orbit statement of Theorem 6.10. By Lemma 6.8, the group $W(K,T)$ is finite. So conjugacy in K reduces to the action of a finite group on T.

The Weyl group also has an algebraic description. For $H \in \mathfrak{t}$ the operator $\operatorname{ad} H$ is skew-symmetric for our fixed $\mathrm{Ad}(K)$-invariant inner product. So its complex-linear extension to $\mathfrak{k}_{\mathbb{C}}$ is diagonalisable with imaginary eigenvalues. These operators commute, because $\mathfrak{t}$ is abelian. So they have common eigenvectors. On a common eigenvector, the eigenvalue is i times a real number that depends linearly on H.

A *root* is a nonzero $\alpha \in \mathfrak{t}^*$ such that some nonzero $V \in \mathfrak{k}_{\mathbb{C}}$ satisfies

$$[H, V] = i\alpha(H)V \qquad (H \in \mathfrak{t}).$$

Let R be the set of roots. A root is real on $\mathfrak{t}$. Lectures 7, 8 and 9 keep this convention. The Lie Algebras course writes the same root as the complex-linear form $i\alpha$ on $\mathfrak{t}_{\mathbb{C}}$, and Lecture 7 gives a dictionary. When $\mathfrak{k}$ is semisimple, $\mathfrak{t}_{\mathbb{C}}$ is a Cartan subalgebra of $\mathfrak{k}_{\mathbb{C}}$ in the sense of that course. Lecture 8 proves this.

Define $\alpha^\sharp \in \mathfrak{t}$ by $\langle \alpha^\sharp, H\rangle = \alpha(H)$ for $H \in \mathfrak{t}$. The reflections and the *algebraic Weyl group* are

$$s_\alpha(H) = H - \frac{2\alpha(H)}{\langle \alpha^\sharp, \alpha^\sharp\rangle}\,\alpha^\sharp, \qquad W(R) = \langle s_\alpha : \alpha \in R\rangle \subseteq \mathrm{GL}(\mathfrak{t}).$$

Theorem 6.11 (H §11.7; BD Ch. V)**.** *Let K be compact and connected, with any $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{k}$. Then $nT \mapsto \mathrm{Ad}(n)|_{\mathfrak{t}}$ is an isomorphism from $W(K,T)$ onto $W(R)$.*

6.5 Unitary groups and a noncompact example

Proposition 6.12. *Let $T \subseteq \mathrm{U}(n)$ be the diagonal unitary matrices. For $\sigma \in S_n$ let $P_\sigma e_j = e_{\sigma(j)}$. Then T is a maximal torus,*

$$N_{\mathrm{U}(n)}(T) = \{DP_\sigma : D \in T,\ \sigma \in S_n\},$$

and $DP_\sigma T \mapsto \sigma$ is an isomorphism $W(\mathrm{U}(n), T) \to S_n$. The action on T is

$$P_\sigma \operatorname{diag}(z_1, \dots, z_n) P_\sigma^{-1} = \operatorname{diag}\left(z_{\sigma^{-1}(1)}, \dots, z_{\sigma^{-1}(n)}\right). \tag{6.4}$$

Proof. The group $T \cong \mathrm{U}(1)^n$ is a torus. Fix $t_* = \operatorname{diag}(c_1, \dots, c_n) \in T$ with distinct entries. A matrix commuting with t_* preserves each eigenline $\mathbb{C}e_j$, so it is diagonal. Hence $Z_{\mathrm{U}(n)}(T) = T$. A torus $T' \supseteq T$ is abelian, so $T' \subseteq Z_{\mathrm{U}(n)}(T) = T$. Thus T is maximal.

Formula (6.4) holds for every $t = \operatorname{diag}(z_1, \dots, z_n) \in T$, because

$$P_\sigma t P_\sigma^{-1} e_{\sigma(j)} = P_\sigma t e_j = z_j e_{\sigma(j)}.$$

So each DP_σ normalises T. Conversely, let $g \in N_{\mathrm{U}(n)}(T)$ and $t' = g t_* g^{-1} \in T$. Then

$$t'(g e_j) = g t_* e_j = c_j\, g e_j.$$

The diagonal matrix t' has the distinct eigenvalues $c_1, \dots, c_n$. So its c_j-eigenspace is a coordinate line $\mathbb{C}e_{\sigma(j)}$. Distinct eigenvalues have distinct eigenlines, so σ is a permutation. Thus $g e_j = d_{\sigma(j)} e_{\sigma(j)}$ with $|d_{\sigma(j)}| = 1$. That is, $g = DP_\sigma$ with $D = \operatorname{diag}(d_1, \dots, d_n)$. Since $P_\sigma P_\tau = P_{\sigma\tau}$ and $P_\sigma D P_\sigma^{-1} \in T$, the map $DP_\sigma T \mapsto \sigma$ is a homomorphism. It is onto, since $P_\sigma \in \mathrm{U}(n)$. It is injective because $DP_\sigma \in T$ only when $\sigma = \mathrm{id}$. ∎

We now compare $W(\mathrm{U}(n), T)$ with the algebraic Weyl group, using the Ad-invariant form $-\operatorname{tr}(XY)$. On $\mathfrak{t} = \{i \operatorname{diag}(a_1, \dots, a_n) : a_j \in \mathbb{R}\}$ this form equals $\sum_j a_j b_j$. Here $\mathfrak{t}_{\mathbb{C}} = \mathfrak{gl}(n, \mathbb{C})$. Let E_{jk} be the matrix unit with 1 in row j and column k. For $H = i \operatorname{diag}(a)$ and $j \neq k$,

$$[H, E_{jk}] = i(a_j - a_k) E_{jk}.$$

Let ε_j be the linear form on $\mathfrak{t}$ with $\varepsilon_j\big(i \operatorname{diag}(a_1, \dots, a_n)\big) = a_j$. The roots are $\alpha_{jk} = \varepsilon_j - \varepsilon_k$ for $j \neq k$, with $\alpha_{jk}^\sharp = i(E_{jj} - E_{kk})$ and $\langle \alpha_{jk}^\sharp, \alpha_{jk}^\sharp \rangle = 2$. Then

$$s_{\alpha_{jk}}\big(i \operatorname{diag}(a)\big) = i \operatorname{diag}(a) - (a_j - a_k)\, i(E_{jj} - E_{kk}).$$

This reflection swaps a_j and a_k. By the permutation formula (6.4), the reflection equals $\operatorname{Ad}(P_{(jk)})|_{\mathfrak{t}}$. Transpositions generate S_n. So $W(R)$ is the image of $W(\mathrm{U}(n), T)$, as the algebraic description of the Weyl group in Theorem 6.11 predicts. Hence conjugacy classes of $\mathrm{U}(n)$ correspond to T/S_n. In other words, a unitary matrix is determined up to conjugacy by its eigenvalues with multiplicity.

Corollary 6.13. *Let $n \geq 2$ and $T_1 = T \cap \mathrm{SU}(n)$. Then T_1 is a maximal torus of $\mathrm{SU}(n)$. The normaliser $N_{\mathrm{SU}(n)}(T_1)$ consists of the DP_σ with $D \in T$ and $\det(DP_\sigma) = 1$, and $DP_\sigma T_1 \mapsto \sigma$ is an isomorphism $W(\mathrm{SU}(n), T_1) \to S_n$.*

Proof. The map $(z_1, \dots, z_{n-1}) \mapsto \operatorname{diag}(z_1, \dots, z_{n-1}, (z_1 \cdots z_{n-1})^{-1})$ shows $T_1 \cong \mathrm{U}(1)^{n-1}$. Let $0 < \epsilon < \pi/(n-1)$. The entries $c_j = e^{i\epsilon(2j-n-1)}$ are distinct, and their product is 1 since $\sum_j (2j-n-1) = 0$. The proof of Proposition 6.12, the normaliser computation for $\mathrm{U}(n)$, applies with this t_*. It shows that T_1 is maximal. It also shows that the normaliser consists of the DP_σ with

$$\det(DP_\sigma) = \det D \cdot \operatorname{sgn}\sigma = 1.$$

Every σ occurs, with $D = \operatorname{diag}(\operatorname{sgn}\sigma, 1, \dots, 1)$. The injectivity argument is unchanged. ∎

The roots match again. The Lie algebra $\mathfrak{t}_1$ of T_1 consists of the $i\operatorname{diag}(a)$ with $\sum_j a_j = 0$. Since $\mathfrak{su}(n)_{\mathbb{C}} = \mathfrak{sl}(n, \mathbb{C})$ is spanned by $\mathfrak{t}_{1,\mathbb{C}}$ and the E_{jk}, the roots are the restrictions of the α_{jk}. They are nonzero, because $\alpha_{jk}^\sharp = i(E_{jj} - E_{kk})$ lies in $\mathfrak{t}_1$ and $\alpha_{jk}(\alpha_{jk}^\sharp) = 2$. So $s_{\alpha_{jk}}$ still swaps a_j and a_k. Put $n_{(jk)} = \operatorname{diag}(-1, 1, \dots, 1)P_{(jk)}$, which lies in $N_{\mathrm{SU}(n)}(T_1)$. Diagonal matrices act trivially on $\mathfrak{t}$, so $\operatorname{Ad}(n_{(jk)})|_{\mathfrak{t}_1} = s_{\alpha_{jk}}$. Hence $W(R)$ is the image of $W(\mathrm{SU}(n), T_1) \cong S_n$.

For $\mathrm{SU}(2)$ the transposition is represented by

$$w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \operatorname{diag}(-1, 1)P_{(12)}, \qquad w \operatorname{diag}(z, \bar{z}) w^{-1} = \operatorname{diag}(\bar{z}, z).$$

By the bijection between $T_1/W(\mathrm{SU}(2), T_1)$ and conjugacy classes, every conjugacy class of $\mathrm{SU}(2)$ contains exactly one matrix $\operatorname{diag}(e^{i\theta}, e^{-i\theta})$ with $\theta \in [0, \pi]$. In physics terms, choosing a maximal torus of $\mathrm{SU}(2)$ is like choosing a quantisation axis.

Now compare the connected noncompact group $\mathrm{SL}(2, \mathbb{R})$. Its connectedness was shown in Lecture 2 (Exercise 2.4). By the computation in Lecture 2, the element

$$\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$$

is not in $\exp(\mathfrak{sl}(2, \mathbb{R}))$. So the surjectivity statement of Corollary 6.9 fails for this connected noncompact group. The lines $\mathbb{R}\operatorname{diag}(1, -1)$ and $\mathfrak{so}(2)$ are both maximal abelian. Their nonzero elements have real and imaginary eigenvalues respectively, so they are not Ad-conjugate.

The proof of the conjugacy statement in Proposition 6.6 breaks at the invariant inner product of Lemma 6.1. For $H = \operatorname{diag}(1, -1)$ the operator $\operatorname{ad} H$ has eigenvalues $2, 0, -2$, but a skew-symmetric operator has imaginary eigenvalues. Averaging is unavailable because $\mathrm{SL}(2, \mathbb{R})$ is not compact. Exercise 6.4 gives the details.

6.6 Finite central quotients

The next lecture uses the following global form of the splitting.

Theorem 6.14 (BD §V.8). *Let K be compact and connected. Let $T_0 = Z(K)_0$ be the identity component of the centre. It is a torus with Lie algebra $\mathfrak{z}(\mathfrak{k})$. Let K_{sc} be the simply connected Lie group with Lie algebra $[\mathfrak{k}, \mathfrak{k}]$ (Lecture 4). Then K_{sc} is compact, and the inclusion $[\mathfrak{k}, \mathfrak{k}] \to \mathfrak{k}$ integrates to a homomorphism $\varphi \colon K_{\mathrm{sc}} \to K$. The map*

$$\mu \colon T_0 \times K_{\mathrm{sc}} \to K, \qquad \mu(z, y) = z\,\varphi(y),$$

is a surjective homomorphism with finite central kernel F. Hence $K \cong (T_0 \times K_{\mathrm{sc}})/F$.

Compactness of K_{sc} is Weyl's theorem and is part of the citation.

For example, for $K = \mathrm{U}(n)$ we have $T_0 = \{zI : |z| = 1\}$ and $K_{\mathrm{sc}} = \mathrm{SU}(n)$, which is simply connected (H Ch. 13). Here $\mu(z, A) = zA$. If $zA = I$, then $A = z^{-1}I$ and $\det A = z^{-n} = 1$. Hence

$$F = \{(\zeta, \zeta^{-1}I) : \zeta^n = 1\} \cong \mathbb{Z}/n, \qquad \mathrm{U}(n) \cong \big(\mathrm{U}(1) \times \mathrm{SU}(n)\big)/F.$$

Exercise 6.5 checks surjectivity for $n = 2$.

Self-checks

1. Split
$$X = \begin{pmatrix} 3i & 1 \\ -1 & i \end{pmatrix} \in \mathfrak{u}(2)$$
along $\mathfrak{z} \oplus \mathfrak{su}(2)$.
Answer. The centre component is $\frac{1}{2}(\operatorname{tr} X)I = 2iI$. The remainder is skew-Hermitian with trace zero:
$$X - 2iI = \begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}.$$
2. Let T be the diagonal torus of $\mathrm{U}(n)$. Without using the torus theorem, prove that every element of $\mathrm{U}(n)$ lies in a conjugate of T, as in Theorem 6.5(a).
Answer. A unitary U is normal. The spectral theorem for normal matrices gives a unitary V such that $V^{-1}UV$ is diagonal. The matrix $V^{-1}UV$ is also unitary, so it lies in T. Hence $U \in VTV^{-1}$.
3. Are $\operatorname{diag}(i, -i)$ and $\operatorname{diag}(-i, i)$ conjugate in $\mathrm{SU}(2)$? Are $\operatorname{diag}(i, -i)$ and I?
Answer. Yes. With w as above, $w \operatorname{diag}(i, -i)w^{-1} = \operatorname{diag}(-i, i)$. The second pair is not conjugate. Conjugate matrices have equal traces, but the traces of $\operatorname{diag}(i, -i)$ and I are 0 and 2.

Summary

An averaged inner product gives the central and semisimple splitting. Maximal tori are conjugate. For connected K, the cited global torus theorem makes exp surjective. Together with the cited orbit theorem for $W(K, T)$, it identifies the conjugacy classes of K with $T/W(K, T)$. The Weyl group is finite. For $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ it is isomorphic to S_n. Compactness is essential, as the connected group $\mathrm{SL}(2, \mathbb{R})$ shows.

Reading

- H: Hall, *Lie Groups, Lie Algebras, and Representations*, §§11.1–11.5 and 11.7.
- BD: Bröcker–tom Dieck, *Representations of Compact Lie Groups*, IV.1–3 and V.8.
- S: Sepanski, *Compact Lie Groups*, Chapter 5.

Lecture 7

Lattices, centres and global forms

Objectives. *The groups* $\mathrm{SU}(2)$ *and* $\mathrm{SO}(3)$ *have isomorphic Lie algebras but are not isomorphic. This lecture asks which discrete data determine connected groups with the same Lie algebra. For compact groups, the answer is the kernel lattice* Λ *of a maximal torus, inside its Lie algebra* $\mathfrak{t}$*, read together with the root data.*

The lecture proves this for semisimple Lie algebras, using cited structure results. Different lattices can still give isomorphic groups, through automorphisms of the Lie algebra that preserve $\mathfrak{t}$*. When there is a central torus, the lecture shows how* Λ *determines a finite group* F*.*

By the end of this lecture you will be able to:

- Prove that a compact connected abelian Lie group is $\mathfrak{t}/\Lambda$, with characters the ℓ satisfying $\ell(\Lambda) \subset 2\pi\mathbb{Z}$.
- Place Λ between the coroot and coweight lattices, and compute centres.
- Describe the connected compact groups (global forms) with a given semisimple Lie algebra $\mathfrak{k}$, and the groups $(T_0 \times K_{sc})/F$.
- Decide which representations descend to $\mathrm{SO}(3)$, $\mathrm{PSU}(3)$ and $\mathrm{U}(2)$.

The Lie groups denoted K or T in this lecture, with or without subscripts, are connected. The subgroups C, F and $Z(K)$ that appear later need not be connected. A Lie group and its identity component have the same Lie algebra, so we restrict to connected groups. Even connected groups with isomorphic Lie algebras need not be isomorphic, as $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ show.

7.1 Compact connected abelian groups

Definition 7.1. A *torus* is a Lie group isomorphic to $\mathrm{U}(1)^n$ for some $n \geq 0$. A *lattice* in a real vector space V is a subgroup $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_n$, where $v_1, \ldots, v_n$ is a basis of V.

Lemma 7.2 (Discrete subgroups)**.** *Every discrete subgroup of a finite-dimensional real vector space equals* $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_k$ *for some linearly independent* $v_1, \ldots, v_k$*.*

We cite this standard lemma from BD, Section I.3, Lemma (3.8).

Theorem 7.3. *Let* T *be a compact connected abelian Lie group with Lie algebra* $\mathfrak{t}$ *of dimension* n*. Then* $\exp\colon \mathfrak{t} \to T$ *is a surjective homomorphism, and* $\Lambda = \ker \exp$ *is a lattice*

in $\mathfrak{t}$. *For a basis* $v_1, \dots, v_n$ *of* Λ, *the map*

$$\Phi\colon \mathrm{U}(1)^n \to T, \qquad \Phi\big(e^{2\pi i x_1}, \dots, e^{2\pi i x_n}\big) = \exp(x_1 v_1 + \cdots + x_n v_n),$$

is an isomorphism of Lie groups.

Proof. Since T is abelian, so is $\mathfrak{t}$. Commuting elements satisfy $\exp(X+Y) = \exp X \exp Y$ (Lecture 2), so exp is a homomorphism. Choose an open set $U = -U$ containing 0 that exp maps diffeomorphically onto an open set (Lecture 2).

Surjectivity. The subgroup $\exp(\mathfrak{t})$ contains $g\exp(U)$ for each of its elements g, so it is open. Its complement is a union of open cosets, so it is also closed. Since T is connected, $\exp(\mathfrak{t}) = T$.

Discreteness. As exp is injective on U, we have $\Lambda \cap U = \{0\}$. If $H, H' \in \Lambda$ and $H' \in H + U$, then $H' - H \in \Lambda \cap U$, so $H' = H$. Hence Λ is discrete. The discrete-subgroup lemma, Lemma 7.2, gives independent generators $v_1, \dots, v_k$ of Λ.

Full rank. Suppose $k < n$. Choose a nonzero linear map $p\colon \mathfrak{t} \to \mathbb{R}$ with $p(v_1) = \cdots = p(v_k) = 0$, and define

$$f\colon T \to \mathbb{R}, \qquad f(\exp H) = p(H).$$

It is well defined because $p(\Lambda) = 0$. It is continuous: near $\exp H_1$ we have $f(\exp H_1 \exp Y) = p(H_1) + p(Y)$ for $Y \in U$, and $Y \mapsto \exp H_1 \exp Y$ is a homeomorphism of U onto a neighbourhood of $\exp H_1$. So $\mathbb{R} = p(\mathfrak{t}) = f(T)$ is compact, which is false. Hence $k = n$.

Isomorphism. Changing x_j by an integer changes $\sum_j x_j v_j$ by an element of Λ. So Φ is a well-defined homomorphism. It is continuous, because near any point it is the composite of a local inverse of $x \mapsto (e^{2\pi i x_j})_j$ with the continuous map $x \mapsto \exp(\sum_j x_j v_j)$. It is bijective because $\ker\exp = \bigoplus_j \mathbb{Z} v_j$. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. Continuous homomorphisms of Lie groups are smooth (Lecture 3), so Φ and Φ^{-1} are smooth. ∎

Lecture 6 defined a torus as a compact connected abelian Lie group. The theorem shows that such a group is a torus in the sense above, and the converse is clear. So the two definitions agree. The theorem also proves the statement $T \cong \mathfrak{t}/\Lambda$ cited in Lecture 6, and it uses nothing from that lecture.

7.2 Characters and cocharacters

Definition 7.4. A *character* of a torus T is a continuous homomorphism $\chi\colon T \to \mathrm{U}(1)$. A *cocharacter* is a continuous homomorphism $\lambda\colon \mathrm{U}(1) \to T$. Under pointwise multiplication, the characters form a group $X^*(T)$ and the cocharacters form a group $X_*(T)$.

These characters are one-dimensional. Trace characters return in Lecture 9.

Theorem 7.5 (Torus character criterion). *Let* T *be a torus with* $\Lambda = \ker\exp$, *and let* $\ell \in \mathfrak{t}^*$. *There is a character* χ_ℓ *with*

$$\chi_\ell(\exp H) = e^{i\ell(H)} \qquad (H \in \mathfrak{t})$$

if and only if $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. *Every character equals* χ_ℓ *for exactly one such* ℓ, *and* $d\chi_\ell = i\ell$.

Hence $\ell \mapsto \chi_\ell$ is an isomorphism from $\{\ell \in \mathfrak{t}^* : \ell(\Lambda) \subset 2\pi\mathbb{Z}\}$ onto $X^*(T)$.

Proof. Suppose $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. The homomorphism $H \mapsto e^{i\ell(H)}$ from $\mathfrak{t}$ to $\mathrm{U}(1)$ is constant on cosets of Λ. Since exp is onto with kernel Λ, it defines a homomorphism χ_ℓ on T. Continuity follows as for f above. Conversely, if χ_ℓ exists, then $1 = \chi_\ell(\exp H) = e^{i\ell(H)}$ for $H \in \Lambda$.

Now let χ be any character. It is smooth (Lecture 3). Its differential maps $\mathfrak{t}$ to $\operatorname{Lie}\mathrm{U}(1) = i\mathbb{R}$, so $d\chi = i\ell$ for some $\ell \in \mathfrak{t}^*$. Naturality of exp (Lecture 2) gives

$$\chi(\exp H) = \exp\bigl(d\chi(H)\bigr) = e^{i\ell(H)}.$$

For $H \in \Lambda$ the left side is $\chi(1) = 1$, so $\ell(H) \in 2\pi\mathbb{Z}$. Thus $\chi = \chi_\ell$. The form ℓ is unique, because $i\ell(H)$ is the derivative of $s \mapsto \chi(\exp sH)$ at $s = 0$. Finally, $\chi_\ell\chi_{\ell'} = \chi_{\ell+\ell'}$ by the formula. ∎

For $H \in \Lambda$ define the cocharacter

$$\lambda_H(e^{i\theta}) = \exp\Bigl(\frac{\theta}{2\pi}H\Bigr).$$

Replacing θ by $\theta+2\pi$ multiplies the right side by $\exp H = 1$, so λ_H is well defined. Exercise 7.1 shows that $H \mapsto \lambda_H$ is an isomorphism $\Lambda \to X_*(T)$. Since $\chi_\ell(\lambda_H(e^{i\theta})) = e^{i\theta\ell(H)/2\pi}$,

$$\chi_\ell\bigl(\lambda_H(z)\bigr) = z^{\langle \ell, H\rangle}, \qquad \langle \ell, H\rangle = \frac{\ell(H)}{2\pi} \in \mathbb{Z}.$$

So $X^*(T)$ is the dual lattice of $X_*(T) \cong \Lambda$. In crystal physics terms, $X^*(T)$ is the reciprocal lattice of Λ, in the convention $k \cdot R \in 2\pi\mathbb{Z}$, and χ_ℓ is a plane wave that is periodic on Λ. From now on we identify $X^*(T)$ with $\{\ell : \ell(\Lambda) \subset 2\pi\mathbb{Z}\}$ and $X_*(T)$ with Λ.

A finite-dimensional representation of T is unitary for an averaged inner product (Lecture 5). The operators of the representation commute and are unitary, so they diagonalise simultaneously. Hence the representation space is a direct sum of lines on which $\exp H$ acts by $e^{i\ell(H)}$. These $\ell \in X^*(T)$ are the *weights* of the representation.

Example 7.6. Let $T \subset \mathrm{U}(n)$ be the diagonal torus. Then $\mathfrak{t} = \{\operatorname{diag}(i\theta_1, \dots, i\theta_n) : \theta \in \mathbb{R}^n\}$ and $\Lambda = \{2\pi i \operatorname{diag}(k_1, \dots, k_n) : k \in \mathbb{Z}^n\}$. Put $\ell_m(\operatorname{diag}(i\theta)) = \sum_j m_j\theta_j$. Then $\ell_m \in X^*(T)$ if and only if $m \in \mathbb{Z}^n$, and

$$\chi_{\ell_m}\bigl(\operatorname{diag}(z_1, \dots, z_n)\bigr) = z_1^{m_1} \cdots z_n^{m_n}.$$

For $H = 2\pi i \operatorname{diag}(k)$ we get $\lambda_H(z) = \operatorname{diag}(z^{k_1}, \dots, z^{k_n})$ and $\langle \ell_m, H\rangle = \sum_j m_j k_j$.

7.3 Roots, coroots and the centre

Let K be a compact connected Lie group with maximal torus T (Lecture 6). We use the set $R \subset \mathfrak{t}^*$ of real roots from Lecture 6, and we import the root data. A root α has the root space $\mathfrak{g}_\alpha = \{X \in \mathfrak{k}_{\mathbb{C}} : [H, X] = i\alpha(H)X \text{ for } H \in \mathfrak{t}\}$, and

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \qquad \operatorname{Ad}(\exp H)X = e^{i\alpha(H)}X \quad (X \in \mathfrak{g}_\alpha,\ H \in \mathfrak{t}).$$

Each root α has a coroot H_α. It is the unique element

$$H_\alpha \in \mathfrak{t} \cap [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \qquad \text{with} \qquad \alpha(H_\alpha) = 2.$$

The condition $\alpha(H) = 2$ alone does not fix H once $\dim \mathfrak{t} \geq 2$, because then $\ker \alpha \neq 0$. There is a homomorphism $\varphi_\alpha \colon \mathrm{SU}(2) \to K$ with $d\varphi_\alpha(H_0) = H_\alpha$, where $H_0 = \mathrm{diag}(i, -i)$ is the matrix A of Lecture 1 and $-2X_3$ of Lecture 4. Its differential is the compact form of a root $\mathfrak{sl}_2$-triple (H, Chapter 11). It integrates because $\mathrm{SU}(2)$ is simply connected (Lecture 4).

The Lie Algebras course works with complex-linear forms. Its coroot h_α spans the line $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$, and $\mathfrak{t} \cap \mathbb{C}h_\alpha = \mathbb{R}H_\alpha$. The following table compares the two conventions, for $H \in \mathfrak{t}$.

Object	Lie Algebras course	Lectures 6 to 9
root	complex-linear $\mu = i\alpha$ on $\mathfrak{t}_\mathbb{C}$	real α on $\mathfrak{t}$
eigenvalue of $\mathrm{ad}\, H$ on $\mathfrak{g}_\alpha$	$\mu(H)$	$i\alpha(H)$
coroot	h_α, with $\mu(h_\alpha) = 2$	$H_\alpha = ih_\alpha$, with $\alpha(H_\alpha) = 2$

Definition 7.7. Define subgroups Q and P of $\mathfrak{t}^*$, and $Q^\vee$ and $P^\vee$ of $\mathfrak{t}$, by

$$Q = \mathbb{Z}R, \qquad P = \{\ell \in \mathfrak{t}^* : \ell(H_\alpha) \in \mathbb{Z} \text{ for all } \alpha\},$$
$$Q^\vee = \mathbb{Z}\{2\pi H_\alpha : \alpha \in R\}, \quad P^\vee = \{H \in \mathfrak{t} : \alpha(H) \in 2\pi\mathbb{Z} \text{ for all } \alpha\}.$$

A real form $\ell \in \mathfrak{t}^*$ is the complex weight $\mu = i\ell$ of the Lie Algebras course. With $h_\alpha = -iH_\alpha$ we get $\mu(h_\alpha) = \ell(H_\alpha)$. When $\mathfrak{k}$ is semisimple, P is the weight lattice of that course, written in real form.

The factors 2π in $Q^\vee$ and $P^\vee$ fit the convention $\Lambda = \ker \exp$, in which characters satisfy $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. Some texts use $\{H : \exp(2\pi H) = 1\}$ in place of Λ, and write $\chi(\exp 2\pi H) = e^{2\pi i \ell(H)}$. This divides the lattices Λ, $Q^\vee$ and $P^\vee$ in $\mathfrak{t}$ by 2π. The weight lattices $X^*(T)$, Q and P are unchanged.

For example, take $K = \mathrm{SU}(2)$, with $\mathfrak{t} = \mathbb{R}H_0$. Since $\exp(sH_0) = \mathrm{diag}(e^{is}, e^{-is})$, we have $\Lambda = 2\pi\mathbb{Z}H_0$. Since $[H_0, E_{12}] = 2iE_{12}$, the positive root has $\alpha(H_0) = 2$, so $H_\alpha = H_0$. Hence $Q^\vee = 2\pi\mathbb{Z}H_0$ and $P^\vee = \pi\mathbb{Z}H_0$. Identifying ℓ with $m = \ell(H_0)$, we get $Q = 2\mathbb{Z}$ and $P = \mathbb{Z}$.

The covering of Lecture 4 identifies the Lie algebra of $\mathrm{SO}(3)$ with $\mathfrak{su}(2)$. The element sH_0 then exponentiates to the identity of $\mathrm{SO}(3)$ exactly when $\exp(sH_0) = \pm I$ in $\mathrm{SU}(2)$. So $\mathrm{SO}(3)$ has $\Lambda = \pi\mathbb{Z}H_0$. These are the only two lattices between $Q^\vee$ and $P^\vee$. The torus character criterion gives $X^*(T) = \mathbb{Z}$ for $\mathrm{SU}(2)$ and $X^*(T) = 2\mathbb{Z}$ for $\mathrm{SO}(3)$. So the larger kernel lattice has the smaller character lattice. The table collects these values, and Fig. 7.1 draws them.

Lattice	Lives in	Defined by	$\mathrm{SU}(2)$	$\mathrm{SO}(3)$
$Q^\vee$	$\mathfrak{t}$	$\mathbb{Z}\{2\pi H_\alpha\}$	$2\pi\mathbb{Z}H_0$	$2\pi\mathbb{Z}H_0$
Λ	$\mathfrak{t}$	$\ker \exp$	$2\pi\mathbb{Z}H_0$	$\pi\mathbb{Z}H_0$
$P^\vee$	$\mathfrak{t}$	$\alpha(H) \in 2\pi\mathbb{Z}$ for all α	$\pi\mathbb{Z}H_0$	$\pi\mathbb{Z}H_0$
Q	$\mathfrak{t}^*$	$\mathbb{Z}R$	$m \in 2\mathbb{Z}$	$m \in 2\mathbb{Z}$
$X^*(T)$	$\mathfrak{t}^*$	$\ell(\Lambda) \subset 2\pi\mathbb{Z}$	$m \in \mathbb{Z}$	$m \in 2\mathbb{Z}$
P	$\mathfrak{t}^*$	$\ell(H_\alpha) \in \mathbb{Z}$ for all α	$m \in \mathbb{Z}$	$m \in \mathbb{Z}$

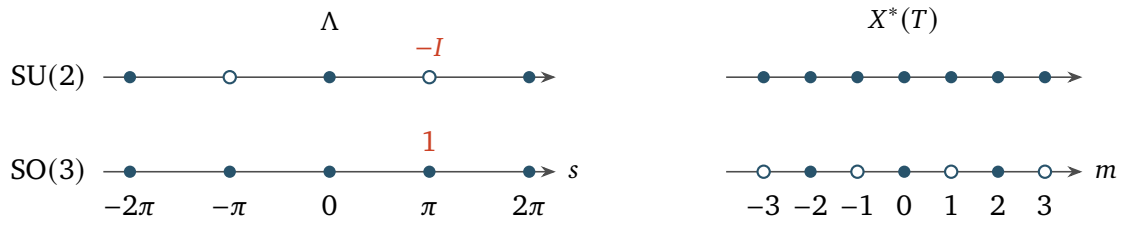


Figure 7.1: The kernel lattice Λ on the left, in the coordinate s of sH_0, and the character lattice $X^*(T)$ on the right, in the coordinate $m = \ell(H_0)$. Filled dots belong to the lattice. A value of m is filled exactly when $ms \in 2\pi\mathbb{Z}$ for every filled s in the same row. The element πH_0 exponentiates to $-I$ in SU(2) and to 1 in SO(3).

Proposition 7.8 (Lattice chain)**.**

$$Q^\vee \subset \Lambda \subset P^\vee, \qquad Q \subset X^*(T) \subset P.$$

Proof. Let $H \in \Lambda$ and $0 \neq X \in \mathfrak{g}_\alpha$. Then $X = \mathrm{Ad}(\exp H)X = e^{i\alpha(H)}X$, so $\alpha(H) \in 2\pi\mathbb{Z}$. Hence $\Lambda \subset P^\vee$. The same computation shows $\alpha \in X^*(T)$, so $Q \subset X^*(T)$. Next, naturality of exp gives

$$\exp(2\pi H_\alpha) = \varphi_\alpha\big(\exp(2\pi H_0)\big) = \varphi_\alpha\begin{pmatrix} e^{2\pi i} & 0 \\ 0 & e^{-2\pi i}\end{pmatrix} = \varphi_\alpha(I) = 1.$$

So $Q^\vee \subset \Lambda$. Finally, if $\ell \in X^*(T)$, then $\ell(2\pi H_\alpha) \in 2\pi\mathbb{Z}$, so $\ell(H_\alpha) \in \mathbb{Z}$ and $\ell \in P$. ∎

Theorem 7.9 (Centre)**.** *Let K be compact connected with maximal torus T. Then $Z(K) \subset T$, and* exp *restricts to a surjective homomorphism $P^\vee \to Z(K)$ with kernel Λ. Thus $Z(K) \cong P^\vee/\Lambda$.*

Proof. Let $z \in Z(K)$. By Lecture 6, $z = gtg^{-1}$ for some $g \in K$ and $t \in T$. Since z is central, $t = g^{-1}zg = z$, so $z \in T$.

Next we show that $g \in K$ is central if and only if $\mathrm{Ad}(g) = \mathrm{id}$. If g is central, differentiating $g\exp(sX)g^{-1} = \exp(sX)$ at $s = 0$ gives $\mathrm{Ad}(g) = \mathrm{id}$. If $\mathrm{Ad}(g) = \mathrm{id}$, then $g\exp(X)g^{-1} = \exp(\mathrm{Ad}(g)X) = \exp X$. The elements $\exp X$ generate the connected group K (Lecture 2), so g is central.

Now write $z = \exp H$ with $H \in \mathfrak{t}$, using the surjectivity of exp in the torus classification, Theorem 7.3. The operator $\mathrm{Ad}(\exp H) = e^{\mathrm{ad}\,H}$ is the identity on $\mathfrak{t}_\mathbb{C}$ and multiplication by $e^{i\alpha(H)}$ on $\mathfrak{g}_\alpha$. Hence, by the criterion just proved, $\exp H$ is central exactly when $H \in P^\vee$. The kernel of exp on $P^\vee$ is $\Lambda \cap P^\vee = \Lambda$. ∎

For the fundamental group we cite the following theorem.

Theorem 7.10 (Fundamental group; H, Chapter 13; BD, Section V.7)**.** *Let K be a compact connected Lie group with maximal torus T. Then*

$$\pi_1(K) \cong \Lambda/Q^\vee = X_*(T)/Q^\vee.$$

Call K *semisimple* if $\mathfrak{k}$ is semisimple. Then R spans $\mathfrak{t}^*$, and the simple roots form a basis of $\mathfrak{t}^*$. The simple coroots form a basis of $\mathfrak{t}$ (the Lie Algebras course, via $h_\alpha = -iH_\alpha$). So all four

subgroups are lattices, and the inclusions above have finite index. Weyl's theorem states that a compact connected semisimple group has finite fundamental group and compact universal cover (H, Chapter 13; Knapp, Chapter IV).

7.4 Global forms

Semisimple groups

By Weyl's theorem, the universal cover of our compact connected semisimple K is compact. Let K_{sc} be a compact, connected, simply connected, semisimple Lie group with Lie algebra $\mathfrak{k}$ and maximal torus T_{sc}. The fundamental group theorem, Theorem 7.10, gives $\Lambda_{sc} = Q^\vee$. The torus character criterion and the centre theorem, Theorems 7.5 and 7.9, then give

$$X^*(T_{sc}) = P, \qquad Z(K_{sc}) \cong P^\vee/Q^\vee.$$

Proposition 7.11 (Forms of a semisimple algebra). *(a) Every connected Lie group with Lie algebra $\mathfrak{k}$ is isomorphic to K_{sc}/C for a subgroup $C \subset Z(K_{sc})$.*

(b) For $K = K_{sc}/C$, the image T of T_{sc} is a maximal torus of K, with kernel lattice

$$\Lambda_C = \{H \in \mathfrak{t} : \exp_{sc} H \in C\}, \qquad \Lambda_C/Q^\vee \cong C.$$

The map $C \mapsto \Lambda_C$ is a bijection from subgroups of $Z(K_{sc})$ to lattices Λ with $Q^\vee \subset \Lambda \subset P^\vee$.

(c) $X^*(T) = \{\ell \in P : \chi_\ell(c) = 1 \text{ for all } c \in C\}$.

Proof. (a) By Lecture 4, such a group is isomorphic to K_{sc}/C with C discrete and normal. Such a subgroup C is central.

(b) First, T is a maximal torus with kernel lattice Λ_C. The covering map $p\colon K_{sc} \to K$ is a local isomorphism. So $\operatorname{Lie} T = \mathfrak{t}$ and $\exp_K = p \circ \exp_{sc}$. Hence $\exp_K H = 1$ exactly when $\exp_{sc} H \in C$. A larger torus would have a larger abelian Lie algebra, but $\mathfrak{t}$ is maximal abelian in $\mathfrak{k}$ (Lecture 6). So T is maximal.

Next, we prove $Q^\vee \subset \Lambda_C \subset P^\vee$ and $\Lambda_C/Q^\vee \cong C$. Since $C \subset Z(K_{sc})$, the centre theorem, Theorem 7.9, gives $Q^\vee \subset \Lambda_C \subset P^\vee$. The map $\exp_{sc}\colon \Lambda_C \to C$ is onto, since $C \subset T_{sc} = \exp_{sc}(\mathfrak{t})$. Its kernel is $Q^\vee$, so $\Lambda_C/Q^\vee \cong C$.

Conversely, let $Q^\vee \subset \Lambda \subset P^\vee$, and put $C = \exp_{sc}(\Lambda)$. By the centre theorem, Theorem 7.9, C is a subgroup of $Z(K_{sc})$. Clearly $\Lambda \subset \Lambda_C$. For the reverse inclusion, let $H \in \Lambda_C$. Then $\exp_{sc} H \in C = \exp_{sc}(\Lambda)$, so $\exp_{sc} H = \exp_{sc} H'$ for some $H' \in \Lambda$. Hence $H - H'$ lies in the kernel of $\exp_{sc}$ on $\mathfrak{t}$, which is $Q^\vee \subset \Lambda$. So $H \in \Lambda$, and $\Lambda_C = \Lambda$. We also showed $\exp_{sc}(\Lambda_C) = C$ for every C. So $C \mapsto \Lambda_C$ and $\Lambda \mapsto \exp_{sc}(\Lambda)$ are inverse to each other.

(c) A character of $T = T_{sc}/C$ is a character of T_{sc} that is trivial on C. ∎

Remark 7.12. Distinct lattices can give isomorphic groups. Let σ be an automorphism of $\mathfrak{k}$ with $\sigma(\mathfrak{t}) = \mathfrak{t}$. It integrates to an automorphism Σ of K_{sc} (Lecture 4). Since $\Sigma(\exp H) = \exp(\sigma H)$, we get $\Sigma(Z(K_{sc})) = Z(K_{sc})$ and $\Lambda_{\Sigma(C)} = \sigma(\Lambda_C)$. Hence $K_{sc}/C \cong K_{sc}/\Sigma(C)$. Exercise 7.5 gives an example. We do not discuss the converse.

The *simply connected form* is $C = 1$, with $\Lambda = Q^\vee$ and $X^*(T) = P$. The *adjoint form* is $K_{ad} = K_{sc}/Z(K_{sc})$, with $\Lambda = P^\vee$. Its centre $P^\vee/P^\vee$ is trivial by the centre theorem, Theorem 7.9. Write T_{ad} for the image of T_{sc} in K_{ad}. Every root is an integer combination of the simple roots α_i (the Lie Algebras course). So the lattice $P^\vee$ is defined by the conditions $\alpha_i(H) \in 2\pi\mathbb{Z}$ for the simple roots alone. Since the simple roots form a basis of $\mathfrak{t}^*$, we get $X^*(T_{ad}) = \mathbb{Z}\{\alpha_i\} = Q$.

Proposition 7.13 (Descent by central character)**.** *Let V be the irreducible representation of K_{sc} with dominant highest weight $\lambda \in P$, and let $C \subset Z(K_{sc})$. Then V descends to K_{sc}/C if and only if*

$$\chi_\lambda(c) = 1 \qquad \text{for all } c \in C.$$

The representation V exists by the highest-weight theorem of the Lie Algebras course and Lecture 4.

Proof. By Schur's lemma, a central element c acts on V by a scalar. On a highest weight vector this scalar is $\chi_\lambda(c)$. By Lecture 4, V descends to K_{sc}/C if and only if C acts trivially. ∎

Groups with a central torus

In U(2) the element $-I$ is both a scalar and an element of SU(2). A representation must act on it consistently. Exercise 4.3 did this by hand. We now state the condition for a general central torus.

If $\mathfrak{k}$ has nonzero centre, then $Z(K)$ is infinite and R does not span $\mathfrak{t}^*$. Moreover, $P^\vee$ contains the centre of $\mathfrak{k}$, so $P^\vee$ is not a lattice. The proposition on forms of a semisimple algebra, Proposition 7.11, does not apply. We use a structure theorem instead.

We recall the global structure theorem of Lecture 6, together with the splitting of that lecture.

Theorem 7.14 (Global structure; recalled from Lecture 6; BD, Section V.8; Knapp, Chapter IV)**.** *Let K be compact connected. Let T_0 be the identity component of $Z(K)$, with Lie algebra $\mathfrak{t}_0$. Let K_{sc} be the simply connected group with Lie algebra $[\mathfrak{k},\mathfrak{k}]$. Then T_0 is a torus, $\mathfrak{k} = \mathfrak{t}_0 \oplus [\mathfrak{k},\mathfrak{k}]$, the ideal $[\mathfrak{k},\mathfrak{k}]$ is semisimple, and K_{sc} is compact. Let $\varphi\colon K_{sc} \to K$ integrate the inclusion of $[\mathfrak{k},\mathfrak{k}]$. Then*

$$T_0 \times K_{sc} \to K, \qquad (t,g) \mapsto t\,\varphi(g),$$

is a surjective homomorphism with finite kernel F. Thus $K \cong (T_0 \times K_{sc})/F$.

Proposition 7.15. *In the global structure theorem, Theorem 7.14, let T_{sc} be a maximal torus of K_{sc} with Lie algebra $\mathfrak{t}_{sc}$, and let T be the image of $T_0 \times T_{sc}$. Identify $\mathfrak{t}^* = \mathfrak{t}_0^* \oplus \mathfrak{t}_{sc}^*$, and let $P_{sc} \subset \mathfrak{t}_{sc}^*$ be the weight lattice of K_{sc}. Then $T \cong (T_0 \times T_{sc})/F$ is a maximal torus of K, and*

$$X^*(T) = \big\{(\ell_0,\ell) \in X^*(T_0) \times P_{sc} : \ \chi_{\ell_0}(t)\,\chi_\ell(g) = 1 \text{ for all } (t,g) \in F\big\}.$$

Proof. A finite normal subgroup of a connected group is central (Lecture 4). So $F \subset T_0 \times Z(K_{sc}) \subset T_0 \times T_{sc}$ by the centre theorem, Theorem 7.9. Hence $T \cong (T_0 \times T_{sc})/F$. Its Lie algebra $\mathfrak{t}_0 \oplus \mathfrak{t}_{sc}$ is maximal abelian in $\mathfrak{k}$, because $\mathfrak{t}_{sc}$ is maximal abelian in $[\mathfrak{k},\mathfrak{k}]$. So T is a maximal torus.

Each root space satisfies $\mathfrak{g}_\alpha = [\mathfrak{t}, \mathfrak{g}_\alpha] \subset [\mathfrak{k}, \mathfrak{k}]_{\mathbb{C}}$. So $H_\alpha \in \mathfrak{t} \cap [\mathfrak{k}, \mathfrak{k}] = \mathfrak{t}_{sc}$, and the coroots of K are those of K_{sc}. Thus $Q^\vee \subset \mathfrak{t}_{sc}$ is the kernel lattice of T_{sc}. Put

$$\Lambda_0 = \ker\bigl(\exp|_{\mathfrak{t}_0}\bigr) = \Lambda \cap \mathfrak{t}_0.$$

The torus $T_0 \times T_{sc}$ has kernel lattice $\Lambda_0 \oplus Q^\vee$. By the torus character criterion, Theorem 7.5, its characters are the pairs with $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$ and $\ell \in P_{sc}$. The characters of T are those trivial on F. ∎

In other words, (ℓ_0, ℓ) is a character of T exactly when $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$, $\ell \in P_{sc}$, and the product $\chi_{\ell_0}\chi_\ell$ is trivial on F.

The group F is determined by Λ. The map $T_0 \times T_{sc} \to T$ has differential the identity of $\mathfrak{t} = \mathfrak{t}_0 \oplus \mathfrak{t}_{sc}$. By the torus classification, Theorem 7.3,

$$T_0 \times T_{sc} \cong \mathfrak{t}/(\Lambda_0 \oplus Q^\vee), \qquad T \cong \mathfrak{t}/\Lambda.$$

The map is induced by the identity of $\mathfrak{t}$, so its kernel is

$$F \cong \Lambda/(\Lambda_0 \oplus Q^\vee).$$

7.5 Examples

SU(2) and SO(3)

We use the values found after the definition of the lattices, and drawn in Fig. 7.1. For SU(2) we have $Q^\vee = \Lambda$. With $m = \ell(H_0)$,

$$P = \mathbb{Z}, \qquad Q = 2\mathbb{Z}, \qquad P^\vee = \pi\mathbb{Z}H_0, \qquad Z(\mathrm{SU}(2)) = \exp(\pi\mathbb{Z}H_0) = \{\pm I\}.$$

The adjoint form is $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ (Lecture 4). It has $\Lambda = \pi\mathbb{Z}H_0$ and

$$X^*(T) = \{m : m\pi \in 2\pi\mathbb{Z}\} = 2\mathbb{Z} = Q, \qquad \pi_1(\mathrm{SO}(3)) \cong \pi\mathbb{Z}/2\pi\mathbb{Z} \cong \mathbb{Z}/2.$$

The module V_m has dimension $m+1$ and weights $m, m-2, \dots, -m$. Physicists call it the spin-$m/2$ representation. Since $-I = \exp(\pi H_0)$, the element $-I$ acts on V_m by $e^{i\pi m} = (-1)^m$. So V_m descends to SO(3) exactly when m is even.

SU(3) and PSU(3)

For the diagonal torus of SU(3),

$$\mathfrak{t} = \Bigl\{\mathrm{diag}(i\theta_1, i\theta_2, i\theta_3) : \sum_j \theta_j = 0\Bigr\}, \qquad \Lambda = \Bigl\{2\pi i\,\mathrm{diag}(k) : k \in \mathbb{Z}^3,\ \sum_j k_j = 0\Bigr\}.$$

The roots are $\alpha_{jk}(H) = \theta_j - \theta_k$ for $j \neq k$, with coroots $H_{jk} = i(E_{jj} - E_{kk})$. Indeed $[E_{jk}, E_{kj}] = E_{jj} - E_{kk}$ and $\alpha_{jk}(H_{jk}) = 2$. Hence $Q^\vee = \Lambda$. Write $\ell_m(H) = \sum_j m_j\theta_j$ with $m \in \mathbb{R}^3$. Adding a multiple of $(1,1,1)$ to m does not change ℓ_m, since $\sum_j \theta_j = 0$. The condition $\ell_m(\Lambda) \subset 2\pi\mathbb{Z}$ means $m_j - m_k \in \mathbb{Z}$, and this is also the condition for P.

Such a form has a representative $m \in \mathbb{Z}^3$, obtained by subtracting $m_3(1,1,1)$. Two integer representatives differ by an element of $\mathbb{Z}(1,1,1)$, which changes $m_1+m_2+m_3$ by a multiple of 3. Using integer representatives,

$$X^*(T) = P \cong \mathbb{Z}^3/\mathbb{Z}(1,1,1),$$
$$Q = \{[m] : m \in \mathbb{Z}^3,\ m_1+m_2+m_3 \equiv 0 \bmod 3\}, \qquad P/Q \cong \mathbb{Z}/3.$$

An element of $P^\vee$ has $\theta_j = \theta + 2\pi k_j$ with $3\theta \in 2\pi\mathbb{Z}$. So $Z(\mathrm{SU}(3)) = \{I, \omega I, \omega^2 I\}$ with $\omega = e^{2\pi i/3}$. The adjoint form $\mathrm{PSU}(3) = \mathrm{SU}(3)/Z(\mathrm{SU}(3))$ has $X^*(T) = Q$, trivial centre and $\pi_1 \cong P^\vee/Q^\vee \cong \mathbb{Z}/3$.

Use the simple roots α_{12}, α_{23}, with simple coroots H_{12}, H_{23}. The fundamental weights are then $\varpi_1 = \ell_{(1,0,0)}$ and $\varpi_2 = \ell_{(1,1,0)}$. So the highest weight $a\varpi_1 + b\varpi_2$ is $\ell_{(a+b,b,0)}$. Write $V(a,b)$ for the irreducible module with this highest weight. Take

$$H_\omega = \mathrm{diag}\Big(\frac{2\pi i}{3}, \frac{2\pi i}{3}, -\frac{4\pi i}{3}\Big), \qquad \exp H_\omega = \mathrm{diag}(\omega, \omega, \omega^{-2}) = \omega I.$$

Then $\ell_{(a+b,b,0)}(H_\omega) = 2\pi(a+2b)/3$, so ωI acts on $V(a,b)$ by ω^{a+2b}. Thus $V(a,b)$ descends to PSU(3) if and only if $a \equiv b \bmod 3$. The adjoint module $V(1,1)$ descends. The module $\mathbb{C}^3 = V(1,0)$ does not. In physics terms, the number $a-b$ modulo 3 is the triality, the module $V(1,1)$ is the octet and $V(1,0)$ is the quark triplet.

Figure 7.2 draws the same data in the plane $\mathfrak{t}$ and in the dual plane $\mathfrak{t}^*$. In $\mathfrak{t}$ the kernel lattice of SU(3) is the smaller lattice $Q^\vee$, and that of PSU(3) is the larger lattice $P^\vee$. In $\mathfrak{t}^*$ the order reverses. The character lattice of SU(3) is the larger lattice P, and that of PSU(3) is the smaller lattice Q. The same three marks appear in both planes.

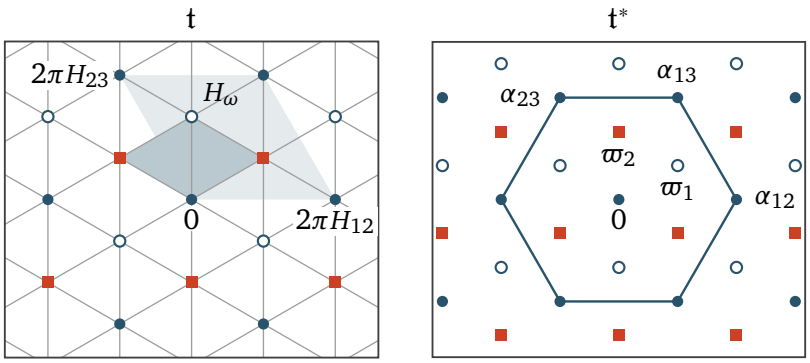


Figure 7.2: The plane $\mathfrak{t}$ of SU(3), with the form $-\mathrm{tr}(XY)$, on the left, and the dual plane $\mathfrak{t}^*$ on the right. Their scales differ. On the left, a disc, circle or square marks a point whose exponential is I, ωI or $\omega^2 I$. Discs form $Q^\vee$, and all marks form $P^\vee$. The thin lines are $\alpha_{jk}(H) \in 2\pi\mathbb{Z}$, and the shaded cells are cells of $Q^\vee$ and of $P^\vee$. On the right, a mark shows the scalar by which ωI acts on a weight: 1, ω or ω^2. Discs form Q, and all marks form P. The hexagon joins the roots, and $\alpha_{13} = \alpha_{12} + \alpha_{23} = \varpi_1 + \varpi_2$.

The group U(2)

Let $T_0 = \{zI : z \in \mathrm{U}(1)\}$ and $K_{sc} = \mathrm{SU}(2)$. The map

$$\Psi\colon\ \mathrm{U}(1) \times \mathrm{SU}(2) \to \mathrm{U}(2), \qquad \Psi(z,g) = zg,$$

is onto. Given u, choose z with $z^2 = \det u$. Then $z^{-1}u \in \mathrm{SU}(2)$. If $zg = I$, then $\det(z^{-1}I) = z^{-2} = 1$, so $z = \pm 1$. Hence

$$\mathrm{U}(2) \cong \big(\mathrm{U}(1) \times \mathrm{SU}(2)\big)/F, \qquad F = \{(1, I), (-1, -I)\}.$$

The lattice statements for semisimple groups fail for U(2) itself. In the semisimple case the roots span $\mathfrak{t}^*$, and Q, P, $Q^\vee$ and $P^\vee$ are lattices. By the computation for diagonal tori, Example 7.6, the diagonal torus of U(2) has $X^*(T) = \mathbb{Z}^2$. The roots are $\pm\alpha$ with $\alpha(\mathrm{diag}(i\theta)) = \theta_1 - \theta_2$, so $Q = \mathbb{Z}(1, -1)$ spans only a line. The set $P^\vee = \{\theta_1 - \theta_2 \in 2\pi\mathbb{Z}\}$ is not a lattice, and $P^\vee/\Lambda \cong \mathrm{U}(1)$ is the group of scalars.

The fundamental group theorem, Theorem 7.10, still applies. Since $[E_{12}, E_{21}] = \mathrm{diag}(1, -1)$, the coroot is $H_\alpha = i\,\mathrm{diag}(1, -1)$, and

$$\pi_1(\mathrm{U}(2)) \cong \mathbb{Z}^2/\mathbb{Z}(1, -1) \cong \mathbb{Z}, \qquad (k_1, k_2) \mapsto k_1 + k_2.$$

Here $\Lambda_0 = 2\pi i\mathbb{Z}I$, and $\Lambda_0 \oplus Q^\vee$ consists of the $2\pi i\,\mathrm{diag}(k_1, k_2)$ with $k_1 \equiv k_2 \bmod 2$. So $\Lambda/(\Lambda_0 \oplus Q^\vee) \cong \mathbb{Z}/2 \cong F$, as predicted.

Now we compute the central compatibility constraint. Recall from Lecture 4 that Π_m denotes the representation of SU(2) on V_m. The element $\mathrm{diag}(w, w^{-1})$, with $w \in \mathrm{U}(1)$, acts on a vector of weight j by w^j. For $k \in \mathbb{Z}$ and $m \geq 0$ define

$$\Pi_{k,m}(z, g) = z^k\,\Pi_m(g) \qquad \text{on } V_m.$$

The representation $\Pi_{k,m}$ is irreducible, already on SU(2). By Lecture 4 it descends to U(2) exactly when it is trivial on F. Since

$$\Pi_{k,m}(-1, -I) = (-1)^k(-1)^m\ \mathrm{id},$$

the constraint is

$$k \equiv m \pmod 2.$$

The descended representation is $\Pi(u) = z^k\Pi_m(z^{-1}u)$ with $z^2 = \det u$. The two choices of z give the same operator exactly because of the constraint. Exercise 7.3 shows that every irreducible representation of U(2) arises in this way.

The description of $X^*(T)$ for a central torus, Proposition 7.15, gives the same constraint. Here k is the exponent on scalars, and j runs over the weights of V_m. Put $\ell_0(isI) = ks$ and $\ell(sH_0) = js$. In T_0, the scalar $z = -1$ is the matrix $-I = \exp(i\pi I)$. So $(-1, -I) = (\exp(i\pi I), \exp(\pi H_0))$, and the pair (ℓ_0, ℓ) is a character of T exactly when $k + j$ is even. Every weight j of V_m satisfies $j \equiv m \bmod 2$, so this condition is the constraint found above.

Assume $k \equiv m \pmod 2$, so that $\Pi_{k,m}$ descends to U(2). Every weight j of V_m satisfies $j \equiv m \pmod 2$, so $k - j$ is even and the exponents below are integers. On $u = \mathrm{diag}(z_1, z_2)$ with $z^2 = z_1z_2$, a vector of weight j is multiplied by

$$z^k\left(\frac{z_1}{z}\right)^j = (z_1z_2)^{(k-j)/2}z_1^{\,j} = z_1^{(k+j)/2}\,z_2^{(k-j)/2}.$$

So $\Pi_{k,m}$ has highest weight $\left(\frac{k+m}{2}, \frac{k-m}{2}\right) \in \mathbb{Z}^2$. For example, $\mathbb{C}^2$ is $(k, m) = (1, 1)$, det is $(2, 0)$, and Ad on the complexification $\mathfrak{su}(2)_\mathbb{C}$ is $(0, 2)$.

Self-checks

1. Does the maximal torus of SO(3) have a character with $\ell(H_0) = 1$?

   Answer. No. Here $\Lambda = \pi\mathbb{Z}H_0$ and $\ell(\pi H_0) = \pi \notin 2\pi\mathbb{Z}$.

2. Compute $Z(\mathrm{PSU}(3))$ and $\pi_1(\mathrm{PSU}(3))$ from the lattices.

   Answer. Here $\Lambda = P^\vee$. So $Z = P^\vee/P^\vee = 1$ and $\pi_1 = P^\vee/Q^\vee \cong \mathbb{Z}/3$.

3. Is there a representation of U(2) on which scalars act by z^2 and SU(2) acts by V_1?

   Answer. No. The element $-I$ is the scalar -1, acting by $(-1)^2 = 1$. The element $-I$ also lies in SU(2), where it acts on V_1 by -1. The two actions disagree.

Summary

A compact connected abelian Lie group is $\mathfrak{t}/\Lambda$, and its characters are the ℓ with $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. For compact connected K, the kernel lattice satisfies $Q^\vee \subset \Lambda \subset P^\vee$. The kernel lattice gives $Z(K) \cong P^\vee/\Lambda$ and, by a cited theorem, $\pi_1(K) \cong \Lambda/Q^\vee$.

For semisimple $\mathfrak{k}$, the connected forms correspond to lattices between $Q^\vee$ and $P^\vee$. Lattices related by an automorphism of $\mathfrak{k}$ preserving $\mathfrak{t}$ give isomorphic groups. With a central torus, write $K \cong (T_0 \times K_{sc})/F$, where $F \cong \Lambda/(\Lambda_0 \oplus Q^\vee)$. A pair (ℓ_0, ℓ) is then a character of the maximal torus exactly when $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$, $\ell \in P_{sc}$, and the product is trivial on F.

Lecture 8 uses $X^*(T)$ to state the classification of representations by highest weights.

Reading

B. Hall, *Lie Groups, Lie Algebras, and Representations* (H), Chapter 13. T. Bröcker and T. tom Dieck, *Representations of Compact Lie Groups* (BD), Sections V.7–8. A. Knapp, *Lie Groups Beyond an Introduction*, Chapter IV. For the discrete-subgroup lemma, Lemma 7.2, see BD, Section I.3, Lemma (3.8).

Lecture 8

Highest weights of representations of compact groups

Objectives. *The Lie Algebras course classified finite-dimensional irreducible modules of semisimple algebras by dominant integral weights. This lecture adds the action of the centre of the Lie algebra and decides which of these modules are representations of a given compact connected Lie group K. These are exactly the modules whose highest weight is a character of a maximal torus.*

Lecture 7 decided, for a given highest-weight module, whether it descends. It did not show that every irreducible representation of K is obtained in this way, or that one is obtained only once. This lecture proves both.

By the end of this lecture you will be able to:

- Prove that highest weights give a bijection $\mathrm{Irr}(K) \to X^*(T)^+$, assuming cited structure results.
- Decide descent to K_{sc}/F by evaluating a highest weight λ on the elements $H \in \mathfrak{t}$ with $\exp H \in F$.
- Treat a central torus by an integral character and a finite compatibility condition.
- Explain why some modules of a reductive Lie algebra come from no compact group.

8.1 Torus characters and dominance

Throughout, K is a compact connected Lie group with Lie algebra $\mathfrak{k}$. Let T be a maximal torus with Lie algebra $\mathfrak{t}$ (Lecture 6). Put $\Lambda = \ker(\exp\colon \mathfrak{t} \to T)$. A *character* of T is a continuous homomorphism $\chi\colon T \to \mathrm{U}(1)$. Its derivative has the form $d\chi = i\ell$ for some $\ell \in \mathfrak{t}^*$, and

$$\chi(\exp H) = e^{i\ell(H)} \quad (H \in \mathfrak{t}), \qquad \ell(\Lambda) \subset 2\pi\mathbb{Z}. \tag{8.1}$$

Conversely, $\exp\colon \mathfrak{t} \to T$ is a surjective homomorphism with kernel Λ. So each ℓ with $\ell(\Lambda) \subset 2\pi\mathbb{Z}$ defines a character. Let $X^*(T)$ be the lattice of these ℓ.

Let Π be a representation of K on V. By Lecture 5, V has a K-invariant inner product. So the operators $\Pi(t)$ with $t \in T$ are commuting unitary operators. Hence these operators have a common orthonormal eigenbasis. For a common eigenvector v, write $\Pi(t)v = c(t)v$. Then c

is a character of T, so $c(\exp H) = e^{i\ell(H)}$ for a unique $\ell \in X^*(T)$. Hence V is the direct sum of the weight spaces

$$V_\ell = \{v \in V : \ \Pi(\exp H)v = e^{i\ell(H)}v \ \text{ for all } H \in \mathfrak{t}\}, \qquad \ell \in X^*(T).$$

Differentiating $\Pi(\exp tH)v = e^{it\ell(H)}v$ at $t = 0$ gives $d\Pi(H)v = i\ell(H)v$ for $v \in V_\ell$. The complex-linear extension of $i\ell$ to $\mathfrak{t}_\mathbb{C}$ is a weight in the sense of the Lie Algebras course. We identify ℓ with this weight.

Roots are the nonzero weights of Ad on $\mathfrak{k}_\mathbb{C}$. Thus $[H, X] = i\alpha(H)X$ for $H \in \mathfrak{t}$ and X in the root space of α. In particular, roots lie in $X^*(T)$.

Fix positive roots R^+ as in [I]. Put

$$\mathbb{N}R^+ = \Big\{ \sum_{\alpha \in R^+} n_\alpha \alpha : \ n_\alpha \in \mathbb{Z}_{\geq 0} \Big\}, \qquad \ell' \preceq \ell \iff \ell - \ell' \in \mathbb{N}R^+.$$

There is $H_+ \in \mathfrak{t}$ with $\alpha(H_+) > 0$ for all $\alpha \in R^+$. So every nonzero element of $\mathbb{N}R^+$ is positive at H_+. Hence $\ell' \preceq \ell$ and $\ell \preceq \ell'$ force $\ell = \ell'$. A *highest weight* of Π is a weight ℓ of Π with $\ell' \preceq \ell$ for every weight ℓ' of Π. It is unique if it exists.

Write $\mathfrak{k} = \mathfrak{z} \oplus \mathfrak{k}_s$, with $\mathfrak{z}$ the centre and $\mathfrak{k}_s = [\mathfrak{k}, \mathfrak{k}]$ semisimple. Then $\mathfrak{t} = \mathfrak{z} \oplus \mathfrak{t}_s$ with $\mathfrak{t}_s = \mathfrak{t} \cap \mathfrak{k}_s$ (Lecture 6). Roots vanish on $\mathfrak{z}$, since $\operatorname{ad} Z = 0$ for $Z \in \mathfrak{z}$.

The highest-weight theorem of the Lie Algebras course is stated for a Cartan subalgebra of $\mathfrak{k}_{s,\mathbb{C}}$. We now check that $\mathfrak{t}_{s,\mathbb{C}}$ is one, so that the theorem applies here.

The algebra $\mathfrak{t}_s$ is maximal abelian in $\mathfrak{k}_s$. Indeed, let $\mathfrak{a} \supset \mathfrak{t}_s$ be an abelian subalgebra of $\mathfrak{k}_s$. Since $\mathfrak{z}$ is central, $\mathfrak{z} \oplus \mathfrak{a}$ is abelian and contains $\mathfrak{t}$. The Lie algebra of a maximal torus is maximal abelian (Lecture 6), so $\mathfrak{z} \oplus \mathfrak{a} = \mathfrak{t}$. Hence $\mathfrak{a} \subset \mathfrak{t} \cap \mathfrak{k}_s = \mathfrak{t}_s$.

The adjoint operators of elements of $\mathfrak{t}_s$ commute and are skew-adjoint, hence simultaneously diagonalisable over $\mathbb{C}$. So the complexification $\mathfrak{t}_{s,\mathbb{C}}$ is toral. It is also maximal toral. If $X \in \mathfrak{k}_s$ commutes with $\mathfrak{t}_s$, then $\mathfrak{t}_s + \mathbb{R}X$ is abelian, so $X \in \mathfrak{t}_s$. Now let $X, Y \in \mathfrak{k}_s$ with $[H, X + iY] = 0$ for all $H \in \mathfrak{t}_s$. The brackets $[H, X]$ and $[H, Y]$ lie in $\mathfrak{k}_s$, so both vanish, and $X, Y \in \mathfrak{t}_s$. Hence $\mathfrak{t}_{s,\mathbb{C}}$ is its own centraliser in $\mathfrak{k}_{s,\mathbb{C}}$. A toral subalgebra $\mathfrak{h} \supset \mathfrak{t}_{s,\mathbb{C}}$ is abelian [I], so $\mathfrak{h}$ lies in this centraliser and $\mathfrak{h} = \mathfrak{t}_{s,\mathbb{C}}$. Thus $\mathfrak{t}_{s,\mathbb{C}}$ is a maximal toral, or Cartan, subalgebra of $\mathfrak{k}_{s,\mathbb{C}}$. This is the Cartan subalgebra used in [I].

For a root α, [I] supplies a coroot $h_\alpha \in \mathfrak{t}_{s,\mathbb{C}}$. The corresponding root in the sense of the Lie Algebras course is the complex-linear extension of $i\alpha$. This extension takes the value 2 at h_α. By Lecture 7, the real coroot is $H_\alpha = ih_\alpha \in \mathfrak{t}_s$. Since $h_\alpha = -iH_\alpha$, the value of the complex-linear extension of $i\alpha$ at h_α equals $\alpha(H_\alpha)$. So

$$\alpha(H_\alpha) = 2, \qquad H_\alpha \in \mathfrak{t}_s.$$

In the same way, the complex-linear extension of $i\ell$ takes the value $\ell(H_\alpha)$ at h_α. So we put

$$\langle \ell, \alpha^\vee \rangle = \ell(H_\alpha), \qquad P = \{\lambda \in \mathfrak{t}_s^* : \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z} \ \forall \alpha\},$$

$$X^*(T)^+ = \{\ell \in X^*(T) : \langle \ell, \alpha^\vee \rangle \geq 0 \ \forall \alpha \in R^+\}.$$

By Lecture 7, $2\pi H_\alpha \in \Lambda$. Hence every $\ell \in X^*(T)$ satisfies

$$\langle \ell, \alpha^\vee \rangle = \tfrac{1}{2\pi}\, \ell(2\pi H_\alpha) \in \mathbb{Z},$$

so $\ell|_{\mathfrak{t}_s} \in P$. Let Q be the root lattice, the lattice generated by the roots. Roots lie in $X^*(T)$, so for semisimple K we obtain $Q \subset X^*(T) \subset P$. If K is also simply connected, then Λ is generated as a lattice by the $2\pi H_\alpha$, and $X^*(T) = P$ (Lecture 7).

For SU(2), put $h = \operatorname{diag}(1, -1)$, so that ih is the matrix H_0 of Lecture 7. Then $\mathfrak{t} = \mathbb{R}\, ih$, with

$$\exp(\theta\, ih) = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}, \qquad \Lambda = 2\pi\mathbb{Z}\, ih.$$

Since $[\theta\, ih, E_{12}] = 2i\theta E_{12}$, we get $\alpha(ih) = 2$ and $h_\alpha = h$. So $X^*(T) = P = \{\ell : \ell(ih) \in \mathbb{Z}\}$, and dominance says that $m = \ell(ih) \geq 0$. The classification theorem below says that this m labels V_m. For SO(3) only the even m remain.

8.2 The classification theorem

We prove the theorem by passing to a finite cover of K. First let V be irreducible for K. By Schur's lemma, the centre $\mathfrak{z}$ acts by a scalar $i\mu$. So V stays irreducible under $\mathfrak{k}_s$, and the highest-weight theorem of the Lie Algebras course gives a highest weight λ. Thus $\ell = \mu + \lambda$ is defined. A representation of a connected group is determined by its differential, so ℓ determines V.

For existence, start from ℓ. Integrate the module $V(\lambda)$ to the simply connected group K_s with Lie algebra $\mathfrak{k}_s$. Multiply by the character χ_μ of the central torus $Z_0 = \exp \mathfrak{z}$ given by $\chi_\mu(\exp Z) = e^{i\mu(Z)}$. This gives a representation of the finite cover $Z_0 \times K_s$ of K. It descends if and only if the finite kernel F acts trivially. By the lifting lemma, Lemma 8.2, each element of F is the exponential of a point $H \in \Lambda$. It acts on a highest weight vector by $e^{i\ell(H)} = 1$, since $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. By Schur's lemma it then acts trivially.

We use the following structure inputs.

(S1) (Torus theorem) In a compact connected Lie group, every element lies in a maximal torus, and all maximal tori are conjugate (Lecture 6). The image of exp generates the group (Lecture 2). A compact connected abelian Lie group is the image of its Lie algebra under exp, and it is isomorphic to $\mathrm{U}(1)^n$ (Lecture 7). The Lie algebra of a maximal torus is a maximal abelian subalgebra (Lecture 6).

(S2) (Lecture 7; H Chapter 13; covering theorem) Let $Z_0 = \exp \mathfrak{z}$, a torus. Let K_s be the simply connected Lie group with Lie algebra $\mathfrak{k}_s$. Then K_s is compact, by Weyl's theorem on the fundamental group (Lecture 7). The inclusion $\mathfrak{k}_s \subset \mathfrak{k}$ integrates to $\iota\colon K_s \to K$. The map

$$p\colon \widetilde{K} = Z_0 \times K_s \to K, \qquad p(z, s) = z\, \iota(s),$$

is a surjective homomorphism with finite central kernel F, and $dp = \mathrm{id}_{\mathfrak{k}}$. Hence p is a covering homomorphism by Lecture 4. In the notation of Lecture 7, $Z_0 = T_0$, $K_s = K_{sc}$ and $\iota = \varphi$. The group $\widetilde{K}$ is a finite cover of K. It is not the universal cover when $\mathfrak{z} \neq 0$, because Z_0 is a torus. For this reason a character of Z_0 must satisfy an integrality condition.

(S3) (Lecture 4; integration and descent) Finite-dimensional representations of $\mathfrak{k}_s$ integrate to K_s. Let $\kappa\colon G_1 \to G_2$ be a covering homomorphism with kernel N. A representation $\widetilde{\Pi}$ of G_1 equals $\Pi \circ \kappa$ for a representation Π of G_2 if and only if $\widetilde{\Pi}(N) = \{I\}$.

(S4) ([I], highest-weight theorem) Finite-dimensional $\mathfrak{k}_{s,\mathbb{C}}$-modules are sums of weight spaces with weights in P. An irreducible such module has a unique highest weight λ. Its weight space is a line, every weight lies in $\lambda - \mathbb{N}R^+$, and $\langle \lambda, \alpha^\vee \rangle \geq 0$ for $\alpha \in R^+$. Each dominant $\lambda \in P$ is the highest weight of exactly one irreducible module $V(\lambda)$, up to isomorphism.

Lemma 8.1. *Let G be a compact connected Lie group with a maximal torus S and $\mathfrak{s} = \operatorname{Lie} S$. Every central element f of G equals $\exp H$ for some $H \in \mathfrak{s}$.*

Proof. By the torus theorem (S1), $f \in gSg^{-1}$ for some $g \in G$. As f is central, $f = g^{-1}fg \in S$. Since $\exp\colon \mathfrak{s} \to S$ is onto, $f = \exp H$ with $H \in \mathfrak{s}$. ∎

Lemma 8.2. *Assume the torus theorem (S1) and the covering theorem (S2). Then $\widetilde{T} = \exp_{\widetilde{K}} \mathfrak{t}$ is a maximal torus of $\widetilde{K}$. Every $f \in F$ equals $\exp_{\widetilde{K}} H$ for some $H \in \Lambda$.*

Proof. Since $dp = \mathrm{id}_{\mathfrak{k}}$, naturality of exp (Lecture 2) gives

$$p(\exp_{\widetilde{K}} X) = \exp_K X \qquad (X \in \mathfrak{k}). \tag{8.2}$$

Let T' be the identity component of the closed subgroup $p^{-1}(T)$. Its Lie algebra consists of the X with $\exp_K(tX) \in T$ for all t. This set is $\mathfrak{t}$. As $\mathfrak{t}$ is abelian, $\exp_{\widetilde{K}}|_{\mathfrak{t}}$ is a homomorphism. Its image is a subgroup of T' containing a neighbourhood of 1 (Lecture 2). So the image is open, hence also closed, in the connected group T'. Thus $\widetilde{T} = T'$.

So $\widetilde{T}$ is compact, connected and abelian, hence a torus. Let $S \supset \widetilde{T}$ be a torus. Then $\operatorname{Lie} S$ is abelian and contains $\mathfrak{t}$. Since $\mathfrak{t}$ is maximal abelian in $\mathfrak{k}$, $\operatorname{Lie} S = \mathfrak{t}$ and $S = \widetilde{T}$.

Finally, $f \in F$ is central in the compact connected group $\widetilde{K}$. By the central-element lemma, Lemma 8.1, $f = \exp_{\widetilde{K}} H$ with $H \in \mathfrak{t}$. By naturality (8.2), $\exp_K H = p(f) = 1$, so $H \in \Lambda$. ∎

Theorem 8.3. *Assume the structure inputs (S1)–(S4). Each irreducible representation of K has a unique highest weight. It lies in $X^*(T)^+$, and this gives a bijection $\operatorname{Irr}(K) \to X^*(T)^+$.*

Proof. Step 1: the highest weight. Let Π be irreducible on V, and let $Z \in \mathfrak{z}$. Since $[Z, X] = 0$ for every $X \in \mathfrak{k}$, the element $\exp tZ$ commutes with every $\exp X$. The elements $\exp X$ generate K by the torus theorem (S1), so $\exp tZ$ is central. Hence $\Pi(\exp tZ)$ commutes with $\Pi(K)$. Differentiating at $t = 0$, so does $d\Pi(Z)$. By Schur's lemma, $d\Pi(Z)$ is a scalar. The operator $d\Pi(Z)$ is skew-adjoint, so the scalar is imaginary. The scalar depends linearly on Z. Hence

$$d\Pi(Z) = i\mu(Z)I \qquad \text{for some } \mu \in \mathfrak{z}^*.$$

Let $W \subset V$ be a complex subspace invariant under $d\Pi(\mathfrak{k}_s)$. Then W is invariant under $d\Pi(\mathfrak{k})$, and so under every $\Pi(\exp X) = e^{d\Pi(X)}$. The elements $\exp X$ generate K by the torus theorem (S1), so W is K-invariant. Hence V is irreducible under $\mathfrak{k}_s$. The complexification $\mathfrak{k}_{s,\mathbb{C}}$ has the same invariant complex subspaces, so V is also irreducible under $\mathfrak{k}_{s,\mathbb{C}}$.

By the highest-weight theorem (S4), the $\mathfrak{k}_{s,\mathbb{C}}$-module V has a highest weight λ and a highest weight vector v. For $H = Z + H_s$ with $Z \in \mathfrak{z}$ and $H_s \in \mathfrak{t}_s$,

$$d\Pi(H)v = i\big(\mu(Z) + \lambda(H_s)\big)v = i\ell(H)v, \qquad \ell := \mu + \lambda \in \mathfrak{t}^*.$$

Exponentiating gives $\Pi(\exp H)v = e^{i\ell(H)}v$. So v is a common eigenvector of $\Pi(T)$, and $\ell \in X^*(T)$ by the description of characters in § 8.1.

Now let ℓ' be any T-weight of V. The restriction of ℓ' to $\mathfrak{z}$ equals μ. The restriction of ℓ' to $\mathfrak{t}_s$ lies in $\lambda - \mathbb{N}R^+$ by the highest-weight theorem (S4). Roots vanish on $\mathfrak{z}$, so $\ell - \ell' \in \mathbb{N}R^+$. Thus ℓ is the highest weight of Π. It is dominant, since $\langle \ell, \alpha^\vee \rangle = \langle \lambda, \alpha^\vee \rangle \geq 0$.

Step 2: injectivity. Let Π_1, Π_2 be irreducible with the same highest weight ℓ. They share μ and λ. By the uniqueness part of the highest-weight theorem (S4), there is an isomorphism $A\colon V_1 \to V_2$ of $\mathfrak{k}_{s,\mathbb{C}}$-modules. It also intertwines $d\Pi_j(Z) = i\mu(Z)I$. Hence for every $X \in \mathfrak{k}$

$$A\,\Pi_1(\exp X) = A\,e^{d\Pi_1(X)} = e^{d\Pi_2(X)}A = \Pi_2(\exp X)\,A.$$

The elements $\exp X$ generate K by the torus theorem (S1), so A is K-equivariant.

Step 3: existence. Let $\ell \in X^*(T)^+$, with $\mu = \ell|_{\mathfrak{z}}$ and $\lambda = \ell|_{\mathfrak{t}_s}$. By the integrality computation in § 8.1, $\lambda \in P$, and λ is dominant. The module $V(\lambda)$ is only known to integrate to K_s. Since the map p has kernel F, we must show that F acts trivially. This step uses the hypothesis $\ell \in X^*(T)$, not only $\lambda \in P$.

The kernel $\Lambda_{\mathfrak{z}} = \Lambda \cap \mathfrak{z}$ of $\exp|_{\mathfrak{z}}$ lies in Λ. So $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, and

$$\chi_\mu(\exp Z) = e^{i\mu(Z)} \qquad (Z \in \mathfrak{z})$$

is a well-defined character of Z_0. By the highest-weight theorem (S4) and integration (S3), $V(\lambda)$ integrates to an irreducible representation Π_λ of K_s. Define on $V(\lambda)$

$$\widetilde{\Pi}(z,s) = \chi_\mu(z)\,\Pi_\lambda(s).$$

It is irreducible, because Π_λ already is. Let v be a highest weight vector. For $H = Z + H_s$ as above,

$$\widetilde{\Pi}(\exp_{\widetilde{K}} H)v = e^{i\mu(Z)}e^{d\Pi_\lambda(H_s)}v = e^{i\mu(Z)}e^{i\lambda(H_s)}v = e^{i\ell(H)}v.$$

Let $f \in F$. By the lifting lemma, Lemma 8.2, $f = \exp_{\widetilde{K}} H$ with $H \in \Lambda$. As f is central, Schur's lemma gives $\widetilde{\Pi}(f) = cI$. Evaluating on v,

$$cv = \widetilde{\Pi}(\exp_{\widetilde{K}} H)v = e^{i\ell(H)}v = v,$$

because $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. So F acts trivially. By the descent criterion (S3), $\widetilde{\Pi} = \Pi \circ p$ for a representation Π of K. It is irreducible, since p is onto.

It remains to identify the highest weight of Π. Since $dp = \mathrm{id}_{\mathfrak{k}}$, we have $d\Pi = d\widetilde{\Pi}$. So

$$d\Pi(Z) = i\mu(Z)I \quad (Z \in \mathfrak{z}), \qquad d\Pi|_{\mathfrak{k}_s} = d\Pi_\lambda.$$

Thus V is $V(\lambda)$ as an $\mathfrak{k}_{s,\mathbb{C}}$-module, with highest weight vector v. These are exactly the two data used in Step 1. Step 1 therefore gives the highest weight $\mu + \lambda = \ell$. ∎

Compactness gave the invariant inner product. This yields the weight decomposition of V. It also makes $d\Pi(Z)$ skew-adjoint, so μ is real. Compactness of $\widetilde{K}$ allowed the torus theorem (S1) inside the lifting lemma, Lemma 8.2. Schur's lemma needs only irreducibility. Connectedness entered through the torus theorem (S1). Connectedness is needed. The trivial character and det of $\mathrm{O}(2)$ have the same differential and differ on a reflection (Lecture 13).

H §§12.1–12.3 phrase the result with *analytically integral* elements. Hall's kernel lattice is our Λ divided by 2π. Hall obtains existence by analytic methods in Chapter 12. Our proof passes through the covering p.

8.3 Central characters and global forms

Let K_{sc} be compact, connected, simply connected and semisimple, with maximal torus T. Let F be a subgroup of its finite centre.

Corollary 8.4. *Assume the torus theorem (S1), integration and descent (S3), and the highest-weight theorem (S4). Let $\lambda \in P$ be dominant. Then $V(\lambda)$ descends to K_{sc}/F if and only if F acts trivially. Equivalently,*

$$\lambda(H) \in 2\pi\mathbb{Z} \qquad \text{whenever } H \in \mathfrak{t} \text{ and } \exp H \in F.$$

Proof. The first statement is the descent criterion (S3). Let Π be the representation of K_{sc} integrating $V(\lambda)$, which exists by integration (S3). Let $f \in F$. By the central-element lemma, Lemma 8.1, $f = \exp H$ with $H \in \mathfrak{t}$. By Schur's lemma, f acts by a scalar. Evaluating on a highest weight vector gives $\Pi(f) = e^{i\lambda(H)} I$. This is I exactly when $\lambda(H) \in 2\pi\mathbb{Z}$. ∎

Since $f \mapsto \Pi(f)$ is a homomorphism, it suffices to test generators of F.

Now allow a central torus. Here $F = \ker p$ again, as in the covering theorem (S2). Since F is central in $\widetilde{K}$, each $(z, s) \in F$ has s central in K_s.

We recall the notation of Step 3 in the proof of the classification theorem. The kernel of $\exp|_{\mathfrak{z}}$ is $\Lambda_{\mathfrak{z}} = \Lambda \cap \mathfrak{z}$. For $\mu \in \mathfrak{z}^*$ with $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, the character χ_μ of Z_0 is given by $\chi_\mu(\exp Z) = e^{i\mu(Z)}$. For dominant $\lambda \in P$, the representation of K_s integrating $V(\lambda)$ is Π_λ.

Let Π be an irreducible representation of a compact group. By Schur's lemma, each central element acts by a scalar. The map sending a central element to its scalar is a homomorphism from the centre to $\mathbb{C}^\times$, the *central character* of Π. Let c_λ be the central character of Π_λ. So $c_\lambda(s)$ is the scalar by which an element s of the centre of K_s acts on $V(\lambda)$.

Corollary 8.5. *Assume the structure inputs (S1)–(S4). Let $\mu \in \mathfrak{z}^*$ and $\lambda \in \mathfrak{t}_s^*$ satisfy*

(a) $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, so χ_μ is a character of Z_0 (integral character);

(b) $\lambda \in P$ is dominant;

(c) $\chi_\mu(z)\, c_\lambda(s) = 1$ for all $(z, s) \in F$ (finite compatibility).

Then $\rho_{\mu,\lambda}(z, s) = \chi_\mu(z)\Pi_\lambda(s)$ descends to an irreducible representation of K. Every irreducible representation of K arises in this way from exactly one pair (μ, λ).

Proof. By the integrality condition (a), the dominance condition (b) and integration (S3), $\rho_{\mu,\lambda}$ is an irreducible representation of $\widetilde{K}$. Let $f = (z, s) \in F$. As noted before the corollary, s is central in K_s. So $\rho_{\mu,\lambda}(f) = \chi_\mu(z)c_\lambda(s)I$. By the descent criterion (S3), $\rho_{\mu,\lambda}$ descends if and only if the compatibility condition (c) holds. The descended representation is irreducible because p is onto. The computation in Step 3 of the proof of the classification theorem shows that the descended representation has highest weight $\mu + \lambda$.

Conversely, let Π be irreducible with highest weight ℓ. Put $\mu = \ell|_{\mathfrak{z}}$ and $\lambda = \ell|_{\mathfrak{t}_s}$. Step 3 proves the integrality condition (a) and the dominance condition (b). It also gives Π' with $\Pi' \circ p = \rho_{\mu,\lambda}$ and highest weight ℓ. By the injectivity in Step 2, $\Pi \cong \Pi'$. Since $\Pi' \circ p$ is trivial on F, the compatibility condition (c) holds. The pair is unique because $\mathfrak{t} = \mathfrak{z} \oplus \mathfrak{t}_s$ recovers (μ, λ) from $\mu + \lambda$. ∎

This corollary and the classification theorem, Theorem 8.3, show that the integrality, dominance and compatibility conditions together are equivalent to $\mu + \lambda \in X^*(T)^+$. The compatibility condition is the description of $X^*(T)$ in Lecture 7. Indeed, a central s lies in a maximal torus of K_s by the central-element lemma, Lemma 8.1. Evaluating on a highest weight vector shows that $c_\lambda(s)$ is the value at s of the character χ_λ. The U(2) example, Example 8.9, checks this directly.

A module of the reductive algebra $\mathfrak{k}_{\mathbb{C}}$ can fail to come from a representation of K, because compactness constrains the action of the centre.

Proposition 8.6. *Let Π be a representation of K and $Z \in \mathfrak{z}$. Then $d\Pi(Z)$ is diagonalisable with eigenvalues in $i\mathbb{R}$. If $\exp Z = 1$, these eigenvalues lie in $2\pi i\mathbb{Z}$.*

Proof. With respect to an invariant inner product, $\Pi(\exp tZ)$ is unitary, so $d\Pi(Z)$ is skew-adjoint. Hence it is diagonalisable with eigenvalues $i\theta_j$, $\theta_j \in \mathbb{R}$. If $\exp Z = 1$, then $e^{d\Pi(Z)} = \Pi(1) = I$, so $e^{i\theta_j} = 1$. ∎

For $\mathfrak{k} = \mathfrak{u}(1) = i\mathbb{R}$, neither of the following modules comes from U(1):

$$\rho_1(it) = t\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \rho_2(it) = \tfrac{i}{2}t.$$

The first is not semisimple. The second has the wrong period: $\exp(2\pi i) = 1$, but $\rho_2(2\pi i) = \pi i \notin 2\pi i\mathbb{Z}$. Both integrate to the noncompact simply connected group $\mathbb{R}$, where no averaging is available. In physics terms, a compact U(1) symmetry has integer charges, and a noncompact $\mathbb{R}$ allows any charge.

The same happens for $\mathfrak{gl}(2,\mathbb{C}) = \mathbb{C}I \oplus \mathfrak{sl}(2,\mathbb{C})$. Letting I act on V_m by any $s \in \mathbb{C}$ gives a module. It comes from U(2) if and only if $s \in \mathbb{Z}$ and $s \equiv m \pmod 2$ (Exercise 8.5). On the semisimple part, Weyl's complete reducibility theorem already gives semisimplicity and weights in P. The remaining constraint is $X^*(T)$, which can be smaller.

8.4 Global forms in examples

Each irreducible representation of K is one dot of the lattice $X^*(T)$ in the dominant region. Passing to a quotient of K removes dots. Figure 8.1 draws this for SU(2) and for SU(3). The labels (a, b) are defined in Example 8.8, and the marks are those of Lecture 7.

Example 8.7 (SU(2) and SO(3))**.** We continue the computation at the end of § 8.1, with $h = \operatorname{diag}(1, -1)$. Thus $\langle \ell, \alpha^\vee \rangle = \ell(ih)$, and $Q = \{\ell : \ell(ih) \in 2\mathbb{Z}\}$. The highest weight of V_m has $\ell(ih) = m$. Now $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ (Lecture 4). The element $-I$ is central, and

$$\exp(\pi i h) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

By the quotient criterion, Corollary 8.4, $-I$ acts on V_m by

$$e^{i\ell(\pi i h)} = e^{i\pi m} = (-1)^m.$$

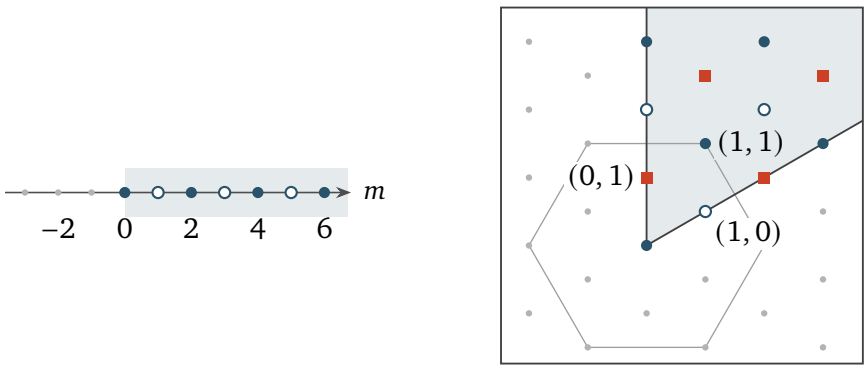


Figure 8.1: Characters of a maximal torus of SU(2), with coordinate $m = \ell(ih)$, on the left, and of SU(3) on the right, in the plane $\mathfrak{t}^*$. The shaded regions are the dominant regions, and the grey dots are not dominant. Filled dots in a shaded region are the highest weights of representations of SO(3) on the left and of PSU(3) on the right. On the right, a circle or a square marks the central value ω or ω^2. The thin hexagon joins the roots.

So V_m descends to SO(3) if and only if m is even. The irreducible representations of SO(3) have dimensions $1, 3, 5, \ldots$. In physics, V_m describes spin $m/2$. Hence the representations of integer spin descend to the rotation group SO(3). The representations of half-integer spin are representations of SU(2) that do not descend.

Let $r\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ be the covering. Its derivative is an isomorphism, and we identify the Lie algebras through it. Naturality of exp (Lecture 2) gives $\exp_{\mathrm{SO}(3)}(\theta\, ih) = r(\exp(\theta\, ih))$. This element is 1 if and only if $\exp(\theta\, ih) = \pm I$, that is, if and only if $\theta \in \pi\mathbb{Z}$. Hence the maximal torus $r(T)$ of SO(3) has $X^*(r(T)) = \{\ell : \pi\ell(ih) \in 2\pi\mathbb{Z}\} = Q$. This lattice has index 2 in P.

Example 8.8 (PSU(3))**.** Let $\mathfrak{t} = \{\mathrm{diag}(i\theta_1, i\theta_2, i\theta_3) : \sum \theta_j = 0\}$ and $\varepsilon_j(H) = \theta_j$. The roots are $\varepsilon_j - \varepsilon_k$ with $j \neq k$, with root vectors E_{jk}. We take the positive roots $\varepsilon_j - \varepsilon_k$ with $j < k$. The simple roots $\varepsilon_1 - \varepsilon_2$ and $\varepsilon_2 - \varepsilon_3$ have the real coroots $H_{12} = i\,\mathrm{diag}(1, -1, 0)$ and $H_{23} = i\,\mathrm{diag}(0, 1, -1)$. Then

$$\varepsilon_1(H_{12}) = 1, \quad \varepsilon_1(H_{23}) = 0, \quad (\varepsilon_1 + \varepsilon_2)(H_{12}) = 0, \quad (\varepsilon_1 + \varepsilon_2)(H_{23}) = 1.$$

The fundamental weights ϖ_1 and ϖ_2 are dual to the simple coroots. Thus ϖ_1 takes the values 1 and 0 at H_{12} and H_{23}, and ϖ_2 takes the values 0 and 1. So $\varpi_1 = \varepsilon_1$ and $\varpi_2 = \varepsilon_1 + \varepsilon_2$. The label (a, b) denotes the highest weight

$$\ell = a\varpi_1 + b\varpi_2 = (a + b)\varepsilon_1 + b\varepsilon_2.$$

The centre is $\{I, \omega I, \omega^2 I\}$ with $\omega = e^{2\pi i/3}$, and

$$\omega I = \exp H_\omega, \qquad H_\omega = \tfrac{2\pi i}{3}\,\mathrm{diag}(1, 1, -2) \in \mathfrak{t}.$$

Then

$$\ell(H_\omega) = (a + b)\tfrac{2\pi}{3} + b\tfrac{2\pi}{3} = \tfrac{2\pi}{3}(a + 2b),$$

so ωI acts on the irreducible module with highest weight (a, b) by ω^{a+2b}. Hence this module descends to $\mathrm{PSU}(3) = \mathrm{SU}(3)/\langle \omega I \rangle$ if and only if $3 \mid a + 2b$. Since $(a + 2b) - (a - b) = 3b$, this means $a \equiv b \pmod 3$. The dimension formula $\frac{1}{2}(a + 1)(b + 1)(a + b + 2)$ [I] gives the following values.

(a,b)	$(1,0)$	$(0,1)$	$(2,0)$	$(1,1)$	$(3,0)$	$(0,3)$	$(2,2)$
dimension	3	3	6	8	10	10	27
ωI acts by	ω	ω^2	ω^2	1	1	1	1

The adjoint representation $(1,1)$ descends, since $\mathrm{Ad}(\omega I) = \mathrm{id}$.

Example 8.9 (U(2)). Let $p\colon \mathrm{U}(1)\times\mathrm{SU}(2)\to\mathrm{U}(2)$ be given by $p(z,g)=zg$. The map p is onto. Indeed, given u, choose z with $z^2=\det u$ and put $g=z^{-1}u$. For the kernel,

$$zg=I \implies g=z^{-1}I,\quad 1=\det g=z^{-2},\qquad \text{so}\qquad F=\{(1,I),(-1,-I)\}.$$

Here $\Lambda_{\mathfrak{z}}=2\pi i\mathbb{Z}\,I$, so the integral characters of $Z_0\cong\mathrm{U}(1)$ are $z\mapsto z^q$ with $q\in\mathbb{Z}$. Put $\Pi_{q,m}(z,g)=z^q\Pi_m(g)$. The compatibility condition (c) is

$$\Pi_{q,m}(-1,-I)=(-1)^q(-1)^m I=I \iff q+m \text{ is even.}$$

We cross-check with the classification theorem, Theorem 8.3. The torus $\mathrm{diag}(e^{i\theta_1},e^{i\theta_2})$ has $X^*(T)=\mathbb{Z}^2$, and (n_1,n_2) is dominant if and only if $n_1\ge n_2$. The torus element equals $p(e^{i\psi},\mathrm{diag}(e^{i\phi},e^{-i\phi}))$ when $\theta_1=\psi+\phi$ and $\theta_2=\psi-\phi$. On a highest weight vector of $\Pi_{q,m}$, this torus element acts by $e^{i(q\psi+m\phi)}$, while

$$n_1\theta_1+n_2\theta_2=(n_1+n_2)\psi+(n_1-n_2)\phi.$$

So $q=n_1+n_2$ and $m=n_1-n_2$. The numbers $n_{1,2}=(q\pm m)/2$ are integers exactly when $q+m$ is even. For example, $\mathbb{C}^2$ corresponds to $(q,m)=(1,1)$, and $\det$ corresponds to $(2,0)$.

8.5 Tensor products and consistency checks

Lemma 8.10. *Let f be central in K, acting by scalars c_1 on V_1 and c_2 on V_2. Then f acts by c_1c_2 on $V_1\otimes V_2$ and on each subrepresentation, and by c_1^{-1} on V_1^*.*

Proof. First, $(\Pi_1\otimes\Pi_2)(f)=c_1I\otimes c_2I=c_1c_2\,I$. Invariant subspaces inherit this scalar. On the dual, $\Pi_1^*(f)=\Pi_1(f^{-1})^T=c_1^{-1}I$. ∎

Hence the decompositions from [I] must respect central characters. We check this in the following cases.

- Clebsch–Gordan: $V_m\otimes V_n=\bigoplus_{k=0}^{\min(m,n)}V_{m+n-2k}$. Every summand has parity $m+n$, matching $(-1)^m(-1)^n$. In particular, $V_1\otimes V_1=V_2\oplus V_0$ is a sum of $\mathrm{SO}(3)$-representations, although V_1 is not an $\mathrm{SO}(3)$-representation.
- $\mathrm{SU}(3)$: $(1,0)\otimes(1,0)=(2,0)\oplus(0,1)$, with $9=6+3$ and $\omega\cdot\omega=\omega^2$ on both summands. Also $(1,0)\otimes(0,1)=(1,1)\oplus(0,0)$, with $9=8+1$ and $\omega\cdot\omega^2=1$.
- Duals: $V(a,b)^*\cong V(b,a)$ [I]. The lemma predicts $\omega^{b+2a}=\omega^{-(a+2b)}$. Indeed $(b+2a)+(a+2b)=3(a+b)$.
- Branching from $\mathrm{U}(2)$ to $\mathrm{SU}(2)$: $\Pi_{q,m}$ restricts to V_m. On the torus of $\mathrm{U}(2)$ it has the $m+1$ weights (n_1-j,n_2+j), $0\le j\le m$. Characters z^q multiply, so $\Pi_{1,1}\otimes\Pi_{1,1}=\Pi_{2,2}\oplus\Pi_{2,0}$. Both summands have $q+m$ even.

Self-checks

1. Does $V_3 \otimes V_5$ descend to $\mathrm{SO}(3)$? Decompose it.

   Answer. Yes. The element $-I$ acts by $(-1)^3(-1)^5 = 1$. By Clebsch–Gordan,

   $$V_3 \otimes V_5 = V_8 \oplus V_6 \oplus V_4 \oplus V_2, \qquad 9 + 7 + 5 + 3 = 24 = 4 \cdot 6.$$

2. Which irreducible representations of $\mathrm{PSU}(3)$ have dimension at most 10?

   Answer. The $\mathrm{SU}(3)$-irreducibles of dimension at most 10 are $(0,0)$, $(1,0)$, $(0,1)$, $(2,0)$, $(0,2)$, $(1,1)$, $(3,0)$, $(0,3)$. The next ones, such as $(2,1)$ and $(4,0)$, have dimension 15. The ones that descend to $\mathrm{PSU}(3)$ are those with $a \equiv b \pmod 3$, namely

   $$(0,0),\ (1,1),\ (3,0),\ (0,3),$$

   of dimensions $1, 8, 10, 10$.

3. Let $I \in \mathfrak{gl}(2,\mathbb{C})$ act by 1 on V_2. Does this module come from $\mathrm{U}(2)$?

   Answer. No. It would be $\Pi_{1,2}$, and $1 + 2$ is odd. Directly, compute $\Pi(-I)$ in two ways:

   $$-I = \exp(i\pi I) \implies \Pi(-I) = e^{i\pi \cdot 1} = -1, \qquad -I \in \mathrm{SU}(2) \implies \Pi(-I) = (-1)^2 = 1.$$

   These disagree.

Summary

Let K be a compact connected Lie group, and assume the cited structure inputs. An irreducible representation of K is determined by its highest weight. The possible highest weights are the dominant characters of a maximal torus T. For K_{sc}/F, test F on one highest weight vector. With a central torus, one adds an integral character of the central torus and a compatibility condition on F. In the examples, the descent conditions are m even for $\mathrm{SO}(3)$, $3 \mid a + 2b$ for $\mathrm{PSU}(3)$, and $q + m$ even for $\mathrm{U}(2)$.

Reading

H denotes Hall, *Lie Groups, Lie Algebras, and Representations*. BD denotes Bröcker–tom Dieck, *Representations of Compact Lie Groups*.

- H §§12.1–12.3: representations of compact groups, analytically integral elements, and the setting of the highest-weight classification for compact connected groups.
- H Chapter 13: fundamental groups of compact groups, and the comparison of algebraic and analytic integrality. This supports (S2).
- BD VI.2: the dominant weights of irreducible representations of compact connected Lie groups.
- Lie Algebras [I]: highest-weight theorem, Weyl's complete reducibility theorem, the dimension formula, Clebsch–Gordan, duals and $\mathfrak{sl}_3$ tensor products.

Lecture 9

Weyl integration and character calculations

Objectives. *A* class function *on a group is a function that is constant on conjugacy classes. On a compact connected group, a class function is determined by its values on a maximal torus. This lecture shows that the Haar integral of a class function can also be computed on the torus. The density is a product over root planes. A factor* $1/|W|$ *corrects for repeated conjugates. We then use the formula to check character norms and tensor-product multiplicities.*

By the end of this lecture you will be able to:

- State the Weyl integration formula with its measure conventions and its imported steps.
- Derive the density $\det(1 - \mathrm{Ad}_t)$ on $\mathfrak{k}/\mathfrak{t}$ and the factor $1/|W|$.
- Compute class integrals on $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ and compare them.
- Check character norms and multiplicities with the Weyl character formula on T.

9.1 Measures, roots and the statement

Throughout, K is a compact connected Lie group and $T \subset K$ is a maximal torus. Their Lie algebras are $\mathfrak{t} \subset \mathfrak{k}$. By Lecture 6, every element of K is conjugate to an element of T. Also by Lecture 6, the group $W = N_K(T)/T$ is finite and acts on $X^*(T)$ as the Weyl group of the root system. Fix an $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{k}$ and set

$$\mathfrak{m} = \mathfrak{t}^{\perp} \subset \mathfrak{k}.$$

For $t \in T$ the map Ad_t is orthogonal and fixes $\mathfrak{t}$ pointwise. Hence Ad_t preserves $\mathfrak{m}$, and $\mathfrak{m} \cong \mathfrak{k}/\mathfrak{t}$ as representations of T.

We use three probability measures.

- dk is the normalised Haar measure on K (Lecture 5).
- dt is the normalised Haar measure on T.
- $d(gT)$ is the unique K-invariant probability measure on K/T.

The existence and uniqueness of $d(gT)$ are cited together with the theorem below. All three measures are given by smooth positive densities.

We keep the real weights and real roots of Lecture 6 and Lecture 7. For $\ell \in X^*(T)$, write t^ℓ for the value at t of the character of T with weight ℓ. Thus $t^\ell = e^{i\ell(H)}$ when $t = \exp H$ with $H \in \mathfrak{t}$, and $t^\ell t^{\ell'} = t^{\ell+\ell'}$.

With this convention, the root decomposition is

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \qquad \mathrm{Ad}_t E = t^\alpha E \quad (E \in \mathfrak{g}_\alpha,\ t \in T). \tag{9.1}$$

Each $\mathfrak{g}_\alpha$ is a complex line. Each root is a weight of the adjoint representation of K. So $t \mapsto t^\alpha$ is a character of T, that is, $\alpha \in X^*(T)$. If $t = \exp H$ with $H \in \mathfrak{t}$, then $t^\alpha = e^{i\alpha(H)}$ with $\alpha(H) \in \mathbb{R}$.

Fix a positive system R^+ and let ρ be half the sum of the positive roots. In general $\rho \notin X^*(T)$. § 9.4 shows how to avoid ρ as a function on T.

The Lie Algebras course states the Weyl character formula and the denominator identity for complex-linear weights on $\mathfrak{t}_{\mathbb{C}}$. In that course, a real weight ℓ appears as the complex weight $i\ell$, a root α as $i\alpha$, and ρ as $i\rho$. A complex weight $\mu = i\ell$ has the exponential $e^{\mu(H)} = e^{i\ell(H)} = t^\ell$ at $t = \exp H$. The map $\ell \mapsto i\ell$ is additive and commutes with W. So those identities keep their form when we write them with real exponents. They then hold in the ring $\mathbb{Z}[X^*(T)]$ of finite integer combinations of formal exponentials e^ℓ, where $e^\ell e^{\ell'} = e^{\ell+\ell'}$. Evaluating e^ℓ at t gives t^ℓ.

Definition 9.1. An element $t \in T$ is *regular* if $t^\alpha \neq 1$ for every $\alpha \in R$. The regular elements of T form T_{reg}. The regular elements of K form $K_{\mathrm{reg}} = \{gtg^{-1} : g \in K,\ t \in T_{\mathrm{reg}}\}$.

The conjugation map is

$$\psi \colon K/T \times T \to K, \qquad \psi(gT, t) = gtg^{-1}. \tag{9.2}$$

It is well defined because T is abelian. The domain and the codomain of ψ both have dimension $\dim K$.

Theorem 9.2 (Weyl integration formula)**.** *Let K, T, W and the measures be as above. Put*

$$J(t) = \det\bigl(1 - \mathrm{Ad}_t|_{\mathfrak{m}}\bigr) \qquad (t \in T).$$

Then $J \geq 0$, and for every continuous $f \colon K \to \mathbb{C}$,

$$\int_K f(k)\,dk = \frac{1}{|W|} \int_T J(t) \int_{K/T} f(gtg^{-1})\,d(gT)\,dt. \tag{9.3}$$

If f is a class function, then

$$\int_K f(k)\,dk = \frac{1}{|W|} \int_T f(t)\,J(t)\,dt. \tag{9.4}$$

The theorem is cited from H §11.6, BD IV.1 (1.11) and Knapp, Beyond, VIII.5. We prove the formula below from two imported analytic steps and the existence of $d(gT)$. The class-function version follows from the general formula (9.3). Indeed $f(gtg^{-1}) = f(t)$, and $d(gT)$ has total mass one.

We prove the formula by changing variables along ψ. Every element of K has the form gtg^{-1}. The Jacobian of ψ is a constant multiple of J, and the constant is fixed at the end. A regular

element has $|W|$ preimages, and this gives the factor $1/|W|$. The remaining elements form a set of Haar measure zero.

For example, take $K = \mathrm{SU}(2)$ and $t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$. The space K/T is the sphere S of Lecture 3, and the coset gT corresponds to $\xi = g\,\mathrm{diag}(1,-1)g^{-1}$. Then

$$g t_\theta g^{-1} = \cos\theta\, I + i \sin\theta\, \xi.$$

The regular elements of T are the t_θ with $\theta \notin \pi\mathbb{Z}$, and K_{reg} is SU(2) without $\pm I$. So the remaining elements are two points. A regular element has the two preimages (ξ, θ) and $(-\xi, 2\pi - \theta)$.

9.2 The root-plane Jacobian and the factor $|W|$

Lemma 9.3 (Root planes). *For $\alpha \in R^+$ put $\mathfrak{m}_\alpha = \mathfrak{k} \cap (\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha})$. Then $\mathfrak{m} = \bigoplus_{\alpha \in R^+} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ has a real basis consisting of X_α and Y_α, in which*

$$\mathrm{Ad}_t\,|_{\mathfrak{m}_\alpha} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \text{where } t^\alpha = e^{i\theta}.$$

Consequently, writing $t^\alpha = e^{i\theta_\alpha}$,

$$J(t) = \prod_{\alpha \in R^+} \left|1 - t^\alpha\right|^2 = \prod_{\alpha \in R} (1 - t^\alpha) = \prod_{\alpha \in R^+} 4 \sin^2 \frac{\theta_\alpha}{2}. \tag{9.5}$$

Proof. Let σ be complex conjugation on $\mathfrak{k}_{\mathbb{C}}$ with respect to $\mathfrak{k}$. It commutes with the real map Ad_t. For $E \in \mathfrak{g}_\alpha$ we get

$$\mathrm{Ad}_t(\sigma E) = \sigma(t^\alpha E) = \overline{t^\alpha}\,\sigma E = t^{-\alpha}\sigma E,$$

because $|t^\alpha| = 1$. So $\sigma\mathfrak{g}_\alpha = \mathfrak{g}_{-\alpha}$. Choose $E \neq 0$ in $\mathfrak{g}_\alpha$ and set

$$X_\alpha = E + \sigma E, \qquad Y_\alpha = i(E - \sigma E).$$

Both are fixed by σ, so they lie in $\mathfrak{k}$. An element $aE + b\,\sigma E$ is fixed by σ exactly when $b = \bar a$. If $a = x + iy$, then $aE + \bar a\,\sigma E = xX_\alpha + yY_\alpha$. Hence X_α and Y_α form a basis of $\mathfrak{m}_\alpha$. With $t^\alpha = e^{i\theta}$,

$$\begin{aligned} \mathrm{Ad}_t\, X_\alpha &= e^{i\theta}E + e^{-i\theta}\sigma E = \cos\theta\, X_\alpha + \sin\theta\, Y_\alpha, \\ \mathrm{Ad}_t\, Y_\alpha &= ie^{i\theta}E - ie^{-i\theta}\sigma E = -\sin\theta\, X_\alpha + \cos\theta\, Y_\alpha. \end{aligned}$$

Taking σ-fixed points in (9.1) gives $\mathfrak{k} = \mathfrak{t} \oplus \bigoplus_{\alpha>0} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ is orthogonal to $\mathfrak{t}$. To see this, pick $H \in \mathfrak{t}$ with $\alpha(H) \neq 0$. Then $[H, X_\alpha] = \alpha(H)Y_\alpha$ and $[H, Y_\alpha] = -\alpha(H)X_\alpha$, so $\mathfrak{m}_\alpha \subset [\mathfrak{t}, \mathfrak{k}]$. Invariance of the inner product gives $\langle [H, X], H' \rangle = -\langle X, [H, H'] \rangle = 0$ for $H' \in \mathfrak{t}$. Counting dimensions, $\mathfrak{m} = \bigoplus_{\alpha>0} \mathfrak{m}_\alpha$.

For one plane,

$$\det \begin{pmatrix} 1 - \cos\theta & \sin\theta \\ -\sin\theta & 1 - \cos\theta \end{pmatrix} = (1 - \cos\theta)^2 + \sin^2\theta = 2 - 2\cos\theta = |1 - e^{i\theta}|^2.$$

This equals $4\sin^2(\theta/2)$. Multiply over $\alpha \in R^+$. Finally $(1 - t^\alpha)(1 - t^{-\alpha}) = |1 - t^\alpha|^2$, since $t^{-\alpha} = \overline{t^\alpha}$. ∎

Thus $J \geq 0$, and $J(t) > 0$ exactly when $t \in T_{\mathrm{reg}}$. Also $J(t^{-1}) = J(t)$.

For $\mathrm{U}(n)$, with $t = \mathrm{diag}(z_1, \dots, z_n)$, the roots of Lecture 6 give the density $\prod_{j<k} |z_j - z_k|^2$. It vanishes when two eigenvalues coincide. So a Haar-random unitary matrix is unlikely to have two close eigenvalues.

Next we write the differential of ψ in translated coordinates. For $X \in \mathfrak{m}$ and $H \in \mathfrak{t}$ put

$$L_{(gT,t)}(X,H) = \frac{d}{ds}\Big|_{s=0} \psi(gT,t)^{-1}\,\psi\big(g\exp(sX)T,\ t\exp(sH)\big) \in \mathfrak{k}.$$

We identify the tangent space of K/T at gT with $\mathfrak{m}$ by $X \mapsto \frac{d}{ds}|_{s=0}\, g\exp(sX)T$. We identify the tangent spaces of T at t and of K at x with $\mathfrak{t}$ and $\mathfrak{k}$ by left translation. In these identifications, each invariant density is given at every point by its density at the base point. Let $\kappa > 0$ be the ratio of the base-point density of dk to the product of the base-point densities of $d(gT)$ and dt, all evaluated on orthonormal bases. Hence the Jacobian of ψ relative to $d(gT)\,dt$ and dk equals $\kappa\,|\det L_{(gT,t)}|$. Here the determinant is taken with respect to orthonormal bases of $\mathfrak{m} \oplus \mathfrak{t}$ and $\mathfrak{k}$.

The value $|\det L_{(gT,t)}|$ does not depend on the representative g of the coset gT. Indeed, replacing g by gs with $s \in T$ composes L with $\mathrm{Ad}_s|_{\mathfrak{m}}$. This map has determinant 1.

The next proposition is the Jacobian computation of Lecture 6, now at a general point. The map ψ is the map Φ there.

Proposition 9.4. *For all $g \in K$ and $t \in T$, $|\det L_{(gT,t)}| = J(t)$.*

Proof. Multiplying out,

$$\psi(gT,t)^{-1}\psi(ge^{sX}T, te^{sH}) = gt^{-1}e^{sX}t\,e^{sH}e^{-sX}g^{-1} = g\ \exp(s\,\mathrm{Ad}_{t^{-1}}X)\exp(sH)\exp(-sX)\,g^{-1}.$$

The velocity at $s = 0$ of a product of one-parameter subgroups is the sum of their velocities (Lecture 2). Hence

$$L_{(gT,t)}(X,H) = \mathrm{Ad}_g\big((\mathrm{Ad}_{t^{-1}} - 1)X + H\big).$$

Relative to $\mathfrak{k} = \mathfrak{m} \oplus \mathfrak{t}$ this is Ad_g composed with the block map

$$\begin{pmatrix} \mathrm{Ad}_{t^{-1}}|_{\mathfrak{m}} - 1 & 0 \\ 0 & 1_{\mathfrak{t}} \end{pmatrix}.$$

The map Ad_g is orthogonal, so $|\det \mathrm{Ad}_g| = 1$. Therefore $|\det L| = |\det(1 - \mathrm{Ad}_{t^{-1}}|_{\mathfrak{m}})| = J(t^{-1}) = J(t)$. ∎

If K is nonabelian, then $J(t)$ tends to zero as t approaches a central element. Indeed, every root character is 1 at that element. If K is a torus, the root product is empty and J is identically 1.

Proposition 9.5. *(a) For $t \in T$, $\dim Z_K(t) = \dim T + 2\,\#\{\alpha \in R^+ : t^\alpha = 1\}$. Hence $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$.*

(b) If $t, t' \in T_{\mathrm{reg}}$ and $gtg^{-1} = t'$, then $g \in N_K(T)$.

(c) Every $x \in K_{\mathrm{reg}}$ has exactly $|W|$ preimages under ψ.

Proof. (a) The Lie algebra of the closed subgroup $Z_K(t)$ is $\{X \in \mathfrak{k} : \mathrm{Ad}_t X = X\}$ (Lecture 3). A plane rotation by θ fixes a nonzero vector only when $e^{i\theta} = 1$. By the root-plane lemma (Lemma 9.3), the fixed space of Ad_t is $\mathfrak{t} \oplus \bigoplus_{t^\alpha = 1} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ has dimension 2, which gives the dimension formula. Centraliser dimensions are unchanged by conjugation. So if t is conjugate to a regular t', then t is regular.

Now let $(gT, t) \in \psi^{-1}(K_{\mathrm{reg}})$. Then gtg^{-1} is conjugate to some $t' \in T_{\mathrm{reg}}$, so t is conjugate to t', and hence $t \in T_{\mathrm{reg}}$. Conversely, $\psi(gT, t) \in K_{\mathrm{reg}}$ for every $t \in T_{\mathrm{reg}}$, by the definition of K_{reg}. These two inclusions give $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$.

(b) By the dimension formula in (a), the identity component $Z_K(t')_0$ has Lie algebra $\mathfrak{t}$. It contains the connected group T with the same Lie algebra, so $Z_K(t')_0 = T$ (Lecture 2). The torus gTg^{-1} is connected and centralises $gtg^{-1} = t'$. So $gTg^{-1} \subset T$. Then $\mathrm{Ad}_g \mathfrak{t} \subset \mathfrak{t}$, with equality by dimension. Hence $gTg^{-1} \supset \exp \mathfrak{t} = T$, and $g \in N_K(T)$.

(c) Let $x = \psi(gT, t)$ with t regular. Suppose $\psi(hT, s) = x$. By the description of $\psi^{-1}(K_{\mathrm{reg}})$ in (a), s is regular. Put $n = g^{-1}h$. Then $nsn^{-1} = t$. Since s and t are regular, the normaliser statement (b) gives $n \in N_K(T)$. Thus the preimages are the pairs $(gnT,\ n^{-1}tn)$ with $nT \in W$. Each such pair is a preimage. Different classes nT give different cosets gnT. So there are exactly $|W|$ preimages. ∎

Lemma 9.6. $\displaystyle\int_T J(t)\, dt = |W|.$

Proof. The Weyl denominator identity [I] holds in $\mathbb{Z}[X^*(T)]$:

$$\prod_{\alpha > 0} (1 - e^{-\alpha}) = \sum_{w \in W} \mathrm{sgn}(w)\, e^{w\rho - \rho}.$$

Here $\rho - w\rho$ is a sum of positive roots. So $w\rho - \rho \in X^*(T)$ even when $\rho \notin X^*(T)$. Evaluate both sides at t, and write $D(t)$ for the value of the left side. Then $|D(t)|^2 = J(t)$ by (9.5), so

$$J(t) = \sum_{w, w' \in W} \mathrm{sgn}(w)\, \mathrm{sgn}(w')\, t^{w\rho - w'\rho}.$$

Distinct characters of T are orthonormal in $L^2(T)$ (Lecture 5). Thus $\int_T t^\ell\, dt = 0$ for $\ell \neq 0$. Since ρ is regular [I], $w\rho = w'\rho$ only when $w = w'$. Hence only the diagonal terms survive, and the integral is $|W|$. ∎

The proof of the formula uses two imported analytic steps. The section on sources and hypotheses at the end of the lecture derives them from the cited results.

(C1) The set K_{reg} is open, and $K \setminus K_{\mathrm{reg}}$ has Haar measure zero. The null-set claim is cited from BD V.2, Proposition (2.6), together with Lee, Corollary 6.11.

(C2) Let M, N be compact manifolds of equal dimension. Give them finite measures defined by smooth positive densities. Let $\psi\colon M \to N$ be smooth and $U \subset N$ open. Assume $N \setminus U$ and $M \setminus \psi^{-1}(U)$ are null. Assume $\psi\colon \psi^{-1}(U) \to U$ is a surjective local diffeomorphism whose fibres all have d points. Then for continuous F on N,

$$\int_M F(\psi(x))\, J_\psi(x)\, dx = d \int_N F(y)\, dy,$$

where J_ψ is the Jacobian of ψ relative to the two densities. We derive this from Lee, Chapter 16, Propositions 16.40–16.42.

The second step is the change of variables for a map that covers its target d times. For $z \mapsto z^2$ on the circle, we have $d = 2$ and the Jacobian is 2, and

$$\frac{1}{2\pi}\int_0^{2\pi} F(e^{2i\theta})\,2\,d\theta = 2\cdot\frac{1}{2\pi}\int_0^{2\pi} F(e^{i\varphi})\,d\varphi.$$

Proof of the Weyl integration formula (Theorem 9.2) from (C1) and (C2). Null sets. The set $K\setminus K_{\mathrm{reg}}$ is null by the regular-set import (C1). The complement of $\psi^{-1}(K_{\mathrm{reg}})$ is $K/T\times(T\setminus T_{\mathrm{reg}})$. Now $T\setminus T_{\mathrm{reg}}$ is the union of the kernels T_α of $t\mapsto t^\alpha$. Each T_α has infinitely many cosets, because t^α maps the connected group T onto the circle. The cosets are disjoint and have equal Haar measure. Since T has finite measure, T_α is null.

Covering. On $K/T\times T_{\mathrm{reg}}$ the Jacobian is $\kappa J(t)>0$, so ψ is a local diffeomorphism there. By the fibre count in Proposition 9.5, its fibres have $|W|$ points.

Change of variables. Both $K/T\times T$ and K are compact. The change-of-variables import (C2) gives

$$\kappa\int_T J(t)\int_{K/T} f(gtg^{-1})\,d(gT)\,dt = |W|\int_K f(k)\,dk.$$

Take $f=1$. Then $\kappa\int_T J\,dt = |W|$. Since $\int_T J\,dt = |W|$ by Lemma 9.6, we get $\kappa = 1$. This is (9.3). ∎

9.3 The model: class integrals on $\mathrm{SU}(2)$

Let $K=\mathrm{SU}(2)$ with maximal torus $T=\{t_\theta\}$, where

$$t_\theta = \begin{pmatrix} e^{i\theta} & 0\\ 0 & e^{-i\theta}\end{pmatrix},\qquad dt = \frac{d\theta}{2\pi}\ \text{ on } [0,2\pi).$$

Use the invariant inner product $\langle X,Y\rangle = -\operatorname{tr}(XY)$ on $\mathfrak{su}(2)$. The plane $\mathfrak{m}$ has basis

$$B=\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix},\qquad C=\begin{pmatrix}0 & i\\ i & 0\end{pmatrix}.$$

A direct multiplication gives

$$t_\theta B t_\theta^{-1} = \begin{pmatrix}0 & e^{2i\theta}\\ -e^{-2i\theta} & 0\end{pmatrix} = \cos 2\theta\, B + \sin 2\theta\, C,\qquad t_\theta C t_\theta^{-1} = -\sin 2\theta\, B + \cos 2\theta\, C.$$

So $t_\theta^\alpha = e^{2i\theta}$ and

$$J(t_\theta) = |1-e^{2i\theta}|^2 = 4\sin^2\theta.$$

The Weyl group has order 2. Its nontrivial element is represented by

$$n=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix},\qquad n t_\theta n^{-1} = t_{-\theta}.$$

Now (9.4) gives, for a continuous class function f,

$$\int_{\mathrm{SU}(2)} f\,dk = \frac{1}{2}\cdot\frac{1}{2\pi}\int_0^{2\pi} f(t_\theta)\,4\sin^2\theta\,d\theta = \frac{1}{\pi}\int_0^{2\pi} f(t_\theta)\sin^2\theta\,d\theta.$$

Since $f(t_{2\pi-\theta}) = f(t_{-\theta}) = f(t_\theta)$ and $\sin^2$ is symmetric about π,

$$\int_{\mathrm{SU}(2)} f\,dk = \frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\,d\theta. \tag{9.6}$$

Every $g \in \mathrm{SU}(2)$ has eigenvalues $e^{\pm i\theta}$ for a unique $\theta \in [0,\pi]$, and $\operatorname{tr} g = 2\cos\theta$. So a class function is a function of $\theta \in [0,\pi]$. For $f = 1$ the right side is $\frac{2}{\pi}\cdot\frac{\pi}{2} = 1$.

Exercise 9.2 shows what this density means on S^3. For $0 < \theta < \pi$, the elements with trace $2\cos\theta$ form a 2-sphere of radius $\sin\theta$, the conjugacy class of t_θ. Its area is proportional to $\sin^2\theta$. At $\theta = 0$ and at $\theta = \pi$ the class is the single point I or $-I$. So $J(t)$ measures the size of the class of t. Elements near $\pm I$ are rare, since those classes are small.

Character norms. Let a representation V decompose as $\bigoplus_i m_i V_i$, with the V_i irreducible and pairwise inequivalent. This is possible by complete reducibility (Lecture 5). Orthonormality of characters (Lecture 5) gives $\int_K \chi_V\overline{\chi_i}\,dk = m_i$ and $\int_K |\chi_V|^2\,dk = \sum_i m_i^2$. So a character of norm one is irreducible.

By the Weyl character formula [I], the representation V_m of dimension $m+1$ has

$$\chi_m(t_\theta) = \sum_{k=0}^{m} e^{i(m-2k)\theta} = \frac{\sin(m+1)\theta}{\sin\theta} \qquad (\theta \notin \pi\mathbb{Z}).$$

Then (9.6) gives

$$\int_{\mathrm{SU}(2)} \chi_m\overline{\chi_n}\,dk = \frac{2}{\pi}\int_0^\pi \sin(m+1)\theta\,\sin(n+1)\theta\,d\theta = \delta_{mn}.$$

The factor $\sin^2\theta$ cancels both denominators. This confirms that each V_m is irreducible and that the representations V_m are pairwise inequivalent.

A tensor-product multiplicity. The multiplicity of V_n in V is $\int_K \chi_V\overline{\chi_n}\,dk$. Take $V = V_1^{\otimes 4}$, so $\chi_V(t_\theta) = (2\cos\theta)^4$. For $n = 0$ we need the Wallis integrals

$$\int_0^\pi \cos^4\theta\,d\theta = \frac{3\pi}{8}, \qquad \int_0^\pi \cos^6\theta\,d\theta = \frac{5\pi}{16}.$$

Write $\sin^2 = 1 - \cos^2$. Then

$$\int_0^\pi \cos^4\theta\sin^2\theta\,d\theta = \frac{3\pi}{8} - \frac{5\pi}{16} = \frac{\pi}{16}, \qquad \dim V^{\mathrm{SU}(2)} = \frac{2}{\pi}\cdot 16\cdot\frac{\pi}{16} = 2.$$

So the trivial representation V_0 occurs twice in $V_1^{\otimes 4}$. Exercise 9.3 finds the full decomposition from weights and checks the multiplicity of V_2 by integration.

Remark 9.7 (Indicator functions)**.** Probabilities of conjugation-invariant sets need discontinuous f. Let $c\colon K \to \mathbb{R}$ be a continuous class function and put $E = \{k \in K : c(k) \geq 0\}$. Then

$$\int_K 1_E\,dk = \frac{1}{|W|}\int_T 1_E(t)\,J(t)\,dt.$$

To prove this, note that the Haar integral on $C(K)$ extends to bounded measurable functions. This is the Haar measure mentioned in Lecture 5, and the same holds on T. We import one

fact about it, the dominated convergence theorem. If measurable functions f_n converge pointwise to f and $|f_n| \le g$ for an integrable g, then $\int f_n \to \int f$.

For $n \ge 1$ let $h_n : \mathbb{R} \to [0,1]$ be continuous with

$$h_n(x) = 1 \ \ (x \ge 0), \qquad h_n(x) = 0 \ \ (x \le -1/n), \qquad h_n \text{ linear on } [-1/n, 0].$$

Each $h_n \circ c$ is a continuous class function. For every k, the values $h_n(c(k))$ decrease to the indicator $1_E(k)$. All these functions are bounded by 1, and J is bounded on T. Applying dominated convergence to both sides of the class-function formula (9.4) gives the claim.

9.4 Characters on the torus and nonsimply connected groups

Let $\lambda \in X^*(T)$ be dominant. By the classification theorem of Lecture 8, K has an irreducible representation V_λ with highest weight λ. Write χ_λ for its character.

Since K is connected, V_λ is determined by its differential. The complex-linear extension of the differential makes V_λ an irreducible $\mathfrak{k}_{\mathbb{C}}$-module, with complex highest weight $i\lambda$. The centre of $\mathfrak{k}_{\mathbb{C}}$ acts by a scalar. Hence the semisimple Weyl character formula [I] applies to the derived algebra, with the central weight retained. Written with real exponents, as explained in § 9.1, the resulting formula is an identity in $\mathbb{Z}[X^*(T)]$. We use it in the form

$$\chi_\lambda(t)\, D(t) = \sum_{w \in W} \operatorname{sgn}(w)\, t^{w(\lambda+\rho)-\rho}, \qquad D(t) = \prod_{\alpha>0} (1 - t^{-\alpha}). \tag{9.7}$$

Each exponent equals $w\lambda - (\rho - w\rho)$, so it lies in $X^*(T)$. Thus (9.7) is an identity of functions on T for every K. If $\rho \in X^*(T)$, multiplying by t^ρ gives the usual quotient form.

Proposition 9.8. *Let $\lambda, \mu \in X^*(T)$ be dominant. Then $\int_K \chi_\lambda \overline{\chi_\mu}\, dk = \delta_{\lambda\mu}$.*

Proof. Since $|D|^2 = J$, (9.4) gives

$$\int_K \chi_\lambda \overline{\chi_\mu}\, dk = \frac{1}{|W|} \sum_{w,w'} \operatorname{sgn}(ww') \int_T t^{w(\lambda+\rho) - w'(\mu+\rho)}\, dt.$$

A term survives only if $w(\lambda+\rho) = w'(\mu+\rho)$. Both $\lambda+\rho$ and $\mu+\rho$ are strictly dominant. Each W-orbit meets the dominant chamber exactly once, and strictly dominant weights have trivial stabiliser [I]. So a term survives only if $\lambda = \mu$ and $w = w'$. There are $|W|$ such terms. ∎

This agrees with Schur orthogonality. It is a consistency check, not a new proof.

Comparison with SO(3). Let $T = \{R_\varphi\}$, where R_φ is the rotation through φ about the z-axis. Exercise 9.4 shows that Ad_{R_φ} rotates $\mathfrak{m}$ through φ. So $R_\varphi^\alpha = e^{i\varphi}$, $J(R_\varphi) = 4\sin^2(\varphi/2)$ and $|W| = 2$. Formal exponentiation of the weight $\rho = \alpha/2$ would give $e^{i\varphi/2}$. This is not a function on T, because φ and $\varphi + 2\pi$ give the same rotation.

The density J is still defined. Conjugation by the rotation $\mathrm{diag}(1,-1,-1)$ reverses the z-axis, so it maps R_φ to $R_{-\varphi} = R_{2\pi-\varphi}$. Hence a class function satisfies $f(R_{2\pi-\varphi}) = f(R_\varphi)$. The

factor $\sin^2\frac{\varphi}{2}$ is also unchanged under $\varphi \mapsto 2\pi - \varphi$, and $2\sin^2\frac{\varphi}{2} = 1 - \cos\varphi$. Therefore the class-function formula (9.4) gives

$$\int_{\mathrm{SO}(3)} f\,dk = \frac{1}{\pi}\int_0^{2\pi} f(R_\varphi)\sin^2\frac{\varphi}{2}\,d\varphi = \frac{1}{\pi}\int_0^{\pi} f(R_\varphi)(1-\cos\varphi)\,d\varphi. \tag{9.8}$$

We compare this with $\mathrm{SU}(2)$. The image of Haar measure under the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ is the Haar measure of $\mathrm{SO}(3)$, by uniqueness (Lecture 5). The element t_θ maps to a rotation through $\pm 2\theta$. So the substitution $\varphi = 2\theta$ turns (9.6) into the first form of (9.8). Exercise 9.4 shows that small rotations are rare.

The irreducible representations of $\mathrm{SO}(3)$ are the V_{2j} (Lecture 8). Here $\lambda = j\alpha$, and (9.7) reads

$$\chi_{2j}(R_\varphi)\,(1 - e^{-i\varphi}) = e^{ij\varphi} - e^{-i(j+1)\varphi}.$$

Hence

$$\int |\chi_{2j}|^2 = \frac{1}{2}\int_T |e^{ij\varphi} - e^{-i(j+1)\varphi}|^2 dt = \frac{1}{2}\cdot 2 = 1.$$

Self-checks

1. Find J for $\mathrm{U}(2)$ with $t = \mathrm{diag}(e^{ia}, e^{ib})$. Is $\rho \in X^*(T)$?
   Answer. There is one positive root, $t^\alpha = e^{i(a-b)}$. So $J = |e^{ia} - e^{ib}|^2 = 4\sin^2\frac{a-b}{2}$. Identify $X^*(T)$ with $\mathbb{Z}^2$, where (p,q) labels the weight ℓ with $t^\ell = e^{i(pa+qb)}$. Then $\alpha = (1,-1)$ and $\rho = (\frac{1}{2}, -\frac{1}{2}) \notin X^*(T)$.
2. What is the Haar probability that $g \in \mathrm{SU}(2)$ has $\operatorname{tr} g \geq 0$?
   Answer. The remark on indicator functions (Remark 9.7) with $c(g) = \operatorname{tr} g$ lets us use (9.6) for this indicator. The condition on the trace means $\theta \in [0, \pi/2]$. The probability is
   $$\frac{2}{\pi}\int_0^{\pi/2} \sin^2\theta\,d\theta = \frac{2}{\pi}\cdot\frac{\pi}{4} = \frac{1}{2}.$$
   This matches the symmetry $g \mapsto -g$.
3. Why can $\int_{\mathrm{SU}(2)} |g_{11}|^2 dk$ not be computed from (9.6) with $f(t_\theta) = |e^{i\theta}|^2$?
   Answer. The function $g \mapsto |g_{11}|^2$ is not a class function. Schur orthogonality (Lecture 5) gives the correct value $1/\dim V_1 = \frac{1}{2}$. Using $f(t_\theta) = 1$ in (9.6) would give the wrong value 1. To compute this integral with the Weyl integration formula, one must use the general form (9.3).

Summary

Let K be a compact connected Lie group with maximal torus T. Every regular element of K has exactly $|W|$ preimages under the conjugation map $K/T \times T \to K$. Relative to the normalised measures, the Jacobian of this map at (gT, t) is $\det(1 - \mathrm{Ad}_t)$ on $\mathfrak{k}/\mathfrak{t}$. This determinant is a product of root-plane factors $|1 - t^\alpha|^2$ over the positive roots α. The Weyl denominator identity fixes the constant. For $\mathrm{SU}(2)$ and a continuous class function f, the Haar integral of f is $\frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\,d\theta$. With the character formula in the form (9.7), norms and multiplicities become Fourier calculations on T. This form never needs ρ to be a character.

Sources and hypotheses

The regular set, (C1). BD V.2, Proposition (2.6), covers $K \setminus K_{\mathrm{reg}}$ by the image of a compact manifold of dimension at most $\dim K - 3$. Such an image is null, because a smooth map from a manifold of lower dimension has a null image (Lee, Corollary 6.11). Haar measure has a smooth positive density, so a set that is null in charts has Haar measure zero. BD's regular elements agree with ours by the centraliser-dimension formula of Proposition 9.5.

Openness also follows from that proposition. The map ψ is onto (Lecture 6), and $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$. Hence $K \setminus K_{\mathrm{reg}} = \psi\big(K/T \times (T \setminus T_{\mathrm{reg}})\big)$. The set $T \setminus T_{\mathrm{reg}}$ is a finite union of kernels of characters, so it is closed in T. So $K \setminus K_{\mathrm{reg}}$ is a continuous image of a compact set. Hence it is compact, and therefore closed in K.

The change of variables, (C2). The source is Lee, Chapter 16, Propositions 16.40–16.42. Proposition 16.40 is the absolute-Jacobian rule for pulling back a density in coordinates. Lee defines integrals of densities through charts and a partition of unity. Lee does not state (C2) itself, so we derive it from these results.

First, each point $y \in U$ has an open neighbourhood $O \subset U$ whose inverse image consists of d disjoint open sets, each mapped diffeomorphically onto O. To see this, let $x_1, \dots, x_d$ be the preimages of y. Choose disjoint open sets $V_i \ni x_i$ in $\psi^{-1}(U)$ that ψ maps diffeomorphically onto open sets $\psi(V_i)$. The set $C = M \setminus \bigcup_i V_i$ is closed in the compact manifold M, so it is compact. It contains no preimage of y. So $\psi(C)$ is compact, hence closed, and $y \notin \psi(C)$. Put $O = \bigcap_i \psi(V_i) \setminus \psi(C)$. This set is open and contains y. Every preimage of a point of O lies outside C, hence in some V_i. So the inverse image of O is the disjoint union of the sets $V_i \cap \psi^{-1}(O)$. Since $O \subset \psi(V_i)$, the map ψ sends each of these sets diffeomorphically onto O.

Thus each point of U has a neighbourhood that is evenly covered by d sheets, as in the definition of a covering map in Lecture 4. The domain need not be connected. For SU(2) it has two components, and each maps diffeomorphically onto K_{reg}.

Next, suppose F is supported in such a set O. Apply the change-of-variables rule on each of the d sets $V_i \cap \psi^{-1}(O)$ and add the results. A partition of unity on U subordinate to these sets O gives the general case: the partition is countable; sum the pieces by dominated convergence, since the partial sums are bounded by $|F \circ \psi| J_\psi$ and $|F|$. The null sets $N \setminus U$ and $M \setminus \psi^{-1}(U)$ do not contribute. BD I.5, (5.19), states the oriented version through the mapping degree.

Reading

- H: Hall, *Lie Groups, Lie Algebras, and Representations*, 2nd ed., §11.6.
- BD: Bröcker–tom Dieck, *Representations of Compact Lie Groups*, IV.1, V.2 and VI.1.
- Knapp Beyond: *Lie Groups Beyond an Introduction*, VIII.5.
- Lee: *Introduction to Smooth Manifolds*, 2nd ed., Corollary 6.11, and Chapter 16, Propositions 16.40–16.42.

Lecture 10

Peter–Weyl and completeness

Objectives. *Lecture 5 proved that matrix coefficients of inequivalent irreducible representations are orthogonal. This lecture proves that finite-dimensional matrix coefficients span a dense subspace of the square-integrable functions on a compact Lie group. It then deduces that every compact Lie group is a closed subgroup of a unitary group.*

By the end of this lecture you will be able to:

- Average a rank-one projection to obtain a nonzero positive compact intertwiner, and use it to prove that the finite-dimensional invariant subspaces span a dense subspace in every strongly continuous unitary representation of a compact Lie group.
- Identify the finite-dimensional subspaces of $L^2(K)$ invariant under the right translations R with spaces of matrix coefficients, and state the Peter–Weyl theorem with multiplicities.
- Prove that a compact Lie group has a faithful finite-dimensional unitary representation.
- Compute the coefficient blocks of $\mathrm{U}(1)$ and $\mathrm{SU}(2)$.

Two results from earlier courses are special cases of the theorem of this lecture. For a finite group G, the regular module gives $\mathbb{C}[G] \cong \bigoplus_U \mathrm{End}(U)$, so that $\sum_U (\dim U)^2 = |G|$ (Finite Groups). For the circle $\mathrm{U}(1)$, a square-integrable function has a Fourier series in the functions $e^{in\theta}$. The Peter–Weyl theorem contains both. Each irreducible representation Π of a compact Lie group K gives a block of d_Π^2 matrix coefficients, and the span of these blocks is dense in $L^2(K)$. For $\mathrm{SU}(2)$ the blocks have $1, 4, 9, \dots$ functions. Compared with a finite group, the new difficulty is the infinite sum.

The proof has three parts. First we average a rank-one projection over the group. The result commutes with the group, and it is a compact operator. Next, its eigenspaces for nonzero eigenvalues are finite-dimensional and invariant, so every unitary representation contains finite-dimensional pieces. Finally we apply this to $L^2(K)$, where a finite-dimensional invariant piece turns out to consist of matrix coefficients.

10.1 Analytic preliminaries

Throughout, K is a compact Lie group. We write $\int_K f(g)\,dg$ for its normalised Haar integral on $C(K)$. We import four facts from Lecture 5.

(H1) The integral is left and right invariant. It is invariant under $g \mapsto g^{-1}$, and $\int_K 1\,dg = 1$.

(H2) If $f \in C(K)$ is nonnegative and not identically zero, then $\int_K f\,dg > 0$.

(H3) A finite-dimensional continuous representation has an invariant inner product. It is then an orthogonal sum of irreducible subrepresentations.

(H4) Schur orthogonality, in the form (10.1) below.

We also import Fubini's theorem for continuous functions on $K \times K$ from Bröcker–tom Dieck, Section I.5. It states that the two iterated Haar integrals agree.

Let (Π, V) be a finite-dimensional continuous unitary representation. Its *matrix coefficients* are the continuous functions

$$c^{\Pi}_{u,v}(x) = \langle u, \Pi(x)v\rangle, \qquad u, v \in V.$$

The *coefficient space* $\mathscr{R}(K)$ is the span of all such functions, for all such (Π, V). Let $\mathrm{Irr}(K)$ be a set of representatives of the isomorphism classes of irreducible finite-dimensional continuous unitary representations. Write $d_\Pi = \dim V_\Pi$. For $\Pi, \Pi' \in \mathrm{Irr}(K)$, Schur orthogonality (H4) reads

$$\int_K \overline{c^{\Pi}_{u,v}(x)}\, c^{\Pi'}_{u',v'}(x)\,dx = \begin{cases} d_\Pi^{-1}\langle u', u\rangle\langle v, v'\rangle, & \Pi' = \Pi, \\ 0, & \Pi' \neq \Pi. \end{cases} \tag{10.1}$$

Definition 10.1. Let $(H_i)_{i\in I}$ be mutually orthogonal closed subspaces of a Hilbert space H. Their *Hilbert sum* $\widehat{\bigoplus}_i H_i$ is the closure of the algebraic sum $\bigoplus_i H_i$.

The elements of the Hilbert sum are the convergent series $\sum_i v_i$ with $v_i \in H_i$ and $\sum_i \|v_i\|^2 < \infty$. For Hilbert spaces H_i not given inside one space, the *external Hilbert sum* $\widehat{\bigoplus}_i H_i$ is the space of families (v_i) with $v_i \in H_i$ and $\sum_i \|v_i\|^2 < \infty$.

A bounded operator on a Hilbert space H is *compact* if it is an operator-norm limit of finite-rank operators. A self-adjoint operator T on H is *positive* if $\langle v, Tv\rangle \geq 0$ for all $v \in H$.

We import the spectral theorem for compact self-adjoint operators from Linear Algebra for Physics, Weeks 13–14, in the following form. If T is compact and self-adjoint, then

$$H = \ker T \;\widehat{\oplus}\; \widehat{\bigoplus_{\mu\neq 0}} \ker(T - \mu).$$

Moreover, each $\ker(T-\mu)$ with $\mu \neq 0$ is finite-dimensional. If $T \neq 0$, then T has a nonzero eigenvalue.

Definition 10.2. The space $L^2(K)$ is the Hilbert completion of $C(K)$ for the sesquilinear form $\langle f, h\rangle = \int_K \overline{f(g)}\, h(g)\,dg$.

By positivity (H2), this sesquilinear form is an inner product on $C(K)$. Hence the map $C(K) \to L^2(K)$ is injective. An element of $L^2(K)$ is *continuous* if it lies in $C(K)$. Only continuous elements have values $f(x)$. For continuous functions, $\|f\|_2 \leq \|f\|_\infty = \sup_x |f(x)|$.

Definition 10.3. A *strongly continuous unitary representation* of K on a Hilbert space H is a homomorphism Π from K to the unitary group of H such that $g \mapsto \Pi(g)v$ is continuous for every $v \in H$.

We do not ask for continuity in operator norm. Already right translation on $L^2(\mathrm{U}(1))$ is not norm continuous (Exercise 10.2).

Lemma 10.4. *Let $F\colon K \to H$ be continuous. There is a unique vector $\int_K F(g)\,dg$ in H with*

$$\left\langle w, \int_K F(g)\,dg \right\rangle = \int_K \langle w, F(g)\rangle\,dg \qquad (w \in H).$$

Moreover $\|\int_K F\| \le \int_K \|F(g)\|\,dg$, and $A\int_K F = \int_K AF$ for bounded A. If each $F(g)$ lies in a closed subspace W, so does $\int_K F$. The integral is unchanged if $F(g)$ is replaced by $F(hg)$.

Proof. The scalar integrand is continuous. The map $w \mapsto \overline{\int_K \langle w, F\rangle}$ is linear with bound $\|w\| \int_K \|F\|$. The Riesz representation theorem for Hilbert spaces, from Linear Algebra for Physics, gives the vector and the norm bound. For bounded A, $\langle w, A\int F\rangle = \int \langle A^* w, F\rangle = \int \langle w, AF\rangle$. If $w \perp W$, the integrand vanishes, so $\int F \in W^{\perp\perp} = W$. Invariance follows from the left invariance in (H1). ∎

10.2 An averaged compact operator

Theorem 10.5. *Let Π be a strongly continuous unitary representation of K on $H \neq 0$.*

(a) The space H contains a nonzero finite-dimensional invariant subspace.

(b) The sum H_{fin} of all finite-dimensional invariant subspaces is dense in H.

We look for a nonzero compact self-adjoint intertwiner. An intertwiner has invariant eigenspaces. A compact self-adjoint operator has finite-dimensional eigenspaces for nonzero eigenvalues. Averaging a rank-one projection over K gives an intertwiner, and compactness of K keeps it a compact operator.

Proof. Fix a unit vector $u \in H$. Let E_u be the rank-one projection given by $E_u v = \langle u, v\rangle u$. Then $\Pi(g)E_u\Pi(g)^{-1}v = \langle \Pi(g)u, v\rangle \Pi(g)u$, which is continuous in g. Define the average

$$Tv = \int_K \langle \Pi(g)u, v\rangle\, \Pi(g)u\,dg. \tag{10.2}$$

Step 1: T is bounded, positive and nonzero. The vector-integral lemma, Lemma 10.4, gives $\|Tv\| \le \|v\|$ and

$$\langle w, Tv\rangle = \int_K \langle \Pi(g)u, v\rangle \langle w, \Pi(g)u\rangle\,dg.$$

Exchanging v, w and conjugating gives the same integral, so $T^* = T$. For $w = v$ the integrand is $|\langle \Pi(g)u, v\rangle|^2 \ge 0$, so T is positive. For $v = w = u$ this integrand is continuous with value 1 at e. By positivity (H2), $\langle u, Tu\rangle > 0$. Hence T is nonzero.

Step 2: T commutes with Π. For $h \in K$,

$$T\Pi(h)v = \int_K \langle \Pi(h^{-1}g)u, v\rangle\, \Pi(g)u\,dg = \int_K \langle \Pi(g)u, v\rangle\, \Pi(h)\Pi(g)u\,dg = \Pi(h)Tv.$$

The first equality uses unitarity. The second replaces g by hg. The third takes $A = \Pi(h)$ in the vector-integral lemma, Lemma 10.4.

Step 3: T is compact. The orbit $O = \{\Pi(g)u : g \in K\}$ is compact, since it is a continuous image of K. Let $\varepsilon > 0$. Since O is compact, finitely many open balls of radius ε with centres in O cover O.

Let P be the orthogonal projection onto the span of the centres. Put $x_g = \Pi(g)u$ and $y_g = Px_g$. Each x_g lies within ε of some centre, and y_g is the nearest point of the span of the centres to x_g. Hence $\|x_g - y_g\| < \varepsilon$ for all g. Now

$$\langle x_g, v\rangle\langle w, x_g\rangle - \langle y_g, v\rangle\langle w, y_g\rangle = \langle x_g - y_g, v\rangle\langle w, x_g\rangle + \langle y_g, v\rangle\langle w, x_g - y_g\rangle.$$

Because $\|x_g\| = 1$ and $\|y_g\| \le 1$, the Cauchy–Schwarz inequality shows that this difference has modulus at most $2\varepsilon\|v\|\|w\|$. On the other hand,

$$\langle w, PTPv\rangle = \langle Pw, TPv\rangle = \int_K \langle y_g, v\rangle\langle w, y_g\rangle\, dg.$$

Subtracting gives $|\langle w, (T - PTP)v\rangle| \le 2\varepsilon\|v\|\|w\|$, so $\|T - PTP\| \le 2\varepsilon$. The operator PTP has finite rank, since P has finite rank. Hence T is compact.

Step 4: proof of (a). By Steps 1 and 3, T is compact, self-adjoint and nonzero. Hence the spectral theorem gives an eigenvalue $\mu \neq 0$ of T with finite-dimensional eigenspace $H_\mu \neq 0$. If $Tv = \mu v$, then Step 2 gives $T\Pi(h)v = \Pi(h)Tv = \mu\Pi(h)v$. So H_μ is invariant.

Step 5: proof of (b). A sum of two finite-dimensional invariant subspaces is again one, so H_{fin} is a subspace. Its closure M is invariant. Hence $M^\perp$ is closed and invariant, since $\langle \Pi(g)w, m\rangle = \langle w, \Pi(g)^{-1}m\rangle$. Restricting Π gives a strongly continuous unitary representation on $M^\perp$. If $M^\perp \neq 0$, the existence statement (a), applied to $M^\perp$, gives a finite-dimensional invariant subspace $W \neq 0$ in $M^\perp$. But $W \subset H_{\mathrm{fin}} \subset M$. Thus $W \subset M \cap M^\perp = 0$, a contradiction. ∎

Corollary 10.6. *Let Π be a strongly continuous unitary representation of K on H. Suppose that Π is irreducible, meaning that $H \neq 0$ and that the only closed invariant subspaces are 0 and H. Then* $\dim H < \infty$.

Proof. Since $H \neq 0$, the existence statement, part (a) of Theorem 10.5, gives a nonzero finite-dimensional invariant subspace W. Finite-dimensional subspaces are closed, so $W = H$ by irreducibility. ∎

This corollary uses compactness. A noncompact Lie group can have irreducible unitary representations of infinite dimension, and Part II treats them.

10.3 The regular representation

We begin with a tool for continuous functions on K, which we use throughout this section.

Lemma 10.7. *(a) For $\varphi \in C(K)$ and $\varepsilon > 0$ there is an open $U \ni e$ with $|\varphi(yx) - \varphi(x)| < \varepsilon$ and $|\varphi(xy) - \varphi(x)| < \varepsilon$ for all $x \in K$, $y \in U$.*

(b) For each open $U \ni e$ there is a continuous $\varphi \ge 0$ vanishing outside U with $\varphi(e) > 0$ and $\int_K \varphi = 1$.

Proof. (a) The function $F(y,x) = |\varphi(yx) - \varphi(x)| + |\varphi(xy) - \varphi(x)|$ is continuous on $K \times K$ and vanishes on $\{e\} \times K$. For each x there are open sets $U_x \ni e$ and $N_x \ni x$ with $F < \varepsilon$ on $U_x \times N_x$. Finitely many $N_{x_1}, \dots, N_{x_r}$ cover K. Take $U = \bigcap_j U_{x_j}$.

(b) Fix a norm $|\cdot|$ on $\mathfrak{k}$, and let B_r be the open ball of radius r about 0. By Lecture 2, exp maps B_r homeomorphically onto an open neighbourhood of e for all sufficiently small $r > 0$. Shrink r so that $\exp(B_r) \subset U$. Put $\psi(\exp X) = \max(0, r/2 - |X|)$ on $\exp(B_r)$ and $\psi = 0$ elsewhere. Then ψ is continuous on $\exp(B_r)$.

The closed ball $\overline{B}_{r/2}$ is compact, so its image under exp is compact and hence closed in the Hausdorff space K. The function ψ vanishes on the open complement $K \setminus \exp(\overline{B}_{r/2})$. The two open sets $\exp(B_r)$ and $K \setminus \exp(\overline{B}_{r/2})$ cover K, so ψ is continuous on K. Since $\psi(e) = r/2 > 0$, positivity (H2) gives $\int \psi > 0$. Take $\varphi = \psi / \int \psi$. Then $\varphi(e) > 0$. ∎

For $f \in C(K)$ define

$$(L_g f)(x) = f(g^{-1}x), \qquad (R_g f)(x) = f(xg).$$

Lemma 10.8. *The translations L and R extend to commuting strongly continuous unitary representations on $L^2(K)$.*

Proof. We treat R. The proof for L is the same, except that it uses left invariance and replaces U below by $U \cap U^{-1}$.

First, R is a homomorphism on $C(K)$. For $f \in C(K)$ and $g, h, x \in K$,

$$(R_g R_h f)(x) = (R_h f)(xg) = f(xgh) = (R_{gh} f)(x).$$

By the right invariance in (H1), R_g preserves $\|\cdot\|_2$. So R_g extends to an isometry of $L^2(K)$ with inverse $R_{g^{-1}}$.

Next, we prove strong continuity at e. Let $f \in L^2(K)$ and $\varepsilon > 0$. Choose $f_0 \in C(K)$ with $\|f - f_0\|_2 < \varepsilon$. Uniform continuity, Lemma 10.7(a), gives an open $U \ni e$ with $\|R_g f_0 - f_0\|_2 \le \|R_g f_0 - f_0\|_\infty < \varepsilon$ for $g \in U$. For $g \in U$, the triangle inequality and the isometry property give

$$\|R_g f - f\|_2 \le \|R_g(f - f_0)\|_2 + \|R_g f_0 - f_0\|_2 + \|f_0 - f\|_2 < 3\varepsilon.$$

Continuity at g_0 follows from $R_g f - R_{g_0} f = R_{g_0}\big(R_{g_0^{-1}g} f - f\big)$, since R_{g_0} is an isometry.

Finally, for $f \in C(K)$ both $L_g R_h f$ and $R_h L_g f$ are the function $x \mapsto f(g^{-1}xh)$. The two bounded operators therefore agree on the dense subspace $C(K)$, so they commute on $L^2(K)$. ∎

For $\varphi \in C(K)$ and $f \in L^2(K)$ put

$$R_\varphi f = \int_K \varphi(g) R_g f \, dg.$$

Elements of $L^2(K)$ have no values at points, but a matrix coefficient must have them. Averaging removes this difficulty. The element $R_\varphi f$ is a weighted average of right translates of f, and it is continuous.

Lemma 10.9 (Smoothing). *For $\varphi \in C(K)$ and $f \in L^2(K)$, the element $R_\varphi f$ is continuous, and*

$$(R_\varphi f)(x) = \langle k_x, f \rangle, \qquad k_x(h) = \overline{\varphi(x^{-1}h)}.$$

Proof. Put $(Sf)(x) = \langle k_x, f\rangle$. Fix $\varepsilon > 0$. Choose U for φ by uniform continuity, Lemma 10.7(a). If $x' \in xU^{-1}$, then $y = x'^{-1}x \in U$. Writing $t = x^{-1}h$ gives $|k_x(h) - k_{x'}(h)| = |\varphi(t) - \varphi(yt)| < \varepsilon$. By the Cauchy–Schwarz inequality, $|Sf(x) - Sf(x')| \le \varepsilon\|f\|_2$. So $Sf \in C(K)$ and $\|Sf\|_2 \le \|Sf\|_\infty \le \|\varphi\|_\infty\|f\|_2$. For $f, w \in C(K)$, Fubini and the substitution $h = xg$ give

$$\langle w, R_\varphi f\rangle = \int_K \varphi(g) \int_K \overline{w(x)} f(xg)\,dx\,dg = \int_K \overline{w(x)} \int_K \varphi(x^{-1}h) f(h)\,dh\,dx = \langle w, Sf\rangle.$$

As $C(K)$ is dense, $R_\varphi f = Sf$ for $f \in C(K)$. Both operators are bounded, so they agree on $L^2(K)$. ∎

Proposition 10.10. *(a) Let $W \subset L^2(K)$ be a finite-dimensional R-invariant subspace. Then $W \subset C(K)$. There is $u_W \in W$ with $f(x) = \langle u_W, R_x f\rangle$ for all $f \in W$ and $x \in K$.*

(b) The coefficient space $\mathscr{R}(K)$ is the sum of all finite-dimensional R-invariant subspaces. It is dense in $L^2(K)$.

Proof. (a) Choose an orthonormal basis $f_1, \dots, f_n$ of W. Strong continuity gives an open $U \ni e$ with $\|R_g f_i - f_i\| < 1/(2n)$ for all i and $g \in U$. By Lemma 10.7(b), there is a continuous $\varphi \ge 0$ vanishing outside U with $\int_K \varphi = 1$. This normalisation gives $R_\varphi f - f = \int_K \varphi(g)\big(R_g f - f\big)\,dg$. For a unit vector $f = \sum a_i f_i$ we have $|a_i| \le 1$. Hence

$$\|R_\varphi f - f\| \le \sum_i |a_i| \int_K \varphi(g)\|R_g f_i - f_i\|\,dg \le n \cdot \frac{1}{2n} = \frac{1}{2}.$$

So $\|R_\varphi f\| \ge \tfrac12$ for every unit vector $f \in W$, and R_φ is injective on W. Since W is closed and R-invariant, the vector-integral lemma, Lemma 10.4, shows that R_φ maps W into W. As W is finite-dimensional, $R_\varphi W = W$. Every element of W is therefore of the form $R_\varphi f$, and the smoothing lemma, Lemma 10.9, gives $W \subset C(K)$.

Now evaluation $f \mapsto f(e)$ is a linear functional on W. The Riesz representation theorem in W gives u_W with $f(e) = \langle u_W, f\rangle$. For $f \in W$, $R_x f$ lies in W and is the function $h \mapsto f(hx)$. Hence $f(x) = (R_x f)(e) = \langle u_W, R_x f\rangle$.

(b) The restriction of R to W is a finite-dimensional continuous unitary representation. By the reproducing formula in (a), each $f \in W$ is a matrix coefficient of this representation. Hence $W \subset \mathscr{R}(K)$. Conversely, $R_h c^{\Pi}_{u,w} = c^{\Pi}_{u,\Pi(h)w}$, so each $c^{\Pi}_{u,v}$ lies in the finite-dimensional R-invariant space $\{c^{\Pi}_{u,w} : w \in V\}$. Density follows from the density statement Theorem 10.5(b), applied to the representation R of the regular-representation lemma, Lemma 10.8. ∎

10.4 The Peter–Weyl theorem

The group $K \times K$ acts on $L^2(K)$ by

$$((g,h)\cdot f)(x) = f(g^{-1}xh), \qquad \text{that is, } (g,h) \mapsto L_g R_h.$$

For $\Pi \in \mathrm{Irr}(K)$, give $\mathrm{End}(V_\Pi)$ the Hilbert–Schmidt inner product $\langle A, B\rangle_{\mathrm{HS}} = \mathrm{tr}(A^*B)$ and the action $(g,h)\cdot A = \Pi(h) A \Pi(g)^{-1}$.

Let $\overline{V}_\Pi$ be the conjugate space, with $\overline{\Pi}(g)\bar{u} = \overline{\Pi(g)u}$ and inner product $\langle \bar{u}, \bar{v}\rangle = \langle v, u\rangle$. For this inner product, $\overline{\Pi}$ is unitary. The representation $\overline{\Pi}$ is irreducible, since $W \mapsto \overline{W}$ matches the invariant subspaces of Π and $\overline{\Pi}$. The coefficients of $\overline{\Pi}$ are the conjugate functions:

$$c^{\overline{\Pi}}_{\bar{u},\bar{v}}(x) = \big\langle \bar{u}, \overline{\Pi(x)v}\big\rangle = \langle \Pi(x)v, u\rangle = \overline{c^{\Pi}_{u,v}(x)}.$$

The space $\overline{V}_\Pi$ is isomorphic to V_Π^* by $\bar{u} \mapsto \langle u, \cdot\rangle$. The map $\bar{u} \otimes v \mapsto v\langle u, \cdot\rangle$ identifies $\overline{V}_\Pi \otimes V_\Pi$ with $\mathrm{End}(V_\Pi)$. For $g, h \in K$, unitarity of $\Pi(g)$ gives $\Pi(h)v\langle u, \Pi(g)^{-1}\cdot\rangle = \Pi(h)v\langle \Pi(g)u, \cdot\rangle$. The left side is the action of (g,h) on $v\langle u,\cdot\rangle$. Hence the pair (g,h) acts on $\overline{V}_\Pi \otimes V_\Pi$ as $\overline{\Pi}(g) \otimes \Pi(h)$. Thus the first tensor factor carries the left action L.

In Dirac notation, $v\langle u,\cdot\rangle$ is the operator $|v\rangle\langle u|$, and $c^\Pi_{u,v}(x) = \langle u|\Pi(x)|v\rangle$ is a matrix element. Under (g,h) the ket moves by $\Pi(h)$ and the bra by $\Pi(g)$, and bras form the conjugate space $\overline{V}_\Pi$.

We define

$$\Phi_\Pi \colon \mathrm{End}(V_\Pi) \to C(K), \qquad \Phi_\Pi(A)(x) = \sqrt{d_\Pi}\ \mathrm{tr}\big(\Pi(x)A\big),$$

and let E_Π be its image. For $A = v\langle u,\cdot\rangle$ we get $\Phi_\Pi(A) = \sqrt{d_\Pi}\, c^\Pi_{u,v}$.

Fix an orthonormal basis $e_1, \dots, e_{d_\Pi}$ of V_Π, and write $\Pi_{ij} = c^\Pi_{e_i,e_j}$. This is the entry of the matrix of $\Pi(x)$ in row i and column j. The operator $e_j\langle e_i,\cdot\rangle$ has matrix with a single entry 1, in row j and column i. Since $\mathrm{tr}\big(\Pi(x)e_j\langle e_i,\cdot\rangle\big) = \Pi_{ij}(x)$, the map Φ_Π sends this operator to $\sqrt{d_\Pi}\,\Pi_{ij}$. So a column of the matrix of A corresponds to a row of the array (Π_{ij}).

Theorem 10.11 (Peter–Weyl). *(a) Each Φ_Π is a $K \times K$-equivariant isometry from $(\mathrm{End}\, V_\Pi, \mathrm{HS})$ onto E_Π, and $E_\Pi \perp E_{\Pi'}$ for $\Pi \neq \Pi'$.*

(b) The coefficient space satisfies $\mathscr{R}(K) = \bigoplus_{\Pi\in\mathrm{Irr}(K)} E_\Pi$, an algebraic direct sum.

(c) The Hilbert completion of $\mathscr{R}(K)$ is $L^2(K) = \widehat{\bigoplus}_{\Pi\in\mathrm{Irr}(K)} E_\Pi$. The map $\bigoplus_\Pi \Phi_\Pi$ extends to a unitary $K \times K$-isomorphism

$$\widehat{\bigoplus_\Pi}\,\mathrm{End}(V_\Pi) \cong \widehat{\bigoplus_\Pi}\,\overline{V}_\Pi \otimes V_\Pi \longrightarrow L^2(K).$$

(d) Under R, $E_\Pi \cong V_\Pi^{\oplus d_\Pi}$ and $\dim \mathrm{Hom}_K(V_\Pi, L^2(K)) = d_\Pi$. Under L, $E_\Pi \cong \overline{V}_\Pi^{\oplus d_\Pi}$, and again each irreducible occurs with multiplicity equal to its dimension.

The theorem says that functions on K are organised in square blocks, one for each irreducible representation. In an orthonormal basis, block Π holds the d_Π^2 matrix entries Π_{ij}, up to the factor $\sqrt{d_\Pi}$. Figure 10.1 draws these blocks as arrays of functions. Since $\Pi(xh) = \Pi(x)\Pi(h)$, we have $\Pi_{ij}(xh) = \sum_k \Pi_{ik}(x)\Pi_{kj}(h)$, so R_h keeps the row i. Since $\Pi(g^{-1}x) = \Pi(g^{-1})\Pi(x)$, we have $\Pi_{ij}(g^{-1}x) = \sum_k \Pi_{ik}(g^{-1})\Pi_{kj}(x)$, so L_g keeps the column j.

Proof. (a) For $A = v\langle u,\cdot\rangle$ and $B = v'\langle u',\cdot\rangle$ we have $A^*B = u\langle v, v'\rangle\langle u',\cdot\rangle$, so $\mathrm{tr}(A^*B) = \langle u', u\rangle\langle v, v'\rangle$. By Schur orthogonality (10.1),

$$\langle \Phi_\Pi(A), \Phi_\Pi(B)\rangle = d_\Pi \cdot d_\Pi^{-1}\langle u', u\rangle\langle v, v'\rangle = \langle A, B\rangle_{\mathrm{HS}}.$$

Both sides are sesquilinear and rank-one operators span $\mathrm{End}(V_\Pi)$. The same argument gives $E_\Pi \perp E_{\Pi'}$. Equivariance is the calculation

$$\Phi_\Pi(A)(g^{-1}xh) = \sqrt{d_\Pi}\,\mathrm{tr}\big(\Pi(g)^{-1}\Pi(x)\Pi(h)A\big) = \Phi_\Pi\big(\Pi(h)A\Pi(g)^{-1}\big)(x).$$

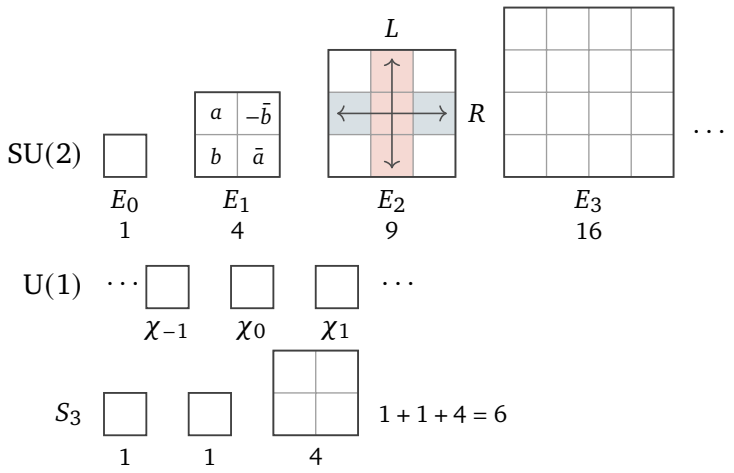


Figure 10.1: Blocks of functions in $L^2(K)$ for SU(2), U(1) and the finite group S_3. A cell is one entry Π_{ij}, with row i and column j, and a number below a block counts its cells. The SU(2) block E_1 shows the entries of the defining representation. In the block E_2, right translation R mixes the entries of the shaded row and left translation L mixes the entries of the shaded column.

(b) Let (σ, V) be a finite-dimensional continuous unitary representation. By complete reducibility (H3), $V = \bigoplus_j V_j$ orthogonally with each V_j irreducible. Write $u = \sum u_j$ and $v = \sum v_j$. Since $\sigma(x)v_k \in V_k$, we get $c^{\sigma}_{u,v} = \sum_j c^{\sigma_j}_{u_j,v_j}$.

Suppose that $\sigma_j \cong \Pi \in \mathrm{Irr}(K)$, and let S be an isomorphism. Since σ_j and Π are unitary, S^* is also an intertwiner. By Schur's lemma, $S^*S = \kappa \cdot 1$, and $\kappa > 0$ because S is injective. Hence $U = \kappa^{-1/2}S$ is a unitary intertwiner, and $c^{\sigma_j}_{u_j,v_j} = c^{\Pi}_{Uu_j,Uv_j} \in E_\Pi$. Thus $\mathscr{R}(K) = \sum_\Pi E_\Pi$. The sum is direct by the orthogonality in (a).

(c) Each E_Π is finite-dimensional, hence closed. By density of coefficients, Proposition 10.10(b), their sum is dense. So $L^2(K)$ is their Hilbert sum. The map $\bigoplus \Phi_\Pi$ is isometric with dense image, so it extends to a unitary. Equivariance passes to the closure by continuity.

(d) Under R alone, $A \mapsto \Pi(h)A$ acts on each column of A, so $E_\Pi \cong V_\Pi^{\oplus d_\Pi}$. Let $J : V_\Pi \to L^2(K)$ intertwine R. By the Hilbert-sum decomposition (c), $Jv = \sum_{\Pi'} P_{\Pi'} Jv$ in $L^2(K)$. The orthogonal projection $P_{\Pi'}$ onto $E_{\Pi'}$ commutes with R, because $E_{\Pi'}$ and $E_{\Pi'}^{\perp}$ are invariant. For $\Pi' \neq \Pi$, Schur's lemma gives $P_{\Pi'} J = 0$. So J maps into E_Π, and $\dim \mathrm{Hom}_K(V_\Pi, L^2(K)) = \dim \mathrm{Hom}_K(V_\Pi, V_\Pi^{\oplus d_\Pi}) = d_\Pi$.

Under L, $A \mapsto A\Pi(g)^{-1}$ acts on rows, giving $\overline{V}_\Pi^{\oplus d_\Pi}$. If $\overline{\Pi} \cong \overline{\Pi'}$, then conjugating the intertwiner gives $\Pi \cong \Pi'$. Also $\overline{\overline{\Pi}} = \Pi$. So conjugation induces a dimension-preserving bijection of the isomorphism classes in $\mathrm{Irr}(K)$. The projections $P_{\Pi'}$ also commute with L, so the same Hilbert-sum and Schur arguments apply. ∎

Theorem 10.12. *The coefficient space $\mathscr{R}(K)$ is dense in $C(K)$ for $\|\cdot\|_\infty$.*

Proof. The constant 1 is a coefficient of the trivial representation. Products are coefficients, since $c^{\Pi}_{u,v} c^{\Pi'}_{u',v'} = c^{\Pi\otimes\Pi'}_{u\otimes u', v\otimes v'}$. Conjugates are coefficients, since $\overline{c^{\Pi}_{u,v}} = c^{\overline{\Pi}}_{\bar{u},\bar{v}}$ for the inner product

on $\overline{V}_\Pi$ fixed above. Hence $\mathscr{R}(K)$ is a subalgebra of $C(K)$ that contains the constants and is closed under conjugation. It remains to prove that $\mathscr{R}(K)$ separates points.

Let $x \neq y$ and suppose $f(x) = f(y)$ for all $f \in \mathscr{R}(K)$. Since $L_g c^{\Pi}_{u,v} = c^{\Pi}_{\Pi(g)u,v}$, the space $\mathscr{R}(K)$ is L-invariant, so $f(g^{-1}x) = f(g^{-1}y)$ for all g. Put $s = x^{-1}y \neq e$ and $g = xt^{-1}$. Then $f(t) = f(ts)$ for all t. Hence $R_s f = f$ on $\mathscr{R}(K)$. Since $\mathscr{R}(K)$ is dense in $L^2(K)$ by Proposition 10.10(b), and R_s is bounded, $R_s = 1$ on $L^2(K)$.

Since K is Hausdorff, some open $U \ni e$ omits s. By Lemma 10.7(b), some continuous $\varphi \geq 0$ vanishes outside U and satisfies $\varphi(e) > 0 = \varphi(s)$. Then $(R_s\varphi)(e) = \varphi(s) \neq \varphi(e)$, so $R_s\varphi \neq \varphi$ in $C(K)$. Since $C(K) \to L^2(K)$ is injective, $R_s\varphi \neq \varphi$ in $L^2(K)$, which contradicts $R_s = 1$. Hence $\mathscr{R}(K)$ separates points.

We import the complex Stone–Weierstrass theorem. For compact Hausdorff X, it states that a subalgebra of $C(X)$ is uniformly dense if it contains the constants, is closed under conjugation and separates points. We apply it to $\mathscr{R}(K) \subset C(K)$. ∎

Separation of points has a structural consequence. A compact Lie group is defined abstractly, but it is always a group of matrices.

Theorem 10.13 (Faithful representation)**.** *A compact Lie group K has an injective finite-dimensional continuous unitary representation $\sigma\colon K \to \mathrm{U}(n)$. Its image is a closed subgroup of $\mathrm{U}(n)$, and σ is an isomorphism of Lie groups from K onto this image.*

Proof. We add summands to a representation until its kernel is trivial, and we show that this process stops.

Step 1: a smaller kernel. Let (σ, V) be a finite-dimensional continuous unitary representation with kernel N. For example, the trivial representation on $\mathbb{C}$ has kernel K. The kernel N is a closed subgroup of K, so it is a compact Lie group by the closed-subgroup theorem of Lecture 3. Suppose $N \neq \{e\}$, and choose $s \in N$ with $s \neq e$. By the proof of Theorem 10.12, the functions in $\mathscr{R}(K)$ separate e and s. These functions are spanned by matrix coefficients. So some finite-dimensional continuous unitary representation Π has a matrix coefficient with $c^{\Pi}_{u,v}(s) \neq c^{\Pi}_{u,v}(e)$. Hence $\Pi(s) \neq I$. The direct sum $\sigma \oplus \Pi$ has kernel $N' = N \cap \ker \Pi$. This is a closed subgroup of N, and $N' \neq N$ because $s \notin N'$.

Step 2: comparing N' with N. Since $N' \subseteq N$, we have $\dim N' \leq \dim N$. Suppose $\dim N' = \dim N$. The inclusion $N' \to N$ is an injective immersion between manifolds of equal dimension, so it is a local diffeomorphism. Hence N' is open in N. Let N_0 be the identity component of N. Then $N' \cap N_0$ is an open subgroup of the connected group N_0. An open subgroup is also closed, since its complement is a union of its cosets. So $N' \cap N_0 = N_0$, that is, $N_0 \subseteq N'$. The connected set N_0 is the largest connected subset of N containing e, so it is also the identity component of N'.

The components of N are the cosets of N_0. They are open and they cover the compact group N, so there are finitely many. Since $N_0 \subseteq N'$ and $N' \neq N$, the group N' consists of fewer of these cosets. So N' has fewer components than N.

Step 3: the process stops. By Step 2, each application of Step 1 either lowers the dimension of the kernel, or keeps the dimension and lowers the finite number of components. The dimension can decrease only finitely often. Between two decreases, the number of components decreases at each step. So after finitely many steps the kernel is $\{e\}$.

Step 4: the image. We now have an injective representation, and an orthonormal basis of its space gives $\sigma\colon K \to \mathrm{U}(n)$. The image $\sigma(K)$ is compact, so it is closed in $\mathrm{U}(n)$, and it is a Lie group by the closed-subgroup theorem. A continuous bijection from a compact space onto a Hausdorff space is a homeomorphism. So σ and its inverse are continuous homomorphisms between Lie groups. By the automatic smoothness theorem of Lecture 3, both are smooth. ∎

The proof applies to disconnected K. Step 3 uses that K is a Lie group. Compactness is also needed, since a covering group of a matrix group need not be a matrix group (Lecture 4).

10.5 Examples

Example 10.14 (The circle group)**.** Let $K = \mathrm{U}(1)$. Here $\int f = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})\,d\theta$. Irreducible representations of an abelian group are one-dimensional by Schur's lemma. By Lecture 7 they are $\chi_n(e^{i\theta}) = e^{in\theta}$, $n \in \mathbb{Z}$. Then $d_{\chi_n} = 1$ and $E_n = \mathbb{C}\chi_n$. Hence

$$\mathscr{R}(\mathrm{U}(1)) = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}\chi_n, \qquad L^2(\mathrm{U}(1)) = \widehat{\bigoplus}_{n\in\mathbb{Z}} \mathbb{C}\chi_n.$$

Every $f \in L^2$ equals $\sum_n \hat{c}_n\chi_n$ with $\hat{c}_n = \langle \chi_n, f\rangle$ and $\|f\|_2^2 = \sum |\hat{c}_n|^2$. This is L^2 completeness of Fourier series.

For the $K \times K$ action,

$$((g,h)\cdot\chi_n)(x) = \chi_n(g^{-1}xh) = \overline{\chi_n(g)}\,\chi_n(h)\,\chi_n(x).$$

So $\mathbb{C}\chi_n \cong \overline{\chi}_n \otimes \chi_n$. Under L the line $\mathbb{C}\chi_n$ carries χ_{-n}. Under R it carries χ_n.

We can now see the operator (10.2) in this case. Let $\Pi = R$ on $L^2(\mathrm{U}(1))$, and let u be a unit vector with coefficients $a_n = \langle \chi_n, u\rangle$. Then $\langle \chi_k, R_{e^{it}}u\rangle = e^{ikt}a_k$. Hence

$$\langle \chi_k, T\chi_m\rangle = \frac{1}{2\pi}\int_0^{2\pi} e^{ikt}a_k\,\overline{e^{imt}a_m}\,dt = \delta_{km}|a_m|^2.$$

So $T\chi_m = |a_m|^2\chi_m$. Since $\sum_n |a_n|^2 = 1$, for each $\mu > 0$ only finitely many n have $|a_n|^2 = \mu$. So each eigenspace of T with $\mu \neq 0$ is finite-dimensional, as the spectral theorem says.

Example 10.15 (The group SU(2))**.** Let $K = \mathrm{SU}(2)$ and write

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

By Lecture 8, $\mathrm{Irr}(\mathrm{SU}(2)) = \{V_m : m \geq 0\}$, where V_m has dimension $m+1$. Realise V_m as symmetric tensors in $(\mathbb{C}^2)^{\otimes m}$. Let $e_1^{m-k}e_2^k$ be the average of the $\binom{m}{k}$ words with $m-k$ letters e_1 and k letters e_2. The group acts by $g^{\otimes m}$, and the tensor inner product gives

$$\langle e_1^{m-k}e_2^k, e_1^{m-j}e_2^j\rangle = \delta_{jk}\binom{m}{k}^{-1}.$$

By the Peter–Weyl theorem, Theorem 10.11,

$$L^2(\mathrm{SU}(2)) = \widehat{\bigoplus}_{m\geq 0} E_m, \qquad \dim E_m = (m+1)^2 = 1, 4, 9, 16, \ldots$$

Under R, V_m occurs $m+1$ times. In physics, the matrix coefficients of V_m are called Wigner D-functions.

The block $m = 1$. For $V_1 = \mathbb{C}^2$, $c_{e_i,e_j}(g) = g_{ij}$. So $E_1 = \operatorname{span}\{a, -\bar{b}, b, \bar{a}\}$, and the functions $\sqrt{2}\, g_{ij}$ are orthonormal. In particular $\int |a|^2\, dg = \frac{1}{2}$. Under R, the row spans $\operatorname{span}\{a, -\bar{b}\}$ and $\operatorname{span}\{b, \bar{a}\}$ are copies of V_1.

The block m. Take the unit vector $u = e_1^m$. Since $ge_1 = ae_1 + be_2$ and the vectors $e_1^{m-k}e_2^k$ are orthogonal, only the $k = 0$ term of the binomial expansion has nonzero inner product with u:

$$c_{u,u}(g) = \left\langle e_1^m, (ae_1 + be_2)^m \right\rangle = a^m.$$

Then Schur orthogonality (10.1) gives $\int |a|^{2m}\, dg = 1/(m+1)$, matching $m = 1$.

Self-checks

1. Why is the operator (10.2) nonzero, although no invariant vector is known?

Answer. By the definition (10.2), $\langle u, Tu\rangle = \int |\langle \Pi(g)u, u\rangle|^2\, dg$. The integrand is continuous, nonnegative and equal to 1 at e. By positivity (H2), the integral is positive. Hence T is nonzero.

2. Give $\dim E_2 \subset L^2(\mathrm{SU}(2))$ and the multiplicity of V_2 under R.

Answer. The block has dimension $\dim E_2 = 9$. The representation V_2 occurs 3 times. Under L, $E_2 \cong \overline{V}_2^{\oplus 3}$.

3. Why does the coefficient proposition, Proposition 10.10, prove continuity before evaluating at e?

Answer. Only continuous elements of $L^2(K)$ have values. Evaluation is not even bounded on $\mathscr{R}(\mathrm{U}(1))$. For example, the function $f_N = (2N+1)^{-1/2} \sum_{|n|\le N} \chi_n$ has $\|f_N\|_2 = 1$ and $f_N(1) = \sqrt{2N+1}$.

Summary

Let K be a compact Lie group. In a strongly continuous unitary representation of K on $H \neq 0$, an averaged rank-one projection is a nonzero positive compact intertwiner. Its eigenspaces for nonzero eigenvalues are finite-dimensional and invariant. The finite-dimensional invariant subspaces span a dense subspace. In $L^2(K)$, smoothing identifies the finite-dimensional R-invariant subspaces with spaces of matrix coefficients. Schur orthogonality then gives the Peter–Weyl theorem. Under R, and also under L, each $\Pi \in \mathrm{Irr}(K)$ occurs in $L^2(K)$ with multiplicity d_Π. Density in $L^2(K)$ shows that $\mathscr{R}(K)$ separates points. By the Stone–Weierstrass theorem, $\mathscr{R}(K)$ is therefore uniformly dense in $C(K)$. Adding finitely many representations that separate points gives a faithful finite-dimensional unitary representation, so K is isomorphic as a Lie group to a closed subgroup of some $\mathrm{U}(n)$.

Reading

- Sepanski, *Compact Lie Groups*, §§3.2–3.3, including Theorem 3.25.
- Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Chapter III, Sections 1–4.
- Etingof, MIT 18.755 lecture notes, Lectures 9–10.

Lecture 11

Fourier analysis and convolution

Objectives. *On the circle, a function has scalar Fourier coefficients. On a nonabelian compact group, the coefficient at an irreducible representation Π is an operator $\hat{f}(\Pi)$. We derive Plancherel from Peter–Weyl. We compute how convolution acts on these operators. We identify the convolution operators that commute with all translations.*

By the end of this lecture you will be able to:

- Compute $\hat{f}(\Pi)$ for matrix coefficients and derive the Plancherel theorem from Peter–Weyl.
- Prove $(f * h)^{\wedge} = \hat{h}\,\hat{f}$ and check this order by a direct $\mathrm{SU}(2)$ computation.
- Separate L^2 convergence of spectral partial sums from pointwise or uniform convergence.
- Characterise central convolution operators and write the isotypic projectors.

11.1 The operator-valued transform

Throughout, K is a compact Lie group with normalised Haar measure dg, so $\int_K dg = 1$. From Lecture 5 we import the following facts. The measure is invariant under left translation, right translation and inversion $g \mapsto g^{-1}$. A continuous function $\varphi \geq 0$ with $\int_K \varphi\, dg = 0$ vanishes. This property is strict positivity, which we call *positivity* below. We also import Fubini's theorem for continuous functions on $K \times K$ from Bröcker–tom Dieck, Section I.5, as in Lecture 10.

Every finite-dimensional representation is unitarisable and is a direct sum of irreducible ones. Finally, Schur orthogonality holds in the following form. For irreducible unitary representations Π, Π', vectors $u, v \in V_\Pi$ and vectors $u', v' \in V_{\Pi'}$,

$$\int_K \overline{\langle u, \Pi(g)v\rangle}\, \langle u', \Pi'(g)v'\rangle\, dg = \begin{cases} d_\Pi^{-1}\langle u', u\rangle\langle v, v'\rangle, & \Pi' = \Pi, \\ 0, & \Pi' \not\cong \Pi, \end{cases} \tag{11.1}$$

where $d_\Pi = \dim V_\Pi$.

From Lecture 10 we import two facts. The space $L^2(K)$ is the Hilbert completion of $C(K)$ with respect to the inner product $\langle \varphi, \psi\rangle = \int_K \overline{\varphi}\,\psi\, dg$. By the Peter–Weyl theorem, the span $\mathscr{R}(K)$ of matrix coefficients of finite-dimensional representations is dense in $L^2(K)$.

Fix a set $\widehat{K}$ containing exactly one irreducible unitary representation (Π, V_Π) from each equivalence class. On $\mathrm{End}(V_\Pi)$ we use the Hilbert–Schmidt inner product $\langle A, B\rangle_{\mathrm{HS}} = \mathrm{tr}(A^* B)$. For $u, v \in V_\Pi$, the operator uv^* is given by $w \mapsto \langle v, w\rangle u$.

Definition 11.1. Let $f \in C(K)$ and $\Pi \in \widehat{K}$. The *Fourier transform* of f at Π is

$$\hat{f}(\Pi) = \int_K f(g)\, \Pi(g^{-1})\, dg \in \mathrm{End}(V_\Pi),$$

integrated entry by entry in any basis. For $A \in \mathrm{End}(V_\Pi)$ the *matrix coefficient* c_A is the function given by $c_A(g) = \mathrm{tr}(A\,\Pi(g))$. The space of matrix coefficients of Π is $M_\Pi = \{c_A : A \in \mathrm{End}(V_\Pi)\}$.

For $K = \mathrm{U}(1)$ and $\Pi = \chi_n$ this is the classical coefficient $\frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta}) e^{-in\theta}\, d\theta$. The inverse g^{-1} in the definition gives the minus sign.

A rank-one operator gives the familiar coefficient $c_{uv^*}(g) = \langle v, \Pi(g)u\rangle$. Let Π' be equivalent to Π, say $\Pi'(g) = T\Pi(g)T^{-1}$ with T invertible. Then

$$\mathrm{tr}\big(A\,\Pi'(g)\big) = \mathrm{tr}\big(T^{-1}AT\,\Pi(g)\big) = c_{T^{-1}AT}(g).$$

Hence the matrix coefficients of Π' also lie in M_Π. So M_Π does not depend on the choice of representative. By complete reducibility, $\mathscr{R}(K)$ is the algebraic sum of the spaces M_Π.

These objects are those of Lecture 10 under other names. There $\widehat{K}$ is $\mathrm{Irr}(K)$, the space M_Π is E_Π, and $c_A = d_\Pi^{-1/2}\Phi_\Pi(A)$. The operator uv^* is $u\langle v, \cdot\rangle$, so the letters u and v are exchanged. The Peter–Weyl theorem already gives that the spaces M_Π are orthogonal and that $L^2(K)$ is their Hilbert sum. It also gives $\|f\|^2 = \sum_\Pi \|A_\Pi\|_{\mathrm{HS}}^2$ when $f = \sum_\Pi \Phi_\Pi(A_\Pi)$, and the coefficient lemma below gives $A_\Pi = \sqrt{d_\Pi}\,\hat{f}(\Pi)$. So what is new in this lecture is the formula for the projection and the operator form of the coefficient.

Lemma 11.2. *For $f \in C(K)$, $\Pi \in \widehat{K}$ and $B \in \mathrm{End}(V_\Pi)$,*

$$\langle c_B, f\rangle = \langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}}. \tag{11.2}$$

Proof. For every operator X we have $\overline{\mathrm{tr}\, X} = \mathrm{tr}\, X^*$. Since $\Pi(g)^* = \Pi(g^{-1})$,

$$\overline{c_B(g)} = \mathrm{tr}\big(\Pi(g)^* B^*\big) = \mathrm{tr}\big(B^*\,\Pi(g^{-1})\big).$$

Multiply by $f(g)$ and integrate. The trace is linear, so it passes through the entrywise integral. This gives $\mathrm{tr}(B^*\hat{f}(\Pi))$. ∎

We use (11.2) to extend the transform to $L^2(K)$. The left side of (11.2) is L^2-continuous in f. For fixed f, it is an antilinear functional of B. Hence the finite-dimensional Riesz theorem gives a unique operator $\hat{f}(\Pi)$ satisfying (11.2). For continuous f, this operator agrees with the integral of Definition 11.1, by the pairing lemma, Lemma 11.2.

Lemma 11.3. *Let $\Pi, \Pi' \in \widehat{K}$, $A \in \mathrm{End}(V_\Pi)$ and $B \in \mathrm{End}(V_{\Pi'})$. Then*

$$\langle c_B, c_A\rangle = \begin{cases} d_\Pi^{-1}\langle B, A\rangle_{\mathrm{HS}}, & \Pi' = \Pi, \\ 0, & \Pi' \neq \Pi. \end{cases} \tag{11.3}$$

Consequently $\hat{c}_A(\Pi) = d_\Pi^{-1}A$, and $\hat{c}_A(\Pi') = 0$ for $\Pi' \neq \Pi$.

Proof. Both sides are linear in A and antilinear in B. Rank-one operators span, so take $A = uv^*$ and $B = u'v'^*$. Distinct elements of $\widehat{K}$ are inequivalent. By (11.1),

$$\langle c_B, c_A\rangle = \int_K \overline{\langle v', \Pi'(g)u'\rangle}\,\langle v, \Pi(g)u\rangle\, dg = \delta_{\Pi\Pi'}\, d_\Pi^{-1}\langle v, v'\rangle\langle u', u\rangle.$$

On the other hand $(u'v'^*)^* = v'u'^*$, so

$$\langle B, A\rangle_{\mathrm{HS}} = \operatorname{tr}\big(v'u'^*uv^*\big) = \langle u', u\rangle\operatorname{tr}(v'v^*) = \langle u', u\rangle\langle v, v'\rangle.$$

This proves (11.3). By (11.2), $\langle B, \hat{c}_A(\Pi')\rangle_{\mathrm{HS}} = \langle c_B, c_A\rangle$ for every B. Comparing with (11.3) gives the transforms. ∎

11.2 Plancherel from Peter–Weyl

For $f \in L^2(K)$ and a finite set $F \subset \widehat{K}$, the *spectral partial sum* is

$$S_F f(g) = \sum_{\Pi\in F} d_\Pi\, \operatorname{tr}\big(\hat{f}(\Pi)\,\Pi(g)\big).$$

Theorem 11.4 (Plancherel). *Let $f, h \in L^2(K)$.*

(a) The orthogonal projection of $L^2(K)$ onto M_Π is $P_\Pi f = d_\Pi\, c_{\hat{f}(\Pi)}$.

(b) For every $\varepsilon > 0$ there is a finite $F_0 \subset \widehat{K}$ with $\|f - S_F f\| < \varepsilon$ for every finite $F \supseteq F_0$.

(c) $\|f\|^2 = \sum_\Pi d_\Pi \|\hat{f}(\Pi)\|_{\mathrm{HS}}^2$ and $\langle f, h\rangle = \sum_\Pi d_\Pi \langle \hat{f}(\Pi), \hat{h}(\Pi)\rangle_{\mathrm{HS}}$.

(d) If $\hat{f}(\Pi) = 0$ for every Π, then $f = 0$.

Proof. Step 1. By (11.3), $M_\Pi \perp M_{\Pi'}$ for $\Pi \neq \Pi'$, and $\|c_A\|^2 = d_\Pi^{-1}\|A\|_{\mathrm{HS}}^2$. Each M_Π is finite-dimensional, hence closed.

Step 2. Put $f_0 = d_\Pi c_{\hat{f}(\Pi)} \in M_\Pi$. For every $B \in \operatorname{End}(V_\Pi)$, (11.2) and (11.3) give

$$\langle c_B, f - f_0\rangle = \langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}} - d_\Pi\, d_\Pi^{-1}\langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}} = 0.$$

So $f - f_0 \perp M_\Pi$, that is, $f_0 = P_\Pi f$. Step 1 gives $\|P_\Pi f\|^2 = d_\Pi\|\hat{f}(\Pi)\|_{\mathrm{HS}}^2$.

Step 3. This step uses the Peter–Weyl theorem. Let $M_F = \bigoplus_{\Pi\in F} M_\Pi$. By Steps 1 and 2, $S_F = \sum_{\Pi\in F} P_\Pi$ is the orthogonal projection onto M_F. Hence $S_F f$ is the point of M_F nearest to f. By the Peter–Weyl theorem, there is $\varphi \in \mathscr{R}(K)$ with $\|f - \varphi\| < \varepsilon$. Since $\mathscr{R}(K)$ is the algebraic sum of the spaces M_Π, there is a finite F_0 with $\varphi \in M_{F_0}$. If $F \supseteq F_0$, then $\varphi \in M_F$, so $\|f - S_F f\| \le \|f - \varphi\| < \varepsilon$.

Step 4. Pythagoras and Step 2 give

$$\|f\|^2 = \|S_F f\|^2 + \|f - S_F f\|^2, \qquad \|S_F f\|^2 = \sum_{\Pi\in F} d_\Pi\|\hat{f}(\Pi)\|_{\mathrm{HS}}^2.$$

Hence every finite partial sum of the series in (c) is at most $\|f\|^2$. By Step 3, the partial sum over F exceeds $\|f\|^2 - \varepsilon^2$ once $F \supseteq F_0$. A sum of nonnegative terms is the supremum of its finite partial sums, so the norm identity holds. Polarisation gives the inner product identity. Finally, if $\hat{f}(\Pi) = 0$ for every Π, then the norm identity gives $\|f\| = 0$. This is the uniqueness statement (d). ∎

On $\mathrm{U}(1)$ we have $d_\Pi = 1$, and the norm identity is Parseval's identity.

Corollary 11.5 (Inversion for finite coefficient sums)**.** *Let $f \in \mathscr{R}(K)$. Then $\hat f(\Pi) \neq 0$ for only finitely many Π, and for every $g \in K$*

$$f(g) = \sum_{\Pi \in \widehat{K}} d_\Pi \operatorname{tr}\big(\hat f(\Pi)\Pi(g)\big).$$

Proof. Write $f = \sum_{\Pi \in F} c_{A_\Pi}$ with F finite. By the coefficient lemma, Lemma 11.3, $\hat f(\Pi) = d_\Pi^{-1} A_\Pi$ for $\Pi \in F$ and $\hat f(\Pi) = 0$ otherwise. Substituting returns $\sum_{\Pi \in F} c_{A_\Pi}(g) = f(g)$. ∎

For general $f \in L^2(K)$ we only have the L^2 convergence of spectral partial sums in the Plancherel theorem, Theorem 11.4(b). This statement says nothing about the value of f at a single point.

11.3 Convolution reverses the order

For a finite group G, the product of the group algebra $\mathbb{C}[G]$ (Finite Groups) is

$$\Big(\sum_{y} f(y)y\Big)\Big(\sum_{z} h(z)z\Big) = \sum_{x}\Big(\sum_{y} f(y)h(y^{-1}x)\Big)x.$$

The next definition is the same formula, with the sum over y replaced by the normalised Haar integral. Finite Groups also proved $\mathbb{C}[G] \cong \bigoplus_U \operatorname{End}(U)$, by sending $\sum f(g)g$ to the operators $\sum f(g)U(g)$. Under the transform, a function becomes a family of matrices, one for each irreducible, and convolution multiplies these matrices one block at a time. The transform $\hat f$ uses $\Pi(g^{-1})$, so it reverses the order.

Definition 11.6. For $f, h \in C(K)$ the *convolution* $f * h$ is

$$(f * h)(x) = \int_K f(y)\, h(y^{-1}x)\, dy.$$

Fix x. The map $y \mapsto y^{-1}x$ is inversion followed by right translation, and both preserve dy. So $y \mapsto h(y^{-1}x)$ has L^2 norm $\|h\|$, and Cauchy–Schwarz gives $|(f * h)(x)| \le \|f\|\,\|h\|$.

The function $f * h$ is also continuous. By the uniform-continuity lemma of Lecture 10, h is uniformly continuous on the compact group K. In other words, for every $\varepsilon > 0$ there is a neighbourhood N of e with $|h(z) - h(zw)| < \varepsilon$ for all $z \in K$ and $w \in N$. Hence $|(f * h)(x) - (f * h)(xw)| \le \varepsilon \int_K |f|\, dy$ for $w \in N$.

Theorem 11.7 (Convolution theorem)**.** *For $f, h \in C(K)$ and $\Pi \in \widehat{K}$,*

$$(f * h)^{\wedge}(\Pi) = \hat h(\Pi)\, \hat f(\Pi).$$

Proof. Exchange the iterated integrals entrywise:

$$(f * h)^{\wedge}(\Pi) = \int_K f(y)\Big(\int_K h(y^{-1}x)\, \Pi(x^{-1})\, dx\Big)dy.$$

In the inner integral substitute $x = yz$. Left invariance gives $dx = dz$, and $x^{-1} = z^{-1}y^{-1}$. So the inner integral is

$$\int_K h(z)\,\Pi(z^{-1})\Pi(y^{-1})\,dz = \hat{h}(\Pi)\,\Pi(y^{-1}).$$

The factor $\Pi(y^{-1})$ coming from f stands on the right, because $x^{-1} = z^{-1}y^{-1}$. Integrating against $f(y)$ and using the definition of $\hat{f}(\Pi)$ gives $\hat{h}(\Pi)\hat{f}(\Pi)$. ∎

Some texts use $\tilde{f}(\Pi) = \int_K f(g)\Pi(g)\,dg$. The same computation then gives $(f * h)^{\sim} = \tilde{f}\,\tilde{h}$. Check the convention before importing formulas.

Corollary 11.8. *Let $A, B \in \mathrm{End}(V_\Pi)$. Then $c_A * c_B = d_\Pi^{-1} c_{BA}$. If instead $B \in \mathrm{End}(V_{\Pi'})$ with $\Pi' \neq \Pi$, then $c_A * c_B = 0$.*

Proof. By the coefficient lemma and the convolution theorem, Lemma 11.3 and Theorem 11.7, $(c_A * c_B)^\wedge(\Pi) = d_\Pi^{-2} BA$. This is the transform of $d_\Pi^{-1} c_{BA}$. At every other element of $\widehat{K}$ both transforms vanish. The difference φ of the two sides is continuous, and $\|\varphi\| = 0$ by the uniqueness statement of the Plancherel theorem, Theorem 11.4(d). Positivity applied to $|\varphi|^2$ gives $\varphi = 0$. In the second case one of $\hat{c}_A, \hat{c}_B$ vanishes at each element of $\widehat{K}$, so the same argument gives $c_A * c_B = 0$. ∎

For rank-one operators, $(u'v'^*)(uv^*) = \langle v', u\rangle\, u'v^*$. Hence the coefficient-convolution corollary, Corollary 11.8, gives

$$\langle v, \Pi(\cdot)u\rangle * \langle v', \Pi(\cdot)u'\rangle = d_\Pi^{-1}\langle v', u\rangle\,\langle v, \Pi(\cdot)u'\rangle.$$

Fix an orthonormal basis $e_1, \dots, e_{d_\Pi}$ of V_Π. Write $E_{ij} = e_i e_j^*$ for the matrix unit. Its matrix in this basis has 1 in row i and column j, and 0 elsewhere. By the last display, the matrix coefficients $D_{ij}(g) = \langle e_i, \Pi(g)e_j\rangle$ satisfy $D_{ij} * D_{kl} = d_\Pi^{-1}\delta_{jk} D_{il}$. So the functions multiply in the natural order. By the coefficient lemma, Lemma 11.3, their transforms are the transposed matrix units $d_\Pi^{-1} E_{ji}$, and these multiply in reverse.

Example 11.9 (Direct check on SU(2))**.** Write elements of SU(2) as

$$U = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

Let Π be the defining representation on $\mathbb{C}^2$, with standard basis e_1, e_2, so $d_\Pi = 2$. Put $f(U) = U_{21} = b$ and $h(U) = U_{12} = -\bar{b}$. Since $\mathrm{tr}(E_{12}U) = U_{21}$ and $\mathrm{tr}(E_{21}U) = U_{12}$, we have $f = c_{E_{12}}$ and $h = c_{E_{21}}$. The convolution theorem predicts the transform of $f * h$. The coefficient-convolution corollary, Corollary 11.8, then gives the function, since $c_{E_{12}} * c_{E_{21}} = \tfrac{1}{2} c_{E_{21}E_{12}} = \tfrac{1}{2} c_{E_{22}}$. Hence

$$(f * h)^\wedge(\Pi) = \hat{h}\hat{f} = \tfrac{1}{4} E_{21}E_{12} = \tfrac{1}{4} E_{22}, \qquad \text{so} \quad (f * h)(U) = \tfrac{1}{2} U_{22} = \tfrac{1}{2}\bar{a}.$$

The opposite order would give the transform $\tfrac{1}{4}E_{11}$ and hence the function $\tfrac{1}{2}a$.

Now compute directly. Since $a = \langle e_1, Ue_1\rangle$ and $b = \langle e_2, Ue_1\rangle$, (11.1) gives

$$\int_{\mathrm{SU}(2)} |b|^2\,dU = \tfrac{1}{2}, \qquad \int_{\mathrm{SU}(2)} \bar{a}\,b\,dU = \tfrac{1}{2}\langle e_2, e_1\rangle = 0.$$

Let y have entries a, b and let x have entries α, β in the same pattern. Since y is unitary, $y^{-1} = y^{*}$. Hence

$$y^{-1}x = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix} \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} = \begin{pmatrix} \bar{a}\alpha + \bar{b}\beta & \bar{b}\bar{\alpha} - \bar{a}\bar{\beta} \\ -b\alpha + a\beta & a\bar{\alpha} + b\bar{\beta} \end{pmatrix}.$$

Since $h(y^{-1}x)$ is the $(1,2)$ entry of this matrix, we get

$$(f * h)(x) = \int b\,(\bar{b}\bar{\alpha} - \bar{a}\bar{\beta})\,dy = \bar{\alpha}\int |b|^2\,dy - \bar{\beta}\int \bar{a}\,b\,dy = \tfrac{1}{2}\bar{\alpha}.$$

This matches the prediction. Using $(y^{-1}x)_{21} = -b\alpha + a\beta$, the same method gives $(h * f)(x) = \tfrac{1}{2}\alpha$. So $f * h \neq h * f$.

11.4 Pointwise and uniform inversion

Lecture 10 also shows that $\mathscr{R}(K)$ is uniformly dense in $C(K)$. This density does not say that the approximants are the partial sums $S_F f$. Uniform convergence of $S_F f$ needs an extra hypothesis.

Proposition 11.10. *Let $f \in C(K)$. Assume*

$$\sum_{\Pi \in \widehat{K}} d_\Pi^{3/2}\, \|\hat{f}(\Pi)\|_{\mathrm{HS}} < \infty.$$

Then the series

$$\sum_{\Pi \in \widehat{K}} d_\Pi\, \operatorname{tr}\big(\hat{f}(\Pi)\,\Pi(g)\big)$$

converges absolutely and uniformly on K. Its sum is $f(g)$ for every $g \in K$.

Proof. We need two facts about the Hilbert–Schmidt norm. The first is Cauchy–Schwarz for $\langle \cdot, \cdot \rangle_{\mathrm{HS}}$, applied to X^{*} and Y. The second uses unitarity of $\Pi(g)$:

$$|\operatorname{tr}(XY)| \le \|X\|_{\mathrm{HS}}\,\|Y\|_{\mathrm{HS}}, \qquad \|\Pi(g)\|_{\mathrm{HS}}^2 = \operatorname{tr}\big(\Pi(g)^{*}\Pi(g)\big) = d_\Pi.$$

Hence, for every $g \in K$,

$$\big|d_\Pi \operatorname{tr}\big(\hat{f}(\Pi)\Pi(g)\big)\big| \le b_\Pi, \qquad b_\Pi = d_\Pi^{3/2}\|\hat{f}(\Pi)\|_{\mathrm{HS}}.$$

The numbers b_Π have finite sum by hypothesis. Hence, for each n, only finitely many b_Π exceed $1/n$, so only countably many terms are nonzero. By the Weierstrass test the series converges absolutely and uniformly to a continuous function Φ.

Explicitly, given $\varepsilon > 0$, choose a finite $F_1 \subset \widehat{K}$ with $\sum_{\Pi \notin F_1} b_\Pi < \varepsilon$. Then for every finite $F \supseteq F_1$,

$$\sup_{g \in K} \big|\Phi(g) - S_F f(g)\big| \le \sum_{\Pi \notin F} b_\Pi < \varepsilon.$$

Since dg has mass one, $\|\Phi - S_F f\| < \varepsilon$. Enlarge F to contain the set F_0 given for this ε by the L^2 convergence statement of the Plancherel theorem, Theorem 11.4(b). By the triangle inequality, $\|\Phi - f\| < 2\varepsilon$. Since $\varepsilon > 0$ was arbitrary, $\|\Phi - f\| = 0$. Positivity applied to $|\Phi - f|^2$ then gives $\Phi = f$. ∎

Remark 11.11 (Cited results). (i) Let K be compact and connected, with maximal torus T of dimension r. Fix a Weyl-invariant inner product on $\mathfrak{t}^*$. By Lecture 8, each $\Pi \in \widehat{K}$ has a highest weight λ_Π. Distinct Π have distinct highest weights, and these lie in the weight lattice of T, a lattice of rank r. Now let $f \in C^\infty(K)$. Then for every N there is C_N with

$$\|\hat{f}(\Pi)\|_{\mathrm{HS}} \le C_N \, (1 + |\lambda_\Pi|)^{-N} \qquad \text{for all } \Pi \in \widehat{K}.$$

We cite this from M. Sugiura, *Fourier series of smooth functions on compact Lie groups*, Osaka J. Math. 8 (1971), 33–47. The proof uses the Casimir operator.

The Weyl dimension formula, imported from the prerequisite Lie algebras course, gives $d_\Pi \le C(1 + |\lambda_\Pi|)^q$, where q is the number of positive roots. A lattice of rank r has $O(R^r)$ points of norm at most R, so $\sum_\Pi (1 + |\lambda_\Pi|)^{-M} < \infty$ for $M > r$. Taking $N > \tfrac{3}{2}q + r$ gives the hypothesis of the uniform inversion proposition, Proposition 11.10. So the Fourier series of a smooth function on a compact connected Lie group converges absolutely and uniformly to it.

(ii) On $K = \mathrm{U}(1)$ there are continuous functions whose symmetric partial sums diverge at a point. This is du Bois-Reymond's classical example (Grafakos, *Classical Fourier Analysis*, Section 3.3.3, Proposition 3.3.5). So convergence at every point, and hence uniform convergence, can fail for continuous functions.

11.5 Central kernels and isotypic projectors

Convolution acts one block at a time. So we can read a function through the matrices $\hat{f}(\Pi)$. Figure 11.1 draws these matrices for SU(2), including two kernels that this section computes: $k(U) = |\operatorname{tr} U|^2$ of Example 11.15, and the projector kernel e_Π of Corollary 11.14 for $\Pi = V_2$. A central kernel turns out to have scalar blocks.

Left and right translation are given, as in Lecture 10, by $(L_g f)(x) = f(g^{-1}x)$ and $(R_g f)(x) = f(xg)$. The character of Π is $\chi_\Pi = \operatorname{tr} \circ \Pi = c_I$. Unitarity gives $\chi_\Pi(g^{-1}) = \overline{\chi_\Pi(g)}$.

Definition 11.12. A function $k \in C(K)$ is *central*, or a *class function* (Lecture 9), if $k(gxg^{-1}) = k(x)$ for all $g, x \in K$. The *convolution operator* of $k \in C(K)$ is the map $T_k \colon C(K) \to C(K)$ given by $T_k f = f * k$. The operator T_k is *central* if k is central.

Substituting $y = gy'$ in the convolution integral shows that $T_k L_g = L_g T_k$ for every k. The next theorem shows that commutation with right translations characterises the central kernels.

Theorem 11.13. *For $k \in C(K)$ the following are equivalent.*

(a) k is central.

(b) $T_k R_g = R_g T_k$ for every $g \in K$.

(c) Every $\hat{k}(\Pi)$ is a scalar multiple of the identity.

In this case $\hat{k}(\Pi) = \kappa_\Pi I$ with $\kappa_\Pi = d_\Pi^{-1} \langle \chi_\Pi, k \rangle$. For $f \in C(K)$ we have $P_\Pi(f * k) = \kappa_\Pi P_\Pi f$. Hence, for finite $F \subset \widehat{K}$, the sums $\sum_{\Pi \in F} \kappa_\Pi P_\Pi f$ converge to $f * k$ in L^2, in the sense of the convergence statement of the Plancherel theorem, Theorem 11.4(b).

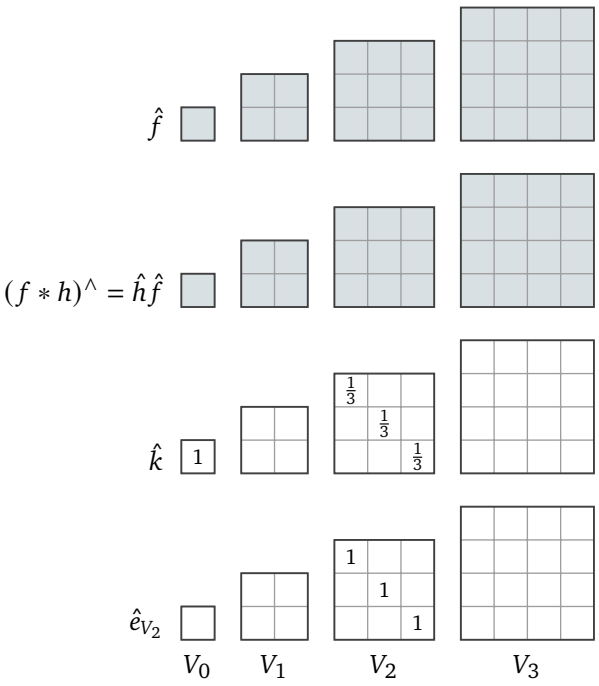


Figure 11.1: The operator blocks $\hat f(V_m)$ on SU(2), for $m = 0, 1, 2, 3$, with a shaded cell for a general entry and a blank cell for zero. The rows show a general f, a convolution $f * h$, the kernel $k(U) = |\operatorname{tr} U|^2$, and the projector kernel e_{V_2}. These are matrices of operators, not the arrays of functions of Lecture 10: since $\hat D_{ij} = d_\Pi^{-1} E_{ji}$, the two grids differ by a transpose.

Proof. *(a)⇔(c).* Fix g and put $k^g(x) = k(gxg^{-1})$. Substitute $x' = gxg^{-1}$, which preserves dx by two-sided invariance:

$$(k^g)^\wedge(\Pi) = \int_K k(x')\,\Pi(g^{-1}x'^{-1}g)\,dx' = \Pi(g)^{-1}\hat k(\Pi)\Pi(g).$$

If k is central, then $k^g = k$. Hence $\hat k(\Pi)$ commutes with every $\Pi(g)$, and by Schur's lemma it is a scalar multiple of the identity. Conversely, if every $\hat k(\Pi)$ is scalar, then k^g and k have equal transforms. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), and positivity, $k^g = k$.

(a)⇔(b). Right invariance with $y = y'g^{-1}$ gives

$$(T_k R_g f)(x) = \int_K f(y)\,k(gy^{-1}x)\,dy, \qquad (R_g T_k f)(x) = \int_K f(y)\,k(y^{-1}xg)\,dy.$$

If k is central, then $k(y^{-1}xg) = k(gy^{-1}x)$, which is (b). Conversely, assume (b) and set $x = e$. Put $\psi(y) = k(y^{-1}g) - k(gy^{-1})$. Then $\int_K f\psi\,dy = 0$ for all f. Taking $f = \bar\psi$ and using positivity gives $\psi = 0$. So $k(zg) = k(gz)$ for all z. Replacing z by zg^{-1} gives $k(z) = k(gzg^{-1})$.

Final claims. We have $\operatorname{tr}\hat k(\Pi) = \int_K k(g)\overline{\chi_\Pi(g)}\,dg = \langle \chi_\Pi, k\rangle$. Since $\hat k(\Pi) = \kappa_\Pi I$, this trace is $d_\Pi\kappa_\Pi$, which gives the formula for κ_Π. By the convolution theorem, Theorem 11.7, $(f * k)^\wedge(\Pi) = \kappa_\Pi \hat f(\Pi)$. Then the projection formula of the Plancherel theorem, Theorem 11.4(a), gives $P_\Pi(f * k) = \kappa_\Pi P_\Pi f$. So $S_F(f * k) = \sum_{\Pi\in F}\kappa_\Pi P_\Pi f$, and the L^2 convergence statement, Theorem 11.4(b), applied to $f * k$ gives the convergence. ∎

Corollary 11.14 (Isotypic projectors). *Let $e_\Pi = d_\Pi \chi_\Pi$. For $f \in C(K)$,*

$$P_\Pi f = f * e_\Pi = e_\Pi * f, \qquad (P_\Pi f)(x) = d_\Pi \int_K f(y)\, \chi_\Pi(y^{-1}x)\, dy.$$

Moreover $e_\Pi * e_{\Pi'} = \delta_{\Pi\Pi'} e_\Pi$.

Proof. First, characters are central, and $e_\Pi = d_\Pi c_I$. By the coefficient lemma, Lemma 11.3, $\hat{e}_\Pi(\Pi) = I$ and $\hat{e}_\Pi(\Pi') = 0$ for $\Pi' \neq \Pi$. So the central kernel theorem, Theorem 11.13, applies to $k = e_\Pi$ with $\kappa_{\Pi'} = \delta_{\Pi\Pi'}$. Its sums therefore equal $P_\Pi f$ once $\Pi \in F$, and they converge to $f * e_\Pi$ in L^2. Hence $\|f * e_\Pi - P_\Pi f\| = 0$. Both functions are continuous, so positivity gives $f * e_\Pi = P_\Pi f$.

Next, by the convolution theorem, Theorem 11.7, the transforms of $f * e_\Pi$ and $e_\Pi * f$ agree, because $\hat{e}_\Pi$ is scalar at every element of $\widehat{K}$. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), the difference of these functions is 0 in $L^2(K)$. Both functions are continuous, so positivity gives $f * e_\Pi = e_\Pi * f$.

Finally, the coefficient-convolution corollary, Corollary 11.8, with $A = B = d_\Pi I$ gives $e_\Pi * e_\Pi = e_\Pi$. Its second case gives $e_\Pi * e_{\Pi'} = 0$ for $\Pi' \neq \Pi$. ∎

As in the Finite Groups course, the kernels e_Π are orthogonal central idempotents. Substituting $y = xg$ in the formula of the corollary gives $P_\Pi f = d_\Pi \int_K \overline{\chi_\Pi(g)}\, R_g f\, dg$. The same formula projects in any finite-dimensional unitary representation σ of K: the operator $d_\Pi \int_K \overline{\chi_\Pi(g)}\, \sigma(g)\, dg$ is the orthogonal projection onto the Π-isotypic part. For trivial Π this is the invariant-vector projector of Lecture 5.

The Π-*isotypic part* of the right regular action R is the sum of all finite-dimensional R-invariant subspaces $W \subset C(K)$ with $R|_W \cong \Pi$.

We now identify M_Π as this isotypic part. By cyclicity of the trace, $c_A(xg) = \operatorname{tr}(\Pi(g)A\Pi(x))$. Hence $R_g c_A = c_{\Pi(g)A}$. The map $A \mapsto c_A$ is injective, because $\|c_A\|^2 = d_\Pi^{-1}\|A\|_{\mathrm{HS}}^2$. Let $e_1, \dots, e_{d_\Pi}$ be an orthonormal basis of V_Π. The map $A \mapsto \Pi(g)A$ sends each column Ae_j to $\Pi(g)Ae_j$. So R on M_Π is equivalent to $\Pi^{\oplus d_\Pi}$. The jth copy of Π consists of the functions c_A with $Ae_i = 0$ for $i \neq j$.

Now let $W \subset C(K)$ be finite-dimensional and R-invariant with $R|_W \cong \Pi$. Fix $\Pi' \neq \Pi$. The map $P_{\Pi'}|_W : W \to M_{\Pi'}$ intertwines R, since $P_{\Pi'} = T_{e_{\Pi'}}$ with $e_{\Pi'}$ central. Its image is 0 or isomorphic to Π, because W is irreducible. Every irreducible subrepresentation of $M_{\Pi'} \cong \Pi'^{\oplus d_{\Pi'}}$ projects nontrivially to some summand, so Schur's lemma makes it isomorphic to Π'. Hence $P_{\Pi'} W = 0$.

So for $f \in W$ we get $S_F f = P_\Pi f$ whenever $\Pi \in F$. By the L^2 convergence statement of the Plancherel theorem, Theorem 11.4(b), $\|f - P_\Pi f\| = 0$. Both functions are continuous, so positivity gives $f = P_\Pi f \in M_\Pi$. Hence every such W lies in M_Π. Conversely, M_Π is the sum of its d_Π copies of Π. Thus M_Π is the Π-isotypic part of R, and P_Π is the orthogonal projector onto it.

Example 11.15 (Central kernels on SU(2) and the circle). Let V_m be the irreducible representation of SU(2) of dimension $m+1$, and write P_m for P_{V_m}. Let $U \in \mathrm{SU}(2)$ have eigenvalues $e^{i\theta}$ and $e^{-i\theta}$ with $0 < \theta < \pi$. Then Lecture 9 gives $\chi_m(U) = \sin((m+1)\theta)/\sin\theta$. By the isotypic projector corollary, Corollary 11.14, $P_m f = (m+1)\, f * \chi_m$.

Take the kernel $k(U) = |\operatorname{tr} U|^2 = 4\cos^2\theta$. It is central, since the trace is invariant under conjugation. Since χ_1 is real and $V_1 \otimes V_1 \cong V_0 \oplus V_2$, we get $k = \chi_0 + \chi_2$. As a check, $\sin 3\theta / \sin\theta = 4\cos^2\theta - 1$.

Characters are orthonormal, so the central kernel theorem, Theorem 11.13, gives $\kappa_0 = 1$, $\kappa_2 = \frac{1}{3}$ and $\kappa_m = 0$ otherwise. Let $f \in C(\mathrm{SU}(2))$. For $F \supseteq \{V_0, V_2\}$, the sums in that theorem equal $P_0 f + \frac{1}{3} P_2 f$, and $P_0 f$ is the constant $\int_{\mathrm{SU}(2)} f\, dU$. These sums converge to $f * k$ in L^2. Both functions are continuous, so positivity gives

$$f * |\operatorname{tr}|^2 = \int_{\mathrm{SU}(2)} f\, dU + \tfrac{1}{3} P_2 f.$$

In particular $U_{11} * k = 0$, because $U_{11} = c_{E_{11}}$ lies in M_1. At $x = I$ this reads $\int a \cdot 4(\operatorname{Re} a)^2\, dy = 0$. Left translation by $-I$ preserves dy and changes the sign of the integrand. Hence this integral vanishes, as predicted.

On $K = \mathrm{U}(1)$ every $\Pi \in \widehat{K}$ is a character χ_n given by $\chi_n(z) = z^n$ for an integer n, with $d_{\chi_n} = 1$. Here $\hat{f}(\chi_n) = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) e^{-in\theta} d\theta$ is the classical coefficient. We write $\hat{f}(n)$ for $\hat{f}(\chi_n)$ and P_n for P_{χ_n}. For $f, h \in C(\mathrm{U}(1))$, the convolution theorem, Theorem 11.7, reads $(f * h)\hat{}(n) = \hat{f}(n)\hat{h}(n)$. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), and positivity, $f * h = h * f$. The group is abelian, so every function is central, and $P_n f = f * \chi_n = \hat{f}(n) z^n$.

Self-checks

1. On $\mathrm{U}(1)$, why is the reversal in the convolution theorem, Theorem 11.7, invisible?
   Answer. Each Π is one-dimensional. The transforms are numbers, and numbers commute.
2. Let $k(U) = U_{11}$ on $\mathrm{SU}(2)$. Is T_k central?
   Answer. No. Here $k = c_{E_{11}}$ for the defining representation Π, so $\hat{k}(\Pi) = \frac{1}{2} E_{11}$ is not a scalar multiple of the identity. Hence, by the central kernel theorem, Theorem 11.13, k is not central, and so T_k is not central. To check this directly, put

   $$w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad t_\theta = \operatorname{diag}(e^{i\theta}, e^{-i\theta}).$$

   Then $k(t_\theta) = e^{i\theta}$, while $k(w t_\theta w^{-1}) = e^{-i\theta}$. These differ, for example, at $\theta = \pi/2$.
3. Compute $(\chi_\Pi * \chi_\Pi)(e)$ in two ways.
   Answer. By definition it is $\int_K \chi_\Pi(y)\chi_\Pi(y^{-1})\, dy = \int_K |\chi_\Pi|^2\, dy = 1$. By the identity $e_\Pi * e_\Pi = e_\Pi$ in the isotypic projector corollary, Corollary 11.14, $\chi_\Pi * \chi_\Pi = d_\Pi^{-1} \chi_\Pi$, whose value at e is 1.

Summary

The Fourier coefficient at Π is the operator $\hat{f}(\Pi) = \int f(g)\Pi(g^{-1})\, dg$. Peter–Weyl gives Plancherel. Hence $L^2(K)$ is the Hilbert direct sum of the spaces M_Π, and the orthogonal projection onto M_Π is $P_\Pi f = d_\Pi \operatorname{tr}(\hat{f}(\Pi)\Pi(\cdot))$. The algebraic sum of these spaces is $\mathscr{R}(K)$, which is dense in $L^2(K)$. The transform of a convolution is $(f * h)\hat{} = \hat{h}\hat{f}$. The inversion

formula holds exactly for finite coefficient sums. For general $f \in L^2(K)$, the spectral partial sums converge to f in L^2 norm. Uniform inversion needs an extra hypothesis, such as the summability condition of the uniform inversion proposition, Proposition 11.10. For compact connected K, smoothness suffices by the cited estimate in Remark 11.11. Central kernels are exactly those with scalar transforms. On $C(K)$, the isotypic projectors are $f \mapsto d_\Pi f * \chi_\Pi$.

The circle and a general compact Lie group compare as follows.

$\mathrm{U}(1)$	compact Lie group K
$\hat{f}(n) \in \mathbb{C}$	$\hat{f}(\Pi) \in \mathrm{End}(V_\Pi)$
$\|f\|^2 = \sum_n \|\hat{f}(n)\|^2$	$\|f\|^2 = \sum_\Pi d_\Pi \|\hat{f}(\Pi)\|_{\mathrm{HS}}^2$
$f = \sum_n \hat{f}(n) z^n$ (in L^2)	$f = \sum_\Pi d_\Pi \operatorname{tr}\big(\hat{f}(\Pi)\Pi(\cdot)\big)$ (in L^2)
$(f * h)\hat{}(n) = \hat{f}(n)\hat{h}(n)$	$(f * h)\hat{}(\Pi) = \hat{h}(\Pi)\hat{f}(\Pi)$

Reading

Sepanski, *Compact Lie Groups*, §3.4, for Fourier theory on compact groups. Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Chapter III, §§2–4, for representative functions and the Peter–Weyl theorem behind the Plancherel theorem, Theorem 11.4. Compare each source's transform and convolution conventions with Definitions 11.1 and 11.6 before using its formulas.

Lecture 12

Homogeneous spaces and compact induction

Objectives. *Let K be a compact Lie group and H a closed subgroup. Which irreducible representations of K occur in functions on K/H, and how often? We show that each irreducible representation V of K occurs $\dim V^H$ times. Induction from H extends this count to vector-valued functions.*

By the end of this lecture you will be able to:

- Construct the invariant integral on K/H and identify $L^2(K/H)$ with the right-H-invariant part of $L^2(K)$.
- Prove that each irreducible V occurs in $L^2(K/H)$ exactly $\dim V^H$ times.
- Define compact induction, describe an element of the induced representation in the tangent-field model on S^2, and prove Frobenius reciprocity with finite multiplicities.
- Decompose $L^2(S^2)$ under $\mathrm{SO}(3)$ and relate the summands to spherical harmonics.

Throughout, K is a compact Lie group, not necessarily connected, and $H \subseteq K$ is a closed subgroup. By Lecture 3, K/H is a compact manifold, and the projection $p\colon K \to K/H$, $p(x) = xH$, is continuous and open. The group K acts on K/H by $g \cdot xH = gxH$.

The Haar measures dx on K and dh on H have total mass 1. They are left, right and inversion invariant (Lecture 5). The space $L^2(K)$ is the completion of $C(K)$ for $\langle f_1, f_2\rangle = \int_K \overline{f_1}\, f_2\, dx$, and $C(K) \to L^2(K)$ is injective (Lecture 10).

We use the left and right translation representations

$$(L_g f)(x) = f(g^{-1}x), \qquad (R_g f)(x) = f(xg).$$

These commuting representations are unitary and strongly continuous (Lecture 10).

12.1 Functions and integration on K/H

By the definition of the quotient topology, a function φ on K/H is continuous exactly when $\varphi \circ p$ is continuous. A right-H-invariant function on K is constant on cosets. Hence the pullback p^* is a linear bijection

$$p^*\colon C(K/H) \to C(K)^H := \{f \in C(K) : f(xh) = f(x) \text{ for all } x \in K,\ h \in H\}, \qquad p^*\varphi = \varphi \circ p. \tag{12.1}$$

Definition 12.1. The *invariant integral* on K/H is the linear functional

$$\int_{K/H} \varphi(\xi)\, d\xi := \int_K \varphi(p(x))\, dx, \qquad \varphi \in C(K/H).$$

Proposition 12.2. *(a) The integral is positive,* $\int_{K/H} 1\, d\xi = 1$, *and* $\int_{K/H} \varphi(g^{-1}\xi)\, d\xi = \int_{K/H} \varphi(\xi)\, d\xi$ *for* $g \in K$.

(b) For $f \in C(K)$, *the* H*-average* $f^H(x) = \int_H f(xh)\, dh$ *lies in* $C(K)^H$, *and*

$$\int_K f(x)\, dx = \int_K f^H(x)\, dx = \int_{K/H} \Big(\int_H f(xh)\, dh \Big)\, d(xH).$$

Proof. (a) Positivity and normalisation hold for dx. Invariance follows from $\varphi(g^{-1}p(x)) = \varphi(p(g^{-1}x))$ and left invariance of dx.

(b) The map $(x, h) \mapsto f(xh)$ is uniformly continuous on the compact space $K \times H$. So f^H is continuous. For $k \in H$, left invariance of dh gives $f^H(xk) = \int_H f(xkh)\, dh = f^H(x)$. Fubini for continuous functions on $K \times H$ and right invariance of dx give

$$\int_K f^H(x)\, dx = \int_H \int_K f(xh)\, dx\, dh = \int_H \int_K f(x)\, dx\, dh = \int_K f(x)\, dx.$$

By the pullback bijection (12.1), the last expression in (b) is $\int_K f^H\, dx$ written on K/H. ∎

A positive normalised invariant integral I on K/H is unique. Indeed, the map $g \mapsto \varphi(g^{-1}\,\cdot\,)$ is continuous in the supremum norm. So averaging the identity $I(\varphi) = I(\varphi(g^{-1}\,\cdot\,))$ over K gives $I(\varphi) = I\big(\int_K \varphi(g^{-1}\,\cdot\,)\, dg\big)$. The function inside is constant, since K acts transitively on K/H, and its value is $\int_{K/H} \varphi\, d\xi$. In particular, for a maximal torus $H = T$ of a compact connected group K, this integral is the measure $d(gT)$ of Lecture 9.

Definition 12.3. The space $L^2(K/H)$ is the completion of $C(K/H)$ for $\langle \varphi, \psi \rangle = \int_{K/H} \overline{\varphi}\, \psi\, d\xi$. Also

$$L^2(K)^H = \{ f \in L^2(K) : R_h f = f \text{ for all } h \in H \}.$$

The form is definite since $\int_{K/H} |\varphi|^2\, d\xi = \|\varphi \circ p\|^2$ and $C(K) \to L^2(K)$ is injective. The same identity shows that p^* is isometric. So p^* extends to an isometry $L^2(K/H) \to L^2(K)$. The subspace $L^2(K)^H$ is closed because each R_h is bounded.

The operator of the next proposition replaces a function by its average over each coset. Averaging does not increase an L^2 norm, and it does not change a function that is constant on cosets. In the sense of the vector-integral lemma of Lecture 10, this operator is the integral $\int_H R_h\, dh$. So it is the projector onto invariants of Lecture 5, on an infinite-dimensional space.

Proposition 12.4. *There is a unique bounded operator* A_H *on* $L^2(K)$ *with* $A_H f = f^H$ *for* $f \in C(K)$. *It satisfies*

$$\langle f_1, A_H f_2 \rangle = \int_H \langle f_1, R_h f_2 \rangle\, dh \qquad (f_1, f_2 \in L^2(K)). \tag{12.2}$$

It is the orthogonal projection onto $L^2(K)^H$. *Consequently* p^* *maps* $L^2(K/H)$ *isometrically onto* $L^2(K)^H$.

Proof. Let $f \in C(K)$. Cauchy–Schwarz on H gives $|f^H(x)|^2 \le (|f|^2)^H(x)$. Integrate and apply the averaging identity of Proposition 12.2(b) to $|f|^2$:

$$\|f^H\|^2 \le \int_K (|f|^2)^H \, dx = \int_K |f|^2 \, dx = \|f\|^2.$$

Since $C(K)$ is dense, A_H exists uniquely, with $\|A_H\| \le 1$.

For continuous f_1, f_2, Fubini shows that both sides of (12.2) equal $\int_K \int_H \overline{f_1(x)} f_2(xh) \, dh \, dx$. Both sides are continuous in (f_1, f_2). On the right, the integrand is continuous in h by strong continuity and bounded by $\|f_1\| \, \|f_2\|$. So (12.2) holds on $L^2(K)$.

Now $R_h^* = R_{h^{-1}}$ and dh is inversion invariant. So the weak formula (12.2) gives $\langle f_1, A_H f_2 \rangle = \langle A_H f_1, f_2 \rangle$, and A_H is self-adjoint.

Next, A_H fixes $L^2(K)^H$. If $f \in L^2(K)^H$, then the weak formula gives $\langle f_1, A_H f \rangle = \langle f_1, f \rangle$ for all f_1, so $A_H f = f$.

Finally, the range of A_H lies in $L^2(K)^H$. Let $k \in H$. For continuous f, $A_H f = f^H$ lies in $C(K)^H$ by the description of the H-average in Proposition 12.2(b), so $R_k A_H f = A_H f$. Since $C(K)$ is dense and both sides are bounded in f, this identity holds for all f.

Hence A_H is a self-adjoint idempotent with range $L^2(K)^H$.

It remains to identify the image of p^*. Since p^* is an isometry, its image is the closure of $p^*(C(K/H)) = C(K)^H$. This closure lies in the closed subspace $L^2(K)^H$. Conversely, let $f \in L^2(K)^H$ and choose $f_n \in C(K)$ with $f_n \to f$. Then $f_n^H \in C(K)^H$ and $f_n^H = A_H f_n \to A_H f = f$. So the image is $L^2(K)^H$. ∎

Each L_g commutes with every R_h, so L_g preserves $L^2(K)^H$. Transported by p^*, the operator L_g acts on $C(K/H)$ by $\varphi \mapsto \varphi(g^{-1} \cdot)$. The resulting unitary representation of K on $L^2(K/H)$ is the *left regular representation*.

12.2 Multiplicities in $L^2(K/H)$

Let (Π, V) be a finite-dimensional unitary representation and $u, v \in V$. The *matrix coefficient* is $c_{u,v}(x) = \langle u, \Pi(x) v \rangle$. Unitarity gives

$$L_g c_{u,v} = c_{\Pi(g)u,\, v}, \qquad R_g c_{u,v} = c_{u,\, \Pi(g)v}. \tag{12.3}$$

For the first identity, $c_{u,v}(g^{-1}x) = \langle u, \Pi(g)^{-1} \Pi(x) v \rangle = \langle \Pi(g) u, \Pi(x) v \rangle$.

The map $u \mapsto c_{u,v}$ is antilinear. The *conjugate representation* $\overline{V}$ is the set V with scalar multiplication $t \cdot u = \bar{t} u$, the same operators $\Pi(g)$, and inner product $\langle u, u' \rangle_{\overline{V}} = \langle u', u \rangle$. It has the same invariant subspaces as V. So $\overline{V}$ is irreducible when V is, and $(\overline{V})^H = V^H$ as sets.

For each $\sigma \in \mathrm{Irr}(K)$ fix a unitary representative (Π_σ, V_σ) of dimension d_σ. Write $\bar{\sigma}$ for the class of $\overline{V_\sigma}$, and let $E_\sigma = \mathrm{span}\{c_{u,v} : u, v \in V_\sigma\} \subseteq C(K)$. We use two earlier results:

(PW1) Schur orthogonality (Lecture 5): $\langle c_{u,v}, c_{u',v'} \rangle = d_\sigma^{-1} \langle u', u \rangle \langle v, v' \rangle$ for $u, v, u', v' \in V_\sigma$.

(PW2) Peter–Weyl (Lecture 10): the E_σ are mutually orthogonal and $L^2(K) = \widehat{\bigoplus}_\sigma E_\sigma$.

By Schur orthogonality (PW1), the linear map $\overline{V_\sigma} \otimes V_\sigma \to E_\sigma$, $u \otimes v \mapsto c_{u,v}$, is $d_\sigma^{-1/2}$ times an isometry. So it is bijective. By the coefficient identities (12.3), it carries $\bar{\Pi}_\sigma \otimes 1$ to L and $1 \otimes \Pi_\sigma$ to R. Let Q_σ be the orthogonal projection onto E_σ. A unitary operator that preserves a finite-dimensional subspace also preserves its orthogonal complement. Hence Q_σ commutes with all L_g and R_g. This Q_σ is the projector onto the matrix coefficients of Π_σ, called P_{Π_σ} in Lecture 11.

For $\sigma \in \mathrm{Irr}(K)$ let $E_\sigma^H = \mathrm{span}\{c_{u,v} : u \in V_\sigma,\ v \in V_\sigma^H\}$, a subspace of E_σ.

We read the count from the matrix coefficient $c_{u,v}$. Right translation by H acts only on v, so $c_{u,v}$ is right-H-invariant when $v \in V_\sigma^H$. Left translation acts only on u, and u enters antilinearly. So the right-H-invariant part of E_σ is $\dim V_\sigma^H$ copies of $\overline{V_\sigma}$. Block σ therefore contributes to the multiplicity of $\bar{\sigma}$. Since $\dim V_\sigma^H = \dim V_{\bar{\sigma}}^H$, the count for σ is still $\dim V_\sigma^H$. For SO(3) every irreducible is self-conjugate, and no shift occurs.

Figure 12.1 draws this count in the arrays of functions of Lecture 10. In a block, the entry c_{e_i,e_j} has its second index on the side of R. So the condition $f(xh) = f(x)$ keeps the columns whose vector e_j lies in V_σ^H.

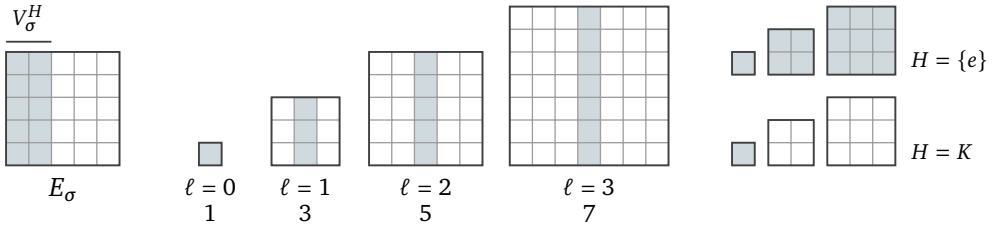


Figure 12.1: Blocks of functions in $L^2(K)$, as in Lecture 10, with the columns that remain on K/H shaded. From left to right: a general H, in a basis of V_σ whose first vectors span V_σ^H; the sphere SO(3)/SO(2), with blocks of side $2\ell + 1$ and one column each, of $2\ell + 1$ functions, for the weight-zero vector of § 12.4; and the extreme cases $H = \{e\}$ and $H = K$. Left translation mixes the entries of a shaded column.

Proposition 12.5. *Let $P_\sigma = \int_H \Pi_\sigma(h)\, dh$ be the orthogonal projection of V_σ onto V_σ^H (Lecture 5). Then $A_H c_{u,v} = c_{u,P_\sigma v}$. Moreover*

$$L^2(K/H) \cong L^2(K)^H = \overbrace{\bigoplus_{\sigma \in \mathrm{Irr}(K)}} E_\sigma^H, \qquad E_\sigma^H \cong \overline{V_\sigma} \otimes V_\sigma^H,$$

where K acts on $\overline{V_\sigma} \otimes V_\sigma^H$ by $\bar{\Pi}_\sigma \otimes 1$.

Proof. The finite-dimensional integral commutes with the functional $w \mapsto \langle u, \Pi_\sigma(x) w \rangle$. So

$$(A_H c_{u,v})(x) = \int_H \langle u, \Pi_\sigma(x)\Pi_\sigma(h) v \rangle\, dh = \left\langle u, \Pi_\sigma(x) \int_H \Pi_\sigma(h) v\, dh \right\rangle = c_{u,P_\sigma v}(x).$$

Thus $A_H(E_\sigma) = E_\sigma^H$, and A_H fixes E_σ^H. Hence $E_\sigma^H \subseteq L^2(K)^H$. Conversely, let $f \in L^2(K)^H$. By Peter–Weyl (PW2), $f = \sum_\sigma Q_\sigma f$ in L^2. Since A_H is bounded,

$$f = A_H f = \sum_\sigma A_H Q_\sigma f, \qquad A_H Q_\sigma f \in E_\sigma^H.$$

Since E_σ^H lies in E_σ, the spaces E_σ^H are mutually orthogonal by Peter–Weyl (PW2). This proves the decomposition. The isomorphism with $\overline{V_\sigma} \otimes V_\sigma^H$ is the restriction of the bijection above. ∎

Induction in § 12.3 uses vector-valued functions and evaluates them at the identity e. An L^2 class has no value at a point. The next lemma shows that equivariant images of V_σ consist of continuous functions, so these evaluations make sense.

Let W be a finite-dimensional Hilbert space with orthonormal basis $w_1, \dots, w_n$. Then $L^2(K) \otimes W$ is the Hilbert space of functions $\sum_j f_j w_j$ with $f_j \in L^2(K)$. It contains $C(K, W) = C(K) \otimes W$. Let K act by $L \otimes 1$.

Lemma 12.6. *Let $\sigma \in \operatorname{Irr}(K)$. Let $T : V_\sigma \to L^2(K) \otimes W$ be linear with $T\Pi_\sigma(g) = (L_g \otimes 1)T$ for all $g \in K$. Then $T(V_\sigma) \subseteq E_{\bar\sigma} \otimes W$. In particular, every vector in $T(V_\sigma)$ is a continuous function.*

Proof. Fix $\eta \in \operatorname{Irr}(K)$. The operator $Q_\eta \otimes 1$ commutes with $L_g \otimes 1$. So $(Q_\eta \otimes 1)T$ is a K-map from V_σ into

$$E_\eta \otimes W \cong \overline{V_\eta} \otimes (V_\eta \otimes W),$$

where K acts only on the first factor. This space is a direct sum of $d_\eta \dim W$ copies of $\overline{V_\eta}$. By Schur's lemma, $(Q_\eta \otimes 1)T = 0$ unless $V_\sigma \cong \overline{V_\eta}$, that is, unless $\eta = \bar\sigma$. Apply Peter–Weyl (PW2) in each coordinate. For $v \in V_\sigma$,

$$Tv = \sum_\eta (Q_\eta \otimes 1)Tv = (Q_{\bar\sigma} \otimes 1)Tv \in E_{\bar\sigma} \otimes W.$$ ∎

Thus the values $(Tv)(x)$ are defined for all $v \in V_\sigma$ and $x \in K$.

Let (E, Π) be a unitary representation of K and $\sigma \in \operatorname{Irr}(K)$. The *multiplicity* of σ in E is $m_\sigma(E) = \dim \operatorname{Hom}_K(V_\sigma, E)$. Here Hom_K consists of the linear maps T with $T\Pi_\sigma(g) = \Pi(g)T$. These maps are bounded because V_σ is finite-dimensional. The *σ-isotypic subspace* of E is the sum of the images of these maps.

Theorem 12.7. *For every $\sigma \in \operatorname{Irr}(K)$,*

$$m_\sigma\big(L^2(K/H)\big) = \dim V_\sigma^H.$$

The σ-isotypic subspace is $E_{\bar\sigma}^H \subseteq C(K/H)$, of dimension $d_\sigma \dim V_\sigma^H$. The space $L^2(K/H)$ is the Hilbert sum of its isotypic subspaces.

Proof. Identify $L^2(K/H)$ with $L^2(K)^H$ through p^*, as in Proposition 12.4. Let $T : V_\sigma \to L^2(K)^H$ be a K-map. By Lemma 12.6 on equivariant images, with $W = \mathbb{C}$, $T(V_\sigma) \subseteq E_{\bar\sigma}$. A vector of $E_{\bar\sigma}$ fixed by A_H lies in $A_H(E_{\bar\sigma}) = E_{\bar\sigma}^H$. Hence

$$\operatorname{Hom}_K\left(V_\sigma, L^2(K/H)\right) = \operatorname{Hom}_K\left(V_\sigma, E_{\bar\sigma}^H\right).$$

Since $V_{\bar\sigma} \cong \overline{V_\sigma}$, we have $\overline{V_{\bar\sigma}} \cong V_\sigma$. So Proposition 12.5 on the decomposition of $L^2(K)^H$ gives

$$E_{\bar\sigma}^H \cong \overline{V_{\bar\sigma}} \otimes V_{\bar\sigma}^H \cong V_\sigma \otimes V_{\bar\sigma}^H,$$

with K acting on the first factor only. By Schur's lemma,

$$\mathrm{Hom}_K\left(V_\sigma, V_\sigma \otimes V_{\bar\sigma}^H\right) \cong V_{\bar\sigma}^H.$$

A K-isomorphism $V_{\bar\sigma} \to \overline{V_\sigma}$ carries $V_{\bar\sigma}^H$ onto $(\overline{V_\sigma})^H$. This is the set V_σ^H with conjugated scalars, so its dimension is $\dim V_\sigma^H$. The images of the K-maps $V_\sigma \to E_{\bar\sigma}^H$ span $E_{\bar\sigma}^H \cong V_\sigma \otimes V_{\bar\sigma}^H$. Finally, $\sigma \mapsto \bar\sigma$ is a bijection of $\mathrm{Irr}(K)$, so the decomposition in Proposition 12.5 gives the Hilbert sum. ∎

Thus the summand E_σ^H of Proposition 12.5 is indexed by σ but is $\bar\sigma$-isotypic. The count is unchanged because $\dim V_\sigma^H = \dim V_{\bar\sigma}^H$. Exercise 12.1 shows this label shift for $K = \mathrm{U}(1)$. Without any bar, the σ-isotypic subspace is spanned by the functions $x \mapsto \langle u, \Pi_\sigma(x)^{-1} v\rangle$ with $u \in V_\sigma^H$ and $v \in V_\sigma$.

12.3 Compact induction and Frobenius reciprocity

Let (τ, W) be a finite-dimensional unitary representation of H. Averaging over H gives every finite-dimensional representation of H an invariant inner product (Lecture 5).

A tangent vector field on S^2 is the model for what follows. Let $K = \mathrm{SO}(3)$, and let H be the subgroup of rotations $R_3(\theta)$ about e_3, as in Lecture 4. Let $x \in \mathrm{SO}(3)$. The columns xe_1 and xe_2 form an oriented orthonormal frame of the tangent plane at $p = xe_3$, and the coset xH is the set of all such frames at p. Let $F(x) \in \mathbb{C}^2$ hold the components of the field at p in this frame. Replacing x by $xR_3(\theta)$ rotates the frame through θ, and the components change by the inverse rotation. So

$$F(xh) = \tau(h)^{-1} F(x), \qquad h \in H,$$

where τ is the rotation action of H on $\mathbb{C}^2$. For example, the tangential part of a fixed vector $v \in \mathbb{R}^3$ has the components $F(x) = (xe_1 \cdot v,\ xe_2 \cdot v)$. Figure 12.2 shows one tangent vector in two frames.

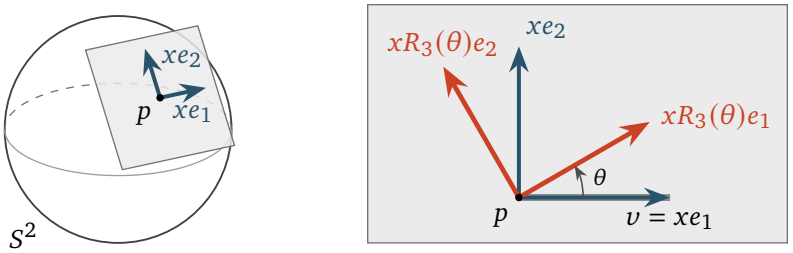


Figure 12.2: On the left, a point $p = xe_3$ of the sphere, with its tangent plane and the frame (xe_1, xe_2). On the right, the same plane seen from outside the sphere, with a second frame for $\theta = \pi/6$, turned anticlockwise. The tangent vector $v = xe_1$ has components $F(x) = (1, 0)$ in the first frame and $F(xR_3(\pi/6)) = (\sqrt{3}/2, -1/2)$ in the second frame.

Definition 12.8. The *induced representation* is

$$\mathrm{Ind}_H^K W = \{F \in C(K, W) : F(xh) = \tau(h)^{-1} F(x) \text{ for all } x \in K,\ h \in H\},$$

with K acting by $(L_g F)(x) = F(g^{-1}x)$.

Left translation preserves the covariance condition because left and right translations commute. For the trivial representation $W = \mathbb{C}$ of H, we recover $C(K)^H \cong C(K/H)$.

On a coset xH the function F is fixed by its value at x. So an element of $\operatorname{Ind}_H^K W$ assigns to each coset a vector of W, once a point of the coset is chosen. In the model above, with $W = \mathbb{C}^2$, the vector $F_1(x)xe_1 + F_2(x)xe_2$ does not depend on the point x of the coset. So $\operatorname{Ind}_H^K W$ is the space of continuous tangent fields on S^2 with complex components. Under this identification, L_g sends a field X to the rotated field $p \mapsto gX(g^{-1}p)$.

For finite K this is the induced representation of the Finite Groups course. The element $x \otimes w$ of $\mathbb{C}[K] \otimes_{\mathbb{C}[H]} W$ corresponds to the function F with $F(xh) = \tau(h)^{-1}w$ for $h \in H$ and $F = 0$ off xH.

Theorem 12.9 (Frobenius reciprocity, continuous form). *Let (Π, V) be a finite-dimensional representation of K. The map*

$$\Phi\colon \operatorname{Hom}_K\left(V, \operatorname{Ind}_H^K W\right) \to \operatorname{Hom}_H(V|_H, W), \qquad \Phi(T)v = (Tv)(e),$$

is a linear bijection. Its inverse sends A to $\Psi(A)$, where $(\Psi(A)v)(x) = A\big(\Pi(x)^{-1}v\big)$.

Proof. First, $\Phi(T)$ is H-equivariant. For $h \in H$, equivariance of T and covariance give

$$\Phi(T)\big(\Pi(h)v\big) = (L_h Tv)(e) = (Tv)(h^{-1}) = \tau(h^{-1})^{-1}(Tv)(e) = \tau(h)\,\Phi(T)v.$$

Next, $\Psi(A)v$ is continuous because $x \mapsto \Pi(x)^{-1}v$ is continuous. Since A is an H-map, $\Psi(A)v$ is covariant:

$$(\Psi(A)v)(xh) = A\big(\Pi(h)^{-1}\Pi(x)^{-1}v\big) = \tau(h)^{-1}A\big(\Pi(x)^{-1}v\big).$$

The map $\Psi(A)$ is a K-map, since $\Pi(x)^{-1}\Pi(g) = \Pi(g^{-1}x)^{-1}$ gives

$$(\Psi(A)\Pi(g)v)(x) = (\Psi(A)v)(g^{-1}x) = (L_g\Psi(A)v)(x).$$

Finally, $\Phi(\Psi(A))v = A(v)$. In the other order,

$$(\Psi(\Phi(T))v)(x) = \big(T\Pi(x^{-1})v\big)(e) = (L_{x^{-1}}Tv)(e) = (Tv)(x). \qquad \blacksquare$$

For finite K, the equality of dimensions is part of Frobenius reciprocity in the Finite Groups course. That course built its bijection on maps out of Ind. Here we build it on maps into Ind, where evaluation at e gives the map directly.

Now pass to L^2. For $h \in H$ define the operator $\widetilde{R}_h = R_h \otimes \tau(h)$ on $L^2(K) \otimes W$. These operators are unitary and commute with $L_g \otimes 1$. For continuous F, $(\widetilde{R}_h F)(x) = \tau(h)F(xh)$. Set

$$L^2\text{-}\operatorname{Ind}_H^K W = \{F \in L^2(K) \otimes W : \widetilde{R}_h F = F \text{ for all } h \in H\}.$$

This closed K-invariant subspace is a unitary representation of K. For the trivial representation $W = \mathbb{C}$, this subspace is $L^2(K)^H$.

Theorem 12.10 (finite multiplicities). *For $\sigma \in \operatorname{Irr}(K)$,*

$$m_\sigma\big(L^2\text{-}\operatorname{Ind}_H^K W\big) = \dim \operatorname{Hom}_H(V_\sigma|_H, W) \le d_\sigma \dim W.$$

Proof. Let $T\colon V_\sigma \to L^2\text{-}\operatorname{Ind}_H^K W$ be a K-map. By Lemma 12.6 on equivariant images, each $F = Tv$ is continuous. Then $\widetilde{R}_h F$ and F are continuous and equal in L^2. Since $C(K)\otimes W \to L^2(K)\otimes W$ is injective, $\tau(h)F(xh) = F(x)$ for all x. So $F \in \operatorname{Ind}_H^K W$. Conversely, continuous covariant functions satisfy $\widetilde{R}_h F = F$. Hence

$$\operatorname{Hom}_K\left(V_\sigma, L^2\text{-}\operatorname{Ind}_H^K W\right) = \operatorname{Hom}_K\left(V_\sigma, \operatorname{Ind}_H^K W\right).$$

Frobenius reciprocity (Theorem 12.9) gives a linear bijection from this space onto $\operatorname{Hom}_H(V_\sigma|_H, W)$, so the dimensions agree. The bound holds because $\operatorname{Hom}_H(V_\sigma|_H, W)$ is a subspace of $\operatorname{Hom}(V_\sigma, W)$, which has dimension $d_\sigma \dim W$. ∎

Remark 12.11 (duals and covariance)**.** For $W = \mathbb{C}$ the count is $\dim \operatorname{Hom}_H(V_\sigma, \mathbb{C}) = \dim(V_\sigma^*)^H$. Since $\langle u, \Pi_\sigma(h)v\rangle = \langle \Pi_\sigma(h)^{-1}u, v\rangle$, the functional $\langle u, \cdot\rangle$ is H-invariant exactly when $u \in V_\sigma^H$. So $u \mapsto \langle u, \cdot\rangle$ is an antilinear bijection $V_\sigma^H \to (V_\sigma^*)^H$. Hence the count agrees with the multiplicity theorem (Theorem 12.7).

The inverse in $\tau(h)^{-1}$ makes the covariance condition compatible with products in H:

$$F(xh_1h_2) = \tau(h_2)^{-1}F(xh_1) = \tau(h_2)^{-1}\tau(h_1)^{-1}F(x) = \tau(h_1h_2)^{-1}F(x).$$

Remark 12.12 (scope)**.** For compact K, we obtained a finite invariant integral. The irreducible representations were finite-dimensional, and Peter–Weyl applied. We proved a count of finite multiplicities, not a general adjunction for Hilbert representations. For noncompact G, the space G/H may carry no invariant measure. Lecture 17 treats that case.

12.4 The sphere $\mathrm{SO}(3)/\mathrm{SO}(2)$

Let $K = \mathrm{SO}(3)$, let e_1, e_2, e_3 be the standard basis of $\mathbb{R}^3$, and let, as in Lecture 4,

$$R_3(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad H = \{R_3(\theta) : \theta \in \mathbb{R}\} \cong \mathrm{SO}(2).$$

The subgroup H is the stabiliser of e_3. Every unit vector is the third column of some rotation, so the orbit map $q(x) = xe_3$ sends K onto S^2. By Lecture 3, q induces a K-equivariant diffeomorphism $K/H \to S^2$. So $F \mapsto F \circ q$ identifies $C(S^2)$ with $C(K)^H$, and $L_g(F\circ q) = (F\circ g^{-1})\circ q$. We define $L^2(S^2)$ through this identification. The invariant integral on S^2 is the usual area integral divided by 4π, because both are rotation-invariant and normalised, and such an integral is unique.

Let E be a finite-dimensional representation of H. An integer k is a *weight* of E if some nonzero $v \in E$ satisfies $R_3(\theta)v = e^{ik\theta}v$ for all $\theta \in \mathbb{R}$. These vectors and 0 form the *weight-k space*. Suppose E has a basis (v_j) with $R_3(\theta)v_j = e^{ik_j\theta}v_j$ for all θ. Then the weight-k space is spanned by the v_j with $k_j = k$. Indeed, a weight-k vector $\sum_j c_j v_j$ satisfies

$$c_j\left(e^{ik_j\theta} - e^{ik\theta}\right) = 0 \qquad \text{for all } \theta,$$

and this forces $c_j = 0$ when $k_j \neq k$.

From Lecture 8, the irreducibles of SO(3) are V_ℓ, $\ell = 0, 1, 2, \ldots$, of dimension $2\ell + 1$ and spin ℓ. In this section an unqualified V_ℓ means the SO(3)-module. The SU(2)-modules of Lecture 8 are always named as such. Choose the covering $\phi = \mathrm{Ad}$ in the ordered basis (X_1, X_2, X_3) of Lecture 4. The explicit calculation there gives

$$\phi\big(\mathrm{diag}(e^{i\theta/2}, e^{-i\theta/2})\big) = R_3(-\theta), \qquad \ker\phi = \{\pm I\}.$$

The pullback $\Pi_\ell \circ \phi$ is the SU(2)-module $V_{2\ell}$. The diagonal element in this display acts on a basis of $V_{2\ell}$ by the scalars $e^{i(\ell-j)\theta}$, $0 \le j \le 2\ell$. Replacing θ by $-\theta$ therefore gives

$$V_\ell \text{ has a basis } (v_k)_{-\ell\le k\le\ell} \text{ with } \Pi_\ell(R_3(\theta))v_k = e^{ik\theta}v_k \text{ for all } \theta \in \mathbb{R}. \tag{12.4}$$

Theorem 12.13. *Let $\mathcal{Y}_\ell \subseteq L^2(S^2)$ be the V_ℓ-isotypic subspace. Then*

$$L^2(S^2) = \widehat{\bigoplus_{\ell\ge 0}} \mathcal{Y}_\ell, \qquad \mathcal{Y}_\ell \cong V_\ell, \qquad \mathcal{Y}_\ell \subseteq C(S^2).$$

Each integral-spin module occurs exactly once.

Proof. The space V_ℓ^H is the weight-zero space. By the weight basis (12.4) it is the line $\mathbb{C}v_0$, so $\dim V_\ell^H = 1$. Apply the multiplicity theorem (Theorem 12.7). ∎

The same weight basis counts tangent fields. In the tangent model, τ has the weights 1 and -1. By Theorem 12.10, V_ℓ occurs twice for $\ell \ge 1$ and not at all for $\ell = 0$. So S^2 carries no nonzero tangent field fixed by all rotations. This statement concerns invariant fields only, and it is not the hairy-ball theorem about a single continuous field.

Let $\mathcal{P}_\ell$ be the space of complex homogeneous polynomials of degree ℓ on $\mathbb{R}^3$, with $(L_g f)(v) = f(g^{-1}v)$. Let $\Delta = \partial_x^2 + \partial_y^2 + \partial_z^2$ and $\mathcal{H}_\ell = \{f \in \mathcal{P}_\ell : \Delta f = 0\}$. The chain rule gives $\Delta(f \circ g) = (\Delta f) \circ g$ for orthogonal g, so $\mathcal{H}_\ell$ is K-invariant. Restriction to S^2 is K-equivariant. Restriction is injective on $\mathcal{P}_\ell$, since $f(rv) = r^\ell f(v)$. Elements of $\mathcal{H}_\ell|_{S^2}$ are called *spherical harmonics* of degree ℓ.

Put $z_\pm = x \pm iy$. Since $R_3(\theta)^{-1} = R_3(-\theta)$,

$$(L_{R_3(\theta)} z_+)(x, y, z) = (\cos\theta\, x + \sin\theta\, y) + i(-\sin\theta\, x + \cos\theta\, y) = e^{-i\theta} z_+.$$

Similarly $L_{R_3(\theta)} z_- = e^{i\theta} z_-$ and $L_{R_3(\theta)} z = z$. The monomials $z_+^a z_-^b z^c$ with $a + b + c = \ell$ form a basis of $\mathcal{P}_\ell$, and for all θ,

$$L_{R_3(\theta)}\, z_+^a z_-^b z^c = e^{i(b-a)\theta}\, z_+^a z_-^b z^c, \qquad |b - a| \le \ell. \tag{12.5}$$

So every weight of $\mathcal{P}_\ell$ satisfies $|k| \le \ell$.

Proposition 12.14. *The space $\mathcal{Y}_\ell$ is spanned by the left translates of $f_\ell = z_+^\ell|_{S^2}$. Hence every element of $\mathcal{Y}_\ell$ is a spherical harmonic of degree ℓ.*

Proof. The polynomial z_+^ℓ is harmonic:

$$\partial_x^2 z_+^\ell + \partial_y^2 z_+^\ell = \ell(\ell-1)z_+^{\ell-2} + i^2\ell(\ell-1)z_+^{\ell-2} = 0.$$

Let U be the span of all $L_g f_\ell$. Then $U \subseteq \mathcal{H}_\ell|_{S^2}$ is finite-dimensional and K-invariant. Decompose U into irreducibles. A copy of V_j in U lies in the V_j-isotypic subspace $\mathcal{Y}_j$. Since $\mathcal{Y}_j \cong V_j$, this copy equals $\mathcal{Y}_j$. Thus $U = \bigoplus_{j \in S} \mathcal{Y}_j$ for a finite set S. Restriction is injective and equivariant, so the weights of U are weights of $\mathcal{P}_\ell$. Since V_j has weight j, this gives $S \subseteq \{0, \dots, \ell\}$.

Write $f_\ell = \sum_{j \in S} y_j$ with $y_j \in \mathcal{Y}_j$. Each $L_{R_3(\theta)}$ preserves each $\mathcal{Y}_j$, and $L_{R_3(\theta)} f_\ell = e^{-i\ell\theta} f_\ell$. Since the sum is direct, comparing components gives $L_{R_3(\theta)} y_j = e^{-i\ell\theta} y_j$ for all θ. For $j < \ell$ the weight $-\ell$ does not occur in V_j, so $y_j = 0$. Hence $f_\ell \in \mathcal{Y}_\ell$ and $U \subseteq \mathcal{Y}_\ell$. Since $U \neq 0$ and $\mathcal{Y}_\ell$ is irreducible, $U = \mathcal{Y}_\ell$. ∎

Figure 12.3 shades the harmonics of the next example by their sign.

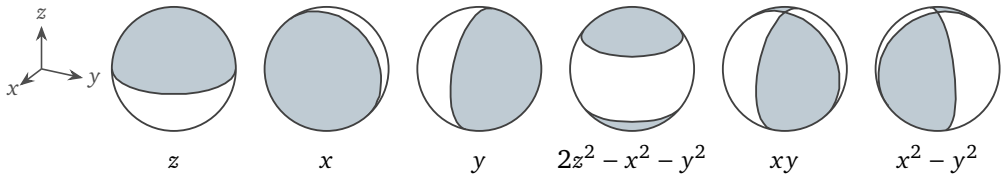


Figure 12.3: Real spherical harmonics of degrees one and two on the unit sphere, shaded where they are positive, with their zero sets drawn. The functions z and $2z^2 - x^2 - y^2$ depend on the latitude only, so they are fixed by H. The second of these vanishes on the latitudes $z = \pm 1/\sqrt{3}$.

Example 12.15 (degrees one and two)**.** We compute $\mathcal{Y}_1$ and $\mathcal{Y}_2$ and their H-fixed lines. Every linear polynomial is harmonic, so $\dim \mathcal{H}_1 = 3$. The map $\Delta\colon \mathcal{P}_2 \to \mathbb{C}$ is onto, since $\Delta(x^2) = 2$. So $\dim \mathcal{H}_2 = 6 - 1 = 5$.

By Proposition 12.14 on spherical harmonics, $\mathcal{Y}_\ell \subseteq \mathcal{H}_\ell|_{S^2}$. Restriction is injective, so $\mathcal{H}_\ell|_{S^2}$ has dimension $2\ell + 1$ for $\ell = 1, 2$. By the sphere theorem (Theorem 12.13), also $\dim \mathcal{Y}_\ell = 2\ell + 1$. So the inclusion is an equality for $\ell = 1, 2$:

$$\mathcal{Y}_1 = \operatorname{span}\{x, y, z\}|_{S^2}, \qquad \mathcal{Y}_2 = \operatorname{span}\{xy,\ yz,\ zx,\ x^2 - y^2,\ 2z^2 - x^2 - y^2\}|_{S^2}.$$

By the monomial weights (12.5), the weight-zero space of $\mathcal{P}_2$ is spanned by $z_+ z_- = x^2 + y^2$ and z^2. Since $\Delta\big(a(x^2 + y^2) + bz^2\big) = 4a + 2b$, its harmonic line is spanned by $2z^2 - x^2 - y^2$. This function spans $\mathcal{Y}_2^H$, the line predicted by $\dim V_2^H = 1$. Similarly z spans $\mathcal{Y}_1^H$. In atomic physics, $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are the angular parts of the p and d orbitals, and on S^2 the H-fixed function $2z^2 - x^2 - y^2 = 3z^2 - 1$ is a multiple of the Legendre polynomial $P_2(z)$.

Exercise 12.5 proves $\mathcal{H}_\ell|_{S^2} = \mathcal{Y}_\ell$ for all ℓ. That exercise defines the Laplacian of the sphere and computes its eigenvalue on $\mathcal{H}_\ell$. Nothing in this lecture depends on it.

12.5 Self-checks

1. Let K be a finite group, regarded as a compact Lie group of dimension zero, and let $H \leq K$ be a subgroup. What does the multiplicity theorem (Theorem 12.7) say?

Answer. Here $L^2(K/H) = \mathbb{C}[K/H]$ is the permutation representation. An irreducible V occurs $\dim V^H$ times. This is finite-group Frobenius reciprocity for the trivial character of H.

2. Why not define $\Phi(T)v = (Tv)(e)$ directly for $T\colon V_\sigma \to L^2\text{-}\operatorname{Ind}_H^K W$?
   Answer. An L^2 class has no value at e. Lemma 12.6 on equivariant images first shows that T takes continuous values.
3. What are the multiplicities for $H = \{e\}$ and for $H = K$?
   Answer. For $H = \{e\}$, σ occurs d_σ times in $L^2(K)$. For $H = K$, a nontrivial irreducible has $V^K = 0$, so only the trivial representation occurs, once, as the constants.

Summary

The invariant integral on K/H is the Haar integral of pullbacks. The averaging operator A_H projects $L^2(K)$ onto $L^2(K)^H \cong L^2(K/H)$. Peter–Weyl then shows that each $\sigma \in \operatorname{Irr}(K)$ occurs in $L^2(K/H)$ exactly $\dim V_\sigma^H$ times.

Let V be a finite-dimensional representation of K and W a finite-dimensional unitary representation of H. An element of $\operatorname{Ind}_H^K W$ is a function on K fixed on each coset by its value at one point, and on S^2 with $W = \mathbb{C}^2$ it is a tangent field with complex components, written in moving frames. Evaluation at e gives $\operatorname{Hom}_K(V, \operatorname{Ind}_H^K W) \cong \operatorname{Hom}_H(V|_H, W)$. Isotypic vectors in $L^2\text{-}\operatorname{Ind}_H^K W$ are continuous. So σ occurs in $L^2\text{-}\operatorname{Ind}_H^K W$ exactly $\dim \operatorname{Hom}_H(V_\sigma|_H, W)$ times.

On $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$ each $\mathrm{SO}(3)$-module V_ℓ occurs once in $L^2(S^2)$, realised by spherical harmonics of degree ℓ.

Reading

- Bröcker and tom Dieck, *Representations of Compact Lie Groups* (1985), Section III.6: induction from closed subgroups and Frobenius reciprocity.
- Sepanski, *Compact Lie Groups* (2007), §2.3 and §§3.3–3.4: the assigned background route for this lecture and the sphere example.
- Knapp, *Lie Groups Beyond an Introduction*, digital second edition (2023), Section IX.2: induced representations of compact groups and Frobenius reciprocity.

Lecture 13

Spin groups and projective representations

Objectives. *A 2π rotation is the identity in* $\mathrm{SO}(3)$. *On an electron's state vector it acts by* -1. *We explain this sign using the double cover* $\mathrm{Spin}(n) \to \mathrm{SO}(n)$. *We then study actions on rays and their lifts to vectors.*

By the end of this lecture you will be able to:

- For $n \geq 2$, construct $\mathrm{Spin}(n)$ and prove that $\rho\colon\ \mathrm{Spin}(n) \to \mathrm{SO}(n)$ is surjective with kernel $\{\pm 1\}$.
- Identify $\mathrm{Spin}(3)$ with $\mathrm{SU}(2)$ and decide which V_m descend to $\mathrm{SO}(3)$.
- Define projective unitary representations, multipliers and central extensions, and lift finite-dimensional projective representations of $\mathrm{SO}(3)$ to $\mathrm{SU}(2)$.
- Use $\mathbb{R}^2$ to explain why a simply connected group can have a nontrivial multiplier.

13.1 Clifford algebras and spin groups

A rotation is a product of an even number of reflections. A reflection is determined by a unit vector u, up to the sign of u. We want an algebra in which conjugation by u gives the reflection in $u^{\perp}$ up to sign. For a product of an even number of unit vectors the signs cancel. The Clifford algebra has this property. For $n = 2$, Exercise 13.1 shows the result on a circle.

Fix $n \geq 2$ and an orthonormal basis $e_1, \dots, e_n$ of Euclidean $\mathbb{R}^n$. Recall from Lie Algebras that the *Clifford algebra* $Cl(n)$ is the real unital associative algebra generated by $\mathbb{R}^n$ subject to $v^2 = -|v|^2$. Replacing v by $u + v$ gives

$$uv + vu = -2\langle u, v\rangle \qquad (u, v \in \mathbb{R}^n). \tag{13.1}$$

This is the prerequisite convention $uv + vu = 2q(u, v)$ with the negative definite symmetric bilinear form $q(u, v) = -\langle u, v\rangle$.

The products $e_I = e_{i_1} \cdots e_{i_k}$ with $i_1 < \cdots < i_k$ form a basis of $Cl(n)$. The *even part* $Cl^0(n)$ is the span of the e_I with $|I|$ even. The *transpose* is the linear map $x \mapsto x^t$ with $(v_1 \cdots v_k)^t = v_k \cdots v_1$. The complexification $Cl(n)_{\mathbb{C}}$ has an irreducible *spinor module* S of dimension $2^{\lfloor n/2 \rfloor}$.

The units $Cl(n)^{\times}$ form an open set and a Lie group. A unit vector u has $u^2 = -1$ and $u^{-1} = -u$. Let $r_u(v) = v - 2\langle u, v\rangle u$ be reflection in $u^{\perp}$.

Definition 13.1. $\mathrm{Spin}(n)$ is the set of all products $u_1u_2\cdots u_{2k}$ with $k \geq 0$ and unit vectors $u_i \in \mathbb{R}^n$.

Lemma 13.2. *(a)* $\mathrm{Spin}(n)$ *is a subgroup of* $Cl^0(n)^\times$, *and* $x^{-1} = x^t$ *for* $x \in \mathrm{Spin}(n)$.
(b) For a unit vector u *and* $v \in \mathbb{R}^n$ *we have* $uvu^{-1} = -r_u(v)$.

Proof. (a) Even products of vectors lie in $Cl^0(n)$, and the set is closed under products. Let $x = u_1\cdots u_{2k}$, so $x^t = u_{2k}\cdots u_1$. In

$$x^t x = u_{2k}\cdots u_2\,(u_1u_1)\,u_2\cdots u_{2k}$$

the middle factor is the scalar -1. Next the two factors u_2 become adjacent and give $u_2u_2 = -1$, and so on. Hence $x^tx = (-1)^{2k} = 1$, and in the same way $xx^t = 1$. Since $x^t \in \mathrm{Spin}(n)$, the set is a subgroup.

(b) By the anticommutation relation (13.1), $uv = -vu - 2\langle u, v\rangle$. Using also $u^{-1} = -u$ and $u^2 = -1$, we get

$$uvu^{-1} = -uvu = (vu + 2\langle u, v\rangle)u = -v + 2\langle u, v\rangle u = -r_u(v). \qquad \blacksquare$$

13.2 The double covering of the rotation group

For $x \in \mathrm{Spin}(n)$ and $v \in \mathbb{R}^n$ put $\rho(x)v = xvx^{-1}$, an element of $Cl(n)$. Write $x = u_1\cdots u_{2k}$ with unit vectors $u_i \in \mathbb{R}^n$. Apply the conjugation formula of Lemma 13.2(b) once for each of the $2k$ factors of x. The $2k$ signs cancel, and we get

$$\rho(x)v = xvx^{-1} = r_{u_1}\circ\cdots\circ r_{u_{2k}}(v) \qquad (v \in \mathbb{R}^n). \tag{13.2}$$

So $\rho(x)$ maps $\mathbb{R}^n$ to itself, and it is a product of an even number of reflections. Hence $\rho(x) \in \mathrm{SO}(n)$, and ρ is a map from $\mathrm{Spin}(n)$ to $\mathrm{SO}(n)$.

The formula (13.2) explains the double cover. A reflection r_u does not change if we replace u by $-u$, but the element u of $Cl(n)$ does change. So x and $-x$ give the same rotation.

Two mirrors show this in a plane. Let e_a and e_b be orthonormal, let $u = \cos t\, e_a + \sin t\, e_b$, and put $x = -ue_a = u(-e_a)$. By (13.2), $\rho(x) = r_u \circ r_{e_a}$, so we reflect first in $e_a^\perp$ and then in $u^\perp$. These two mirrors meet at angle t. Example 13.5 below shows that the composition is the rotation through $2t$ from e_a towards e_b. Figure 13.1 draws the two reflections. In its second row, t runs from 0 to π. The mirror of u returns to its start, the rotation runs once around, and the product x runs from 1 to -1. At $t = \pi$ we have $u = -e_a$, with the same mirror and the same rotation, but with product -1.

Lemma 13.3 (Cartan–Dieudonné, weak form)**.** *Every* $g \in \mathrm{O}(n)$ *is a product of at most* n *reflections. If* $\det g = 1$, *the number of reflections is even.*

Proof. Induct on n. For $n = 1$ we have $\mathrm{O}(1) = \{1, r_{e_1}\}$. Let $n \geq 2$. If $ge_1 = e_1$, then g preserves $e_1^\perp \cong \mathbb{R}^{n-1}$. By induction $g|_{e_1^\perp}$ is a product of at most $n-1$ reflections of $e_1^\perp$, and each extends

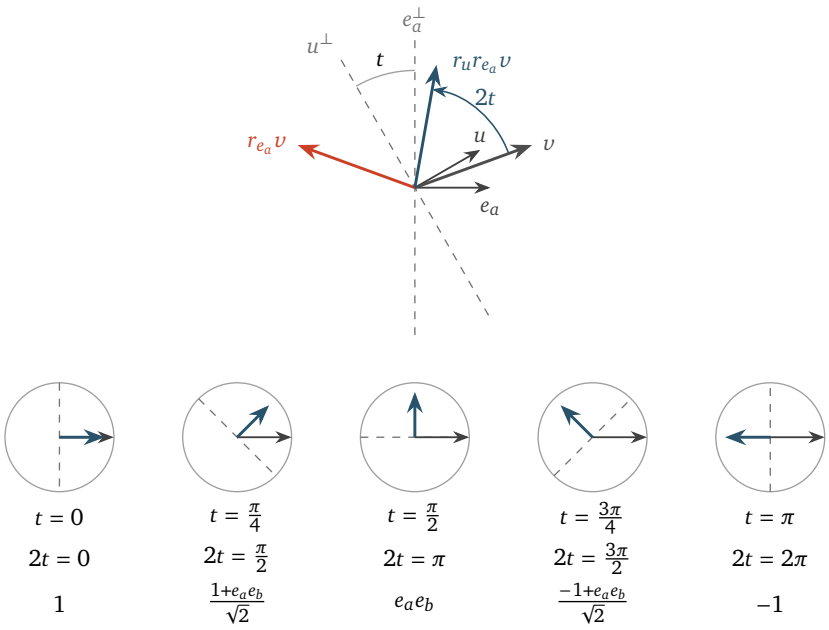


Figure 13.1: Two mirrors in the plane of e_a and e_b. The first panel has $t = \pi/6$: a vector v at angle $\pi/9$ goes to $r_{e_a} v$, and then to $r_u r_{e_a} v$, which is v turned through $2t$. The second row shows e_a, the vector u in colour, and the dashed mirror $u^\perp$, for five values of t. Beneath each are the rotation angle $2t$ and the product $x = -ue_a$.

to a reflection of $\mathbb{R}^n$ fixing e_1. If $a = ge_1 \neq e_1$, put $u = (a - e_1)/|a - e_1|$. Since $|a| = 1$, we have $\langle a - e_1, a\rangle = \frac{1}{2}|a - e_1|^2$. Therefore

$$r_u(a) = a - \frac{2\langle a - e_1, a\rangle}{|a - e_1|^2}(a - e_1) = e_1.$$

So $r_u g$ fixes e_1, and $g = r_u(r_u g)$ uses at most n reflections. Each reflection has determinant -1, which gives the parity. ∎

Theorem 13.4. *Let $n \geq 2$.*

(a) $\rho\colon \mathrm{Spin}(n) \to \mathrm{SO}(n)$ *is a continuous surjective homomorphism.*

(b) $\ker \rho = \{\pm 1\}$.

(c) $\mathrm{Spin}(n)$ *is a compact connected Lie group, and ρ is a two-sheeted covering homomorphism.*

(d) For $n \geq 3$, using the cited fact that $\pi_1(\mathrm{SO}(n))$ has order two, the group $\mathrm{Spin}(n)$ is simply connected. So $\mathrm{Spin}(n)$ is the universal covering group of $\mathrm{SO}(n)$.

Proof. (a) Clearly $\rho(xy) = \rho(x)\rho(y)$. Since $x^{-1} = x^t$, the entries of $\rho(x)$ are polynomials in the coordinates of x. Given $g \in \mathrm{SO}(n)$, write $g = r_{w_1} \cdots r_{w_{2k}}$ with $2k \leq n$ by the Cartan–Dieudonné lemma, Lemma 13.3. By the reflection formula (13.2), $y = w_1 \cdots w_{2k}$ satisfies $\rho(y) = g$.

(b) Note $-1 = e_1 e_1 \in \mathrm{Spin}(n)$, and ± 1 act trivially. Conversely, let $\rho(x) = \mathrm{id}$, so x commutes with every e_i. Fix i. Sorting the basis elements of x by whether they contain e_i, and moving e_i to the front, gives

$$x = a + e_i b,$$

where a, b are combinations of e_J with $i \notin J$, a even and b odd. By the anticommutation relation (13.1), e_i commutes with a and anticommutes with b. Hence

$$xe_i = e_i a - e_i e_i b, \qquad e_i x = e_i a + e_i e_i b.$$

Equality forces $2e_i^2 b = -2b = 0$. So no basis element containing e_i occurs in x. As i was arbitrary, $x = \lambda 1$ with $\lambda \in \mathbb{R}$. Then $1 = x^t x = \lambda^2$, so $\lambda = \pm 1$.

(c) Let $N = 2\lfloor n/2 \rfloor \geq 2$ and $x \in \mathrm{Spin}(n)$. The surjectivity proof in part (a) gives $y = w_1 \cdots w_{2k}$ with $2k \leq N$ and $\rho(y) = \rho(x)$. Append pairs $e_1(-e_1) = 1$ until there are N factors. By the kernel statement (b), $x = \pm y$, so we may replace w_1 by $-w_1$ to get $x = y$. Hence $\mathrm{Spin}(n)$ is the image of the compact connected space $(S^{n-1})^N$ under the continuous map $(w_1, \ldots, w_N) \mapsto w_1 \cdots w_N$. So $\mathrm{Spin}(n)$ is compact and connected.

Being compact, the group $\mathrm{Spin}(n)$ is closed in $Cl(n)^\times$, hence a Lie group by the closed subgroup theorem (Lecture 3).

The smooth transitive action of $\mathrm{Spin}(n)$ on $\mathrm{SO}(n)$ through ρ has stabiliser $\{\pm 1\}$. The homogeneous-space theorem (Lecture 3) identifies $\mathrm{Spin}(n)/\{\pm 1\}$ with $\mathrm{SO}(n)$. The quotient by the discrete subgroup $\{\pm 1\}$ is a local diffeomorphism, so $d\rho$ is an isomorphism. The covering criterion of Lecture 4 applies and gives two sheets.

(d) We cite that $\pi_1(\mathrm{SO}(n))$ has order two for $n \geq 3$ (Hall, Chapter 13). So the universal covering $q \colon \tilde{G} \to \mathrm{SO}(n)$ has kernel of order two. Integrate $(d\rho)^{-1} dq$ by the integration theorem of Lecture 4 to a homomorphism $\phi \colon \tilde{G} \to \mathrm{Spin}(n)$. The homomorphisms $\rho \circ \phi$ and q have the same differential on connected $\tilde{G}$, so they agree.

Since dq and $d\rho$ are isomorphisms, so is $d\phi$. Thus $\phi(\tilde{G})$ is an open, hence closed, subgroup, and ϕ is onto by connectedness. Now $\ker q = \phi^{-1}(\{\pm 1\})$ has order $2|\ker \phi|$, so $\ker \phi$ is trivial. A bijective homomorphism of Lie groups is an isomorphism (Lecture 3). ∎

For $n = 2$, Exercise 13.1 shows that $\mathrm{Spin}(2)$ is a circle and ρ is $z \mapsto z^2$. So $\mathrm{Spin}(2)$ is not simply connected.

Example 13.5 (Rotations in a coordinate plane)**.** Fix $a \neq b$ and put

$$x_{ab}(t) = \cos t + \sin t\, e_a e_b = -(\cos t\, e_a + \sin t\, e_b)\, e_a \in \mathrm{Spin}(n).$$

Since $(e_a e_b)^2 = -1$, we have $x_{ab}(t) = \exp(t\, e_a e_b)$, so $e_a e_b \in \mathrm{Lie}(\mathrm{Spin}(n))$. The vectors e_a, e_b anticommute with $e_a e_b$, so $e_a x_{ab}(t)^{-1} = x_{ab}(t) e_a$, and similarly for e_b. Every e_c with $c \neq a, b$ commutes with $e_a e_b$. Using $e_a e_b e_a = e_b$ and $e_a e_b e_b = -e_a$ we get

$$\rho(x_{ab}(t))e_a = x_{ab}(2t)e_a = \cos 2t\, e_a + \sin 2t\, e_b, \qquad \rho(x_{ab}(t))e_b = -\sin 2t\, e_a + \cos 2t\, e_b,$$

and $\rho(x_{ab}(t))e_c = e_c$. So $\rho(x_{ab}(t))$ is the rotation through $2t$ in the (e_a, e_b)-plane. For $t \in [0, \pi]$ the rotations run once around a closed loop. The lift runs from 1 to $x_{ab}(\pi) = -1$, so it is not closed.

Since $\mathrm{Spin}(n) \subset Cl(n)_{\mathbb{C}}$, the spinor module gives a representation Π_S of $\mathrm{Spin}(n)$. The element $-1 \in \mathrm{Spin}(n)$ is the scalar -1 of $Cl(n)$. Hence

$$\Pi_S(-1) = -\mathrm{id}_S, \qquad \rho(-1) = \mathrm{id}\,.$$

So Π_S is not of the form $\Pi' \circ \rho$. This gives the opening sign change.

13.3 Spin(3), SU(2) and descent

In $Cl^0(3)$ put $\mathbf{i} = e_2e_3$, $\mathbf{j} = e_3e_1$, $\mathbf{k} = e_1e_2$. For instance $\mathbf{ij} = e_2e_3e_3e_1 = -e_2e_1 = \mathbf{k}$. In the same way

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1, \qquad \mathbf{ij} = -\mathbf{ji} = \mathbf{k}, \quad \mathbf{jk} = -\mathbf{kj} = \mathbf{i}, \quad \mathbf{ki} = -\mathbf{ik} = \mathbf{j}.$$

So $Cl^0(3) = \operatorname{span}\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is the quaternion algebra $\mathbb{H}$. Since $\mathbf{i}^t = e_3e_2 = -\mathbf{i}$, the transpose is quaternion conjugation $q \mapsto \bar{q}$. For $q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$ we have $q\bar{q} = |q|^2 = a^2 + b^2 + c^2 + d^2$. Let $\mathrm{Sp}(1) = \{q \in \mathbb{H} : |q| = 1\}$ be the group of unit quaternions.

Define a real linear map $\Phi\colon \mathbb{H} \to M_2(\mathbb{C})$ by

$$\Phi(1) = I, \quad \Phi(\mathbf{i}) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \Phi(\mathbf{j}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \Phi(\mathbf{k}) = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}.$$

Direct multiplication shows that these matrices satisfy the quaternion relations. For example, $\Phi(\mathbf{i})\Phi(\mathbf{j}) = \Phi(\mathbf{k})$. Hence Φ is an algebra homomorphism, and

$$\Phi(a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = \begin{pmatrix} a + bi & -c - di \\ c - di & a - bi \end{pmatrix}, \qquad \det \Phi(q) = |q|^2. \tag{13.3}$$

Proposition 13.6. *(a)* $\mathrm{Spin}(3) = \mathrm{Sp}(1)$.
(b) Φ *restricts to an isomorphism of Lie groups* $\mathrm{Sp}(1) \to \mathrm{SU}(2)$.

Proof. (a) If $q \in \mathrm{Spin}(3)$, then $q\bar{q} = qq^t = 1$ by the inverse formula of Lemma 13.2(a), so $|q| = 1$. Conversely, let $|q| = 1$ and write $q = a + p$ with $p \in \operatorname{span}\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$. If $p = 0$, then $q = \pm 1 \in \mathrm{Spin}(3)$. Otherwise put $w = p/|p|$. Cross terms in w^2 cancel by anticommutation, so $w^2 = -1$. Choose θ with $\cos\theta = a$ and $\sin\theta = |p|$. Then

$$q = \cos\theta + \sin\theta\, w = \exp(\theta w).$$

By the plane-rotation example, Example 13.5, $\mathbf{i}, \mathbf{j}, \mathbf{k} \in \mathrm{Lie}(\mathrm{Spin}(3))$. The Lie algebra of a closed subgroup is a linear subspace, and exp maps it into the subgroup (Lecture 3). So $\theta w \in \mathrm{Lie}(\mathrm{Spin}(3))$ and $q \in \mathrm{Spin}(3)$.

(b) By the matrix formula (13.3), $\Phi(\bar{q}) = \Phi(q)^*$. So $\Phi(q)\Phi(q)^* = \Phi(q\bar{q}) = I$ for $|q| = 1$, and $\det\Phi(q) = 1$. Conversely, let $g \in \mathrm{SU}(2)$ have first column $(\alpha, \beta)^T$. The second column is a unit vector orthogonal to the first, and $\det g = 1$. This forces it to be $(-\bar{\beta}, \bar{\alpha})^T$. So $g = \Phi(q)$ with $\alpha = a + bi$ and $\beta = c - di$. Thus Φ is a bijective continuous homomorphism. Its inverse is linear, hence continuous. ∎

From now on we identify Spin(3) with SU(2) through Φ. Then $\rho\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ has kernel $\{\pm I\}$.

This ρ is the cover Ad of Lecture 4, up to a rotation of coordinates. Indeed, Exercise 13.2(a) shows that ρ is conjugation in the basis $(\mathbf{i}, \mathbf{j}, \mathbf{k})$, and in the basis (X_1, X_2, X_3) of Lecture 4 we have $\Phi(\mathbf{i}) = -2X_3$, $\Phi(\mathbf{j}) = 2X_2$ and $\Phi(\mathbf{k}) = 2X_1$. So $\rho = O\ \mathrm{Ad}\ O^{-1}$, where O is the rotation with $Oe_1 = e_3$, $Oe_2 = e_2$ and $Oe_3 = -e_1$. Hence the diagonal matrices rotate about e_1 here, and about the third axis in Lecture 4 and Lecture 12. A spinor module of $Cl(3)_{\mathbb{C}}$ is $\mathbb{C}^2$ with

e_j acting by $i\sigma_j$, for the Pauli matrices σ_j of Lecture 4. It sends $\mathbf{i}, \mathbf{j}, \mathbf{k}$ to $-i\sigma_1, -i\sigma_2, -i\sigma_3$, and with this isomorphism in place of Φ, the map ρ is exactly Ad.

For $m \geq 0$ let V_m be the space of homogeneous polynomials of degree m in $z = (z_1, z_2)$, with

$$(\Pi_m(g)f)(z) = f(g^{-1}z) \qquad (g \in \mathrm{SU}(2)).$$

The space V_m has dimension $m+1$ and spin $m/2$. Every finite-dimensional irreducible representation of SU(2) is isomorphic to exactly one V_m (Lecture 8). Fix an SU(2)-invariant inner product on each V_m (Lecture 5).

The spinor module of $Cl(3)_{\mathbb{C}}$ has dimension two. If its restriction to SU(2) were reducible, complete reducibility would make it $V_0 \oplus V_0$, since V_0 is the only one-dimensional irreducible. That restriction would be trivial, contrary to $\Pi_S(-1) = -\mathrm{id}_S$. Thus the restriction is isomorphic to V_1.

By the Descent Theorem of Lecture 4, a representation Π of SU(2) has the form $\Pi' \circ \rho$ exactly when $\Pi(\pm I) = \mathrm{id}$. The hypotheses hold by Theorem 13.4.

Proposition 13.7. *Π_m factors through $\rho\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ if and only if m is even.*

Proof. We have $\ker\rho = \{\pm I\}$ and $(\Pi_m(-I)f)(z) = f(-z) = (-1)^m f(z)$. Apply the Descent Theorem of Lecture 4. ∎

So the irreducible representations of SO(3) are the descents of the V_{2l}, of dimension $2l+1$. The spinor V_1 is not among them.

Example 13.8 (A full turn on V_m)**.** Put

$$u(\theta) = \Phi(\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\,\mathbf{i}) = \mathrm{diag}(e^{i\theta/2}, e^{-i\theta/2}).$$

Since $\mathbf{i} = e_2e_3$, the plane-rotation example, Example 13.5, shows that $\rho(u(\theta))$ is the rotation $R(\theta)$ through θ in the (e_2, e_3)-plane. On the monomial $f_k = z_1^{m-k}z_2^k$ we get

$$\Pi_m(u(\theta))f_k = (e^{-i\theta/2}z_1)^{m-k}(e^{i\theta/2}z_2)^k = e^{i(2k-m)\theta/2}f_k.$$

At $\theta = 2\pi$ every eigenvalue is $e^{i(2k-m)\pi} = (-1)^m$. Hence

$$R(2\pi) = \mathrm{id}, \qquad \Pi_m(u(2\pi)) = (-1)^m\,\mathrm{id}\,.$$

For odd m every vector changes sign. Each ray $\mathbb{C}f$ is unchanged, and so is every transition probability $|\langle f, h\rangle|^2/(\|f\|^2\|h\|^2)$. In physics, the sign is visible only relative to another part of a state, for example in the interference of a beam, one part of which has been turned through 2π.

Remark 13.9. Connectedness matters when a representation is recovered from its differential. The trivial character and det of O(2) both have differential 0 on $\mathfrak{so}(2)$, but they differ on a reflection. So for a disconnected group, $d\Pi$ need not determine Π (Exercise 13.4).

13.4 Projective unitary representations

In quantum mechanics a pure state is a ray $\mathbb{C}f$ in a Hilbert space, with $f \neq 0$. Transition probabilities depend only on rays. So a symmetry group acts on rays. Lifting this action to vectors may require choices of phase.

Let $\mathcal{H}$ be a complex Hilbert space of finite dimension $d \geq 1$, and let $\mathbb{T} = \{\lambda \in \mathbb{C} : |\lambda| = 1\}$.

Definition 13.10. The *projective unitary group* is $\mathrm{PU}(\mathcal{H}) = \mathrm{U}(\mathcal{H})/\mathbb{T}$ id with the quotient topology. We write $[U]$ for the class of U.

It is a Lie group (Lecture 3). It acts on rays by $[U] \cdot \mathbb{C}f = \mathbb{C}Uf$.

Definition 13.11. A *projective unitary representation* of a Lie group G on $\mathcal{H}$ is a continuous homomorphism $\bar{\pi} \colon G \to \mathrm{PU}(\mathcal{H})$.

We call $\bar{\pi}$ *irreducible* if 0 and $\mathcal{H}$ are the only subspaces invariant under every U with $[U] \in \bar{\pi}(G)$. A *lift* is any function $U \colon G \to \mathrm{U}(\mathcal{H})$ with $[U(g)] = \bar{\pi}(g)$. No continuity is assumed. Since $[U(g)U(h)] = [U(gh)]$, there is a function $c \colon G \times G \to \mathbb{T}$ with

$$U(g)U(h) = c(g,h)\,U(gh). \tag{13.4}$$

Definition 13.12. A *multiplier* of G is a function $c \colon G \times G \to \mathbb{T}$ such that

$$c(g,h)\,c(gh,k) = c(h,k)\,c(g,hk) \qquad (g,h,k \in G).$$

Multipliers c, c' are *equivalent* if $c'(g,h) = c(g,h)\,b(g)b(h)/b(gh)$ for some function $b \colon G \to \mathbb{T}$. A multiplier is *trivial* if it is equivalent to 1.

Lemma 13.13. *The function c in (13.4) is a multiplier. Its equivalence class does not depend on the lift. It is trivial if and only if $\bar{\pi}$ has a lift that is a homomorphism.*

Proof. Expand $U(g)U(h)U(k)$ in two ways:

$$(U(g)U(h))U(k) = c(g,h)c(gh,k)U(ghk), \qquad U(g)(U(h)U(k)) = c(h,k)c(g,hk)U(ghk).$$

This gives the identity. Any other lift is bU with $b \colon G \to \mathbb{T}$, and its multiplier is $c(g,h)\,b(g)b(h)/b(gh)$. So the class is well defined. The lift bU is a homomorphism exactly when this multiplier equals 1. ∎

Example 13.14 (Spin one half)**.** Let $\bar{\pi}_1$ be the action of $\mathrm{SO}(3)$ on rays of $\mathbb{C}^2$ of Corollary 13.18 below. For each rotation g choose one of its two preimages $s(g) \in \mathrm{SU}(2)$, and put $U(g) = \Pi_1(s(g))$. Since $s(g)s(h) = \pm s(gh)$ and $\Pi_1(-I) = -\mathrm{id}$, the multiplier c takes the values ± 1.

No choice of s is continuous. A continuous choice along the loop $t \mapsto \rho(x_{ab}(t))$, $t \in [0, \pi]$, would be a closed lift of this loop. Lifts are unique given their start (Lecture 4), and Example 13.5 shows that the lift starting at 1 runs to -1. Multiplying by -1 shows that the lift starting at -1 runs to 1. So no lift of the loop is closed.

Definition 13.15. A *central extension* of G by $\mathbb{T}$ is a Lie group E with a closed central subgroup $Z \cong \mathbb{T}$ and an isomorphism of Lie groups $E/Z \cong G$.

Proposition 13.16. *Let $\bar\pi$ be a projective unitary representation of G on $\mathcal{H}$. Then*

$$E_{\bar\pi} = \{(g, U) \in G \times \mathrm{U}(\mathcal{H}) : [U] = \bar\pi(g)\}$$

is a central extension of G by $Z = \{(1, \lambda\,\mathrm{id}) : \lambda \in \mathbb{T}\}$. The map $(g, U) \mapsto U$ is a unitary representation of $E_{\bar\pi}$ in which $(1, \lambda\,\mathrm{id})$ acts by λ.

Proof. The set $E_{\bar\pi}$ is a subgroup of $G \times \mathrm{U}(\mathcal{H})$. It is the preimage of the diagonal under the continuous map $(g, U) \mapsto (\bar\pi(g), [U])$ into the Hausdorff space $\mathrm{PU}(\mathcal{H})^2$. So it is closed, hence a Lie group (Lecture 3). Clearly Z is closed and central. The projection to G is surjective with kernel Z. The projection defines a smooth transitive action of $E_{\bar\pi}$ on G, with stabiliser Z. The homogeneous-space theorem of Lecture 3 therefore identifies $E_{\bar\pi}/Z$ smoothly with G. ∎

Conversely, a unitary representation of a central extension E in which Z acts by $\lambda \mapsto \lambda\,\mathrm{id}$ defines a projective representation of E/Z. In this sense, projective representations of G are ordinary representations of central extensions of G.

For $\bar\pi_1$ of Example 13.14, the map $(g, U) \mapsto U$ is an isomorphism $E_{\bar\pi_1} \to \mathrm{U}(2)$, with $Z = \mathbb{T}\,\mathrm{id}$, because $\bar\pi_1$ is a bijection $\mathrm{SO}(3) \to \mathrm{PU}(\mathbb{C}^2)$. So $\mathrm{U}(2)$ is a central extension of $\mathrm{SO}(3)$ by $\mathbb{T}$, and spin one half is an ordinary representation of it.

13.5 Lifting projective representations

Theorem 13.17. *Let G be a connected Lie group with universal covering homomorphism $q\colon \tilde G \to G$ (Lecture 4). Let $\bar\pi$ be a projective unitary representation of G on a finite-dimensional $\mathcal{H}$. Then there is a unique continuous homomorphism $U\colon \tilde G \to \mathrm{SU}(\mathcal{H})$ with $[U(x)] = \bar\pi(q(x))$.*

Proof. Let $d = \dim \mathcal{H}$. Every $V \in \mathrm{U}(\mathcal{H})$ equals λV_0 with $\lambda^d = \det V$ and $V_0 \in \mathrm{SU}(\mathcal{H})$. So the restriction $p\colon \mathrm{SU}(\mathcal{H}) \to \mathrm{PU}(\mathcal{H})$ of the quotient map is surjective. Its kernel $\{\lambda\,\mathrm{id} : \lambda^d = 1\}$ is finite, and $\mathrm{SU}(\mathcal{H})$ is connected.

The differential of p has kernel $\mathfrak{su}(d) \cap i\mathbb{R}I = 0$. Both $\mathfrak{su}(d)$ and the tangent algebra of $\mathrm{U}(d)/\mathbb{T}I$ have dimension $d^2 - 1$, so dp is an isomorphism. By the covering criterion of Lecture 4, p is a covering homomorphism. For $d = 1$ both groups are trivial.

Since $\tilde G$ is simply connected, the based lifting theorem of Lecture 4 gives a unique continuous lift U with $U(e) = I$ and $p \circ U = \bar\pi \circ q$. It remains to check that U is multiplicative. The two maps $(x, y) \mapsto U(xy)$ and $(x, y) \mapsto U(x)U(y)$ lift the same map on the connected space $\tilde G \times \tilde G$ and agree at (e, e). Uniqueness of covering lifts makes them equal. Thus U is a homomorphism. Any homomorphic lift sends e to I, so uniqueness of the based lift also gives the stated uniqueness. ∎

Corollary 13.18. *Let $\mathcal{H}$ be finite-dimensional.*

(a) For each projective unitary representation $\bar{\pi}$ of $\mathrm{SO}(3)$ on $\mathcal{H}$, there is a unitary representation U of $\mathrm{SU}(2)$ with $\bar{\pi}(\rho(x)) = [U(x)]$. Moreover $U(-I) = \pm\,\mathrm{id}$.

(b) If $\bar{\pi}$ is irreducible, then U is unitarily equivalent to Π_m with $m = \dim \mathcal{H} - 1$.

(c) For each $m \geq 0$, the rule $\bar{\pi}_m(\rho(x)) = [\Pi_m(x)]$ defines a projective unitary representation of $\mathrm{SO}(3)$. Its multiplier is trivial if and only if m is even.

Proof. (a) The group $\mathrm{SU}(2) \cong S^3$ is simply connected (Lecture 4). Apply the lifting theorem, Theorem 13.17. Since $[U(-I)] = \bar{\pi}(\mathrm{id})$, $U(-I)$ is a scalar. Its square is $U(I) = \mathrm{id}$, so the scalar is ± 1.

(b) Scalars preserve every subspace, so U is irreducible. By the classification, $U \cong \Pi_m$ with $m + 1 = \dim \mathcal{H}$. By Schur's lemma the isomorphism becomes unitary after rescaling (Lecture 5).

(c) If $\rho(x) = \rho(x')$, then $x' = \pm x$ and $\Pi_m(x') = \pm \Pi_m(x)$. So $\bar{\pi}_m$ is well defined. It is continuous by the Descent Theorem of Lecture 4, with target $\mathrm{PU}(V_m)$. For even m, the descent criterion Proposition 13.7 gives a lift that is a homomorphism.

Let m be odd. Put $A = \rho(\Phi(\mathbf{i}))$ and $B = \rho(\Phi(\mathbf{j}))$. Since $\mathbf{i} = x_{23}(\pi/2)$ and $\mathbf{j} = x_{31}(\pi/2)$, the plane-rotation example, Example 13.5, gives

$$A = \mathrm{diag}(1, -1, -1), \qquad B = \mathrm{diag}(-1, 1, -1), \qquad AB = BA.$$

Put $P = \Pi_m(\Phi(\mathbf{i}))$ and $Q = \Pi_m(\Phi(\mathbf{j}))$. Since $\mathbf{ij} = -\mathbf{ji}$ and $\Pi_m(-I) = -\,\mathrm{id}$,

$$PQ = \Pi_m(-I)\, QP = -QP.$$

Suppose $W\colon\ \mathrm{SO}(3) \to \mathrm{U}(V_m)$ is a homomorphism lifting $\bar{\pi}_m$, with no continuity assumed. Then $W(A) = \alpha P$ and $W(B) = \beta Q$ with $\alpha, \beta \in \mathbb{T}$. So $W(A)W(B) = -W(B)W(A)$. But $AB = BA$ gives $W(A)W(B) = W(B)W(A)$. Hence $2W(A)W(B) = 0$, which is impossible for invertible operators. By the triviality criterion of Lemma 13.13, the multiplier is nontrivial. ∎

So every finite-dimensional irreducible projective representation of $\mathrm{SO}(3)$ is equivalent to some $\bar{\pi}_m$. Here equivalence means that a unitary $T : V_m \to \mathcal{H}$ satisfies

$$\bar{\pi}(\rho(x)) = [T \Pi_m(x) T^{-1}] \qquad (x \in \mathrm{SU}(2)).$$

All spins $m/2$ occur, but only integer spins come from genuine representations. This is the mathematical form of the half-integer spin of the electron.

For separable Hilbert spaces we cite the following result.

Theorem 13.19 (Bargmann; cited)**.** *Let G be a connected Lie group with semisimple Lie algebra $\mathfrak{g}$ and universal covering $q\colon \tilde{G} \to G$. Let $\mathcal{H}$ be a separable Hilbert space, and let $\bar{\pi}$ be a homomorphism from G to $\mathrm{U}(\mathcal{H})/\mathbb{T}\,\mathrm{id}$. Assume that $g \mapsto |\langle f, U(g)h\rangle|$ is continuous for all $f, h \in \mathcal{H}$ and one (hence every) lift U. Then there is a strongly continuous unitary representation $\tilde{U}$ of $\tilde{G}$ with $[\tilde{U}(x)] = \bar{\pi}(q(x))$ for $x \in \tilde{G}$.*

Example 13.20 ($\mathbb{R}^2$ and the Heisenberg extension)**.** On $G = \mathbb{R}^2$ put

$$\omega(x, y) = x_1 y_2 - x_2 y_1, \qquad c(x, y) = e^{i\omega(x,y)/2}.$$

The exponent is bilinear, so c is a multiplier. The factor $b(x)b(y)/b(x+y)$ is symmetric in x, y. So the ratio

$$c(x, y)/c(y, x) = e^{i\omega(x,y)}$$

depends only on the class of c. The ratio is $e^i \neq 1$ for $x = (1, 0)$ and $y = (0, 1)$, so c is nontrivial. Yet $\mathbb{R}^2$ is its own universal cover.

On $E = \mathbb{R}^2 \times \mathbb{T}$ put $(x, \lambda)(y, \mu) = (x+y, \lambda\mu\, c(x, y))$. The multiplier identity gives associativity, and Exercise 13.5 shows that E is a group. Its multiplication is smooth. Projection onto $\mathbb{R}^2$ identifies the quotient $E/(\{0\} \times \mathbb{T})$ with $\mathbb{R}^2$, so E is a central extension. The commutator of (x, λ) and (y, μ) in E is $(0, e^{i\omega(x,y)})$.

This does not contradict the lifting theorem, Theorem 13.17. A lift on $\mathbb{C}^d$ with multiplier equivalent to c would satisfy $U(x)U(y) = e^{i\omega(x,y)}U(y)U(x)$. Taking determinants gives $e^{id\,\omega(x,y)} = 1$ for all x, y, which is false. So c occurs only on infinite-dimensional spaces (Lecture 18). In physics, this is the group form of the fact that $\mathrm{tr}[x, p] = 0$ rules out $[x, p] = i\hbar$ on a finite-dimensional space. Bargmann's theorem does not apply to $\mathbb{R}^2$, because its Lie algebra is abelian rather than semisimple.

Self-check questions

1. In Spin(4), what is $\rho(e_1e_2e_3e_4)$?
   Answer. The element $e_1e_2e_3e_4$ equals $x_{12}(\pi/2)\, x_{34}(\pi/2)$. By the plane-rotation example, Example 13.5, its image under ρ rotates through π in both coordinate planes, so $\rho(e_1e_2e_3e_4) = -\mathrm{id}$. The other lift of $-\mathrm{id}$ is $-e_1e_2e_3e_4$.
2. For odd m, does $\Pi_m(u(2\pi))$ change any transition probability?
   Answer. No. It equals $-\mathrm{id}$. Hence it fixes every ray, and $|\langle -f, -h\rangle| = |\langle f, h\rangle|$.
3. Is there a projective unitary representation of $\mathbb{R}^2$ on $\mathbb{C}^3$ with multiplier $e^{i\omega/2}$?
   Answer. No. The determinant argument of the Heisenberg example, Example 13.20, with $d = 3$ gives $e^{3i\omega(x,y)} = 1$ for all x, y, which is false.

Summary

The group $\mathrm{Spin}(n)$ consists of even products of unit vectors in $Cl(n)$. For $n \geq 2$, conjugation maps it onto $\mathrm{SO}(n)$ with kernel $\{\pm 1\}$. For $n \geq 3$, the cited value of $\pi_1(\mathrm{SO}(n))$ shows that it is the universal cover. Spinor modules send -1 to $-\mathrm{id}$ and do not factor through ρ.

For $n = 3$, $\mathrm{Spin}(3) = \mathrm{Sp}(1) \cong \mathrm{SU}(2)$, and V_m descends to $\mathrm{SO}(3)$ exactly for even m. The lift $u(2\pi)$ of a full turn acts on V_m by $(-1)^m$, which fixes every ray.

In finite dimensions, projective unitary representations of connected Lie groups lift to unitary representations of the universal cover. On separable Hilbert spaces, Bargmann's cited theorem gives strongly continuous lifts for connected groups with semisimple Lie algebra, under its continuity hypothesis. The multiplier on the simply connected group $\mathbb{R}^2$ shows that semisimplicity cannot be dropped.

Reading

- BD, Section I.6: Clifford algebras and Spin(n). Note the sign convention for v^2.
- Sepanski, Section 1.3: the Clifford-algebra construction of the double cover of SO(n).
- Kirillov, Elements, §14: projective representations and their multipliers.
- Hall, Chapter 13: fundamental groups of compact Lie groups, including SO(n).
- V. Bargmann, On unitary ray representations of continuous groups, Ann. of Math. 59 (1954).

Lecture 14

Representative functions and tensor reconstruction

Objectives. *This lecture asks what information about a compact Lie group K is contained in all its representations. We collect all matrix coefficients into one algebra $\mathscr{R}(K)$. The group law becomes three maps on this algebra. We then state how K is recovered, and what is recovered when the unitary structure is forgotten.*

By the end of this lecture you will be able to:

- Prove that $\mathscr{R}(K)$ is a $*$-algebra whose products come from tensor products and whose conjugation comes from conjugate representations.
- Define Δ, ε, S and prove the Hopf algebra identities, also in matrix form.
- State Tannaka–Krein reconstruction with the unitary fibre functor, and contrast it with the complex reconstruction of $K_{\mathbb{C}}$.
- Compute these structures for $\mathrm{U}(1)$ and $\mathrm{SU}(2)$, and explain why $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category when K is infinite.

A space can be replaced by the algebra of functions on it. A group has more: its product appears as a map on these functions. This description still makes sense for algebras that are not commutative, and Lecture 28 returns to them. Nothing here depends on that.

Throughout, K is a compact Lie group. The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ has as objects the finite-dimensional continuous complex representations (Π_V, V). Its morphisms are the K-equivariant linear maps.

14.1 Matrix coefficients

Definition 14.1. Let V be an object of $\mathrm{Rep}_{\mathrm{fd}}(K)$, $\lambda \in V^*$ and $v \in V$. The *matrix coefficient* $c^V_{\lambda,v} \colon K \to \mathbb{C}$ is

$$c^V_{\lambda,v}(g) = \lambda\big(\Pi_V(g)v\big).$$

The space $\mathscr{R}(K)$ of *representative functions* is the linear span of all matrix coefficients of all objects V.

This is the coefficient space of Lecture 10, since every finite-dimensional continuous representation is unitarisable (Lecture 5).

Fix a basis (e_j) of V with dual basis (e^i). The matrix of $\Pi_V(g)$ in this basis is $D^V(g)$, with entries $D^V_{ij}(g) = e^i(\Pi_V(g)e_j)$. Expanding $\lambda = \sum_i \lambda(e_i)e^i$ and $v = \sum_j e^j(v)e_j$ gives

$$c^V_{\lambda,v} = \sum_{i,j} \lambda(e_i)\, e^j(v)\, D^V_{ij}.$$

So every coefficient of V is a linear combination of the D^V_{ij}. For a K-invariant inner product on V (Lecture 5) we may take $\lambda = \langle u, \cdot\rangle$, which gives $c^V_{\lambda,v}(g) = \langle u, \Pi_V(g)v\rangle$.

The *conjugate representation* $\overline{V}$ is the additive group V with scalar multiplication $t \cdot v = \bar{t}v$ and the same operators $\Pi_V(g)$. These operators are linear on $\overline{V}$:

$$\Pi_V(g)(t \cdot v) = \Pi_V(g)(\bar{t}v) = \bar{t}\,\Pi_V(g)v = t \cdot \Pi_V(g)v.$$

For $\lambda \in V^*$ put $\bar{\lambda}(v) = \overline{\lambda(v)}$. The functional $\bar{\lambda}$ is linear on $\overline{V}$:

$$\bar{\lambda}(t \cdot v) = \overline{\bar{t}\,\lambda(v)} = t\,\bar{\lambda}(v).$$

A basis (e_j) of V is also a basis of $\overline{V}$, with dual basis $(\overline{e^i})$. Hence

$$D^{\overline{V}}_{ij}(g) = \overline{e^i}\big(\Pi_V(g)e_j\big) = \overline{D^V_{ij}(g)}, \qquad \text{so} \qquad D^{\overline{V}}(g) = \overline{D^V(g)}.$$

Proposition 14.2. *Let $\lambda \in V^*$, $v \in V$, $\mu \in W^*$, $w \in W$ and $h \in K$. Then:*

(a) $c^V_{\lambda,v} + c^W_{\mu,w} = c^{V\oplus W}_{(\lambda,\mu),(v,w)}$;

(b) $c^V_{\lambda,v}\, c^W_{\mu,w} = c^{V\otimes W}_{\lambda\otimes\mu,\, v\otimes w}$;

(c) $\overline{c^V_{\lambda,v}} = c^{\overline{V}}_{\bar{\lambda},v}$;

(d) $g \mapsto c^V_{\lambda,v}(gh)$ *equals* $c^V_{\lambda,\Pi_V(h)v}$, *and* $g \mapsto c^V_{\lambda,v}(hg)$ *equals* $c^V_{\lambda\circ\Pi_V(h),v}$.

Hence $\mathscr{R}(K)$ is the set of matrix coefficients. It is a subalgebra of $C(K)$ containing the constants, closed under conjugation and translations.

Proof. Matrix coefficients are continuous because $g \mapsto \Pi_V(g)$ is continuous. The sum rule (a) holds because $\Pi_{V\oplus W}(g)$ acts componentwise. For the product rule (b), the tensor product representation gives

$$(\lambda \otimes \mu)\big(\Pi_V(g)v \otimes \Pi_W(g)w\big) = \lambda(\Pi_V(g)v)\,\mu(\Pi_W(g)w).$$

The conjugation rule (c) is the definition of $\bar{\lambda}$. The translation rule (d) uses $\Pi_V(gh) = \Pi_V(g)\Pi_V(h)$ and $\Pi_V(hg) = \Pi_V(h)\Pi_V(g)$.

Scalar multiples are coefficients, since $t\,c^V_{\lambda,v} = c^V_{t\lambda,v}$. By the sum rule (a), sums of coefficients are coefficients. So the span consists of coefficients. By the product, conjugation and translation rules (b)–(d), this set is closed under products, conjugation and translations. The constant 1 is $c^{\mathbb{C}}_{1,1}$ for the trivial representation. ∎

We use the Peter–Weyl theorem for compact Lie groups from Lecture 10. Choose one representative V for each class in $\mathrm{Irr}(K)$ and a basis of each. Then

$$\mathscr{R}(K) = \bigoplus_{[V]\in \mathrm{Irr}(K)} \operatorname{span}\{D^V_{ij}\},$$

an algebraic direct sum whose summand for V has dimension $(\dim V)^2$. Moreover $\mathscr{R}(K)$ is dense in $C(K)$ for the supremum norm and separates points.

14.2 The group law as structure maps

The group law is a map $K \times K \to K$. Pulling functions back along the group law produces functions of two variables. We first show that a tensor of functions is determined by the function of two variables that it defines.

Lemma 14.3. *Let A and B be linear spaces of functions on sets X and Y. The linear map from $A \otimes B$ to functions on $X \times Y$ with $f \otimes h \mapsto \big((x,y) \mapsto f(x)h(y)\big)$ is injective.*

Proof. Every tensor can be written $F = \sum_i f_i \otimes h_i$ with $h_1, \dots, h_n$ linearly independent. Suppose $\sum_i f_i(x)h_i(y) = 0$ for all x, y. Fix x. Then $\sum_i f_i(x)h_i$ is the zero function on Y. The numbers $f_i(x)$ are the coefficients of a vanishing combination of independent functions, so $f_i(x) = 0$ for all i. As x was arbitrary, each $f_i = 0$ and $F = 0$. ∎

Applying the lemma twice handles three factors. So we identify $\mathscr{R}(K)^{\otimes 2}$ and $\mathscr{R}(K)^{\otimes 3}$ with spaces of functions on K^2 and K^3. The identification is an algebra homomorphism for $(f \otimes h)(f' \otimes h') = ff' \otimes hh'$. Indeed both sides give the function $(x,y) \mapsto f(x)f'(x)h(y)h'(y)$.

Definition 14.4. A *Hopf algebra* is a unital algebra A with unital algebra homomorphisms $\Delta \colon A \to A \otimes A$, $\varepsilon \colon A \to \mathbb{C}$ and a linear map $S \colon A \to A$ such that

$$(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta, \qquad (\varepsilon \otimes \mathrm{id})\Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon)\Delta,$$
$$m(S \otimes \mathrm{id})\Delta = \eta\varepsilon = m(\mathrm{id} \otimes S)\Delta,$$

where m is multiplication and $\eta(t) = t1$. A *Hopf $*$-algebra* is a Hopf algebra with an antilinear map $*$ such that $f^{**} = f$, $(fh)^* = h^* f^*$, and Δ, ε preserve $*$. Here $(f \otimes h)^* = f^* \otimes h^*$ and $t^* = \bar{t}$ on $\mathbb{C}$.

For $f \in \mathscr{R}(K)$ and $g, h \in K$ define

$$\Delta f(g,h) = f(gh), \qquad \varepsilon f = f(e), \qquad Sf(g) = f(g^{-1}), \qquad f^*(g) = \overline{f(g)}.$$

For the coordinate function z on $\mathrm{U}(1)$ we have $\Delta z^n = z^n \otimes z^n$, since $(gh)^n = g^n h^n$. For a matrix entry we will find $\Delta D_{ij} = \sum_k D_{ik} \otimes D_{kj}$, which is matrix multiplication. In general, a priori Δf is only a function on $K \times K$. The first task is to show that it is a tensor in $\mathscr{R}(K)^{\otimes 2}$.

Theorem 14.5. *These maps make $\mathscr{R}(K)$ a commutative Hopf $*$-algebra, and $S(f^*) = (Sf)^*$. For every V and basis (e_j),*

$$\Delta D^V_{ij} = \sum_k D^V_{ik} \otimes D^V_{kj}, \qquad \varepsilon D^V_{ij} = \delta_{ij}, \qquad SD^V_{ij}(g) = \big(D^V(g)^{-1}\big)_{ij}. \tag{14.1}$$

For an orthonormal basis of a K-invariant inner product, $SD^V_{ij} = (D^V_{ji})^*$.

Proof. Codomains. Insert the expansion $v = \sum_k e^k(v)e_k$ for the vector $v = \Pi_V(h)e_j$. This gives

$$D^V_{ij}(gh) = e^i\big(\Pi_V(g)\Pi_V(h)e_j\big) = \sum_k e^i(\Pi_V(g)e_k)\, e^k(\Pi_V(h)e_j).$$

So ΔD^V_{ij} is the function of the tensor $\sum_k D^V_{ik} \otimes D^V_{kj}$. This tensor is unique by the tensor injectivity lemma, Lemma 14.3. The same insertion for a general coefficient gives

$$\Delta c^V_{\lambda,v} = \sum_k c^V_{\lambda,e_k} \otimes c^V_{e^k,v}. \tag{14.2}$$

Hence Δ maps $\mathscr{R}(K)$ into $\mathscr{R}(K)^{\otimes 2}$. Let $\Pi_{V^*}(g)\lambda = \lambda \circ \Pi_V(g)^{-1}$ be the dual representation. Let $\hat{e}_j \in V^{**}$ be evaluation at e_j. Then

$$SD^V_{ij}(g) = e^i\big(\Pi_V(g)^{-1}e_j\big) = \hat{e}_j\big(\Pi_{V^*}(g)e^i\big) = c^{V^*}_{\hat{e}_j,e^i}(g).$$

So S preserves $\mathscr{R}(K)$. Also $\varepsilon D^V_{ij} = e^i(e_j) = \delta_{ij}$, and $f^* \in \mathscr{R}(K)$ by the conjugation rule in Proposition 14.2.

Algebra maps. As functions, $\Delta(ff')(g,h) = f(gh)f'(gh) = (\Delta f\,\Delta f')(g,h)$. The identification is an injective algebra homomorphism, so $\Delta(ff') = \Delta f\,\Delta f'$. Also $\Delta 1 = 1 \otimes 1$. Pointwise products show that ε and S are unital algebra homomorphisms.

Identities. Write $\Delta f = \sum_i f_i \otimes f'_i$, so $\sum_i f_i(g)f'_i(h) = f(gh)$ for $g,h \in K$. As functions of $(g,h,k') \in K^3$,

$$(\Delta \otimes \mathrm{id})\Delta f\,(g,h,k') = \textstyle\sum_i f_i(gh)f'_i(k') = f\big((gh)k'\big),$$
$$(\mathrm{id} \otimes \Delta)\Delta f\,(g,h,k') = \textstyle\sum_i f_i(g)f'_i(hk') = f\big(g(hk')\big).$$

These agree by associativity, so the tensors agree by the tensor injectivity lemma, Lemma 14.3. For the counit, $(\varepsilon \otimes \mathrm{id})\Delta f = \sum_i f_i(e)f'_i$ is the function $h \mapsto f(eh) = f(h)$. For the antipode,

$$m(S \otimes \mathrm{id})\Delta f\,(g) = \sum_i f_i(g^{-1})f'_i(g) = f(g^{-1}g) = f(e).$$

The mirror identities use $he = h$ and $gg^{-1} = e$. Each group axiom is used once.

$*$-structure. Conjugation is antilinear and involutive, and $\mathscr{R}(K)$ is commutative. The function $\Delta(f^*)(g,h) = \overline{f(gh)}$ is that of $(\Delta f)^*$. Also $\varepsilon(f^*) = \overline{f(e)}$ and $S(f^*)(g) = \overline{f(g^{-1})}$. Finally, if $D^V(g)$ is unitary then $(D^V(g)^{-1})_{ij} = \overline{D^V_{ji}(g)}$. ∎

On matrix coefficients the axioms become matrix identities. We write $D = D^V$ and evaluate each side at group elements:

$$\begin{aligned}
\text{coassociativity:} &\quad \textstyle\sum_{k,l} D_{ik}(g)D_{kl}(h)D_{lj}(k') = \big(D(g)D(h)D(k')\big)_{ij},\\
\text{counit:} &\quad \textstyle\sum_k \delta_{ik}D_{kj}(g) = \big(D(e)D(g)\big)_{ij} = D_{ij}(g),\\
\text{antipode:} &\quad \textstyle\sum_k (SD_{ik})(g)\,D_{kj}(g) = \big(D(g)^{-1}D(g)\big)_{ij} = \delta_{ij}.
\end{aligned}$$

So the three identities encode $D(ghk') = D(g)D(h)D(k')$, $D(e) = 1$ and $D(g^{-1}) = D(g)^{-1}$. In an orthonormal basis the antipode identity reads

$$\sum_k (D_{ki})^* D_{kj} = \delta_{ij}. \tag{14.3}$$

In other words, the columns of $D(g)$ are orthonormal. Hence S and $*$ together encode the unitarity of $D(g)$. Exercise 14.7 checks these identities on the spin 1 representation of SU(2).

14.3 Recovering K

Definition 14.6. A *character* of $\mathscr{R}(K)$ is a unital algebra homomorphism $\psi\colon \mathscr{R}(K) \to \mathbb{C}$. It is a *$*$-character* if $\psi(f^*) = \overline{\psi(f)}$ for all f.

This algebraic use of the word differs from the trace character of a representation and from the characters $T \to \mathrm{U}(1)$ of a torus T (Lecture 7).

For $g \in K$ the evaluation $\mathrm{ev}_g(f) = f(g)$ is a $*$-character.

Proposition 14.7. *The characters form a group $\mathcal{X}(K)$ with*

$$\psi_1\psi_2 = (\psi_1 \otimes \psi_2) \circ \Delta, \qquad 1 = \varepsilon, \qquad \psi^{-1} = \psi \circ S.$$

The $*$-characters form a subgroup $\mathcal{X}_*(K)$, and $g \mapsto \mathrm{ev}_g$ is an injective homomorphism $K \to \mathcal{X}_*(K)$.

Proof. Since Δ, $\psi_1 \otimes \psi_2$ and S are algebra homomorphisms, $\psi_1\psi_2$ and $\psi \circ S$ are characters. Applying $\psi_1 \otimes \psi_2 \otimes \psi_3$ to coassociativity gives $(\psi_1\psi_2)\psi_3 = \psi_1(\psi_2\psi_3)$. The counit identity gives $\varepsilon\psi = \psi\varepsilon = \psi$. Multiplicativity means $\psi \circ m = m_{\mathbb{C}} \circ (\psi \otimes \psi)$, so

$$(\psi \circ S)\psi = \psi \circ m \circ (S \otimes \mathrm{id}) \circ \Delta = \psi \circ \eta \circ \varepsilon = \varepsilon.$$

The mirror identity gives $\psi(\psi \circ S) = \varepsilon$. The relations $\Delta(f^*) = (\Delta f)^*$ and $S(f^*) = (Sf)^*$ show that products and inverses of $*$-characters are $*$-characters. Finally $(\mathrm{ev}_g \otimes \mathrm{ev}_h)\Delta f = f(gh)$. Injectivity holds because $\mathscr{R}(K)$ separates points. ∎

In other words, Δ, ε and S give the multiplication, the identity and the inverses in $\mathcal{X}(K)$. To recover K we must know that every $*$-character is an evaluation. We now describe the same question in categorical terms.

An element $g \in K$ gives one matrix $\Pi_V(g)$ on each representation V. These matrices are compatible with every intertwiner and with tensor products, and they are unitary for invariant inner products. We now ask whether every family of unitary matrices with these properties comes from an element of K.

Definition 14.8. The *fibre functor* $\omega\colon \mathrm{Rep}_{\mathrm{fd}}(K) \to \mathrm{Vect}_{\mathbb{C}}$ sends each object to its underlying space and each morphism to itself. A *tensor automorphism* of ω is a family $\theta = (\theta_V)$ with $\theta_V \in \mathrm{GL}(V)$ for each object V, such that

$$\theta_W T = T\theta_V \ \ (T\colon V \to W \text{ a morphism}), \qquad \theta_{V\otimes W} = \theta_V \otimes \theta_W, \qquad \theta_{\mathbb{C}} = 1.$$

A tensor automorphism θ is *unitary* if θ_V is unitary for every K-invariant inner product on every V.

On $\mathrm{SU}(2)$, the decomposition $V_1 \otimes V_1 = V_2 \oplus V_0$ expresses θ_{V_2} through θ_{V_1}. So θ_{V_1} determines the family, as Example 14.12 shows.

Tensor automorphisms form a group $\mathrm{Aut}_{\otimes}(\omega)$ under $(\theta\theta')_V = \theta_V\theta'_V$, with identity (id_V) and inverse (θ_V^{-1}). Indeed, products of natural families are natural, and naturality $\theta_W T = T\theta_V$ gives $T\theta_V^{-1} = \theta_W^{-1}T$. Products and inverses also preserve the tensor and unit conditions.

Products and inverses of unitary operators are unitary, so the unitary tensor automorphisms form a subgroup $\mathrm{Aut}^{u}_{\otimes}(\omega)$.

The data of the unitary fibre functor are the following.

(i) The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ with its tensor product and unit object $\mathbb{C}$.
(ii) The functor ω, which remembers the underlying vector spaces and linear maps.
(iii) On each $\omega(V)$, the set of K-invariant inner products. On the algebra side these inner products correspond to the $*$-structure.

Each $g \in K$ gives the family with $\theta_V = \Pi_V(g)$. This family is natural because morphisms are equivariant. It is monoidal by the definition of $V \otimes W$. It is unitary because the inner products are K-invariant.

Tensor automorphisms and characters are related by the map $\theta \mapsto \psi_\theta$ given by

$$\psi_\theta\big(c^{V}_{\lambda,v}\big) = \lambda(\theta_V v). \tag{14.4}$$

Exercises 14.4 and 14.5 show that $\theta \mapsto \psi_\theta$ is a group isomorphism $\mathrm{Aut}_{\otimes}(\omega) \to \mathcal{X}(K)$. They also show that θ is unitary exactly when ψ_θ is a $*$-character. In any basis of V, the matrix of θ_V is $\big(\psi_\theta(D^{V}_{ij})\big)_{ij}$.

So a $*$-character ψ gives one operator θ_V on each V, determined by $\psi = \psi_\theta$. The operator θ_V is unitary for every K-invariant inner product on V. Hence its matrix in an orthonormal basis is unitary, while its matrix in a general basis need not be. The hard step of reconstruction is to find a single $g \in K$ whose operators $\Pi_V(g)$ equal the operators θ_V for all V at once. The $\mathrm{SU}(2)$ model, Example 14.12, finds g directly. In general we cite this step.

Theorem 14.9 (Tannaka–Krein duality; cited from BD III.7). *Let K be a compact Lie group. Every $*$-character of $\mathcal{R}(K)$ is ev_g for a unique $g \in K$. Equivalently, $g \mapsto (\Pi_V(g))_V$ is a group isomorphism $K \to \mathrm{Aut}^{u}_{\otimes}(\omega)$.*

The cases to keep in mind for the next theorem are $\mathrm{U}(1) \subset \mathbb{C}^{\times}$ and $\mathrm{SU}(2) \subset \mathrm{SL}(2,\mathbb{C})$, computed in the next section.

Theorem 14.10 (Complexification; cited from BD III.8). *Let K be a compact Lie group and put $K_{\mathbb{C}} = \mathcal{X}(K)$, with the topology of pointwise convergence on $\mathcal{R}(K)$. Then $K_{\mathbb{C}}$ carries a complex manifold structure that makes it a complex Lie group with Lie algebra $\mathfrak{k}_{\mathbb{C}} = \mathfrak{k} \otimes_{\mathbb{R}} \mathbb{C}$. The map $g \mapsto \mathrm{ev}_g$ identifies K with a maximal compact subgroup of $K_{\mathbb{C}}$. Every object of $\mathrm{Rep}_{\mathrm{fd}}(K)$ extends uniquely to a holomorphic representation of $K_{\mathbb{C}}$. In algebraic language, $K_{\mathbb{C}}$ is a complex algebraic group with reductive identity component and coordinate ring $\mathcal{R}(K)$. For connected K, it is a connected complex reductive group.*

A map between complex manifolds is *holomorphic* if it is complex differentiable in charts. A *maximal compact subgroup* is a compact subgroup contained in no larger compact subgroup.

Thus plain complex tensor automorphisms recover $\mathrm{Aut}_{\otimes}(\omega) \cong K_{\mathbb{C}}$, not K. The unitary condition, equivalently the $*$-structure, cuts $K_{\mathbb{C}}$ down to K.

14.4 Two models

Example 14.11 (U(1)). The irreducible representations of U(1) are the lines $\mathbb{C}_n$, $n \in \mathbb{Z}$, on which z acts by z^n (characters of tori, Lecture 7). The functions z^n are orthonormal for Haar measure (Lecture 5), hence independent. So

$$\mathscr{R}(\mathrm{U}(1)) = \mathbb{C}[z, z^{-1}], \qquad \Delta z = z \otimes z, \quad \varepsilon z = 1, \quad Sz = z^{-1}, \quad z^* = z^{-1}.$$

A character ψ is determined by $w = \psi(z)$. Since $\psi(z)\psi(z^{-1}) = 1$, we have $w \neq 0$. Every $w \in \mathbb{C}^\times$ occurs, via $z^n \mapsto w^n$. The $*$-condition reads $\bar{w} = \psi(z^*) = w^{-1}$, that is $|w| = 1$. Thus $\mathcal{X} = \mathbb{C}^\times$ and $\mathcal{X}_* = \mathrm{U}(1)$.

The categorical computation agrees. Let $\theta \in \mathrm{Aut}_\otimes(\omega)$, and let w_n be the scalar $\theta_{\mathbb{C}_n}$. Every V is the sum of its weight spaces

$$V(n) = \{v \in V : \Pi_V(z)v = z^n v \text{ for all } z \in \mathrm{U}(1)\}.$$

For $v \in V(n)$, the map $T\colon \mathbb{C}_n \to V$, $1 \mapsto v$, is a morphism, so

$$\theta_V v = \theta_V T 1 = T\theta_{\mathbb{C}_n} 1 = w_n v.$$

The isomorphism $\mathbb{C}_m \otimes \mathbb{C}_n \to \mathbb{C}_{m+n}$, $1 \otimes 1 \mapsto 1$, gives $w_{m+n} = w_m w_n$, and $w_0 = 1$. Hence θ_V acts by w^n on $V(n)$, with $w = w_1 \in \mathbb{C}^\times$. Weight spaces are orthogonal for every invariant inner product, so θ is unitary exactly when $|w| = 1$. Conversely, each $w \in \mathbb{C}^\times$ occurs, through the character $z^n \mapsto w^n$ and Exercise 14.5. So $\mathrm{Aut}_\otimes(\omega) \cong \mathbb{C}^\times = \mathrm{U}(1)_\mathbb{C}$ and $\mathrm{Aut}^u_\otimes(\omega) \cong \mathrm{U}(1)$.

Example 14.12 (SU(2)). Define functions a, b, c, d on SU(2) by

$$g = \begin{pmatrix} a(g) & b(g) \\ c(g) & d(g) \end{pmatrix} = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Here a is the $(1,1)$ entry, as before, but b is now the $(1,2)$ entry. Earlier lectures used b for the $(2,1)$ entry. These functions are the coefficients D_{ij} of $\mathbb{C}^2$ in the standard basis, and

$$ad - bc = 1, \qquad a^* = d, \qquad b^* = -c.$$

Since $\det g = 1$, the inverse matrix is

$$g^{-1} = \begin{pmatrix} d(g) & -b(g) \\ -c(g) & a(g) \end{pmatrix}.$$

So the matrix formulas (14.1) give

$$\begin{aligned} \Delta a &= a \otimes a + b \otimes c, & \Delta b &= a \otimes b + b \otimes d, & Sa &= d, & Sb &= -b, \\ \Delta c &= c \otimes a + d \otimes c, & \Delta d &= c \otimes b + d \otimes d, & Sc &= -c, & Sd &= a. \end{aligned}$$

Also ε sends a, d to 1 and b, c to 0. One checks $SD_{ij} = D^*_{ji}$; for example $Sb = -b = c^*$.

The functions a, b, c, d generate $\mathscr{R}(\mathrm{SU}(2))$ as an algebra. Indeed every irreducible representation is $V_m \cong \mathrm{Sym}^m(\mathbb{C}^2)$ (Lecture 8). In it g sends $e_1^p e_2^{m-p}$ to $(ae_1 + ce_2)^p (be_1 + de_2)^{m-p}$. So each $D_{ij}^{V_m}$ is a polynomial of degree m in a, b, c, d. By the Peter–Weyl theorem these coefficients span $\mathscr{R}(\mathrm{SU}(2))$, which proves the claim.

Let ψ be a $*$-character. Collect its values on the generators in the matrix

$$A = \begin{pmatrix} \psi(a) & \psi(b) \\ \psi(c) & \psi(d) \end{pmatrix}.$$

The relations give

$$\psi(d) = \overline{\psi(a)}, \qquad \psi(c) = -\overline{\psi(b)}, \qquad |\psi(a)|^2 + |\psi(b)|^2 = \psi(ad - bc) = 1.$$

So $A \in \mathrm{SU}(2)$. The algebra maps ψ and ev_A agree on generators, so $\psi = \mathrm{ev}_A$. This verifies Tannaka–Krein duality, Theorem 14.9, for $\mathrm{SU}(2)$ directly.

Without the $*$-condition we only learn $\det A = 1$. Conversely every $A \in \mathrm{SL}(2,\mathbb{C})$ gives a character $\psi_A \colon p(a,b,c,d) \mapsto p(A)$ (Exercise 14.3). So $\mathcal{X}(\mathrm{SU}(2)) = \mathrm{SL}(2,\mathbb{C}) = \mathrm{SU}(2)_{\mathbb{C}}$. For $A = \mathrm{diag}(2, \frac{1}{2})$ we get $\psi_A(a) = 2$ but $\psi_A(a^*) = \psi_A(d) = \frac{1}{2} \neq \bar{2}$. The corresponding tensor automorphism has $\theta_{\mathbb{C}^2} = A$, which is not unitary. In physics, this matrix $\mathrm{diag}(2, \frac{1}{2})$ is a boost in the group $\mathrm{SL}(2,\mathbb{C})$, which covers $\mathrm{SO}^+(1,3)$ (Lecture 24).

14.5 The representation category

The Category Theory course studied fusion categories, such as $\mathrm{SU}(2)_k$, with finitely many simple objects. The Clebsch–Gordan rule has the same form. We show that an infinite compact Lie group falls outside this class, and that finiteness is the only axiom that fails.

We use the notions of a rigid and of a symmetric monoidal category from the prerequisite course. Rigidity requires duals, with evaluation and coevaluation maps that satisfy the zigzag identities. A symmetry is a natural isomorphism $\tau_{V,W} \colon V \otimes W \to W \otimes V$ with $\tau_{W,V}\tau_{V,W} = \mathrm{id}$, compatible with associativity.

Definition 14.13. A *fusion category* is a $\mathbb{C}$-linear semisimple rigid monoidal category with finite-dimensional morphism spaces, simple unit object, and finitely many isomorphism classes of simple objects.

The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ is semisimple (Lecture 5) with simple unit $\mathbb{C}$ and finite-dimensional morphism spaces. It is rigid, with the dual representation and

$$\mathrm{ev}_V(\lambda \otimes v) = \lambda(v), \qquad \mathrm{coev}_V(1) = \sum_i e_i \otimes e^i.$$

It is symmetric with $\tau(v \otimes w) = w \otimes v$. Exercise 14.6 checks these claims.

Proposition 14.14. *If K is infinite, then $\mathrm{Irr}(K)$ is infinite and $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category. In particular, both conclusions hold when K is connected and nontrivial.*

Proof. We first show that $\mathscr{R}(K)$ is infinite-dimensional. Let $x_1, \dots, x_n \in K$ be distinct. Since $\mathscr{R}(K)$ separates points, for $i \neq j$ we can choose $f_{ij} \in \mathscr{R}(K)$ with $f_{ij}(x_i) \neq f_{ij}(x_j)$. Then

$$h_i = \prod_{j \neq i} \frac{f_{ij} - f_{ij}(x_j)}{f_{ij}(x_i) - f_{ij}(x_j)} \in \mathscr{R}(K).$$

At x_i every factor equals 1. At x_k with $k \neq i$ the factor $j = k$ vanishes. So $h_i(x_k) = \delta_{ik}$. A relation $\sum_i t_i h_i = 0$ evaluated at x_k gives $t_k = 0$. So $h_1, \dots, h_n$ are independent and $\dim \mathscr{R}(K) \geq n$ for every n. If $\mathrm{Irr}(K)$ were finite, Peter–Weyl would give

$$\dim \mathscr{R}(K) = \sum_{[V] \in \mathrm{Irr}(K)} (\dim V)^2 < \infty.$$

So $\mathrm{Irr}(K)$ is infinite. The simple objects of $\mathrm{Rep}_{\mathrm{fd}}(K)$ are the irreducible representations, and a fusion category has only finitely many isomorphism classes of simple objects. So $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category.

Finally, a finite Hausdorff space is discrete. Hence a connected Hausdorff space with at least two points is infinite, and the first part applies to connected nontrivial K. ∎

For a finite group the representation category is a fusion category (Exercise 14.2).

The algebra $\mathscr{R}(K)$ is commutative. Its coproduct is cocommutative only for abelian K. Indeed cocommutativity means

$$f(gh) = f(hg) \quad \text{for all } f \in \mathscr{R}(K),\ g, h \in K.$$

Since $\mathscr{R}(K)$ separates points, this forces $gh = hg$ for all $g, h \in K$.

A later course on quantum groups deforms this classical structure into noncommutative Hopf $*$-algebras. Lecture 28 returns to this comparison. Nothing in this lecture depends on that course.

Self-checks

1. In $\mathscr{R}(\mathrm{U}(1))$ let $f(z) = \mathrm{Re}\, z$. Compute Δf, εf and Sf.
   Answer. $f = \frac{1}{2}(z + z^{-1})$, so $\Delta f = \frac{1}{2}(z \otimes z + z^{-1} \otimes z^{-1})$, $\varepsilon f = 1$ and $Sf = f$. Note $\Delta f \neq f \otimes f$.
2. On $\mathrm{SU}(2)$ write $g \mapsto |g_{11}|^2$ in terms of a, b, c, d. Evaluate ψ_A on it for $A = \mathrm{diag}(2, \frac{1}{2})$.
   Answer. $|\alpha|^2 = aa^* = ad$, and $\psi_A(ad) = 2 \cdot \frac{1}{2} = 1$, while $|A_{11}|^2 = 4$. A non-$*$ character does not evaluate absolute values.
3. For $\mathrm{U}(1)$ let $\theta \in \mathrm{Aut}_{\otimes}(\omega)$ with $\theta_{\mathbb{C}_1} = 3$. Find $\theta_{\mathbb{C}_{-2}}$ and $\theta_{\mathbb{C}_1 \oplus \mathbb{C}_{-1}}$. Is θ unitary?
   Answer. $\theta_{\mathbb{C}_{-2}} = \frac{1}{9}$ and $\theta_{\mathbb{C}_1 \oplus \mathbb{C}_{-1}} = \mathrm{diag}(3, \frac{1}{3})$. It is not unitary, since $|3| \neq 1$.

Summary

Matrix coefficients of all finite-dimensional continuous representations of the compact Lie group K form the commutative $*$-algebra $\mathscr{R}(K)$. Tensor products give its products and conjugate representations give its involution. The maps $\Delta f(g, h) = f(gh)$, $\varepsilon f = f(e)$ and $Sf(g) = f(g^{-1})$ make it a Hopf $*$-algebra. On coefficients they are $D(gh) = D(g)D(h)$, $D(e) = 1$ and $D(g^{-1}) = D(g)^{-1}$. The $*$-characters, or unitary tensor automorphisms of ω, recover K. All characters, or all tensor automorphisms, give $K_{\mathbb{C}}$. The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ is rigid and symmetric. When K is infinite, it has infinitely many simple objects and is not a fusion category.

Part I in review

Part I asked what a group determines that its Lie algebra does not. A finite-dimensional representation of a Lie algebra integrates to the simply connected group, and it descends to a quotient exactly when the central kernel acts trivially (Lecture 4). This step does not use compactness. For a compact connected group the test reads from the lattice of a maximal torus, and the irreducible representations are labelled by dominant characters (Lecture 7 and Lecture 8). So SU(2) has every V_m, and SO(3) has only those with m even. The lift of a full spatial turn acts on V_m by $(-1)^m$. This sign fixes every ray, so Lecture 13 passes to projective representations.

Compactness gave the rest of Part I. The Haar integral has finite total mass. Averaging gives invariant inner products, complete reducibility and Schur orthogonality (Lecture 5). Averaging a rank-one projection gives a nonzero compact operator, and the Peter–Weyl theorem follows (Lecture 10). So every irreducible unitary representation is finite-dimensional, and $L^2(K)$ is a Hilbert sum of blocks of matrix coefficients. Lecture 11 and Lecture 12 read this as Fourier analysis and as a count of multiplicities on K/H. This lecture recovers K from its coefficient algebra, by a cited theorem.

A noncompact group has no normalised finite Haar integral, so these averaging arguments are no longer available in general. Part II starts from that point.

Reading

BD denotes Bröcker–tom Dieck, *Representations of Compact Lie Groups*.

BD III.1 for representative functions and their algebra structure. BD III.7 for Tannaka–Krein duality. BD III.8 for the complexification of compact Lie groups. The tensor-category terms are from the prerequisite course.